Appendix A: Connections of the component level

Table A-1: Connections of the solid slab in the 9.0 m model length

Solid slab	CCS	DCS	Remarks				
Connection system	–	46.28kg adjust to 47kg	System	Component	Specification	Unit weight (kg)	Total weight (kg)
			Lifting system	4 Lifting loop	B300B Φ25 L=500mm	1.93	7.72
				8 Anchor bar for lifting loop	B400B Φ12 L=400mm	0.36	2.88
				4 M12 sleeve with nut	M12 L=100mm	0.12	0.48
			Slab- to- beam connection	8 M12 sleeve with nut	M12 L=100mm	0.12	0.80
				8 Embedded plate	S235JR 150×150×10mm	1.77	14.16
				16 Anchor bar	B400B Φ14 L=450mm	0.54	8.64
			Slab- to- slab connection	12 M12 sleeve with nut	M12 L=100mm	0.12	1.20
				6 Embedded plate	S235JR 100×100×10mm	0.78	4.68
				12 Anchor bar	B400B Φ10 L=150mm	0.09	1.08
			Local reinforcement	8 U- shaped bar at lifting point	B400B Φ12 L=600mm	0.53	4.24
End solid zone reinforcement	-	11.72kg adjust to 12kg	Component		Specification	Unit weight (kg)	Total weight (kg)
			4 Lifting point mesh		B400B Φ10@150 Single layer	1.15	4.60
			8 Lifting point U-bar		B300B Φ8 L=800mm	0.32	2.56
			4 Lifting point transverse bar		B400B Φ10 L=600mm	0.37	1.48
			8 End longitudinal anti-crack bar		B400B Φ6 L=600mm	0.13	1.04
			6 Additional tie bars		B400B Φ8 L=400mm	0.16	0.96
			4 End stirrup (new)		B400B Φ6 L=1200mm	0.27	1.08

Table A-2: Connections of the hollow core slab in the 9.0 m model length

Hollow core slab	CCS	DCS	Remarks				
Connection system	–	31.00kg adjust to 31kg	System	Component	Specification	Unit weight (kg)	Total weight (kg)
			Lifting system	4 Lifting loop	B300B Φ25 L=500mm	1.93	7.72
				8 Anchor bar for lifting loop	B400B Φ12 L=400mm	0.36	2.88
				4 M12 sleeve with nut	M12 L=100mm	0.12	0.48
			Slab-to-beam connection	8 M12 sleeve with nut	M12 L=100mm	0.12	0.96
				8 Embedded plate	S235JR 100×100×10mm	0.78	6.24
				16 Anchor bar	B400B Φ12 L=350mm	0.31	4.96
			Slab-to-slab connection	8 M12 sleeve with nut	M12 L=80mm	0.10	0.80
				4 Embedded plate	S235JR 80×80×10mm	0.50	2.00
				8 Anchor bar	B400B Φ10 L=150mm	0.09	0.72
			Local reinforcement	8 U-shaped bar at lifting point	B400B Φ12 L=600mm	0.53	4.24
End solid zone reinforcement	-	34.04kg adjust to 34kg	Component		Specification	Unit weight (kg)	Total weight (kg)
			2 End zone mesh (each end)		B400B Φ10@150 Single layer	14.80	29.60
			6 End U-bars		B300B Φ10 L=1200mm	0.74	4.44

Table A-3: Connections of the ribbed slab in the 9.0 m model length

Ribbed slab	CCS	DCS	Remarks				
Connection system	–	21.00kg adjust to 21kg	System	Component	Specification	Unit weight (kg)	Total weight (kg)
			Lifting system	4 Lifting loop	B300B Φ20 L=500mm	1.23	4.92
				8 Anchor bar for lifting loop	B400B Φ12 L=400mm	0.36	2.88
				4 M12 sleeve with nut	M12 L=100mm	0.12	0.48
			Slab- to- beam connection	4 M12 sleeve with nut	M12 L=100mm	0.12	0.48
				4 Embedded plate	S235JR 80×80×10mm	0.50	2.00
				8 Anchor bar	B400B Φ12 L=350mm	0.31	2.48
			Slab- to- slab connection	8 M12 sleeve with nut	M12 L=80mm	0.10	0.80
				4 Embedded plate	S235JR 80×80×10mm	0.50	2.00
				8 Anchor bar	B400B Φ10 L=150mm	0.09	0.72
			Local reinforcement	8 U- shaped bar at lifting point	B400B Φ12 L=600mm	0.53	4.24
End solid zone reinforcement	-	4.02kg adjust to 4kg	Component		Specification	Unit weight (kg)	Total weight (kg)
			6 Rib end U-bars		B400B Φ10 L=700mm	0.43	2.58
			4 Rib end longitudinal short bars		B400B Φ12 L=400mm	0.36	1.44

Table A-4: Connections of the double T slab in the 9.0 m model length

Double T slab	CCS	DCS	Remarks				
Connection system	–	30.36kg adjust to 31kg	System	Component	Specification	Unit weight (kg)	Total weight (kg)
			Lifting system	4 Lifting loop	B300B Φ20 L=500mm	1.23	4.92
				8 Anchor bar for lifting loop	B400B Φ12 L=400mm	0.36	2.88
				4 M12 sleeve with nut	M12 L=100mm	0.12	0.48
			Slab-to-beam connection	8 M12 sleeve with nut	M12 L=100mm	0.12	0.96
				4 Embedded plate	S235JR 200×200×10mm	3.14	12.56
				8 Anchor bar	B400B Φ14 L=450mm	0.54	4.32
			Local reinforcement	8 U-shaped bar at lifting point	B400B Φ12 L=600mm	0.53	4.24
End solid zone reinforcement	-	8.52kg adjust to 9kg	Component		Specification	Unit weight (kg)	Total weight (kg)
			4 Web end U-bars		B400B Φ12 L=600mm	0.53	2.12
			8 Web end anchor bars		B400B Φ12 L=400mm	0.36	2.88
			4 End transverse bars (flange)		B400B Φ12 L=1000mm	0.88	3.52

The internal threaded sleeves and anchorage bars embedded at the ends of detachable floor slabs shall satisfy the load-bearing requirements under transient conditions, including hoisting, transportation, and disassembly. This section conducts load-bearing capacity checks on the embedded components for the four slab types in accordance with EN 1992-4 and EN 1991-1-6/EN 1991-3, encompassing steel tensile failure, concrete local compressive failure, and concrete conical failure.

(1) The basis for determining external loads during disassembly shall comply with Appendix A of EN 1991-1-6 and Appendix A of EN 1991-3. For prefabricated components, the equivalent static load shall be calculated by multiplying the self-weight of the component by the dynamic coefficient ϕ during hoisting, installation, and disassembly operations. For disassembly operations in which mechanical jacks are used for slow lifting, the dynamic coefficient ϕ shall be taken as 1.10. The standard self-weight value G_k of the floor slab shall be calculated by multiplying the actual concrete volume of each slab type by the unit weight of reinforced concrete, 25 kN/m³. Considering the most unfavorable four-point hoisting condition, the design tensile force acting on

each sleeve shall be calculated according to the following formula. In the formula, n denotes the number of sleeves at one end that are subjected to force (taken as the total number of embedded sleeves at that end). The calculation results for various plate profiles are provided in Table A-5.

$$F_{\mathrm{Ed}}=\frac{\phi G_{\mathrm{k}}}{n}$$

(2) The ultimate load capacity of the steel sleeve shall be calculated in accordance with clause 7.2.2 of EN 1992-4.

$$N_{\mathrm{Rd,s}}=\frac{A_{\mathrm{s}}\times f_{\mathrm{yb}}}{\gamma_{\mathrm{Ms}}}$$

In the formula, A_{s} denotes the effective cross-sectional area of the sleeve (for M12, A_{s} = 84.3 mm^2); f_{yb} denotes the yield strength of the steel material (for Grade 10.9 high-strength steel, f_{yb} = 900 MPa); and γ_{Ms} = 1.15. The calculated value of $N_{\mathrm{Rd,s}}$ is: $N_{\mathrm{Rd,s}}=\frac{84.3\times900}{1.15\times1000}$=66.0kN.

(3) The end of the embedded sleeve for the local bearing capacity of concrete is equipped with an embedded steel plate, which resists the pressure resulting from bolt preloading and external loads. According to clause 6.7 of EN 1992-1-1, the lower design value for the local bearing capacity of concrete (ignoring load redistribution and adopting a safety factor) is as shown in:

$$F_{\mathrm{Rd,min}}=A_{\mathrm{c0}}\times f_{\mathrm{cd}}$$

$$f_{\mathrm{cd}}=\frac{\alpha_{\mathrm{cc}}\times f_{\mathrm{ck}}}{\gamma_{\mathrm{c}}}$$

In the formula, A_{c0} denotes the area of the embedded steel plate (see Table A-5), with α_{cc}=0.85 and γ_{c}=1.5 . For solid, hollow, and ribbed steel slabs, the concrete grade is C25/30, with f_{ck}=30MP and f_{cd}=17.0MPa ; for double T steel slabs, the concrete grade is C40/45, with f_{ck}=40MP and f_{cd}=22.67MPa. Additionally, the maximum compressive force is taken as the sum of the bolt pre-tension force and the external load. For the bolt pre-tension force, the values are 100 kN for solid slabs, 70 kN for hollow core slabs, 70 kN for ribbed slabs, and 155 kN for double T slabs. Consequently, the maximum compressive forces are calculated as follows: 100+25.4=125.4kN for solid slabs; 70+23.0=93.0kN for hollow core slabs; 70+57.5=127.5kN for ribbed slabs; and 155+13.0=168.0kN for double T slabs. All these values are significantly smaller than $F_{\mathrm{Rd,min}}$, and thus satisfy the local bearing requirements.

(4) When no reinforcement is provided for the concrete cone failure, the characteristic ultimate bearing capacity of a single anchor bolt for concrete cone failure shall be calculated in accordance with clause 7.2.1.3 of EN 1992-4.

$$N^{0}_{\mathrm{Rk,\,c}}=8.5\times(h_{\mathrm{ef}})^{1.5}\times\sqrt{f_{\mathrm{ck}}}$$

For M12 bars, the anchorage depth h_{ef}=8d=96mm is adopted. For C25/30 concrete, f_{ck}=30MPa is taken, with $N^{0}_{\mathrm{Rk,\,c}}$=1.94kN ; for C40/45 concrete, f_{ck}=50MPa is taken, with $N^{0}_{\mathrm{Rk,\,c}}$=2.50kN . However, all plate types are equipped with U-shaped stirrups and spiral reinforcement around the sleeve. According to Chapter 8 of EN 1992-4, when an adequate number of additional

reinforcement bars are provided, concrete cone failure can be effectively suppressed, and the failure mode transitions to yielding of the reinforcement or concrete splitting, which exhibits superior ductility. Therefore, in this design, only construction measures are considered to ensure that cone failure is not governing; quantitative verification calculations are not performed.

(5) The verification parameters and results for each plate type are presented in Table 3-Y. For all slab types, the steel safety factor exceeds 1.00, thereby complying with the requirements of EN 1992-4. The lower limit of local concrete compressive stress for all cases is significantly greater than the maximum compressive stress, ensuring adequate safety margins. Given the reinforcement effect, concrete cone failure is deemed to comply with the specified requirements.

Table A-5 Load-capacity verification for embedded parts

Slab Type	Sleeve Size	Concrete Grade	Embedded Plate Area (mm^2)	G_k (kN)	F_{Ed} Per Sleeve (kN)	$N_{Rd,s}$ (kN)	Steel Safety Factor	$F_{Rd,min}$ (kN)	Maximum Press (kN)	Concrete cone failure	Result
Solid Slab	M12	C25/30	150×150 =22500	92.3	25.4	66	2.6	382.5	125.4	Suppressed by reinforcement	Pass
Hollow Core Slab	M12	C25/30	100×100 =10000	83.6	23.0	66	2.87	170.0	93.0	Suppressed by reinforcement	Pass
Ribbed Slab	M12	C25/30	80×80 =6400	104.6	57.5	66	1.15	108.8	127.5	Suppressed by reinforcement	Pass
Double T Slab	M12	C40/45	200×200 =40000	47.3	13.0	66	5.08	907.0	168.0	Suppressed by reinforcement	Pass

Appendix B: Specific material lists for slabs of the component level

Table B-1 Specific material lists of the solid slab

Model Length (m)	Design	Number of Beam	Slab Span (m)	Type of Slab	Thickness of Slab (mm)	Concrete Base Quantity (m^3)	Concrete Increment (m^3)	Total Concrete (m^3)	Steel Base Quantity (kg)	Steel Increment (kg)	Total Steel (kg)
9.0	CCS	1	4.50	Two-way	120	3.690	-	3.690	128.832	-	128.832
	DCS	1	4.50	Two-way	130	3.690	0.473	4.163	128.832	59.000	187.832
10.5	CCS	2	3.50	Two-way	120	4.830	-	4.830	217.219	-	217.219
	DCS	2	3.50	Two-way	130	4.830	0.495	5.325	217.219	68.833	286.052
12.0	CCS	2	4.00	Two-way	120	5.520	-	5.520	247.010	-	247.010
	DCS	2	4.00	Two-way	130	5.520	0.517	6.038	247.010	78.667	325.676
13.5	CCS	2	4.50	Two-way	120	6.210	-	6.210	278.774	-	278.774
	DCS	2	4.50	Two-way	130	6.210	0.540	6.750	278.774	88.500	367.274
15.0	CCS	2	5.00	Two-way	120	6.900	-	6.900	308.565	-	308.565
	DCS	2	5.00	Two-way	130	6.900	0.563	7.463	308.565	98.333	406.898
16.5	CCS	3	4.12	Two-way	120	8.415	-	8.415	461.219	-	461.219
	DCS	3	4.12	Two-way	130	8.415	0.585	9.000	461.219	108.167	569.386
18.0	CCS	3	4.50	Two-way	120	9.180	-	9.180	502.503	-	502.503
	DCS	3	4.50	Two-way	130	9.180	0.607	9.787	502.503	118.000	620.503

Table B-2 Specific material lists of the hollow core slab

Model Length (m)	Design	Number of Beam	Slab Span (m)	Type of Slab	Thickness of Slab (mm)	Concrete Base Quantity (m^3)	Concrete Increment (m^3)	Total Concrete (m^3)	Steel Base Quantity (kg)	Steel Increment (kg)	Total Steel (kg)
9.0	CCS	2	3.00	Two-way	120	3.345	-	3.345	185.454	-	185.454
	DCS	2	3.00	Two-way	120	3.345	0.778	4.122	185.454	65.000	250.454
10.5	CCS	2	3.50	Two-way	120	3.902	-	3.902	217.219	-	217.219
	DCS	2	3.50	Two-way	120	3.902	0.778	4.680	217.219	75.833	293.052
12.0	CCS	2	4.00	Two-way	120	4.460	-	4.460	247.010	-	247.010
	DCS	2	4.00	Two-way	120	4.460	0.778	5.237	247.010	86.667	333.676
13.5	CCS	3	3.38	Two-way	120	5.692	-	5.692	377.469	-	377.469
	DCS	3	3.38	Two-way	120	5.692	0.778	6.470	377.469	97.500	474.969
15.0	CCS	3	3.75	Two-way	120	6.325	-	6.325	418.752	-	418.752
	DCS	3	3.75	Two-way	120	6.325	0.778	7.102	418.752	108.333	527.086
16.5	CCS	3	4.12	Two-way	120	6.957	-	6.957	461.219	-	461.219
	DCS	3	4.12	Two-way	120	6.957	0.778	7.735	461.219	119.167	580.386
18.0	CCS	4	3.60	Two-way	120	8.490	-	8.490	640.804	-	640.804
	DCS	4	3.60	Two-way	120	8.490	0.778	9.267	640.804	130.000	770.804

Table B-3 Specific material lists of the ribbed slab

Model Length (m)	Design	Number of Beam	Slab Span (m)	Type of Slab	Thickness of Slab (mm)	Concrete Base Quantity (m^3)	Concrete Increment (m^3)	Total Concrete (m^3)	Steel Base Quantity (kg)	Steel Increment (kg)	Total Steel (kg)
9.0	CCS	3	2.25	Two-way	247	4.182	-	4.182	304.816	-	304.816
	DCS	3	2.25	Two-way	247	4.182	0.116	4.297	304.816	25.000	329.816
10.5	CCS	3	2.62	Two-way	278	5.113	-	5.113	360.157	-	360.157
	DCS	3	2.62	Two-way	278	5.113	0.130	5.243	360.157	29.167	389.323
12.0	CCS	4	2.40	Two-way	260	6.288	-	6.288	499.938	-	499.938
	DCS	4	2.40	Two-way	260	6.288	0.122	6.410	499.938	33.333	533.271
13.5	CCS	4	2.70	Two-way	285	7.317	-	7.317	578.670	-	578.670
	DCS	4	2.70	Two-way	285	7.317	0.133	7.450	578.670	37.500	616.170
15.0	CCS	5	2.50	Two-way	268	8.696	-	8.696	745.764	-	745.764
	DCS	5	2.50	Two-way	268	8.696	0.125	8.822	745.764	41.667	787.430
16.5	CCS	6	2.36	Two-way	256	10.248	-	10.248	950.616	-	950.616
	DCS	6	2.36	Two-way	256	10.248	0.120	10.368	950.616	45.833	996.450
18.0	CCS	6	2.57	Two-way	274	11.413	-	11.413	1038.320	-	1038.320
	DCS	6	2.57	Two-way	274	11.413	0.128	11.542	1038.320	50.000	1088.320

Table B-4 Specific material lists of the double T slab

Model Length (m)	Design	Number of Beam	Slab Span (m)	Type of Slab	Thickness of Slab (mm)	Concrete Base Quantity (m^3)	Concrete Increment (m^3)	Total Concrete (m^3)	Steel Base Quantity (kg)	Steel Increment (kg)	Total Steel (kg)
9.0	CCS	0	-	One-way	350	1.890	-	1.890	111.273	-	111.273
	DCS	0	-	One-way	350	1.890	0.076	1.966	111.273	40.000	151.273
10.5	CCS	0	-	One-way	400	2.365	-	2.365	137.022	-	137.022
	DCS	0	-	One-way	400	2.365	0.086	2.451	137.022	46.667	183.689
12.0	CCS	0	-	One-way	450	2.840	-	2.840	164.828	-	164.828
	DCS	0	-	One-way	450	2.840	0.097	2.937	164.828	53.333	218.162
13.5	CCS	0	-	One-way	525	3.515	-	3.515	198.997	-	198.997
	DCS	0	-	One-way	525	3.515	0.113	3.628	198.997	60.000	258.997
15.0	CCS	0	-	One-way	600	4.190	-	4.190	236.180	-	236.180
	DCS	0	-	One-way	600	4.190	0.130	4.320	236.180	66.667	302.847
16.5	CCS	0	-	One-way	650	4.885	-	4.885	282.677	-	282.677
	DCS	0	-	One-way	650	4.885	0.140	5.025	282.677	73.333	356.010
18.0	CCS	0	-	One-way	700	5.580	-	5.580	333.333	-	333.333
	DCS	0	-	One-way	700	5.580	0.151	5.731	333.333	80.000	413.333

Appendix C: Environmental impacts of 18 across 18 impact categories

Table C-1 Specific environmental impacts of Global warming

Global warming (kg CO_2 eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Solid slab	9.0	1.43E+03	5.10E+02	3.85E+02	2.32E+03	2.08E+03	5.69E+02	4.38E+02	3.95E+02	4.36E+02	3.09E+03	3.53E+03	3.92E+03
	10.5	1.95E+03	6.70E+02	5.05E+02	3.13E+03	2.74E+03	7.31E+02	5.63E+02	5.07E+02	5.59E+02	4.03E+03	4.59E+03	5.10E+03
	12.0	2.23E+03	7.66E+02	5.78E+02	3.57E+03	3.11E+03	8.28E+02	6.38E+02	5.75E+02	6.34E+02	4.57E+03	5.21E+03	5.79E+03
	13.5	2.51E+03	8.62E+02	6.50E+02	4.02E+03	3.48E+03	9.26E+02	7.14E+02	6.43E+02	7.09E+02	5.12E+03	5.83E+03	6.48E+03
	15.0	2.79E+03	9.58E+02	7.22E+02	4.46E+03	3.85E+03	1.02E+03	7.89E+02	7.11E+02	7.84E+02	5.66E+03	6.45E+03	7.16E+03
	16.5	3.54E+03	1.17E+03	8.84E+02	5.60E+03	4.78E+03	1.24E+03	9.55E+02	8.60E+02	9.49E+02	6.97E+03	7.92E+03	8.78E+03
	18.0	3.86E+03	1.28E+03	9.64E+02	6.10E+03	5.20E+03	1.35E+03	1.04E+03	9.35E+02	1.03E+03	7.58E+03	8.62E+03	9.55E+03
Hollow core slab	9.0	1.41E+03	4.66E+02	3.52E+02	2.23E+03	2.17E+03	5.67E+02	4.37E+02	3.93E+02	4.34E+02	3.17E+03	3.61E+03	4.00E+03
	10.5	1.65E+03	5.44E+02	4.10E+02	2.60E+03	2.48E+03	6.44E+02	4.96E+02	4.47E+02	4.93E+02	3.62E+03	4.11E+03	4.56E+03
	12.0	1.88E+03	6.22E+02	4.69E+02	2.97E+03	2.79E+03	7.21E+02	5.56E+02	5.01E+02	5.52E+02	4.06E+03	4.61E+03	5.12E+03
	13.5	2.50E+03	7.97E+02	6.01E+02	3.90E+03	3.55E+03	8.95E+02	6.89E+02	6.21E+02	6.85E+02	5.13E+03	5.82E+03	6.44E+03
	15.0	2.78E+03	8.85E+02	6.68E+02	4.33E+03	3.90E+03	9.82E+02	7.57E+02	6.82E+02	7.52E+02	5.64E+03	6.39E+03	7.08E+03
	16.5	3.06E+03	9.74E+02	7.34E+02	4.77E+03	4.26E+03	1.07E+03	8.25E+02	7.43E+02	8.19E+02	6.15E+03	6.98E+03	7.72E+03
	18.0	3.86E+03	1.19E+03	9.00E+02	5.96E+03	5.23E+03	1.29E+03	9.91E+02	8.93E+02	9.85E+02	7.50E+03	8.50E+03	9.39E+03
Ribbed slab	9.0	1.88E+03	5.87E+02	4.43E+02	2.91E+03	2.38E+03	5.95E+02	4.58E+02	4.13E+02	4.56E+02	3.43E+03	3.89E+03	4.30E+03
	10.5	2.28E+03	7.17E+02	5.41E+02	3.54E+03	2.88E+03	7.25E+02	5.59E+02	5.03E+02	5.55E+02	4.16E+03	4.72E+03	5.22E+03
	12.0	2.90E+03	8.85E+02	6.67E+02	4.46E+03	3.62E+03	8.90E+02	6.86E+02	6.17E+02	6.81E+02	5.19E+03	5.88E+03	6.49E+03
	13.5	3.37E+03	1.03E+03	7.76E+02	5.18E+03	4.20E+03	1.03E+03	7.97E+02	7.18E+02	7.92E+02	6.03E+03	6.82E+03	7.54E+03
	15.0	4.11E+03	1.23E+03	9.25E+02	6.26E+03	5.07E+03	1.23E+03	9.46E+02	8.52E+02	9.40E+02	7.24E+03	8.19E+03	9.04E+03
	16.5	4.96E+03	1.45E+03	1.09E+03	7.50E+03	6.08E+03	1.45E+03	1.11E+03	1.00E+03	1.11E+03	8.64E+03	9.75E+03	1.08E+04
	18.0	5.49E+03	1.61E+03	1.22E+03	8.32E+03	6.73E+03	1.61E+03	1.24E+03	1.12E+03	1.23E+03	9.58E+03	1.08E+04	1.19E+04

Global warming (kg CO_2 eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Double T slab	9.0	1.07E+03	2.33E+02	1.99E+02	1.50E+03	1.17E+03	2.72E+02	2.10E+02	1.89E+02	2.08E+02	1.65E+03	1.86E+03	2.05E+03
	10.5	1.33E+03	2.91E+02	2.49E+02	1.87E+03	1.45E+03	3.39E+02	2.61E+02	2.35E+02	2.60E+02	2.05E+03	2.31E+03	2.55E+03
	12.0	1.60E+03	3.50E+02	2.99E+02	2.25E+03	1.73E+03	4.06E+02	3.13E+02	2.82E+02	3.11E+02	2.45E+03	2.77E+03	3.05E+03
	13.5	1.97E+03	4.33E+02	3.70E+02	2.77E+03	2.13E+03	5.01E+02	3.86E+02	3.48E+02	3.84E+02	3.01E+03	3.40E+03	3.74E+03
	15.0	2.35E+03	5.16E+02	4.41E+02	3.30E+03	2.52E+03	5.97E+02	4.60E+02	4.14E+02	4.57E+02	3.57E+03	4.03E+03	4.45E+03
	16.5	2.75E+03	6.02E+02	5.14E+02	3.87E+03	2.94E+03	6.94E+02	5.35E+02	4.82E+02	5.31E+02	4.16E+03	4.70E+03	5.18E+03
	18.0	3.16E+03	6.88E+02	5.87E+02	4.43E+03	3.36E+03	7.92E+02	6.10E+02	5.50E+02	6.06E+02	4.76E+03	5.37E+03	5.92E+03

Table C-2 Specific environmental impacts of Stratospheric ozone depletion

Stratospheric ozone depletion (kg CFC-11 eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Solid slab	9.0	3.09E-04	2.23E-04	2.62E-04	7.95E-04	5.72E-04	2.68E-04	1.79E-04	1.49E-04	2.97E-04	1.14E-03	1.32E-03	1.47E-03
	10.5	4.28E-04	2.94E-04	3.45E-04	1.07E-03	7.55E-04	3.44E-04	2.30E-04	1.92E-04	3.82E-04	1.48E-03	1.71E-03	1.90E-03
	12.0	4.89E-04	3.36E-04	3.94E-04	1.22E-03	8.57E-04	3.90E-04	2.61E-04	2.17E-04	4.33E-04	1.68E-03	1.94E-03	2.16E-03
	13.5	5.50E-04	3.78E-04	4.43E-04	1.37E-03	9.60E-04	4.37E-04	2.91E-04	2.43E-04	4.84E-04	1.88E-03	2.17E-03	2.41E-03
	15.0	6.11E-04	4.20E-04	4.93E-04	1.52E-03	1.06E-03	4.83E-04	3.22E-04	2.69E-04	5.35E-04	2.08E-03	2.40E-03	2.67E-03
	16.5	7.87E-04	5.14E-04	6.03E-04	1.90E-03	1.32E-03	5.84E-04	3.90E-04	3.25E-04	6.47E-04	2.55E-03	2.94E-03	3.27E-03
	18.0	8.58E-04	5.61E-04	6.58E-04	2.08E-03	1.44E-03	6.35E-04	4.24E-04	3.54E-04	7.04E-04	2.77E-03	3.20E-03	3.55E-03
Hollow core slab	9.0	3.14E-04	2.04E-04	2.40E-04	7.58E-04	5.99E-04	2.67E-04	1.78E-04	1.49E-04	2.96E-04	1.16E-03	1.34E-03	1.49E-03
	10.5	3.67E-04	2.38E-04	2.80E-04	8.85E-04	6.85E-04	3.04E-04	2.03E-04	1.69E-04	3.37E-04	1.32E-03	1.53E-03	1.70E-03
	12.0	4.18E-04	2.72E-04	3.20E-04	1.01E-03	7.69E-04	3.40E-04	2.27E-04	1.89E-04	3.77E-04	1.49E-03	1.71E-03	1.90E-03
	13.5	5.65E-04	3.49E-04	4.10E-04	1.32E-03	9.82E-04	4.22E-04	2.82E-04	2.35E-04	4.67E-04	1.87E-03	2.15E-03	2.39E-03
	15.0	6.27E-04	3.88E-04	4.55E-04	1.47E-03	1.08E-03	4.63E-04	3.09E-04	2.58E-04	5.13E-04	2.06E-03	2.37E-03	2.62E-03

Stratospheric ozone depletion (kg CFC-11 eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
	16.5	6.90E-04	4.27E-04	5.01E-04	1.62E-03	1.18E-03	5.05E-04	3.37E-04	2.81E-04	5.59E-04	2.24E-03	2.58E-03	2.86E-03
	18.0	8.81E-04	5.23E-04	6.14E-04	2.02E-03	1.45E-03	6.06E-04	4.05E-04	3.38E-04	6.72E-04	2.73E-03	3.13E-03	3.47E-03
Ribbed slab	9.0	4.28E-04	2.57E-04	3.02E-04	9.88E-04	6.59E-04	2.80E-04	1.87E-04	1.56E-04	3.11E-04	1.25E-03	1.44E-03	1.59E-03
	10.5	5.18E-04	3.14E-04	3.69E-04	1.20E-03	7.98E-04	3.42E-04	2.28E-04	1.90E-04	3.79E-04	1.52E-03	1.75E-03	1.94E-03
	12.0	6.65E-04	3.88E-04	4.55E-04	1.51E-03	1.00E-03	4.19E-04	2.80E-04	2.34E-04	4.65E-04	1.89E-03	2.17E-03	2.40E-03
	13.5	7.72E-04	4.51E-04	5.30E-04	1.75E-03	1.17E-03	4.87E-04	3.25E-04	2.71E-04	5.40E-04	2.19E-03	2.52E-03	2.79E-03
	15.0	9.46E-04	5.38E-04	6.31E-04	2.11E-03	1.41E-03	5.79E-04	3.86E-04	3.22E-04	6.41E-04	2.63E-03	3.01E-03	3.34E-03
	16.5	1.15E-03	6.35E-04	7.46E-04	2.53E-03	1.69E-03	6.82E-04	4.55E-04	3.80E-04	7.56E-04	3.13E-03	3.58E-03	3.96E-03
	18.0	1.27E-03	7.07E-04	8.30E-04	2.81E-03	1.87E-03	7.59E-04	5.06E-04	4.22E-04	8.41E-04	3.47E-03	3.98E-03	4.40E-03
Double T slab	9.0	2.53E-04	1.07E-04	1.36E-04	4.95E-04	2.81E-04	1.28E-04	8.57E-05	7.15E-05	1.42E-04	5.51E-04	6.37E-04	7.09E-04
	10.5	3.15E-04	1.34E-04	1.70E-04	6.19E-04	3.48E-04	1.60E-04	1.07E-04	8.90E-05	1.77E-04	6.85E-04	7.91E-04	8.80E-04
	12.0	3.79E-04	1.60E-04	2.04E-04	7.43E-04	4.16E-04	1.92E-04	1.28E-04	1.07E-04	2.12E-04	8.19E-04	9.47E-04	1.05E-03
	13.5	4.66E-04	1.98E-04	2.52E-04	9.17E-04	5.08E-04	2.36E-04	1.58E-04	1.32E-04	2.62E-04	1.01E-03	1.16E-03	1.30E-03
	15.0	5.55E-04	2.36E-04	3.01E-04	1.09E-03	6.02E-04	2.81E-04	1.88E-04	1.57E-04	3.12E-04	1.20E-03	1.38E-03	1.54E-03
	16.5	6.51E-04	2.76E-04	3.51E-04	1.28E-03	7.03E-04	3.27E-04	2.18E-04	1.82E-04	3.63E-04	1.39E-03	1.61E-03	1.79E-03
	18.0	7.49E-04	3.15E-04	4.01E-04	1.47E-03	8.05E-04	3.73E-04	2.49E-04	2.08E-04	4.14E-04	1.59E-03	1.84E-03	2.05E-03

Table C-3 Specific environmental impacts of Ionizing radiation

Ionizing radiation (kBq Co-60 eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Solid slab	9.0	4.45E+01	7.21E+00	6.76E+00	5.84E+01	5.90E+01	8.28E+00	5.99E+00	5.22E+00	7.65E+00	7.50E+01	8.10E+01	8.62E+01
	10.5	6.12E+01	9.47E+00	8.88E+00	7.96E+01	7.84E+01	1.06E+01	7.69E+00	6.70E+00	9.83E+00	9.88E+01	1.07E+02	1.13E+02
	12.0	6.99E+01	1.08E+01	1.01E+01	9.09E+01	8.89E+01	1.21E+01	8.72E+00	7.60E+00	1.11E+01	1.12E+02	1.21E+02	1.28E+02

Ionizing radiation (kBq Co-60 eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
	13.5	7.87E+01	1.22E+01	1.14E+01	1.02E+02	9.96E+01	1.35E+01	9.75E+00	8.50E+00	1.25E+01	1.26E+02	1.35E+02	1.44E+02
	15.0	8.73E+01	1.35E+01	1.27E+01	1.14E+02	1.10E+02	1.49E+01	1.08E+01	9.40E+00	1.38E+01	1.39E+02	1.50E+02	1.59E+02
	16.5	1.12E+02	1.66E+01	1.55E+01	1.44E+02	1.38E+02	1.80E+01	1.30E+01	1.14E+01	1.67E+01	1.73E+02	1.86E+02	1.97E+02
	18.0	1.22E+02	1.81E+01	1.69E+01	1.57E+02	1.50E+02	1.96E+01	1.42E+01	1.24E+01	1.81E+01	1.88E+02	2.02E+02	2.14E+02
Hollow core slab	9.0	4.46E+01	6.59E+00	6.18E+00	5.73E+01	6.25E+01	8.26E+00	5.97E+00	5.20E+00	7.63E+00	7.84E+01	8.43E+01	8.95E+01
	10.5	5.20E+01	7.69E+00	7.21E+00	6.69E+01	7.15E+01	9.38E+00	6.78E+00	5.91E+00	8.67E+00	8.95E+01	9.63E+01	1.02E+02
	12.0	5.94E+01	8.79E+00	8.24E+00	7.64E+01	8.03E+01	1.05E+01	7.59E+00	6.62E+00	9.70E+00	1.01E+02	1.08E+02	1.15E+02
	13.5	7.97E+01	1.13E+01	1.06E+01	1.01E+02	1.03E+02	1.30E+01	9.41E+00	8.21E+00	1.20E+01	1.28E+02	1.38E+02	1.46E+02
	15.0	8.85E+01	1.25E+01	1.17E+01	1.13E+02	1.14E+02	1.43E+01	1.03E+01	9.02E+00	1.32E+01	1.41E+02	1.51E+02	1.60E+02
	16.5	9.74E+01	1.38E+01	1.29E+01	1.24E+02	1.24E+02	1.56E+01	1.13E+01	9.82E+00	1.44E+01	1.54E+02	1.65E+02	1.75E+02
	18.0	1.24E+02	1.69E+01	1.58E+01	1.56E+02	1.53E+02	1.87E+01	1.35E+01	1.18E+01	1.73E+01	1.89E+02	2.03E+02	2.15E+02
Ribbed slab	9.0	6.02E+01	8.30E+00	7.78E+00	7.63E+01	6.94E+01	8.66E+00	6.26E+00	5.46E+00	8.00E+00	8.61E+01	9.23E+01	9.78E+01
	10.5	7.29E+01	1.01E+01	9.50E+00	9.25E+01	8.39E+01	1.06E+01	7.63E+00	6.66E+00	9.76E+00	1.04E+02	1.12E+02	1.18E+02
	12.0	9.32E+01	1.25E+01	1.17E+01	1.17E+02	1.06E+02	1.30E+01	9.36E+00	8.17E+00	1.20E+01	1.31E+02	1.40E+02	1.49E+02
	13.5	1.08E+02	1.46E+01	1.36E+01	1.36E+02	1.23E+02	1.51E+01	1.09E+01	9.49E+00	1.39E+01	1.52E+02	1.63E+02	1.72E+02
	15.0	1.32E+02	1.73E+01	1.63E+01	1.66E+02	1.49E+02	1.79E+01	1.29E+01	1.13E+01	1.65E+01	1.84E+02	1.97E+02	2.08E+02
	16.5	1.60E+02	2.05E+01	1.92E+01	2.00E+02	1.80E+02	2.11E+01	1.52E+01	1.33E+01	1.95E+01	2.21E+02	2.36E+02	2.49E+02
	18.0	1.77E+02	2.28E+01	2.14E+01	2.21E+02	1.99E+02	2.34E+01	1.69E+01	1.48E+01	2.16E+01	2.44E+02	2.61E+02	2.76E+02
Double T slab	9.0	3.11E+01	3.34E+00	3.49E+00	3.79E+01	3.46E+01	3.96E+00	2.86E+00	2.50E+00	3.66E+00	4.22E+01	4.51E+01	4.76E+01
	10.5	3.88E+01	4.18E+00	4.37E+00	4.73E+01	4.28E+01	4.94E+00	3.57E+00	3.11E+00	4.56E+00	5.23E+01	5.59E+01	5.90E+01
	12.0	4.66E+01	5.02E+00	5.25E+00	5.69E+01	5.12E+01	5.91E+00	4.28E+00	3.73E+00	5.46E+00	6.25E+01	6.68E+01	7.05E+01
	13.5	5.74E+01	6.21E+00	6.49E+00	7.01E+01	6.25E+01	7.30E+00	5.27E+00	4.60E+00	6.74E+00	7.66E+01	8.19E+01	8.65E+01
	15.0	6.83E+01	7.41E+00	7.74E+00	8.35E+01	7.41E+01	8.68E+00	6.28E+00	5.48E+00	8.02E+00	9.08E+01	9.71E+01	1.03E+02
	16.5	8.01E+01	8.64E+00	9.03E+00	9.78E+01	8.65E+01	1.01E+01	7.30E+00	6.37E+00	9.34E+00	1.06E+02	1.13E+02	1.20E+02
	18.0	9.21E+01	9.88E+00	1.03E+01	1.12E+02	9.91E+01	1.15E+01	8.33E+00	7.27E+00	1.07E+01	1.21E+02	1.30E+02	1.37E+02

Table C-4 Specific environmental impacts of Ozone formation, Human health

Ozone formation, Human health (kg NOx eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Solid slab	9.0	3.29E+00	1.78E+00	5.57E-01	5.62E+00	4.98E+00	1.93E+00	1.74E+00	1.68E+00	6.31E-01	7.54E+00	9.29E+00	1.10E+01
	10.5	4.54E+00	2.33E+00	7.32E-01	7.61E+00	6.60E+00	2.48E+00	2.24E+00	2.15E+00	8.10E-01	9.89E+00	1.21E+01	1.43E+01
	12.0	5.19E+00	2.67E+00	8.37E-01	8.69E+00	7.49E+00	2.81E+00	2.53E+00	2.44E+00	9.19E-01	1.12E+01	1.38E+01	1.62E+01
	13.5	5.84E+00	3.00E+00	9.41E-01	9.78E+00	8.39E+00	3.14E+00	2.83E+00	2.73E+00	1.03E+00	1.26E+01	1.54E+01	1.81E+01
	15.0	6.48E+00	3.34E+00	1.05E+00	1.09E+01	9.28E+00	3.47E+00	3.13E+00	3.02E+00	1.14E+00	1.39E+01	1.70E+01	2.00E+01
	16.5	8.32E+00	4.08E+00	1.28E+00	1.37E+01	1.16E+01	4.21E+00	3.79E+00	3.66E+00	1.37E+00	1.72E+01	2.09E+01	2.46E+01
	18.0	9.07E+00	4.46E+00	1.40E+00	1.49E+01	1.26E+01	4.57E+00	4.12E+00	3.98E+00	1.49E+00	1.87E+01	2.28E+01	2.68E+01
Hollow core slab	9.0	3.32E+00	1.62E+00	5.09E-01	5.45E+00	5.25E+00	1.92E+00	1.74E+00	1.67E+00	6.29E-01	7.80E+00	9.54E+00	1.12E+01
	10.5	3.87E+00	1.89E+00	5.94E-01	6.36E+00	6.01E+00	2.19E+00	1.97E+00	1.90E+00	7.15E-01	8.91E+00	1.09E+01	1.28E+01
	12.0	4.42E+00	2.17E+00	6.79E-01	7.27E+00	6.75E+00	2.45E+00	2.21E+00	2.13E+00	8.00E-01	1.00E+01	1.22E+01	1.43E+01
	13.5	5.95E+00	2.78E+00	8.70E-01	9.59E+00	8.65E+00	3.04E+00	2.74E+00	2.64E+00	9.92E-01	1.27E+01	1.54E+01	1.81E+01
	15.0	6.60E+00	3.08E+00	9.67E-01	1.07E+01	9.52E+00	3.33E+00	3.01E+00	2.90E+00	1.09E+00	1.39E+01	1.69E+01	1.98E+01
	16.5	7.27E+00	3.39E+00	1.06E+00	1.17E+01	1.04E+01	3.63E+00	3.28E+00	3.16E+00	1.19E+00	1.52E+01	1.85E+01	2.16E+01
	18.0	9.25E+00	4.16E+00	1.30E+00	1.47E+01	1.28E+01	4.36E+00	3.94E+00	3.79E+00	1.43E+00	1.86E+01	2.26E+01	2.64E+01
Ribbed slab	9.0	4.50E+00	2.04E+00	6.41E-01	7.19E+00	5.81E+00	2.02E+00	1.82E+00	1.75E+00	6.60E-01	8.49E+00	1.03E+01	1.21E+01
	10.5	5.44E+00	2.50E+00	7.83E-01	8.72E+00	7.03E+00	2.46E+00	2.22E+00	2.14E+00	8.04E-01	1.03E+01	1.25E+01	1.47E+01
	12.0	6.97E+00	3.08E+00	9.67E-01	1.10E+01	8.88E+00	3.02E+00	2.72E+00	2.62E+00	9.87E-01	1.29E+01	1.56E+01	1.82E+01
	13.5	8.10E+00	3.59E+00	1.12E+00	1.28E+01	1.03E+01	3.51E+00	3.16E+00	3.05E+00	1.15E+00	1.50E+01	1.81E+01	2.12E+01
	15.0	9.91E+00	4.27E+00	1.34E+00	1.55E+01	1.25E+01	4.17E+00	3.76E+00	3.62E+00	1.36E+00	1.80E+01	2.18E+01	2.54E+01
	16.5	1.20E+01	5.05E+00	1.58E+00	1.87E+01	1.50E+01	4.91E+00	4.43E+00	4.27E+00	1.60E+00	2.15E+01	2.60E+01	3.02E+01
	18.0	1.33E+01	5.62E+00	1.76E+00	2.07E+01	1.66E+01	5.46E+00	4.93E+00	4.75E+00	1.78E+00	2.39E+01	2.88E+01	3.35E+01
Double T slab	9.0	2.52E+00	8.28E-01	2.88E-01	3.64E+00	2.80E+00	9.24E-01	8.33E-01	8.03E-01	3.02E-01	4.02E+00	4.86E+00	5.66E+00
	10.5	3.15E+00	1.04E+00	3.60E-01	4.54E+00	3.47E+00	1.15E+00	1.04E+00	1.00E+00	3.76E-01	4.99E+00	6.03E+00	7.03E+00
	12.0	3.78E+00	1.24E+00	4.33E-01	5.46E+00	4.14E+00	1.38E+00	1.24E+00	1.20E+00	4.51E-01	5.97E+00	7.22E+00	8.41E+00

Ozone formation, Human health (kg NOx eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
	13.5	4.65E+00	1.54E+00	5.35E-01	6.73E+00	5.07E+00	1.70E+00	1.53E+00	1.48E+00	5.56E-01	7.32E+00	8.86E+00	1.03E+01
	15.0	5.54E+00	1.83E+00	6.38E-01	8.01E+00	6.01E+00	2.02E+00	1.83E+00	1.76E+00	6.62E-01	8.69E+00	1.05E+01	1.23E+01
	16.5	6.50E+00	2.14E+00	7.44E-01	9.38E+00	7.00E+00	2.36E+00	2.12E+00	2.05E+00	7.70E-01	1.01E+01	1.23E+01	1.43E+01
	18.0	7.47E+00	2.45E+00	8.51E-01	1.08E+01	8.02E+00	2.69E+00	2.42E+00	2.34E+00	8.78E-01	1.16E+01	1.40E+01	1.63E+01

Table C-5 Specific environmental impacts of Fine particulate matter formation

Fine particulate matter formation (kg PM2.5 eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Solid slab	9.0	1.59E+00	9.10E-01	3.47E-01	2.84E+00	2.93E+00	9.64E-01	8.47E-01	8.07E-01	3.93E-01	4.29E+00	5.14E+00	5.95E+00
	10.5	2.29E+00	1.20E+00	4.56E-01	3.94E+00	3.95E+00	1.24E+00	1.09E+00	1.04E+00	5.04E-01	5.70E+00	6.78E+00	7.82E+00
	12.0	2.61E+00	1.37E+00	5.21E-01	4.50E+00	4.49E+00	1.40E+00	1.23E+00	1.18E+00	5.72E-01	6.46E+00	7.70E+00	8.87E+00
	13.5	2.94E+00	1.54E+00	5.86E-01	5.06E+00	5.03E+00	1.57E+00	1.38E+00	1.31E+00	6.39E-01	7.24E+00	8.62E+00	9.93E+00
	15.0	3.26E+00	1.71E+00	6.51E-01	5.62E+00	5.57E+00	1.74E+00	1.52E+00	1.45E+00	7.07E-01	8.01E+00	9.53E+00	1.10E+01
	16.5	4.34E+00	2.09E+00	7.97E-01	7.23E+00	7.06E+00	2.10E+00	1.84E+00	1.76E+00	8.56E-01	1.00E+01	1.19E+01	1.36E+01
	18.0	4.73E+00	2.28E+00	8.70E-01	7.88E+00	7.68E+00	2.28E+00	2.01E+00	1.91E+00	9.31E-01	1.09E+01	1.29E+01	1.48E+01
Hollow core slab	9.0	1.73E+00	8.32E-01	3.17E-01	2.88E+00	3.19E+00	9.61E-01	8.44E-01	8.04E-01	3.92E-01	4.54E+00	5.38E+00	6.19E+00
	10.5	2.03E+00	9.71E-01	3.70E-01	3.37E+00	3.66E+00	1.09E+00	9.58E-01	9.14E-01	4.45E-01	5.19E+00	6.15E+00	7.07E+00
	12.0	2.31E+00	1.11E+00	4.23E-01	3.84E+00	4.12E+00	1.22E+00	1.07E+00	1.02E+00	4.98E-01	5.84E+00	6.91E+00	7.93E+00
	13.5	3.22E+00	1.42E+00	5.42E-01	5.18E+00	5.36E+00	1.52E+00	1.33E+00	1.27E+00	6.18E-01	7.49E+00	8.82E+00	1.01E+01
	15.0	3.57E+00	1.58E+00	6.02E-01	5.76E+00	5.91E+00	1.66E+00	1.46E+00	1.39E+00	6.78E-01	8.25E+00	9.71E+00	1.11E+01
	16.5	3.93E+00	1.74E+00	6.62E-01	6.33E+00	6.46E+00	1.81E+00	1.59E+00	1.52E+00	7.39E-01	9.02E+00	1.06E+01	1.21E+01
	18.0	5.14E+00	2.13E+00	8.11E-01	8.08E+00	8.07E+00	2.18E+00	1.91E+00	1.82E+00	8.88E-01	1.11E+01	1.31E+01	1.49E+01
Ribbed	9.0	2.48E+00	1.05E+00	3.99E-01	3.93E+00	3.62E+00	1.01E+00	8.85E-01	8.44E-01	4.11E-01	5.04E+00	5.93E+00	6.77E+00

Fine particulate matter formation (kg PM2.5 eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
slab	10.5	2.98E+00	1.28E+00	4.88E-01	4.75E+00	4.36E+00	1.23E+00	1.08E+00	1.03E+00	5.01E-01	6.09E+00	7.17E+00	8.20E+00
	12.0	3.91E+00	1.58E+00	6.02E-01	6.10E+00	5.58E+00	1.51E+00	1.32E+00	1.26E+00	6.14E-01	7.71E+00	9.03E+00	1.03E+01
	13.5	4.54E+00	1.84E+00	7.00E-01	7.08E+00	6.47E+00	1.75E+00	1.54E+00	1.47E+00	7.14E-01	8.94E+00	1.05E+01	1.19E+01
	15.0	5.65E+00	2.19E+00	8.34E-01	8.67E+00	7.92E+00	2.08E+00	1.83E+00	1.74E+00	8.48E-01	1.08E+01	1.27E+01	1.44E+01
	16.5	6.97E+00	2.59E+00	9.86E-01	1.05E+01	9.62E+00	2.45E+00	2.15E+00	2.05E+00	9.99E-01	1.31E+01	1.52E+01	1.73E+01
	18.0	7.67E+00	2.88E+00	1.10E+00	1.16E+01	1.06E+01	2.73E+00	2.39E+00	2.28E+00	1.11E+00	1.45E+01	1.68E+01	1.91E+01
Double T slab	9.0	1.49E+00	4.08E-01	1.79E-01	2.08E+00	1.71E+00	4.61E-01	4.05E-01	3.86E-01	1.88E-01	2.36E+00	2.76E+00	3.15E+00
	10.5	1.86E+00	5.10E-01	2.24E-01	2.59E+00	2.11E+00	5.75E-01	5.04E-01	4.81E-01	2.34E-01	2.92E+00	3.42E+00	3.90E+00
	12.0	2.24E+00	6.12E-01	2.69E-01	3.12E+00	2.52E+00	6.89E-01	6.04E-01	5.76E-01	2.81E-01	3.49E+00	4.09E+00	4.67E+00
	13.5	2.75E+00	7.57E-01	3.33E-01	3.84E+00	3.07E+00	8.49E-01	7.46E-01	7.11E-01	3.46E-01	4.26E+00	5.01E+00	5.72E+00
	15.0	3.27E+00	9.03E-01	3.97E-01	4.57E+00	3.63E+00	1.01E+00	8.87E-01	8.46E-01	4.12E-01	5.05E+00	5.94E+00	6.79E+00
	16.5	3.84E+00	1.05E+00	4.63E-01	5.36E+00	4.24E+00	1.18E+00	1.03E+00	9.85E-01	4.79E-01	5.89E+00	6.93E+00	7.91E+00
	18.0	4.43E+00	1.20E+00	5.30E-01	6.17E+00	4.86E+00	1.34E+00	1.18E+00	1.12E+00	5.47E-01	6.75E+00	7.93E+00	9.06E+00

Table C-6 Specific environmental impacts of Ozone formation, Terrestrial ecosystem

Ozone formation, Terrestrial ecosystem (kg NOx eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Solid slab	9.0	3.37E+00	1.80E+00	5.88E-01	5.75E+00	5.08E+00	1.96E+00	1.76E+00	1.70E+00	6.66E-01	7.71E+00	9.47E+00	1.12E+01
	10.5	4.65E+00	2.37E+00	7.72E-01	7.79E+00	6.74E+00	2.52E+00	2.26E+00	2.18E+00	8.55E-01	1.01E+01	1.24E+01	1.45E+01
	12.0	5.31E+00	2.70E+00	8.83E-01	8.90E+00	7.65E+00	2.86E+00	2.57E+00	2.47E+00	9.69E-01	1.15E+01	1.40E+01	1.65E+01
	13.5	5.98E+00	3.04E+00	9.93E-01	1.00E+01	8.56E+00	3.19E+00	2.87E+00	2.76E+00	1.08E+00	1.28E+01	1.57E+01	1.85E+01
	15.0	6.64E+00	3.38E+00	1.10E+00	1.11E+01	9.47E+00	3.53E+00	3.17E+00	3.05E+00	1.20E+00	1.42E+01	1.74E+01	2.04E+01
	16.5	8.53E+00	4.14E+00	1.35E+00	1.40E+01	1.18E+01	4.27E+00	3.84E+00	3.69E+00	1.45E+00	1.76E+01	2.14E+01	2.51E+01

Ozone formation, Terrestrial ecosystem (kg NOx eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
	18.0	9.31E+00	4.52E+00	1.47E+00	1.53E+01	1.29E+01	4.65E+00	4.17E+00	4.02E+00	1.58E+00	1.91E+01	2.33E+01	2.73E+01
Hollow core slab	9.0	3.40E+00	1.65E+00	5.37E-01	5.59E+00	5.37E+00	1.96E+00	1.76E+00	1.69E+00	6.63E-01	7.98E+00	9.74E+00	1.14E+01
	10.5	3.97E+00	1.92E+00	6.27E-01	6.52E+00	6.14E+00	2.22E+00	2.00E+00	1.92E+00	7.54E-01	9.11E+00	1.11E+01	1.30E+01
	12.0	4.54E+00	2.19E+00	7.16E-01	7.45E+00	6.90E+00	2.49E+00	2.23E+00	2.15E+00	8.44E-01	1.02E+01	1.25E+01	1.46E+01
	13.5	6.11E+00	2.81E+00	9.18E-01	9.84E+00	8.85E+00	3.08E+00	2.77E+00	2.67E+00	1.05E+00	1.30E+01	1.57E+01	1.84E+01
	15.0	6.78E+00	3.13E+00	1.02E+00	1.09E+01	9.74E+00	3.39E+00	3.04E+00	2.93E+00	1.15E+00	1.43E+01	1.73E+01	2.02E+01
	16.5	7.47E+00	3.44E+00	1.12E+00	1.20E+01	1.06E+01	3.69E+00	3.31E+00	3.19E+00	1.25E+00	1.56E+01	1.89E+01	2.21E+01
	18.0	9.51E+00	4.21E+00	1.37E+00	1.51E+01	1.31E+01	4.44E+00	3.98E+00	3.83E+00	1.50E+00	1.91E+01	2.31E+01	2.69E+01
Ribbed slab	9.0	4.63E+00	2.07E+00	6.76E-01	7.38E+00	5.95E+00	2.05E+00	1.84E+00	1.77E+00	6.96E-01	8.70E+00	1.05E+01	1.23E+01
	10.5	5.60E+00	2.53E+00	8.26E-01	8.95E+00	7.19E+00	2.50E+00	2.25E+00	2.16E+00	8.48E-01	1.05E+01	1.28E+01	1.49E+01
	12.0	7.17E+00	3.12E+00	1.02E+00	1.13E+01	9.09E+00	3.07E+00	2.76E+00	2.65E+00	1.04E+00	1.32E+01	1.60E+01	1.86E+01
	13.5	8.33E+00	3.64E+00	1.19E+00	1.32E+01	1.05E+01	3.56E+00	3.20E+00	3.08E+00	1.21E+00	1.53E+01	1.85E+01	2.16E+01
	15.0	1.02E+01	4.33E+00	1.41E+00	1.59E+01	1.28E+01	4.23E+00	3.80E+00	3.66E+00	1.44E+00	1.85E+01	2.23E+01	2.59E+01
	16.5	1.24E+01	5.12E+00	1.67E+00	1.92E+01	1.54E+01	4.99E+00	4.48E+00	4.31E+00	1.69E+00	2.21E+01	2.66E+01	3.09E+01
	18.0	1.37E+01	5.70E+00	1.86E+00	2.12E+01	1.70E+01	5.55E+00	4.98E+00	4.80E+00	1.88E+00	2.45E+01	2.94E+01	3.42E+01
Double T slab	9.0	2.57E+00	8.41E-01	3.04E-01	3.72E+00	2.86E+00	9.39E-01	8.43E-01	8.11E-01	3.18E-01	4.12E+00	4.96E+00	5.77E+00
	10.5	3.21E+00	1.05E+00	3.80E-01	4.64E+00	3.54E+00	1.17E+00	1.05E+00	1.01E+00	3.97E-01	5.11E+00	6.16E+00	7.17E+00
	12.0	3.86E+00	1.26E+00	4.57E-01	5.57E+00	4.23E+00	1.40E+00	1.26E+00	1.21E+00	4.75E-01	6.11E+00	7.37E+00	8.58E+00
	13.5	4.75E+00	1.56E+00	5.65E-01	6.87E+00	5.18E+00	1.73E+00	1.55E+00	1.49E+00	5.86E-01	7.49E+00	9.04E+00	1.05E+01
	15.0	5.65E+00	1.86E+00	6.73E-01	8.19E+00	6.13E+00	2.06E+00	1.85E+00	1.78E+00	6.98E-01	8.89E+00	1.07E+01	1.25E+01
	16.5	6.63E+00	2.17E+00	7.85E-01	9.58E+00	7.16E+00	2.39E+00	2.15E+00	2.07E+00	8.12E-01	1.04E+01	1.25E+01	1.46E+01
	18.0	7.62E+00	2.48E+00	8.98E-01	1.10E+01	8.20E+00	2.73E+00	2.45E+00	2.36E+00	9.27E-01	1.19E+01	1.43E+01	1.67E+01

Table C-7 Specific environmental impacts of Terrestrial acidification

Terrestrial acidification (kg SO_2 eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Solid slab	9.0	3.20E+00	2.34E+00	7.48E-01	3.20E+00	6.29E+00	6.53E+00	2.44E+00	2.18E+00	2.10E+00	9.81E+00	1.20E+01	1.41E+01
	10.5	4.49E+00	3.08E+00	9.83E-01	4.49E+00	8.55E+00	8.64E+00	3.13E+00	2.80E+00	2.69E+00	1.29E+01	1.57E+01	1.83E+01
	12.0	5.12E+00	3.52E+00	1.12E+00	5.12E+00	9.76E+00	9.81E+00	3.55E+00	3.18E+00	3.05E+00	1.46E+01	1.78E+01	2.08E+01
	13.5	5.77E+00	3.95E+00	1.26E+00	5.77E+00	1.10E+01	1.10E+01	3.96E+00	3.55E+00	3.41E+00	1.63E+01	1.99E+01	2.33E+01
	15.0	6.40E+00	4.39E+00	1.40E+00	6.40E+00	1.22E+01	1.21E+01	4.38E+00	3.93E+00	3.77E+00	1.81E+01	2.20E+01	2.58E+01
	16.5	8.33E+00	5.38E+00	1.72E+00	8.33E+00	1.54E+01	1.52E+01	5.30E+00	4.75E+00	4.57E+00	2.23E+01	2.71E+01	3.16E+01
	18.0	9.09E+00	5.87E+00	1.87E+00	9.09E+00	1.68E+01	1.65E+01	5.77E+00	5.17E+00	4.97E+00	2.43E+01	2.94E+01	3.44E+01
Hollow core slab	9.0	3.33E+00	2.14E+00	6.83E-01	3.33E+00	6.15E+00	6.87E+00	2.43E+00	2.17E+00	2.09E+00	1.01E+01	1.23E+01	1.44E+01
	10.5	3.89E+00	2.50E+00	7.97E-01	3.89E+00	7.18E+00	7.86E+00	2.76E+00	2.47E+00	2.37E+00	1.16E+01	1.40E+01	1.64E+01
	12.0	4.43E+00	2.85E+00	9.11E-01	4.43E+00	8.20E+00	8.83E+00	3.09E+00	2.77E+00	2.66E+00	1.30E+01	1.58E+01	1.84E+01
	13.5	6.05E+00	3.66E+00	1.17E+00	6.05E+00	1.09E+01	1.13E+01	3.83E+00	3.43E+00	3.30E+00	1.65E+01	1.99E+01	2.32E+01
	15.0	6.71E+00	4.06E+00	1.30E+00	6.71E+00	1.21E+01	1.25E+01	4.20E+00	3.77E+00	3.62E+00	1.81E+01	2.19E+01	2.55E+01
	16.5	7.39E+00	4.47E+00	1.43E+00	7.39E+00	1.33E+01	1.36E+01	4.58E+00	4.10E+00	3.94E+00	1.98E+01	2.39E+01	2.78E+01
	18.0	9.50E+00	5.47E+00	1.75E+00	9.50E+00	1.67E+01	1.68E+01	5.51E+00	4.93E+00	4.74E+00	2.42E+01	2.91E+01	3.39E+01
Ribbed slab	9.0	4.61E+00	2.69E+00	8.60E-01	4.61E+00	8.17E+00	7.60E+00	2.55E+00	2.28E+00	2.19E+00	1.10E+01	1.33E+01	1.55E+01
	10.5	5.56E+00	3.29E+00	1.05E+00	5.56E+00	9.90E+00	9.19E+00	3.10E+00	2.78E+00	2.67E+00	1.34E+01	1.62E+01	1.88E+01
	12.0	7.20E+00	4.06E+00	1.30E+00	7.20E+00	1.26E+01	1.16E+01	3.81E+00	3.41E+00	3.28E+00	1.67E+01	2.01E+01	2.34E+01
	13.5	8.35E+00	4.72E+00	1.51E+00	8.35E+00	1.46E+01	1.35E+01	4.43E+00	3.96E+00	3.81E+00	1.94E+01	2.34E+01	2.72E+01
	15.0	1.03E+01	5.63E+00	1.80E+00	1.03E+01	1.77E+01	1.63E+01	5.25E+00	4.71E+00	4.52E+00	2.34E+01	2.81E+01	3.26E+01
	16.5	1.26E+01	6.65E+00	2.13E+00	1.26E+01	2.14E+01	1.96E+01	6.19E+00	5.55E+00	5.33E+00	2.80E+01	3.35E+01	3.88E+01
	18.0	1.39E+01	7.40E+00	2.37E+00	1.39E+01	2.36E+01	2.17E+01	6.89E+00	6.17E+00	5.93E+00	3.10E+01	3.72E+01	4.31E+01
Double T slab	9.0	3.23E+00	1.03E+00	3.87E-01	3.23E+00	4.65E+00	3.59E+00	1.17E+00	1.04E+00	1.00E+00	5.16E+00	6.20E+00	7.20E+00
	10.5	4.03E+00	1.29E+00	4.84E-01	4.03E+00	5.80E+00	4.44E+00	1.45E+00	1.30E+00	1.25E+00	6.40E+00	7.70E+00	8.95E+00
	12.0	4.84E+00	1.55E+00	5.81E-01	4.84E+00	6.97E+00	5.31E+00	1.74E+00	1.56E+00	1.50E+00	7.65E+00	9.21E+00	1.07E+01

Terrestrial acidification (kg SO_2 eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
	13.5	5.96E+00	1.92E+00	7.18E-01	5.96E+00	8.60E+00	6.49E+00	2.15E+00	1.92E+00	1.85E+00	9.38E+00	1.13E+01	1.32E+01
	15.0	7.10E+00	2.29E+00	8.56E-01	7.10E+00	1.02E+01	7.69E+00	2.55E+00	2.29E+00	2.20E+00	1.11E+01	1.34E+01	1.56E+01
	16.5	8.32E+00	2.67E+00	9.99E-01	8.32E+00	1.20E+01	8.97E+00	2.97E+00	2.66E+00	2.56E+00	1.30E+01	1.56E+01	1.82E+01
	18.0	9.57E+00	3.05E+00	1.14E+00	9.57E+00	1.38E+01	1.03E+01	3.39E+00	3.04E+00	2.92E+00	1.48E+01	1.79E+01	2.08E+01

Table C-8 Specific environmental impacts of Freshwater eutrophication

Freshwater eutrophication (kg P eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Solid slab	9.0	4.00E-01	1.78E-01	3.50E-02	6.13E-01	5.66E-01	1.77E-01	1.65E-01	1.61E-01	3.97E-02	7.83E-01	9.48E-01	1.11E+00
	10.5	6.04E-01	2.34E-01	4.61E-02	8.84E-01	8.01E-01	2.27E-01	2.12E-01	2.07E-01	5.10E-02	1.08E+00	1.29E+00	1.50E+00
	12.0	6.89E-01	2.67E-01	5.26E-02	1.01E+00	9.10E-01	2.58E-01	2.40E-01	2.35E-01	5.78E-02	1.23E+00	1.47E+00	1.70E+00
	13.5	7.76E-01	3.01E-01	5.92E-02	1.14E+00	1.02E+00	2.88E-01	2.69E-01	2.62E-01	6.46E-02	1.38E+00	1.64E+00	1.91E+00
	15.0	8.60E-01	3.34E-01	6.58E-02	1.26E+00	1.13E+00	3.19E-01	2.97E-01	2.90E-01	7.14E-02	1.52E+00	1.82E+00	2.11E+00
	16.5	1.19E+00	4.09E-01	8.06E-02	1.68E+00	1.50E+00	3.85E-01	3.60E-01	3.51E-01	8.65E-02	1.97E+00	2.33E+00	2.68E+00
	18.0	1.30E+00	4.46E-01	8.79E-02	1.83E+00	1.63E+00	4.19E-01	3.91E-01	3.82E-01	9.40E-02	2.14E+00	2.53E+00	2.92E+00
Hollow core slab	9.0	4.77E-01	1.63E-01	3.20E-02	6.72E-01	6.68E-01	1.76E-01	1.64E-01	1.61E-01	3.96E-02	8.84E-01	1.05E+00	1.21E+00
	10.5	5.58E-01	1.90E-01	3.74E-02	7.85E-01	7.73E-01	2.00E-01	1.87E-01	1.82E-01	4.49E-02	1.02E+00	1.21E+00	1.39E+00
	12.0	6.36E-01	2.17E-01	4.27E-02	8.95E-01	8.75E-01	2.24E-01	2.09E-01	2.04E-01	5.03E-02	1.15E+00	1.36E+00	1.56E+00
	13.5	9.15E-01	2.78E-01	5.47E-02	1.25E+00	1.19E+00	2.78E-01	2.60E-01	2.53E-01	6.24E-02	1.53E+00	1.79E+00	2.04E+00
	15.0	1.02E+00	3.09E-01	6.08E-02	1.39E+00	1.31E+00	3.06E-01	2.85E-01	2.78E-01	6.85E-02	1.69E+00	1.97E+00	2.25E+00
	16.5	1.12E+00	3.40E-01	6.69E-02	1.52E+00	1.44E+00	3.33E-01	3.10E-01	3.03E-01	7.47E-02	1.85E+00	2.16E+00	2.46E+00
	18.0	1.49E+00	4.16E-01	8.20E-02	1.99E+00	1.85E+00	4.00E-01	3.73E-01	3.64E-01	8.97E-02	2.34E+00	2.71E+00	3.08E+00
Ribbed	9.0	7.18E-01	2.05E-01	4.03E-02	9.63E-01	8.12E-01	1.85E-01	1.73E-01	1.68E-01	4.15E-02	1.04E+00	1.21E+00	1.38E+00

Freshwater eutrophication (kg P eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
slab	10.5	8.57E-01	2.50E-01	4.93E-02	1.16E+00	9.68E-01	2.26E-01	2.10E-01	2.05E-01	5.06E-02	1.24E+00	1.45E+00	1.66E+00
	12.0	1.15E+00	3.09E-01	6.08E-02	1.52E+00	1.28E+00	2.77E-01	2.58E-01	2.52E-01	6.21E-02	1.62E+00	1.88E+00	2.13E+00
	13.5	1.33E+00	3.59E-01	7.07E-02	1.76E+00	1.48E+00	3.22E-01	3.00E-01	2.93E-01	7.21E-02	1.88E+00	2.18E+00	2.47E+00
	15.0	1.68E+00	4.28E-01	8.43E-02	2.19E+00	1.85E+00	3.82E-01	3.56E-01	3.48E-01	8.56E-02	2.32E+00	2.67E+00	3.02E+00
	16.5	2.10E+00	5.06E-01	9.96E-02	2.71E+00	2.29E+00	4.50E-01	4.20E-01	4.10E-01	1.01E-01	2.84E+00	3.26E+00	3.67E+00
	18.0	2.30E+00	5.63E-01	1.11E-01	2.98E+00	2.52E+00	5.01E-01	4.67E-01	4.56E-01	1.12E-01	3.13E+00	3.60E+00	4.05E+00
Double T slab	9.0	3.11E-01	7.54E-02	1.81E-02	4.05E-01	3.83E-01	8.47E-02	7.90E-02	7.71E-02	1.90E-02	4.87E-01	5.66E-01	6.43E-01
	10.5	3.86E-01	9.43E-02	2.27E-02	5.03E-01	4.70E-01	1.05E-01	9.84E-02	9.60E-02	2.37E-02	5.99E-01	6.97E-01	7.93E-01
	12.0	4.64E-01	1.13E-01	2.72E-02	6.05E-01	5.60E-01	1.26E-01	1.18E-01	1.15E-01	2.83E-02	7.14E-01	8.32E-01	9.47E-01
	13.5	5.66E-01	1.40E-01	3.37E-02	7.40E-01	6.74E-01	1.56E-01	1.45E-01	1.42E-01	3.50E-02	8.65E-01	1.01E+00	1.15E+00
	15.0	6.73E-01	1.67E-01	4.01E-02	8.80E-01	7.93E-01	1.86E-01	1.73E-01	1.69E-01	4.16E-02	1.02E+00	1.19E+00	1.36E+00
	16.5	7.97E-01	1.95E-01	4.68E-02	1.04E+00	9.29E-01	2.16E-01	2.01E-01	1.97E-01	4.84E-02	1.19E+00	1.39E+00	1.59E+00
	18.0	9.28E-01	2.23E-01	5.35E-02	1.20E+00	1.07E+00	2.46E-01	2.30E-01	2.24E-01	5.53E-02	1.37E+00	1.60E+00	1.83E+00

Table C-9 Specific environmental impacts of Marine eutrophication

Marine eutrophication (kg N eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Solid slab	9.0	4.88E-02	1.13E-02	2.66E-03	6.28E-02	6.02E-02	1.14E-02	1.05E-02	1.02E-02	3.01E-03	7.46E-02	8.51E-02	9.53E-02
	10.5	6.65E-02	1.49E-02	3.50E-03	8.49E-02	7.95E-02	1.46E-02	1.34E-02	1.31E-02	3.87E-03	9.79E-02	1.11E-01	1.24E-01
	12.0	7.59E-02	1.70E-02	3.99E-03	9.69E-02	9.02E-02	1.66E-02	1.52E-02	1.48E-02	4.39E-03	1.11E-01	1.26E-01	1.41E-01
	13.5	8.54E-02	1.92E-02	4.49E-03	1.09E-01	1.01E-01	1.85E-02	1.70E-02	1.66E-02	4.90E-03	1.24E-01	1.41E-01	1.58E-01
	15.0	9.49E-02	2.13E-02	4.99E-03	1.21E-01	1.12E-01	2.05E-02	1.88E-02	1.83E-02	5.42E-03	1.38E-01	1.56E-01	1.75E-01
	16.5	1.20E-01	2.61E-02	6.11E-03	1.52E-01	1.39E-01	2.48E-02	2.28E-02	2.21E-02	6.56E-03	1.70E-01	1.93E-01	2.15E-01

Marine eutrophication (kg N eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
	18.0	1.31E-01	2.85E-02	6.67E-03	1.66E-01	1.51E-01	2.69E-02	2.48E-02	2.41E-02	7.14E-03	1.85E-01	2.10E-01	2.34E-01
Hollow core slab	9.0	4.79E-02	1.04E-02	2.43E-03	6.07E-02	6.30E-02	1.13E-02	1.04E-02	1.01E-02	3.00E-03	7.74E-02	8.78E-02	9.80E-02
	10.5	5.59E-02	1.21E-02	2.84E-03	7.08E-02	7.20E-02	1.29E-02	1.19E-02	1.15E-02	3.41E-03	8.83E-02	1.00E-01	1.12E-01
	12.0	6.38E-02	1.38E-02	3.24E-03	8.09E-02	8.09E-02	1.44E-02	1.33E-02	1.29E-02	3.82E-03	9.92E-02	1.12E-01	1.25E-01
	13.5	8.47E-02	1.77E-02	4.16E-03	1.07E-01	1.03E-01	1.79E-02	1.65E-02	1.60E-02	4.74E-03	1.26E-01	1.42E-01	1.58E-01
	15.0	9.41E-02	1.97E-02	4.62E-03	1.18E-01	1.14E-01	1.96E-02	1.81E-02	1.76E-02	5.20E-03	1.38E-01	1.57E-01	1.74E-01
	16.5	1.04E-01	2.17E-02	5.08E-03	1.30E-01	1.24E-01	2.14E-02	1.97E-02	1.91E-02	5.67E-03	1.51E-01	1.71E-01	1.90E-01
	18.0	1.30E-01	2.65E-02	6.22E-03	1.63E-01	1.53E-01	2.57E-02	2.37E-02	2.30E-02	6.81E-03	1.85E-01	2.09E-01	2.32E-01
Ribbed slab	9.0	6.37E-02	1.31E-02	3.06E-03	7.98E-02	6.93E-02	1.19E-02	1.09E-02	1.06E-02	3.15E-03	8.44E-02	9.53E-02	1.06E-01
	10.5	7.72E-02	1.60E-02	3.74E-03	9.69E-02	8.39E-02	1.45E-02	1.33E-02	1.30E-02	3.84E-03	1.02E-01	1.16E-01	1.29E-01
	12.0	9.79E-02	1.97E-02	4.61E-03	1.22E-01	1.06E-01	1.78E-02	1.64E-02	1.59E-02	4.71E-03	1.28E-01	1.44E-01	1.60E-01
	13.5	1.14E-01	2.29E-02	5.37E-03	1.42E-01	1.23E-01	2.07E-02	1.90E-02	1.85E-02	5.47E-03	1.49E-01	1.68E-01	1.86E-01
	15.0	1.38E-01	2.73E-02	6.40E-03	1.72E-01	1.48E-01	2.45E-02	2.26E-02	2.19E-02	6.50E-03	1.79E-01	2.02E-01	2.24E-01
	16.5	1.67E-01	3.23E-02	7.56E-03	2.07E-01	1.78E-01	2.89E-02	2.66E-02	2.59E-02	7.66E-03	2.14E-01	2.41E-01	2.67E-01
	18.0	1.85E-01	3.59E-02	8.42E-03	2.29E-01	1.97E-01	3.22E-02	2.96E-02	2.88E-02	8.52E-03	2.38E-01	2.67E-01	2.96E-01
Double T slab	9.0	1.53E-02	4.83E-03	1.38E-03	2.15E-02	1.78E-02	5.44E-03	5.01E-03	4.87E-03	1.44E-03	2.47E-02	2.97E-02	3.45E-02
	10.5	1.90E-02	6.04E-03	1.72E-03	2.68E-02	2.19E-02	6.78E-03	6.24E-03	6.06E-03	1.80E-03	3.05E-02	3.67E-02	4.28E-02
	12.0	2.29E-02	7.26E-03	2.07E-03	3.22E-02	2.62E-02	8.12E-03	7.48E-03	7.26E-03	2.15E-03	3.64E-02	4.39E-02	5.12E-02
	13.5	2.80E-02	8.98E-03	2.56E-03	3.96E-02	3.18E-02	1.00E-02	9.22E-03	8.96E-03	2.65E-03	4.44E-02	5.37E-02	6.26E-02
	15.0	3.34E-02	1.07E-02	3.05E-03	4.71E-02	3.75E-02	1.19E-02	1.10E-02	1.07E-02	3.16E-03	5.26E-02	6.36E-02	7.43E-02
	16.5	3.93E-02	1.25E-02	3.55E-03	5.53E-02	4.39E-02	1.39E-02	1.28E-02	1.24E-02	3.68E-03	6.14E-02	7.42E-02	8.66E-02
	18.0	4.54E-02	1.43E-02	4.06E-03	6.38E-02	5.04E-02	1.58E-02	1.46E-02	1.42E-02	4.19E-03	7.04E-02	8.50E-02	9.92E-02

Table C-10 Specific environmental impacts of Terrestrial ecotoxicity

Terrestrial ecotoxicity (kg 1,4-DCB)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Solid slab	9.0	4.14E+03	1.89E+03	4.15E+03	1.02E+04	5.37E+03	2.58E+03	1.17E+03	6.96E+02	4.70E+03	1.26E+04	1.38E+04	1.45E+04
	10.5	6.02E+03	2.48E+03	5.46E+03	1.40E+04	7.43E+03	3.31E+03	1.50E+03	8.93E+02	6.04E+03	1.68E+04	1.83E+04	1.92E+04
	12.0	6.86E+03	2.83E+03	6.23E+03	1.59E+04	8.44E+03	3.75E+03	1.70E+03	1.01E+03	6.85E+03	1.90E+04	2.07E+04	2.17E+04
	13.5	7.73E+03	3.19E+03	7.01E+03	1.79E+04	9.48E+03	4.19E+03	1.90E+03	1.13E+03	7.65E+03	2.13E+04	2.32E+04	2.44E+04
	15.0	8.58E+03	3.54E+03	7.79E+03	1.99E+04	1.05E+04	4.64E+03	2.10E+03	1.25E+03	8.46E+03	2.36E+04	2.57E+04	2.69E+04
	16.5	1.15E+04	4.34E+03	9.54E+03	2.54E+04	1.36E+04	5.61E+03	2.54E+03	1.51E+03	1.02E+04	2.95E+04	3.20E+04	3.35E+04
	18.0	1.25E+04	4.73E+03	1.04E+04	2.77E+04	1.48E+04	6.10E+03	2.76E+03	1.65E+03	1.11E+04	3.21E+04	3.48E+04	3.65E+04
Hollow core slab	9.0	4.60E+03	1.72E+03	3.79E+03	1.01E+04	6.11E+03	2.57E+03	1.16E+03	6.93E+02	4.69E+03	1.34E+04	1.45E+04	1.52E+04
	10.5	5.38E+03	2.01E+03	4.43E+03	1.18E+04	7.04E+03	2.92E+03	1.32E+03	7.88E+02	5.32E+03	1.53E+04	1.66E+04	1.74E+04
	12.0	6.13E+03	2.30E+03	5.06E+03	1.35E+04	7.95E+03	3.27E+03	1.48E+03	8.82E+02	5.96E+03	1.72E+04	1.87E+04	1.95E+04
	13.5	8.59E+03	2.95E+03	6.49E+03	1.80E+04	1.06E+04	4.05E+03	1.83E+03	1.09E+03	7.39E+03	2.20E+04	2.39E+04	2.50E+04
	15.0	9.53E+03	3.27E+03	7.21E+03	2.00E+04	1.17E+04	4.45E+03	2.01E+03	1.20E+03	8.12E+03	2.43E+04	2.63E+04	2.75E+04
	16.5	1.05E+04	3.60E+03	7.93E+03	2.20E+04	1.28E+04	4.85E+03	2.19E+03	1.31E+03	8.84E+03	2.65E+04	2.87E+04	3.00E+04
	18.0	1.38E+04	4.41E+03	9.71E+03	2.79E+04	1.63E+04	5.82E+03	2.64E+03	1.57E+03	1.06E+04	3.27E+04	3.54E+04	3.70E+04
Ribbed slab	9.0	6.65E+03	2.17E+03	4.78E+03	1.36E+04	7.22E+03	2.69E+03	1.22E+03	7.27E+02	4.92E+03	1.48E+04	1.60E+04	1.68E+04
	10.5	7.97E+03	2.65E+03	5.84E+03	1.65E+04	8.64E+03	3.28E+03	1.49E+03	8.86E+02	5.99E+03	1.79E+04	1.94E+04	2.03E+04
	12.0	1.05E+04	3.27E+03	7.20E+03	2.10E+04	1.13E+04	4.03E+03	1.82E+03	1.09E+03	7.35E+03	2.27E+04	2.45E+04	2.56E+04
	13.5	1.22E+04	3.81E+03	8.38E+03	2.44E+04	1.31E+04	4.68E+03	2.12E+03	1.26E+03	8.55E+03	2.63E+04	2.84E+04	2.97E+04
	15.0	1.52E+04	4.54E+03	9.99E+03	2.97E+04	1.62E+04	5.56E+03	2.52E+03	1.50E+03	1.01E+04	3.19E+04	3.44E+04	3.59E+04
	16.5	1.88E+04	5.36E+03	1.18E+04	3.60E+04	1.99E+04	6.55E+03	2.96E+03	1.77E+03	1.20E+04	3.84E+04	4.14E+04	4.31E+04
	18.0	2.07E+04	5.97E+03	1.31E+04	3.98E+04	2.19E+04	7.29E+03	3.30E+03	1.97E+03	1.33E+04	4.25E+04	4.58E+04	4.77E+04
Double T slab	9.0	2.83E+03	9.57E+02	2.15E+03	5.94E+03	3.38E+03	1.23E+03	5.58E+02	3.33E+02	2.25E+03	6.86E+03	7.42E+03	7.76E+03
	10.5	3.52E+03	1.20E+03	2.69E+03	7.40E+03	4.16E+03	1.54E+03	6.95E+02	4.15E+02	2.80E+03	8.49E+03	9.19E+03	9.60E+03
	12.0	4.23E+03	1.44E+03	3.23E+03	8.89E+03	4.96E+03	1.84E+03	8.32E+02	4.97E+02	3.36E+03	1.02E+04	1.10E+04	1.15E+04

Terrestrial ecotoxicity (kg 1,4-DCB)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
	13.5	5.17E+03	1.78E+03	3.99E+03	1.09E+04	5.99E+03	2.27E+03	1.03E+03	6.13E+02	4.14E+03	1.24E+04	1.34E+04	1.40E+04
	15.0	6.15E+03	2.12E+03	4.75E+03	1.30E+04	7.07E+03	2.70E+03	1.22E+03	7.29E+02	4.93E+03	1.47E+04	1.59E+04	1.67E+04
	16.5	7.26E+03	2.47E+03	5.55E+03	1.53E+04	8.27E+03	3.14E+03	1.42E+03	8.48E+02	5.74E+03	1.71E+04	1.86E+04	1.94E+04
	18.0	8.42E+03	2.83E+03	6.34E+03	1.76E+04	9.52E+03	3.59E+03	1.62E+03	9.68E+02	6.55E+03	1.97E+04	2.13E+04	2.22E+04

Table C-11 Specific environmental impacts of Freshwater ecotoxicity

Freshwater ecotoxicity (kg 1,4-DCB)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Solid slab	9.0	4.37E+01	1.12E+01	7.10E+00	6.20E+01	6.18E+01	1.21E+01	9.69E+00	8.89E+00	8.04E+00	8.20E+01	9.17E+01	1.01E+02
	10.5	6.84E+01	1.48E+01	9.33E+00	9.25E+01	8.96E+01	1.55E+01	1.24E+01	1.14E+01	1.03E+01	1.16E+02	1.28E+02	1.39E+02
	12.0	7.78E+01	1.69E+01	1.07E+01	1.05E+02	1.02E+02	1.76E+01	1.41E+01	1.29E+01	1.17E+01	1.31E+02	1.45E+02	1.58E+02
	13.5	8.78E+01	1.90E+01	1.20E+01	1.19E+02	1.15E+02	1.97E+01	1.58E+01	1.45E+01	1.31E+01	1.48E+02	1.63E+02	1.78E+02
	15.0	9.73E+01	2.11E+01	1.33E+01	1.32E+02	1.27E+02	2.18E+01	1.74E+01	1.60E+01	1.45E+01	1.63E+02	1.81E+02	1.97E+02
	16.5	1.38E+02	2.59E+01	1.63E+01	1.80E+02	1.71E+02	2.64E+01	2.11E+01	1.94E+01	1.75E+01	2.15E+02	2.36E+02	2.56E+02
	18.0	1.51E+02	2.82E+01	1.78E+01	1.97E+02	1.87E+02	2.87E+01	2.30E+01	2.10E+01	1.90E+01	2.34E+02	2.57E+02	2.78E+02
Hollow core slab	9.0	5.54E+01	1.03E+01	6.49E+00	7.22E+01	7.61E+01	1.21E+01	9.66E+00	8.86E+00	8.01E+00	9.61E+01	1.06E+02	1.15E+02
	10.5	6.48E+01	1.20E+01	7.57E+00	8.44E+01	8.84E+01	1.37E+01	1.10E+01	1.01E+01	9.10E+00	1.11E+02	1.22E+02	1.32E+02
	12.0	7.38E+01	1.37E+01	8.65E+00	9.62E+01	1.00E+02	1.53E+01	1.23E+01	1.13E+01	1.02E+01	1.26E+02	1.38E+02	1.49E+02
	13.5	1.09E+02	1.76E+01	1.11E+01	1.37E+02	1.38E+02	1.90E+01	1.52E+01	1.40E+01	1.26E+01	1.70E+02	1.85E+02	1.99E+02
	15.0	1.20E+02	1.95E+01	1.23E+01	1.52E+02	1.53E+02	2.09E+01	1.67E+01	1.53E+01	1.39E+01	1.88E+02	2.05E+02	2.20E+02
	16.5	1.33E+02	2.15E+01	1.36E+01	1.68E+02	1.68E+02	2.28E+01	1.82E+01	1.67E+01	1.51E+01	2.06E+02	2.24E+02	2.41E+02
	18.0	1.80E+02	2.63E+01	1.66E+01	2.23E+02	2.19E+02	2.74E+01	2.19E+01	2.01E+01	1.82E+01	2.64E+02	2.86E+02	3.06E+02
Ribbed	9.0	8.60E+01	1.29E+01	8.17E+00	1.07E+02	9.51E+01	1.27E+01	1.01E+01	9.29E+00	8.41E+00	1.16E+02	1.26E+02	1.36E+02

Freshwater ecotoxicity (kg 1,4-DCB)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
slab	10.5	1.02E+02	1.58E+01	9.98E+00	1.28E+02	1.13E+02	1.54E+01	1.24E+01	1.13E+01	1.02E+01	1.39E+02	1.51E+02	1.62E+02
	12.0	1.39E+02	1.95E+01	1.23E+01	1.71E+02	1.51E+02	1.89E+01	1.52E+01	1.39E+01	1.26E+01	1.83E+02	1.98E+02	2.12E+02
	13.5	1.61E+02	2.27E+01	1.43E+01	1.98E+02	1.75E+02	2.20E+01	1.76E+01	1.61E+01	1.46E+01	2.12E+02	2.29E+02	2.45E+02
	15.0	2.05E+02	2.71E+01	1.71E+01	2.49E+02	2.21E+02	2.61E+01	2.09E+01	1.92E+01	1.73E+01	2.64E+02	2.85E+02	3.04E+02
	16.5	2.58E+02	3.20E+01	2.02E+01	3.10E+02	2.76E+02	3.08E+01	2.46E+01	2.26E+01	2.04E+01	3.27E+02	3.52E+02	3.74E+02
	18.0	2.82E+02	3.56E+01	2.25E+01	3.40E+02	3.02E+02	3.42E+01	2.74E+01	2.51E+01	2.27E+01	3.59E+02	3.86E+02	4.12E+02
Double T slab	9.0	3.43E+01	4.98E+00	3.67E+00	4.30E+01	4.39E+01	5.79E+00	4.64E+00	4.25E+00	3.85E+00	5.35E+01	5.81E+01	6.24E+01
	10.5	4.24E+01	6.22E+00	4.59E+00	5.32E+01	5.36E+01	7.21E+00	5.78E+00	5.30E+00	4.79E+00	6.56E+01	7.13E+01	7.66E+01
	12.0	5.10E+01	7.48E+00	5.51E+00	6.40E+01	6.37E+01	8.64E+00	6.92E+00	6.34E+00	5.74E+00	7.81E+01	8.50E+01	9.14E+01
	13.5	6.20E+01	9.25E+00	6.82E+00	7.80E+01	7.63E+01	1.07E+01	8.54E+00	7.83E+00	7.08E+00	9.40E+01	1.03E+02	1.10E+02
	15.0	7.36E+01	1.10E+01	8.13E+00	9.28E+01	8.96E+01	1.27E+01	1.02E+01	9.32E+00	8.43E+00	1.11E+02	1.21E+02	1.30E+02
	16.5	8.75E+01	1.29E+01	9.49E+00	1.10E+02	1.05E+02	1.48E+01	1.18E+01	1.08E+01	9.81E+00	1.30E+02	1.41E+02	1.52E+02
	18.0	1.02E+02	1.47E+01	1.08E+01	1.28E+02	1.22E+02	1.68E+01	1.35E+01	1.24E+01	1.12E+01	1.50E+02	1.63E+02	1.75E+02

Table C-12 Specific environmental impacts of Marine ecotoxicity

Marine ecotoxicity (kg 1,4-DCB)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Solid slab	9.0	6.20E+01	1.63E+01	1.18E+01	9.00E+01	8.76E+01	1.78E+01	1.38E+01	1.24E+01	1.33E+01	1.19E+02	1.32E+02	1.45E+02
	10.5	9.70E+01	2.14E+01	1.55E+01	1.34E+02	1.27E+02	2.28E+01	1.77E+01	1.60E+01	1.71E+01	1.67E+02	1.85E+02	2.01E+02
	12.0	1.10E+02	2.44E+01	1.77E+01	1.53E+02	1.44E+02	2.59E+01	2.00E+01	1.81E+01	1.94E+01	1.90E+02	2.10E+02	2.28E+02
	13.5	1.25E+02	2.75E+01	1.99E+01	1.72E+02	1.62E+02	2.89E+01	2.24E+01	2.02E+01	2.17E+01	2.13E+02	2.35E+02	2.56E+02
	15.0	1.38E+02	3.05E+01	2.21E+01	1.91E+02	1.80E+02	3.20E+01	2.48E+01	2.24E+01	2.40E+01	2.36E+02	2.61E+02	2.83E+02
	16.5	1.96E+02	3.74E+01	2.71E+01	2.60E+02	2.43E+02	3.87E+01	3.00E+01	2.71E+01	2.90E+01	3.10E+02	3.40E+02	3.67E+02

Marine ecotoxicity (kg 1,4-DCB)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
	18.0	2.14E+02	4.08E+01	2.95E+01	2.84E+02	2.64E+02	4.21E+01	3.26E+01	2.95E+01	3.16E+01	3.38E+02	3.70E+02	4.00E+02
Hollow core slab	9.0	7.86E+01	1.49E+01	1.08E+01	1.04E+02	1.08E+02	1.77E+01	1.37E+01	1.24E+01	1.33E+01	1.39E+02	1.52E+02	1.65E+02
	10.5	9.20E+01	1.73E+01	1.26E+01	1.22E+02	1.25E+02	2.01E+01	1.56E+01	1.41E+01	1.51E+01	1.60E+02	1.76E+02	1.90E+02
	12.0	1.05E+02	1.98E+01	1.43E+01	1.39E+02	1.42E+02	2.25E+01	1.75E+01	1.58E+01	1.69E+01	1.81E+02	1.99E+02	2.15E+02
	13.5	1.54E+02	2.54E+01	1.84E+01	1.98E+02	1.96E+02	2.79E+01	2.16E+01	1.96E+01	2.10E+01	2.45E+02	2.66E+02	2.86E+02
	15.0	1.71E+02	2.82E+01	2.04E+01	2.19E+02	2.17E+02	3.07E+01	2.38E+01	2.15E+01	2.30E+01	2.70E+02	2.94E+02	3.16E+02
	16.5	1.88E+02	3.10E+01	2.25E+01	2.42E+02	2.38E+02	3.34E+01	2.59E+01	2.34E+01	2.51E+01	2.97E+02	3.23E+02	3.46E+02
	18.0	2.55E+02	3.80E+01	2.75E+01	3.20E+02	3.10E+02	4.02E+01	3.11E+01	2.81E+01	3.01E+01	3.80E+02	4.11E+02	4.40E+02
Ribbed slab	9.0	1.22E+02	1.87E+01	1.35E+01	1.54E+02	1.35E+02	1.86E+01	1.44E+01	1.30E+01	1.39E+01	1.67E+02	1.82E+02	1.95E+02
	10.5	1.45E+02	2.28E+01	1.65E+01	1.85E+02	1.60E+02	2.26E+01	1.75E+01	1.58E+01	1.70E+01	2.00E+02	2.17E+02	2.33E+02
	12.0	1.97E+02	2.82E+01	2.04E+01	2.46E+02	2.14E+02	2.78E+01	2.15E+01	1.94E+01	2.09E+01	2.63E+02	2.85E+02	3.04E+02
	13.5	2.28E+02	3.28E+01	2.38E+01	2.85E+02	2.48E+02	3.23E+01	2.50E+01	2.26E+01	2.42E+01	3.05E+02	3.30E+02	3.52E+02
	15.0	2.90E+02	3.91E+01	2.83E+01	3.58E+02	3.13E+02	3.83E+01	2.97E+01	2.68E+01	2.88E+01	3.80E+02	4.09E+02	4.36E+02
	16.5	3.65E+02	4.62E+01	3.35E+01	4.45E+02	3.91E+02	4.52E+01	3.50E+01	3.16E+01	3.39E+01	4.70E+02	5.05E+02	5.36E+02
	18.0	4.00E+02	5.14E+01	3.72E+01	4.89E+02	4.28E+02	5.03E+01	3.89E+01	3.52E+01	3.77E+01	5.16E+02	5.55E+02	5.90E+02
Double T slab	9.0	4.89E+01	7.25E+00	6.09E+00	6.22E+01	6.24E+01	8.50E+00	6.59E+00	5.95E+00	6.38E+00	7.73E+01	8.39E+01	8.98E+01
	10.5	6.04E+01	9.07E+00	7.62E+00	7.71E+01	7.62E+01	1.06E+01	8.21E+00	7.41E+00	7.95E+00	9.47E+01	1.03E+02	1.10E+02
	12.0	7.26E+01	1.09E+01	9.14E+00	9.27E+01	9.07E+01	1.27E+01	9.83E+00	8.88E+00	9.52E+00	1.13E+02	1.23E+02	1.32E+02
	13.5	8.83E+01	1.35E+01	1.13E+01	1.13E+02	1.09E+02	1.57E+01	1.21E+01	1.10E+01	1.17E+01	1.36E+02	1.48E+02	1.59E+02
	15.0	1.05E+02	1.61E+01	1.35E+01	1.34E+02	1.27E+02	1.86E+01	1.44E+01	1.30E+01	1.40E+01	1.60E+02	1.75E+02	1.88E+02
	16.5	1.25E+02	1.87E+01	1.57E+01	1.59E+02	1.50E+02	2.17E+01	1.68E+01	1.52E+01	1.63E+01	1.87E+02	2.04E+02	2.19E+02
	18.0	1.46E+02	2.14E+01	1.80E+01	1.85E+02	1.73E+02	2.47E+01	1.92E+01	1.73E+01	1.86E+01	2.16E+02	2.35E+02	2.53E+02

Table C-13 Specific environmental impacts of Human carcinogenic toxicity

Human carcinogenic toxicity (kg 1,4-DCB)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Solid slab	9.0	1.94E+02	1.56E+01	9.33E+00	2.19E+02	2.81E+02	1.68E+01	1.37E+01	1.26E+01	1.06E+01	3.08E+02	3.22E+02	3.34E+02
	10.5	3.20E+02	2.05E+01	1.23E+01	3.53E+02	4.21E+02	2.16E+01	1.75E+01	1.62E+01	1.36E+01	4.56E+02	4.73E+02	4.90E+02
	12.0	3.64E+02	2.34E+01	1.40E+01	4.02E+02	4.79E+02	2.45E+01	1.99E+01	1.84E+01	1.54E+01	5.19E+02	5.39E+02	5.57E+02
	13.5	4.11E+02	2.64E+01	1.58E+01	4.53E+02	5.40E+02	2.74E+01	2.23E+01	2.05E+01	1.72E+01	5.84E+02	6.06E+02	6.27E+02
	15.0	4.55E+02	2.93E+01	1.75E+01	5.02E+02	5.98E+02	3.03E+01	2.46E+01	2.27E+01	1.90E+01	6.47E+02	6.72E+02	6.94E+02
	16.5	6.69E+02	3.59E+01	2.14E+01	7.27E+02	8.27E+02	3.67E+01	2.98E+01	2.75E+01	2.30E+01	8.87E+02	9.16E+02	9.44E+02
	18.0	7.29E+02	3.92E+01	2.34E+01	7.92E+02	9.01E+02	3.99E+01	3.24E+01	2.99E+01	2.50E+01	9.66E+02	9.98E+02	1.03E+03
Hollow core slab	9.0	2.69E+02	1.43E+01	8.53E+00	2.92E+02	3.65E+02	1.68E+01	1.36E+01	1.26E+01	1.05E+01	3.92E+02	4.06E+02	4.18E+02
	10.5	3.15E+02	1.67E+01	9.95E+00	3.41E+02	4.26E+02	1.91E+01	1.55E+01	1.43E+01	1.20E+01	4.57E+02	4.72E+02	4.87E+02
	12.0	3.58E+02	1.90E+01	1.14E+01	3.89E+02	4.84E+02	2.13E+01	1.73E+01	1.60E+01	1.34E+01	5.19E+02	5.36E+02	5.52E+02
	13.5	5.41E+02	2.44E+01	1.46E+01	5.80E+02	6.83E+02	2.65E+01	2.15E+01	1.98E+01	1.66E+01	7.26E+02	7.48E+02	7.68E+02
	15.0	6.00E+02	2.71E+01	1.62E+01	6.44E+02	7.58E+02	2.91E+01	2.36E+01	2.18E+01	1.82E+01	8.05E+02	8.28E+02	8.50E+02
	16.5	6.61E+02	2.98E+01	1.78E+01	7.09E+02	8.34E+02	3.17E+01	2.57E+01	2.37E+01	1.99E+01	8.85E+02	9.11E+02	9.35E+02
	18.0	9.12E+02	3.65E+01	2.18E+01	9.70E+02	1.10E+03	3.81E+01	3.09E+01	2.85E+01	2.39E+01	1.16E+03	1.19E+03	1.22E+03
Ribbed slab	9.0	4.35E+02	1.80E+01	1.07E+01	4.63E+02	4.73E+02	1.76E+01	1.43E+01	1.32E+01	1.10E+01	5.02E+02	5.16E+02	5.29E+02
	10.5	5.15E+02	2.19E+01	1.31E+01	5.50E+02	5.60E+02	2.15E+01	1.74E+01	1.61E+01	1.35E+01	5.94E+02	6.12E+02	6.28E+02
	12.0	7.10E+02	2.71E+01	1.62E+01	7.53E+02	7.61E+02	2.63E+01	2.14E+01	1.97E+01	1.65E+01	8.04E+02	8.26E+02	8.45E+02
	13.5	8.22E+02	3.15E+01	1.88E+01	8.72E+02	8.80E+02	3.06E+01	2.48E+01	2.29E+01	1.92E+01	9.30E+02	9.55E+02	9.78E+02
	15.0	1.05E+03	3.76E+01	2.24E+01	1.11E+03	1.12E+03	3.63E+01	2.95E+01	2.72E+01	2.28E+01	1.18E+03	1.21E+03	1.24E+03
	16.5	1.34E+03	4.44E+01	2.65E+01	1.41E+03	1.41E+03	4.28E+01	3.48E+01	3.21E+01	2.69E+01	1.48E+03	1.52E+03	1.55E+03
	18.0	1.46E+03	4.94E+01	2.95E+01	1.54E+03	1.54E+03	4.76E+01	3.87E+01	3.57E+01	2.99E+01	1.62E+03	1.66E+03	1.70E+03
Double T slab	9.0	1.63E+02	6.96E+00	4.82E+00	1.75E+02	2.18E+02	8.06E+00	6.54E+00	6.03E+00	5.05E+00	2.31E+02	2.38E+02	2.44E+02
	10.5	2.01E+02	8.71E+00	6.03E+00	2.16E+02	2.65E+02	1.00E+01	8.15E+00	7.52E+00	6.30E+00	2.81E+02	2.89E+02	2.97E+02
	12.0	2.42E+02	1.05E+01	7.25E+00	2.60E+02	3.15E+02	1.20E+01	9.76E+00	9.01E+00	7.54E+00	3.34E+02	3.44E+02	3.53E+02

Human carcinogenic toxicity (kg 1,4-DCB)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
	13.5	2.93E+02	1.29E+01	8.96E+00	3.15E+02	3.75E+02	1.48E+01	1.20E+01	1.11E+01	9.31E+00	3.99E+02	4.11E+02	4.22E+02
	15.0	3.48E+02	1.54E+01	1.07E+01	3.74E+02	4.39E+02	1.77E+01	1.43E+01	1.32E+01	1.11E+01	4.68E+02	4.82E+02	4.95E+02
	16.5	4.15E+02	1.80E+01	1.25E+01	4.46E+02	5.15E+02	2.05E+01	1.67E+01	1.54E+01	1.29E+01	5.49E+02	5.66E+02	5.81E+02
	18.0	4.89E+02	2.06E+01	1.42E+01	5.23E+02	5.98E+02	2.34E+01	1.90E+01	1.76E+01	1.47E+01	6.36E+02	6.55E+02	6.72E+02

Table C-14 Specific environmental impacts of Human non-carcinogenic toxicity

Human non-carcinogenic toxicity (kg 1,4-DCB)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Solid slab	9.0	1.33E+03	3.29E+02	2.57E+02	1.91E+03	1.85E+03	3.63E+02	2.76E+02	2.46E+02	2.91E+02	2.50E+03	2.78E+03	3.03E+03
	10.5	2.07E+03	4.32E+02	3.38E+02	2.84E+03	2.68E+03	4.66E+02	3.54E+02	3.16E+02	3.74E+02	3.52E+03	3.88E+03	4.19E+03
	12.0	2.36E+03	4.94E+02	3.86E+02	3.24E+03	3.05E+03	5.28E+02	4.01E+02	3.59E+02	4.24E+02	4.00E+03	4.40E+03	4.76E+03
	13.5	2.66E+03	5.56E+02	4.34E+02	3.65E+03	3.43E+03	5.91E+02	4.49E+02	4.01E+02	4.74E+02	4.50E+03	4.95E+03	5.35E+03
	15.0	2.95E+03	6.17E+02	4.82E+02	4.05E+03	3.80E+03	6.53E+02	4.96E+02	4.44E+02	5.24E+02	4.98E+03	5.48E+03	5.92E+03
	16.5	4.19E+03	7.56E+02	5.91E+02	5.53E+03	5.13E+03	7.90E+02	6.00E+02	5.37E+02	6.34E+02	6.56E+03	7.16E+03	7.69E+03
	18.0	4.56E+03	8.25E+02	6.44E+02	6.03E+03	5.59E+03	8.60E+02	6.53E+02	5.84E+02	6.90E+02	7.14E+03	7.79E+03	8.38E+03
Hollow core slab	9.0	1.68E+03	3.01E+02	2.35E+02	2.21E+03	2.28E+03	3.62E+02	2.75E+02	2.46E+02	2.90E+02	2.93E+03	3.20E+03	3.45E+03
	10.5	1.96E+03	3.51E+02	2.74E+02	2.59E+03	2.65E+03	4.11E+02	3.12E+02	2.79E+02	3.30E+02	3.39E+03	3.70E+03	3.98E+03
	12.0	2.24E+03	4.01E+02	3.13E+02	2.95E+03	3.00E+03	4.60E+02	3.49E+02	3.12E+02	3.69E+02	3.83E+03	4.18E+03	4.49E+03
	13.5	3.29E+03	5.14E+02	4.01E+02	4.20E+03	4.14E+03	5.71E+02	4.33E+02	3.88E+02	4.58E+02	5.17E+03	5.61E+03	5.99E+03
	15.0	3.65E+03	5.71E+02	4.46E+02	4.66E+03	4.59E+03	6.27E+02	4.76E+02	4.26E+02	5.03E+02	5.72E+03	6.19E+03	6.62E+03
	16.5	4.01E+03	6.28E+02	4.91E+02	5.13E+03	5.04E+03	6.83E+02	5.18E+02	4.64E+02	5.48E+02	6.27E+03	6.79E+03	7.25E+03
	18.0	5.44E+03	7.69E+02	6.01E+02	6.81E+03	6.56E+03	8.21E+02	6.23E+02	5.57E+02	6.58E+02	8.04E+03	8.67E+03	9.22E+03
Ribbed	9.0	2.60E+03	3.79E+02	2.96E+02	3.28E+03	2.85E+03	3.79E+02	2.88E+02	2.58E+02	3.04E+02	3.54E+03	3.82E+03	4.08E+03

Human non-carcinogenic toxicity (kg 1,4-DCB)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
slab	10.5	3.10E+03	4.62E+02	3.61E+02	3.92E+03	3.39E+03	4.63E+02	3.51E+02	3.14E+02	3.71E+02	4.22E+03	4.57E+03	4.89E+03
	12.0	4.21E+03	5.71E+02	4.46E+02	5.22E+03	4.54E+03	5.68E+02	4.31E+02	3.85E+02	4.55E+02	5.56E+03	6.00E+03	6.38E+03
	13.5	4.87E+03	6.64E+02	5.19E+02	6.05E+03	5.25E+03	6.60E+02	5.01E+02	4.48E+02	5.29E+02	6.44E+03	6.94E+03	7.39E+03
	15.0	6.19E+03	7.91E+02	6.18E+02	7.60E+03	6.62E+03	7.83E+02	5.95E+02	5.32E+02	6.28E+02	8.03E+03	8.63E+03	9.16E+03
	16.5	7.80E+03	9.35E+02	7.31E+02	9.46E+03	8.28E+03	9.23E+02	7.01E+02	6.27E+02	7.40E+02	9.94E+03	1.06E+04	1.13E+04
	18.0	8.54E+03	1.04E+03	8.13E+02	1.04E+04	9.07E+03	1.03E+03	7.80E+02	6.97E+02	8.23E+02	1.09E+04	1.17E+04	1.24E+04
Double T slab	9.0	1.03E+03	1.48E+02	1.33E+02	1.31E+03	1.32E+03	1.74E+02	1.32E+02	1.18E+02	1.39E+02	1.63E+03	1.77E+03	1.88E+03
	10.5	1.28E+03	1.85E+02	1.66E+02	1.63E+03	1.61E+03	2.16E+02	1.64E+02	1.47E+02	1.73E+02	2.00E+03	2.17E+03	2.31E+03
	12.0	1.54E+03	2.22E+02	2.00E+02	1.96E+03	1.92E+03	2.59E+02	1.97E+02	1.76E+02	2.08E+02	2.39E+03	2.58E+03	2.76E+03
	13.5	1.87E+03	2.74E+02	2.47E+02	2.39E+03	2.30E+03	3.20E+02	2.43E+02	2.17E+02	2.56E+02	2.87E+03	3.12E+03	3.33E+03
	15.0	2.22E+03	3.27E+02	2.94E+02	2.84E+03	2.70E+03	3.80E+02	2.89E+02	2.58E+02	3.05E+02	3.38E+03	3.67E+03	3.93E+03
	16.5	2.64E+03	3.81E+02	3.43E+02	3.36E+03	3.16E+03	4.43E+02	3.36E+02	3.01E+02	3.55E+02	3.96E+03	4.30E+03	4.60E+03
	18.0	3.08E+03	4.36E+02	3.93E+02	3.91E+03	3.66E+03	5.05E+02	3.84E+02	3.43E+02	4.05E+02	4.57E+03	4.95E+03	5.30E+03

Table C-15 Specific environmental impacts of Land use

Land use (m^2 a crop eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Solid slab	9.0	3.76E+01	7.94E+00	1.33E+01	5.88E+01	4.70E+01	1.01E+01	5.55E+00	4.05E+00	1.50E+01	7.21E+01	7.76E+01	8.17E+01
	10.5	5.09E+01	1.04E+01	1.75E+01	7.88E+01	6.17E+01	1.29E+01	7.13E+00	5.20E+00	1.93E+01	9.39E+01	1.01E+02	1.06E+02
	12.0	5.81E+01	1.19E+01	1.99E+01	9.00E+01	7.00E+01	1.47E+01	8.08E+00	5.89E+00	2.19E+01	1.07E+02	1.15E+02	1.20E+02
	13.5	6.54E+01	1.34E+01	2.24E+01	1.01E+02	7.83E+01	1.64E+01	9.04E+00	6.59E+00	2.45E+01	1.19E+02	1.28E+02	1.35E+02
	15.0	7.27E+01	1.49E+01	2.49E+01	1.13E+02	8.66E+01	1.81E+01	9.99E+00	7.28E+00	2.71E+01	1.32E+02	1.42E+02	1.49E+02
	16.5	9.16E+01	1.83E+01	3.05E+01	1.40E+02	1.07E+02	2.19E+01	1.21E+01	8.82E+00	3.28E+01	1.62E+02	1.74E+02	1.83E+02

Land use (m² a crop eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
	18.0	9.99E+01	1.99E+01	3.33E+01	1.53E+02	1.17E+02	2.38E+01	1.32E+01	9.59E+00	3.56E+01	1.76E+02	1.89E+02	1.99E+02
Hollow core slab	9.0	3.65E+01	7.26E+00	1.21E+01	5.59E+01	4.87E+01	1.00E+01	5.53E+00	4.03E+00	1.50E+01	7.38E+01	7.93E+01	8.33E+01
	10.5	4.26E+01	8.47E+00	1.42E+01	6.52E+01	5.56E+01	1.14E+01	6.29E+00	4.58E+00	1.70E+01	8.41E+01	9.03E+01	9.49E+01
	12.0	4.86E+01	9.68E+00	1.62E+01	7.45E+01	6.24E+01	1.28E+01	7.04E+00	5.13E+00	1.91E+01	9.43E+01	1.01E+02	1.06E+02
	13.5	6.42E+01	1.24E+01	2.07E+01	9.74E+01	7.93E+01	1.58E+01	8.73E+00	6.36E+00	2.37E+01	1.19E+02	1.28E+02	1.34E+02
	15.0	7.13E+01	1.38E+01	2.31E+01	1.08E+02	8.73E+01	1.74E+01	9.58E+00	6.99E+00	2.60E+01	1.31E+02	1.40E+02	1.47E+02
	16.5	7.85E+01	1.52E+01	2.54E+01	1.19E+02	9.53E+01	1.89E+01	1.04E+01	7.61E+00	2.83E+01	1.42E+02	1.53E+02	1.61E+02
	18.0	9.84E+01	1.86E+01	3.11E+01	1.48E+02	1.17E+02	2.28E+01	1.26E+01	9.15E+00	3.40E+01	1.74E+02	1.86E+02	1.95E+02
Ribbed slab	9.0	4.81E+01	9.14E+00	1.53E+01	7.25E+01	5.32E+01	1.05E+01	5.80E+00	4.23E+00	1.57E+01	7.94E+01	8.52E+01	8.95E+01
	10.5	5.84E+01	1.12E+01	1.87E+01	8.82E+01	6.44E+01	1.28E+01	7.08E+00	5.16E+00	1.92E+01	9.64E+01	1.04E+02	1.09E+02
	12.0	7.38E+01	1.38E+01	2.30E+01	1.11E+02	8.08E+01	1.57E+01	8.68E+00	6.33E+00	2.35E+01	1.20E+02	1.29E+02	1.35E+02
	13.5	8.57E+01	1.60E+01	2.68E+01	1.29E+02	9.37E+01	1.83E+01	1.01E+01	7.35E+00	2.73E+01	1.39E+02	1.49E+02	1.57E+02
	15.0	1.04E+02	1.91E+01	3.19E+01	1.55E+02	1.13E+02	2.17E+01	1.20E+01	8.73E+00	3.25E+01	1.67E+02	1.79E+02	1.88E+02
	16.5	1.25E+02	2.26E+01	3.78E+01	1.85E+02	1.35E+02	2.56E+01	1.41E+01	1.03E+01	3.83E+01	1.99E+02	2.13E+02	2.24E+02
	18.0	1.38E+02	2.51E+01	4.20E+01	2.06E+02	1.50E+02	2.85E+01	1.57E+01	1.14E+01	4.26E+01	2.21E+02	2.37E+02	2.48E+02
Double T slab	9.0	1.45E+01	3.85E+00	6.87E+00	2.52E+01	1.63E+01	4.82E+00	2.66E+00	1.94E+00	7.20E+00	2.84E+01	3.10E+01	3.29E+01
	10.5	1.81E+01	4.82E+00	8.59E+00	3.15E+01	2.02E+01	6.00E+00	3.31E+00	2.41E+00	8.97E+00	3.52E+01	3.85E+01	4.09E+01
	12.0	2.17E+01	5.79E+00	1.03E+01	3.79E+01	2.41E+01	7.19E+00	3.96E+00	2.89E+00	1.07E+01	4.21E+01	4.60E+01	4.89E+01
	13.5	2.67E+01	7.16E+00	1.28E+01	4.67E+01	2.95E+01	8.87E+00	4.89E+00	3.57E+00	1.33E+01	5.16E+01	5.65E+01	6.00E+01
	15.0	3.18E+01	8.53E+00	1.52E+01	5.56E+01	3.49E+01	1.06E+01	5.82E+00	4.24E+00	1.58E+01	6.12E+01	6.70E+01	7.13E+01
	16.5	3.74E+01	9.95E+00	1.77E+01	6.51E+01	4.07E+01	1.23E+01	6.77E+00	4.94E+00	1.84E+01	7.13E+01	7.81E+01	8.30E+01
	18.0	4.31E+01	1.14E+01	2.03E+01	7.47E+01	4.67E+01	1.40E+01	7.73E+00	5.63E+00	2.09E+01	8.16E+01	8.94E+01	9.50E+01

Table C-16 Specific environmental impacts of Mineral resource scarcity

Mineral resource scarcity (kg Cu eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Solid slab	9.0	1.82E+01	5.40E-01	8.89E-01	1.97E+01	2.47E+01	6.97E-01	3.95E-01	2.94E-01	1.01E+00	2.64E+01	2.68E+01	2.71E+01
	10.5	2.84E+01	7.09E-01	1.17E+00	3.03E+01	3.59E+01	8.95E-01	5.07E-01	3.78E-01	1.29E+00	3.81E+01	3.86E+01	3.90E+01
	12.0	3.24E+01	8.11E-01	1.34E+00	3.45E+01	4.08E+01	1.02E+00	5.75E-01	4.28E-01	1.47E+00	4.33E+01	4.39E+01	4.43E+01
	13.5	3.65E+01	9.12E-01	1.50E+00	3.89E+01	4.59E+01	1.13E+00	6.43E-01	4.79E-01	1.64E+00	4.87E+01	4.94E+01	4.98E+01
	15.0	4.05E+01	1.01E+00	1.67E+00	4.31E+01	5.09E+01	1.25E+00	7.11E-01	5.30E-01	1.81E+00	5.39E+01	5.46E+01	5.52E+01
	16.5	5.73E+01	1.24E+00	2.05E+00	6.06E+01	6.87E+01	1.52E+00	8.60E-01	6.41E-01	2.19E+00	7.24E+01	7.33E+01	7.39E+01
	18.0	6.25E+01	1.35E+00	2.23E+00	6.61E+01	7.49E+01	1.65E+00	9.36E-01	6.97E-01	2.39E+00	7.89E+01	7.98E+01	8.05E+01
Hollow core slab	9.0	2.30E+01	4.93E-01	8.13E-01	2.43E+01	3.05E+01	6.95E-01	3.94E-01	2.93E-01	1.00E+00	3.22E+01	3.26E+01	3.29E+01
	10.5	2.69E+01	5.76E-01	9.49E-01	2.84E+01	3.55E+01	7.90E-01	4.47E-01	3.33E-01	1.14E+00	3.74E+01	3.78E+01	3.82E+01
	12.0	3.06E+01	6.58E-01	1.08E+00	3.24E+01	4.02E+01	8.84E-01	5.01E-01	3.73E-01	1.28E+00	4.24E+01	4.29E+01	4.32E+01
	13.5	4.49E+01	8.43E-01	1.39E+00	4.71E+01	5.56E+01	1.10E+00	6.21E-01	4.63E-01	1.58E+00	5.82E+01	5.89E+01	5.93E+01
	15.0	4.98E+01	9.37E-01	1.54E+00	5.23E+01	6.15E+01	1.20E+00	6.82E-01	5.08E-01	1.74E+00	6.45E+01	6.52E+01	6.57E+01
	16.5	5.49E+01	1.03E+00	1.70E+00	5.76E+01	6.76E+01	1.31E+00	7.43E-01	5.53E-01	1.90E+00	7.08E+01	7.16E+01	7.21E+01
	18.0	7.43E+01	1.26E+00	2.08E+00	7.76E+01	8.81E+01	1.58E+00	8.93E-01	6.65E-01	2.28E+00	9.19E+01	9.28E+01	9.35E+01
Ribbed slab	9.0	3.56E+01	6.21E-01	1.02E+00	3.72E+01	3.83E+01	7.29E-01	4.13E-01	3.08E-01	1.05E+00	4.00E+01	4.04E+01	4.08E+01
	10.5	4.23E+01	7.59E-01	1.25E+00	4.43E+01	4.54E+01	8.89E-01	5.03E-01	3.75E-01	1.28E+00	4.76E+01	4.81E+01	4.85E+01
	12.0	5.74E+01	9.37E-01	1.54E+00	5.99E+01	6.09E+01	1.09E+00	6.18E-01	4.60E-01	1.58E+00	6.36E+01	6.42E+01	6.47E+01
	13.5	6.65E+01	1.09E+00	1.80E+00	6.94E+01	7.05E+01	1.27E+00	7.18E-01	5.35E-01	1.83E+00	7.36E+01	7.43E+01	7.48E+01
	15.0	8.45E+01	1.30E+00	2.14E+00	8.79E+01	8.89E+01	1.50E+00	8.52E-01	6.35E-01	2.17E+00	9.26E+01	9.34E+01	9.40E+01
	16.5	1.06E+02	1.53E+00	2.53E+00	1.10E+02	1.11E+02	1.77E+00	1.00E+00	7.48E-01	2.56E+00	1.15E+02	1.16E+02	1.17E+02
	18.0	1.16E+02	1.71E+00	2.81E+00	1.21E+02	1.22E+02	1.97E+00	1.12E+00	8.32E-01	2.85E+00	1.27E+02	1.28E+02	1.29E+02
Double T slab	9.0	1.41E+01	2.70E-01	4.60E-01	1.48E+01	1.79E+01	3.34E-01	1.89E-01	1.41E-01	4.82E-01	1.88E+01	1.90E+01	1.91E+01
	10.5	1.74E+01	3.37E-01	5.75E-01	1.83E+01	2.19E+01	4.16E-01	2.35E-01	1.75E-01	6.01E-01	2.29E+01	2.32E+01	2.33E+01
	12.0	2.09E+01	4.05E-01	6.91E-01	2.20E+01	2.61E+01	4.98E-01	2.82E-01	2.10E-01	7.19E-01	2.73E+01	2.76E+01	2.78E+01

Mineral resource scarcity (kg Cu eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
	13.5	2.54E+01	5.01E-01	8.55E-01	2.67E+01	3.12E+01	6.14E-01	3.48E-01	2.59E-01	8.88E-01	3.27E+01	3.31E+01	3.33E+01
	15.0	3.02E+01	5.97E-01	1.02E+00	3.18E+01	3.67E+01	7.31E-01	4.14E-01	3.08E-01	1.06E+00	3.85E+01	3.89E+01	3.92E+01
	16.5	3.59E+01	6.97E-01	1.19E+00	3.77E+01	4.30E+01	8.51E-01	4.82E-01	3.59E-01	1.23E+00	4.51E+01	4.56E+01	4.59E+01
	18.0	4.19E+01	7.96E-01	1.36E+00	4.41E+01	4.97E+01	9.71E-01	5.50E-01	4.10E-01	1.40E+00	5.21E+01	5.27E+01	5.31E+01

Table C-17 Specific environmental impacts of Fossil resource scarcity

Fossil resource scarcity (kg oil eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Solid slab	9.0	2.13E+02	1.47E+02	1.32E+02	4.91E+02	3.34E+02	1.68E+02	1.23E+02	1.08E+02	1.49E+02	6.52E+02	7.75E+02	8.83E+02
	10.5	2.98E+02	1.93E+02	1.73E+02	6.64E+02	4.46E+02	2.16E+02	1.58E+02	1.39E+02	1.92E+02	8.54E+02	1.01E+03	1.15E+03
	12.0	3.40E+02	2.20E+02	1.98E+02	7.59E+02	5.06E+02	2.45E+02	1.79E+02	1.58E+02	2.18E+02	9.69E+02	1.15E+03	1.31E+03
	13.5	3.83E+02	2.48E+02	2.23E+02	8.54E+02	5.67E+02	2.74E+02	2.01E+02	1.76E+02	2.43E+02	1.08E+03	1.28E+03	1.46E+03
	15.0	4.25E+02	2.75E+02	2.48E+02	9.48E+02	6.28E+02	3.02E+02	2.22E+02	1.95E+02	2.69E+02	1.20E+03	1.42E+03	1.62E+03
	16.5	5.53E+02	3.37E+02	3.03E+02	1.19E+03	7.89E+02	3.66E+02	2.68E+02	2.36E+02	3.25E+02	1.48E+03	1.75E+03	1.98E+03
	18.0	6.03E+02	3.68E+02	3.31E+02	1.30E+03	8.58E+02	3.98E+02	2.92E+02	2.57E+02	3.54E+02	1.61E+03	1.90E+03	2.16E+03
Hollow core slab	9.0	2.21E+02	1.34E+02	1.21E+02	4.75E+02	3.57E+02	1.67E+02	1.23E+02	1.08E+02	1.49E+02	6.74E+02	7.96E+02	9.04E+02
	10.5	2.58E+02	1.56E+02	1.41E+02	5.55E+02	4.09E+02	1.90E+02	1.40E+02	1.23E+02	1.69E+02	7.68E+02	9.08E+02	1.03E+03
	12.0	2.94E+02	1.79E+02	1.61E+02	6.34E+02	4.60E+02	2.13E+02	1.56E+02	1.37E+02	1.89E+02	8.62E+02	1.02E+03	1.16E+03
	13.5	4.00E+02	2.29E+02	2.06E+02	8.36E+02	5.94E+02	2.64E+02	1.94E+02	1.70E+02	2.35E+02	1.09E+03	1.29E+03	1.46E+03
	15.0	4.45E+02	2.55E+02	2.29E+02	9.28E+02	6.54E+02	2.90E+02	2.13E+02	1.87E+02	2.58E+02	1.20E+03	1.41E+03	1.60E+03
	16.5	4.89E+02	2.80E+02	2.52E+02	1.02E+03	7.15E+02	3.16E+02	2.32E+02	2.04E+02	2.81E+02	1.31E+03	1.54E+03	1.75E+03
	18.0	6.28E+02	3.43E+02	3.09E+02	1.28E+03	8.87E+02	3.80E+02	2.79E+02	2.45E+02	3.38E+02	1.61E+03	1.88E+03	2.13E+03
Ribbed	9.0	3.05E+02	1.69E+02	1.52E+02	6.26E+02	4.00E+02	1.76E+02	1.29E+02	1.13E+02	1.56E+02	7.32E+02	8.61E+02	9.74E+02

Fossil resource scarcity (kg oil eq)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
slab	10.5	3.68E+02	2.06E+02	1.85E+02	7.60E+02	4.83E+02	2.14E+02	1.57E+02	1.38E+02	1.90E+02	8.88E+02	1.04E+03	1.18E+03
	12.0	4.76E+02	2.54E+02	2.29E+02	9.59E+02	6.14E+02	2.63E+02	1.93E+02	1.69E+02	2.34E+02	1.11E+03	1.30E+03	1.47E+03
	13.5	5.52E+02	2.96E+02	2.66E+02	1.11E+03	7.12E+02	3.05E+02	2.24E+02	1.97E+02	2.72E+02	1.29E+03	1.51E+03	1.71E+03
	15.0	6.80E+02	3.53E+02	3.17E+02	1.35E+03	8.66E+02	3.63E+02	2.66E+02	2.34E+02	3.22E+02	1.55E+03	1.82E+03	2.05E+03
	16.5	8.30E+02	4.17E+02	3.75E+02	1.62E+03	1.05E+03	4.27E+02	3.13E+02	2.75E+02	3.80E+02	1.85E+03	2.17E+03	2.44E+03
	18.0	9.16E+02	4.64E+02	4.17E+02	1.80E+03	1.16E+03	4.75E+02	3.49E+02	3.06E+02	4.23E+02	2.06E+03	2.40E+03	2.71E+03
Double T slab	9.0	1.73E+02	6.81E+01	6.82E+01	3.09E+02	1.94E+02	8.04E+01	5.90E+01	5.18E+01	7.15E+01	3.46E+02	4.05E+02	4.57E+02
	10.5	2.15E+02	8.52E+01	8.53E+01	3.86E+02	2.40E+02	1.00E+02	7.34E+01	6.45E+01	8.91E+01	4.29E+02	5.03E+02	5.67E+02
	12.0	2.59E+02	1.02E+02	1.02E+02	4.64E+02	2.87E+02	1.20E+02	8.80E+01	7.73E+01	1.07E+02	5.14E+02	6.02E+02	6.79E+02
	13.5	3.18E+02	1.27E+02	1.27E+02	5.72E+02	3.50E+02	1.48E+02	1.09E+02	9.54E+01	1.32E+02	6.30E+02	7.39E+02	8.34E+02
	15.0	3.79E+02	1.51E+02	1.51E+02	6.81E+02	4.15E+02	1.76E+02	1.29E+02	1.14E+02	1.57E+02	7.48E+02	8.77E+02	9.90E+02
	16.5	4.45E+02	1.76E+02	1.76E+02	7.97E+02	4.84E+02	2.05E+02	1.50E+02	1.32E+02	1.82E+02	8.71E+02	1.02E+03	1.15E+03
	18.0	5.12E+02	2.01E+02	2.01E+02	9.15E+02	5.55E+02	2.34E+02	1.72E+02	1.51E+02	2.08E+02	9.97E+02	1.17E+03	1.32E+03

Table C-18 Specific environmental impacts of Water consumption

Water consumption (m^3)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Solid slab	9.0	1.65E+01	1.11E+00	1.06E+00	1.87E+01	2.14E+01	1.27E+00	9.15E-01	7.95E-01	1.20E+00	2.39E+01	2.48E+01	2.56E+01
	10.5	2.22E+01	1.46E+00	1.39E+00	2.50E+01	2.79E+01	1.64E+00	1.17E+00	1.02E+00	1.54E+00	3.11E+01	3.23E+01	3.33E+01
	12.0	2.53E+01	1.66E+00	1.59E+00	2.86E+01	3.17E+01	1.86E+00	1.33E+00	1.16E+00	1.75E+00	3.53E+01	3.66E+01	3.78E+01
	13.5	2.85E+01	1.87E+00	1.79E+00	3.22E+01	3.54E+01	2.07E+00	1.49E+00	1.29E+00	1.95E+00	3.95E+01	4.10E+01	4.23E+01
	15.0	3.16E+01	2.08E+00	1.99E+00	3.57E+01	3.92E+01	2.29E+00	1.65E+00	1.43E+00	2.16E+00	4.36E+01	4.53E+01	4.67E+01
	16.5	3.96E+01	2.55E+00	2.43E+00	4.46E+01	4.82E+01	2.78E+00	1.99E+00	1.73E+00	2.61E+00	5.36E+01	5.56E+01	5.73E+01

Water consumption (m³)													
System	Module length (m)	CCS				DCS							
		E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
	18.0	4.32E+01	2.78E+00	2.65E+00	4.86E+01	5.25E+01	3.02E+00	2.17E+00	1.88E+00	2.84E+00	5.83E+01	6.05E+01	6.24E+01
Hollow core slab	9.0	1.58E+01	1.01E+00	9.68E-01	1.78E+01	2.20E+01	1.27E+00	9.12E-01	7.92E-01	1.19E+00	2.44E+01	2.53E+01	2.61E+01
	10.5	1.84E+01	1.18E+00	1.13E+00	2.07E+01	2.50E+01	1.44E+00	1.04E+00	9.00E-01	1.36E+00	2.78E+01	2.89E+01	2.98E+01
	12.0	2.10E+01	1.35E+00	1.29E+00	2.37E+01	2.81E+01	1.62E+00	1.16E+00	1.01E+00	1.52E+00	3.12E+01	3.24E+01	3.34E+01
	13.5	2.76E+01	1.73E+00	1.65E+00	3.10E+01	3.55E+01	2.00E+00	1.44E+00	1.25E+00	1.89E+00	3.93E+01	4.08E+01	4.20E+01
	15.0	3.06E+01	1.92E+00	1.84E+00	3.44E+01	3.90E+01	2.20E+00	1.58E+00	1.37E+00	2.07E+00	4.33E+01	4.48E+01	4.62E+01
	16.5	3.37E+01	2.11E+00	2.02E+00	3.78E+01	4.25E+01	2.40E+00	1.72E+00	1.50E+00	2.26E+00	4.72E+01	4.89E+01	5.04E+01
	18.0	4.21E+01	2.59E+00	2.48E+00	4.71E+01	5.19E+01	2.88E+00	2.07E+00	1.80E+00	2.71E+00	5.75E+01	5.95E+01	6.13E+01
Ribbed slab	9.0	2.06E+01	1.27E+00	1.22E+00	2.31E+01	2.37E+01	1.33E+00	9.57E-01	8.31E-01	1.25E+00	2.63E+01	2.73E+01	2.81E+01
	10.5	2.50E+01	1.56E+00	1.49E+00	2.81E+01	2.88E+01	1.62E+00	1.17E+00	1.01E+00	1.53E+00	3.19E+01	3.31E+01	3.41E+01
	12.0	3.15E+01	1.92E+00	1.84E+00	3.52E+01	3.59E+01	1.99E+00	1.43E+00	1.24E+00	1.88E+00	3.98E+01	4.12E+01	4.24E+01
	13.5	3.66E+01	2.24E+00	2.14E+00	4.09E+01	4.17E+01	2.32E+00	1.66E+00	1.44E+00	2.18E+00	4.62E+01	4.78E+01	4.93E+01
	15.0	4.42E+01	2.66E+00	2.55E+00	4.94E+01	5.00E+01	2.75E+00	1.97E+00	1.72E+00	2.59E+00	5.54E+01	5.73E+01	5.91E+01
	16.5	5.29E+01	3.15E+00	3.01E+00	5.91E+01	5.97E+01	3.24E+00	2.33E+00	2.02E+00	3.05E+00	6.60E+01	6.83E+01	7.03E+01
	18.0	5.87E+01	3.50E+00	3.35E+00	6.55E+01	6.62E+01	3.61E+00	2.59E+00	2.25E+00	3.39E+00	7.32E+01	7.57E+01	7.80E+01
Double T slab	9.0	9.33E+00	5.14E-01	5.47E-01	1.04E+01	1.01E+01	6.10E-01	4.38E-01	3.80E-01	5.74E-01	1.13E+01	1.18E+01	1.21E+01
	10.5	1.16E+01	6.43E-01	6.85E-01	1.30E+01	1.26E+01	7.60E-01	5.45E-01	4.74E-01	7.15E-01	1.40E+01	1.46E+01	1.51E+01
	12.0	1.40E+01	7.72E-01	8.22E-01	1.56E+01	1.50E+01	9.10E-01	6.53E-01	5.68E-01	8.56E-01	1.68E+01	1.75E+01	1.80E+01
	13.5	1.73E+01	9.55E-01	1.02E+00	1.92E+01	1.85E+01	1.12E+00	8.06E-01	7.00E-01	1.06E+00	2.06E+01	2.14E+01	2.21E+01
	15.0	2.06E+01	1.14E+00	1.21E+00	2.29E+01	2.19E+01	1.34E+00	9.59E-01	8.33E-01	1.26E+00	2.45E+01	2.55E+01	2.63E+01
	16.5	2.40E+01	1.33E+00	1.41E+00	2.68E+01	2.55E+01	1.55E+00	1.12E+00	9.70E-01	1.46E+00	2.85E+01	2.97E+01	3.06E+01
	18.0	2.76E+01	1.52E+00	1.62E+00	3.07E+01	2.92E+01	1.77E+00	1.27E+00	1.11E+00	1.67E+00	3.26E+01	3.39E+01	3.50E+01

Appendix D: p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) for different Slab types at 9.0 m

Table D-1 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Global warming

Global warming (kg CO_2 eq)											
Project		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Solid Slab	$E^{DCS}_{expected}$	3.09E+03	2.83E+03	2.57E+03	2.34E+03	2.13E+03	1.95E+03	1.78E+03	1.64E+03	1.51E+03	1.40E+03	1.31E+03
	$E^{CCS}_{expected}$	2.32E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.329	1.216	1.108	1.008	0.918	0.838	0.767	0.705	0.651	0.604	0.563
Hollow Core Slab	$E^{DCS}_{expected}$	3.17E+03	2.90E+03	2.64E+03	2.40E+03	2.19E+03	1.99E+03	1.82E+03	1.68E+03	1.55E+03	1.43E+03	1.33E+03
	$E^{CCS}_{expected}$	2.23E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.424	1.302	1.186	1.078	0.981	0.895	0.819	0.753	0.694	0.643	0.599
Ribbed Slab	$E^{DCS}_{expected}$	3.43E+03	3.14E+03	2.85E+03	2.59E+03	2.36E+03	2.15E+03	1.97E+03	1.81E+03	1.66E+03	1.54E+03	1.43E+03
	$E^{CCS}_{expected}$	2.91E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.177	1.076	0.979	0.890	0.809	0.738	0.675	0.619	0.571	0.529	0.492
Double T Slab	$E^{DCS}_{expected}$	1.65E+03	1.51E+03	1.37E+03	1.25E+03	1.13E+03	1.03E+03	9.41E+02	8.63E+02	7.95E+02	7.35E+02	6.83E+02
	$E^{CCS}_{expected}$	1.50E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.101	1.006	0.915	0.831	0.755	0.687	0.628	0.576	0.530	0.490	0.456

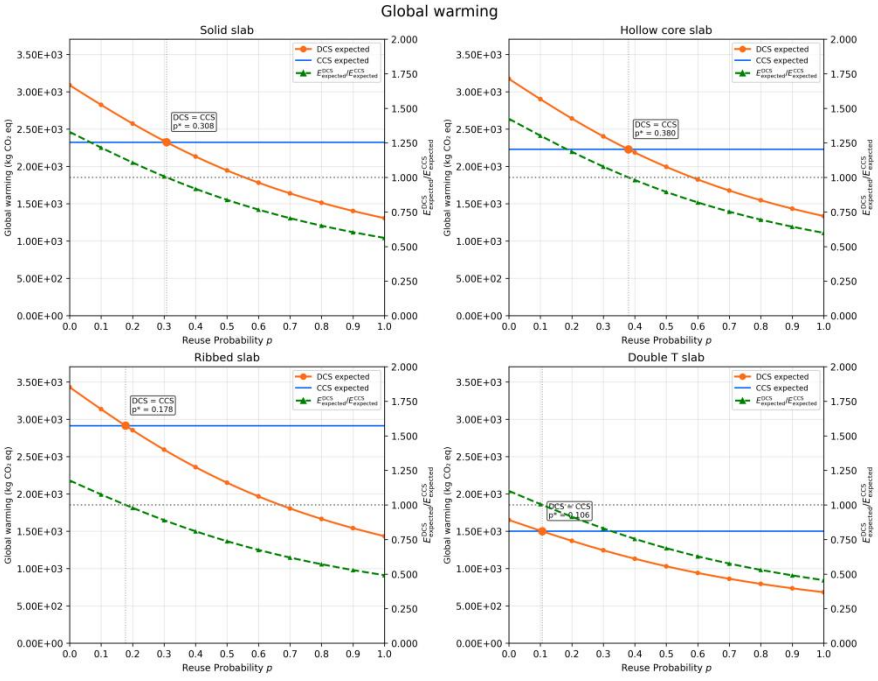


Figure D-1 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Global warming

Table D-2 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Stratospheric ozone depletion

Stratospheric ozone depletion (kg CFC11 eq)											
Project		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Solid Slab	$E^{DCS}_{expected}$	1.14E-03	1.04E-03	9.51E-04	8.67E-04	7.91E-04	7.23E-04	6.63E-04	6.10E-04	5.64E-04	5.24E-04	4.89E-04
	$E^{CCS}_{expected}$	7.95E-04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.432	1.312	1.197	1.091	0.995	0.909	0.834	0.768	0.710	0.659	0.615
Hollow Core Slab	$E^{DCS}_{expected}$	1.16E-03	1.06E-03	9.71E-04	8.85E-04	8.06E-04	7.37E-04	6.75E-04	6.21E-04	5.74E-04	5.33E-04	4.97E-04
	$E^{CCS}_{expected}$	7.58E-04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.534	1.405	1.281	1.167	1.064	0.972	0.891	0.820	0.757	0.703	0.655
Ribbed Slab	$E^{DCS}_{expected}$	1.25E-03	1.14E-03	1.04E-03	9.50E-04	8.66E-04	7.90E-04	7.24E-04	6.66E-04	6.15E-04	5.70E-04	5.31E-04
	$E^{CCS}_{expected}$	9.88E-04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.266	1.159	1.057	0.962	0.876	0.800	0.733	0.674	0.622	0.577	0.538
Double T Slab	$E^{DCS}_{expected}$	5.51E-04	5.05E-04	4.61E-04	4.20E-04	3.83E-04	3.50E-04	3.21E-04	2.95E-04	2.73E-04	2.53E-04	2.36E-04
	$E^{CCS}_{expected}$	4.95E-04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.113	1.020	0.930	0.848	0.773	0.706	0.647	0.596	0.551	0.511	0.477

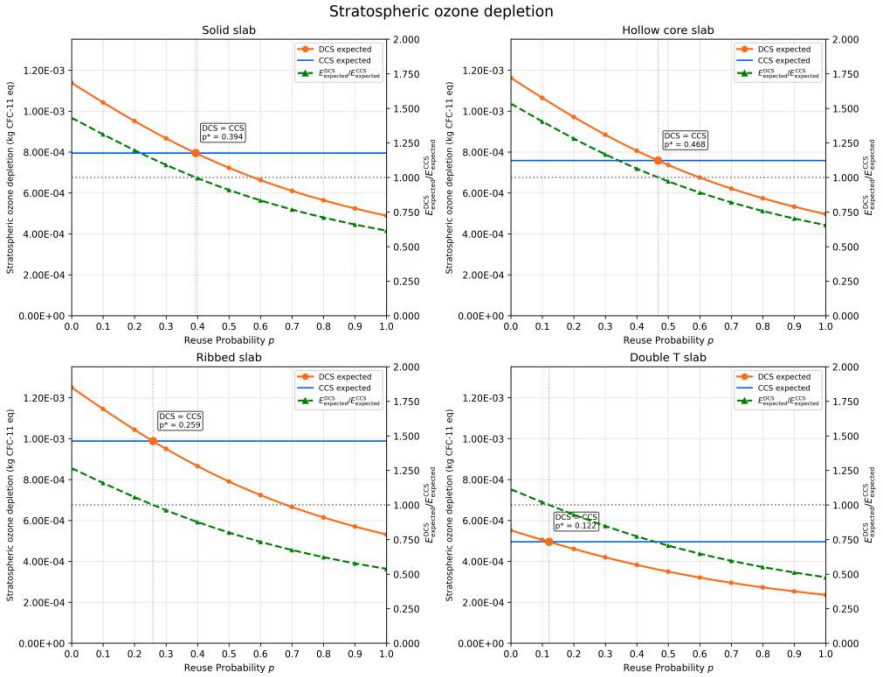


Figure D-2 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Stratospheric ozone depletion

Table D-3 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ionizing radiation

Ionizing radiation (kBq Co-60 eq)											
Project		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Solid Slab	$E^{DCS}_{expected}$	7.50E+01	6.81E+01	6.16E+01	5.56E+01	5.01E+01	4.53E+01	4.10E+01	3.73E+01	3.41E+01	3.12E+01	2.87E+01
	$E^{CCS}_{expected}$	5.84E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.283	1.166	1.054	0.951	0.858	0.775	0.703	0.639	0.583	0.534	0.492
Hollow Core Slab	$E^{DCS}_{expected}$	7.84E+01	7.12E+01	6.43E+01	5.80E+01	5.23E+01	4.72E+01	4.28E+01	3.89E+01	3.54E+01	3.25E+01	2.98E+01
	$E^{CCS}_{expected}$	5.73E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.367	1.242	1.122	1.012	0.912	0.824	0.746	0.678	0.618	0.566	0.521
Ribbed Slab	$E^{DCS}_{expected}$	8.61E+01	7.82E+01	7.06E+01	6.36E+01	5.73E+01	5.18E+01	4.68E+01	4.25E+01	3.88E+01	3.55E+01	3.26E+01
	$E^{CCS}_{expected}$	7.63E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.128	1.024	0.925	0.834	0.751	0.678	0.614	0.557	0.508	0.465	0.427
Double T Slab	$E^{DCS}_{expected}$	4.22E+01	3.83E+01	3.46E+01	3.11E+01	2.80E+01	2.53E+01	2.29E+01	2.07E+01	1.89E+01	1.73E+01	1.59E+01
	$E^{CCS}_{expected}$	3.79E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.112	1.009	0.911	0.821	0.739	0.666	0.603	0.547	0.498	0.455	0.418

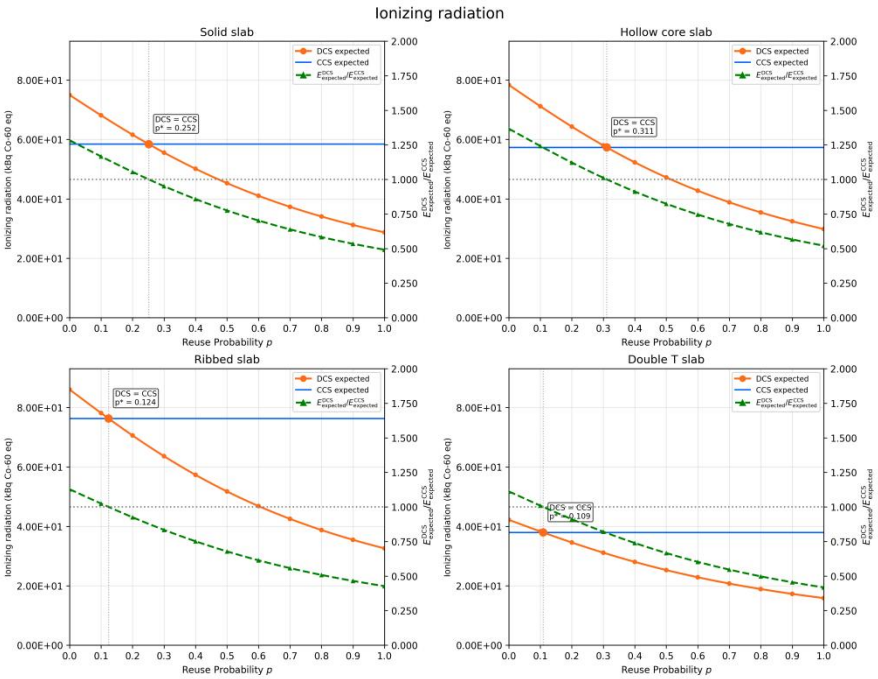


Figure D-3 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ionizing radiation

Table D-4 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation,Human health

Ozone formation,Human health (kg NOx eq)											
Project		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Solid Slab	$E^{DCS}_{expected}$	7.54E+00	6.97E+00	6.42E+00	5.91E+00	5.45E+00	5.05E+00	4.69E+00	4.38E+00	4.10E+00	3.86E+00	3.65E+00
	$E^{CCS}_{expected}$	5.62E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.342	1.239	1.142	1.051	0.970	0.898	0.834	0.778	0.730	0.687	0.650
Hollow Core Slab	$E^{DCS}_{expected}$	7.80E+00	7.20E+00	6.63E+00	6.10E+00	5.62E+00	5.19E+00	4.82E+00	4.49E+00	4.21E+00	3.96E+00	3.74E+00
	$E^{CCS}_{expected}$	5.45E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.432	1.322	1.216	1.119	1.031	0.953	0.884	0.824	0.772	0.726	0.686
Ribbed Slab	$E^{DCS}_{expected}$	8.49E+00	7.83E+00	7.20E+00	6.62E+00	6.09E+00	5.62E+00	5.21E+00	4.85E+00	4.54E+00	4.26E+00	4.02E+00
	$E^{CCS}_{expected}$	7.19E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.181	1.089	1.001	0.920	0.847	0.782	0.725	0.675	0.631	0.593	0.560
Double T Slab	$E^{DCS}_{expected}$	4.02E+00	3.71E+00	3.41E+00	3.13E+00	2.88E+00	2.65E+00	2.46E+00	2.28E+00	2.13E+00	2.00E+00	1.89E+00
	$E^{CCS}_{expected}$	3.64E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.106	1.019	0.936	0.859	0.790	0.729	0.675	0.628	0.586	0.550	0.519

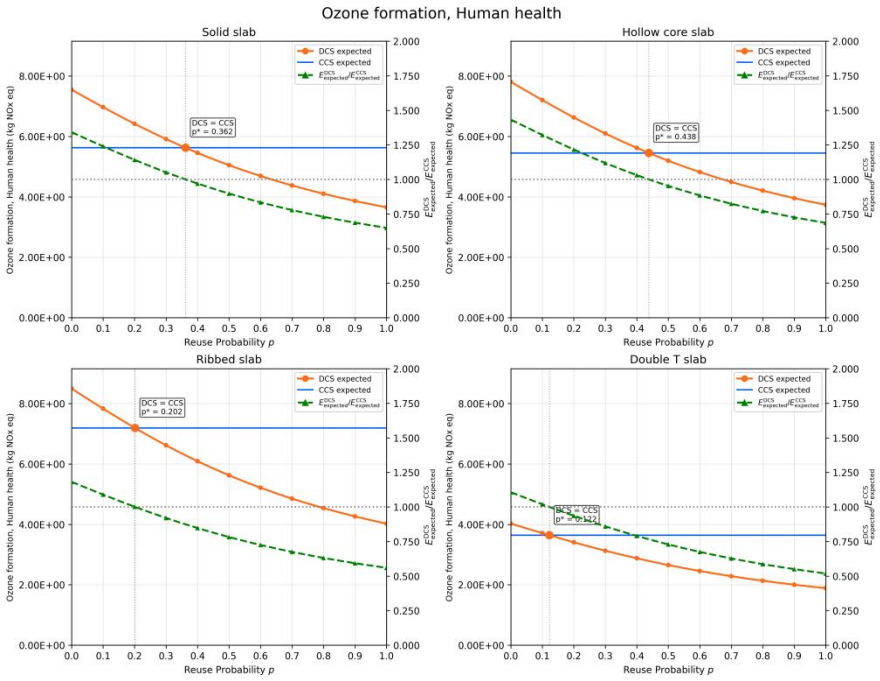


Figure D-4 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation,Human health

Table D-5 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fine particulate matter formation

Fine particulate matter formation (kg PM2.5eq)											
Project		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Solid Slab	$E^{DCS}_{expected}$	4.29E+00	3.95E+00	3.62E+00	3.32E+00	3.05E+00	2.81E+00	2.60E+00	2.41E+00	2.25E+00	2.11E+00	1.98E+00
	$E^{CCS}_{expected}$	2.84E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.509	1.389	1.274	1.169	1.073	0.988	0.913	0.848	0.791	0.741	0.697
Hollow Core Slab	$E^{DCS}_{expected}$	4.54E+00	4.17E+00	3.82E+00	3.50E+00	3.21E+00	2.95E+00	2.72E+00	2.52E+00	2.35E+00	2.20E+00	2.06E+00
	$E^{CCS}_{expected}$	2.88E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.574	1.447	1.326	1.214	1.113	1.023	0.944	0.875	0.814	0.761	0.715
Ribbed Slab	$E^{DCS}_{expected}$	5.04E+00	4.63E+00	4.24E+00	3.87E+00	3.55E+00	3.25E+00	3.00E+00	2.77E+00	2.58E+00	2.41E+00	2.26E+00
	$E^{CCS}_{expected}$	3.93E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.283	1.178	1.078	0.985	0.902	0.828	0.763	0.706	0.656	0.612	0.574
Double T Slab	$E^{DCS}_{expected}$	2.36E+00	2.17E+00	1.98E+00	1.81E+00	1.66E+00	1.52E+00	1.40E+00	1.29E+00	1.20E+00	1.12E+00	1.05E+00
	$E^{CCS}_{expected}$	2.08E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.133	1.040	0.951	0.869	0.795	0.730	0.672	0.621	0.577	0.538	0.504

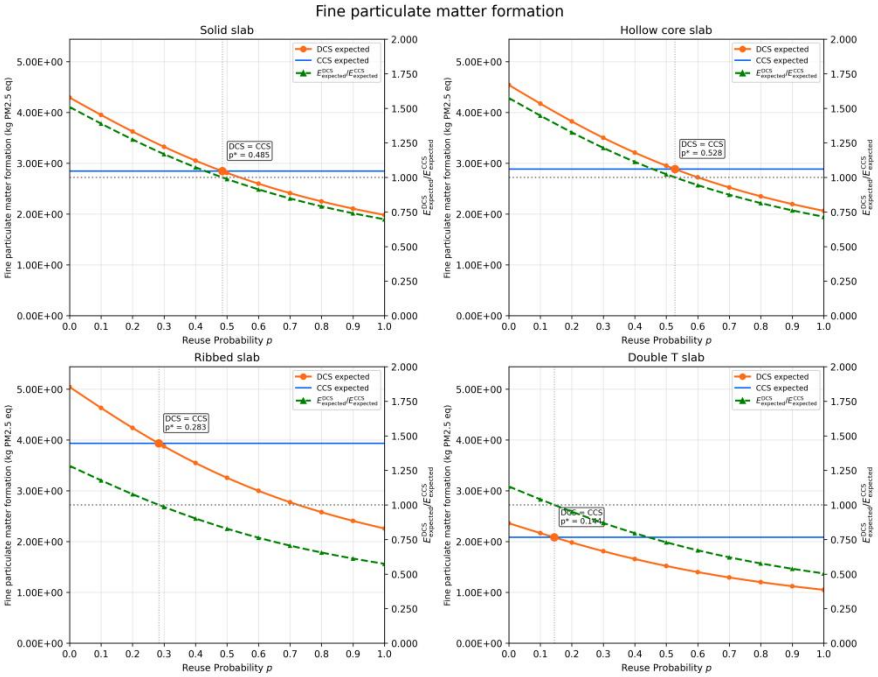


Figure D-5 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fine particulate matter formation

Table D-6 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation,Terrestrial ecosystem

Ozone formation,Terrestrial ecosystem (kg NOx eq)											
Project		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Solid Slab	$E^{DCS}_{expected}$	7.71E+00	7.12E+00	6.56E+00	6.04E+00	5.57E+00	5.15E+00	4.78E+00	4.46E+00	4.18E+00	3.94E+00	3.72E+00
	$E^{CCS}_{expected}$	5.75E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.340	1.237	1.139	1.049	0.968	0.895	0.831	0.776	0.727	0.684	0.647
Hollow Core Slab	$E^{DCS}_{expected}$	7.98E+00	7.37E+00	6.78E+00	6.23E+00	5.74E+00	5.31E+00	4.92E+00	4.58E+00	4.29E+00	4.03E+00	3.81E+00
	$E^{CCS}_{expected}$	5.59E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.429	1.319	1.213	1.116	1.028	0.950	0.881	0.821	0.768	0.722	0.682
Ribbed Slab	$E^{DCS}_{expected}$	8.70E+00	8.02E+00	7.37E+00	6.77E+00	6.23E+00	5.75E+00	5.33E+00	4.96E+00	4.63E+00	4.35E+00	4.10E+00
	$E^{CCS}_{expected}$	7.38E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.179	1.087	0.999	0.917	0.844	0.779	0.722	0.672	0.628	0.590	0.556
Double T Slab	$E^{DCS}_{expected}$	4.12E+00	3.79E+00	3.48E+00	3.20E+00	2.94E+00	2.71E+00	2.51E+00	2.33E+00	2.18E+00	2.04E+00	1.92E+00
	$E^{CCS}_{expected}$	3.72E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.108	1.020	0.937	0.860	0.791	0.729	0.675	0.627	0.585	0.549	0.517

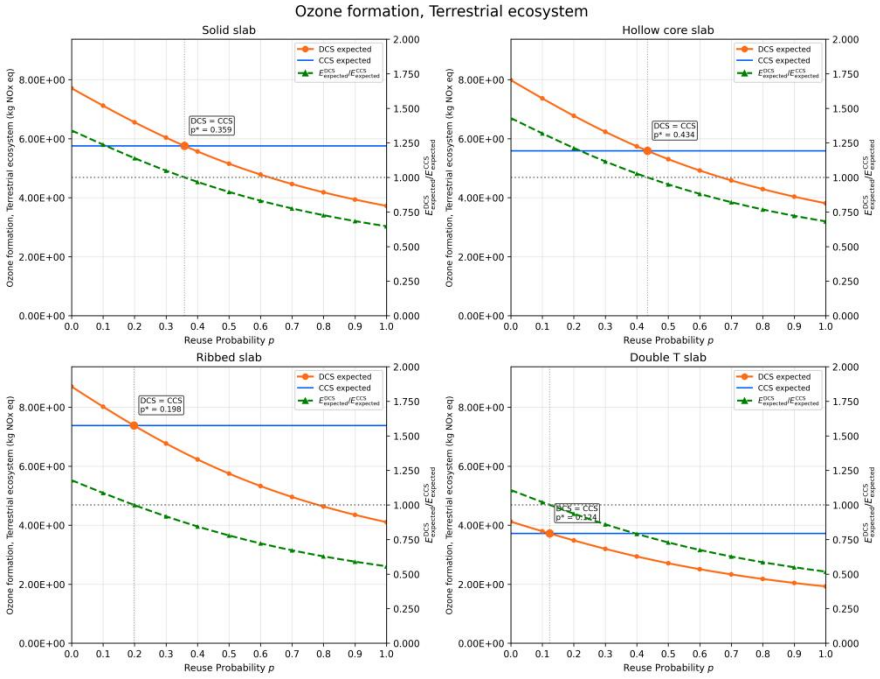


Figure D-6 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation,Terrestrial ecosystem

Table D-7 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial acidification

Terrestrial acidification (kg SO2 eq)											
Project		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Solid Slab	$E^{DCS}_{expected}$	9.81E+00	9.05E+00	8.33E+00	7.66E+00	7.06E+00	6.53E+00	6.06E+00	5.65E+00	5.29E+00	4.97E+00	4.70E+00
	$E^{CCS}_{expected}$	6.29E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.561	1.440	1.325	1.219	1.124	1.039	0.964	0.898	0.841	0.791	0.747
Hollow Core Slab	$E^{DCS}_{expected}$	1.01E+01	9.35E+00	8.60E+00	7.90E+00	7.27E+00	6.72E+00	6.22E+00	5.79E+00	5.42E+00	5.09E+00	4.80E+00
	$E^{CCS}_{expected}$	6.15E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.650	1.521	1.398	1.285	1.183	1.092	1.012	0.942	0.881	0.828	0.781
Ribbed Slab	$E^{DCS}_{expected}$	1.10E+01	1.02E+01	9.34E+00	8.57E+00	7.88E+00	7.27E+00	6.73E+00	6.26E+00	5.85E+00	5.48E+00	5.17E+00
	$E^{CCS}_{expected}$	8.17E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.351	1.245	1.143	1.050	0.965	0.890	0.824	0.766	0.716	0.672	0.633
Double T Slab	$E^{DCS}_{expected}$	5.16E+00	4.75E+00	4.36E+00	4.00E+00	3.68E+00	3.39E+00	3.14E+00	2.91E+00	2.72E+00	2.55E+00	2.40E+00
	$E^{CCS}_{expected}$	4.65E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.109	1.021	0.938	0.860	0.791	0.729	0.674	0.626	0.585	0.548	0.516

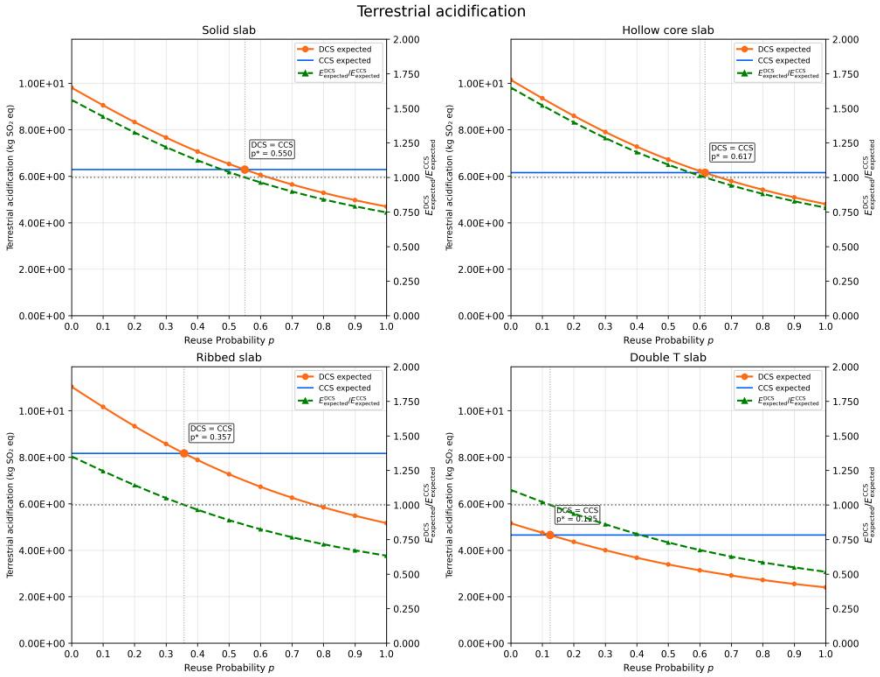


Figure D-7 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial acidification

Table D-8 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater eutrophication

Freshwater eutrophication (kg P eq)											
Project		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Solid Slab	$E^{DCS}_{expected}$	7.83E-01	7.22E-01	6.63E-01	6.09E-01	5.61E-01	5.17E-01	4.79E-01	4.46E-01	4.17E-01	3.92E-01	3.70E-01
	$E^{CCS}_{expected}$	6.13E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.278	1.178	1.082	0.994	0.915	0.845	0.783	0.728	0.681	0.640	0.603
Hollow Core Slab	$E^{DCS}_{expected}$	8.84E-01	8.13E-01	7.45E-01	6.82E-01	6.25E-01	5.75E-01	5.31E-01	4.92E-01	4.58E-01	4.29E-01	4.03E-01
	$E^{CCS}_{expected}$	6.72E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.317	1.210	1.109	1.016	0.931	0.856	0.791	0.733	0.683	0.639	0.600
Ribbed Slab	$E^{DCS}_{expected}$	1.04E+00	9.52E-01	8.71E-01	7.95E-01	7.27E-01	6.67E-01	6.13E-01	5.67E-01	5.26E-01	4.91E-01	4.60E-01
	$E^{CCS}_{expected}$	9.63E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.078	0.989	0.904	0.825	0.755	0.692	0.637	0.589	0.546	0.509	0.477
Double T Slab	$E^{DCS}_{expected}$	4.87E-01	4.47E-01	4.08E-01	3.72E-01	3.40E-01	3.12E-01	2.87E-01	2.65E-01	2.46E-01	2.29E-01	2.14E-01
	$E^{CCS}_{expected}$	4.05E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.203	1.103	1.007	0.920	0.840	0.770	0.708	0.654	0.607	0.565	0.529

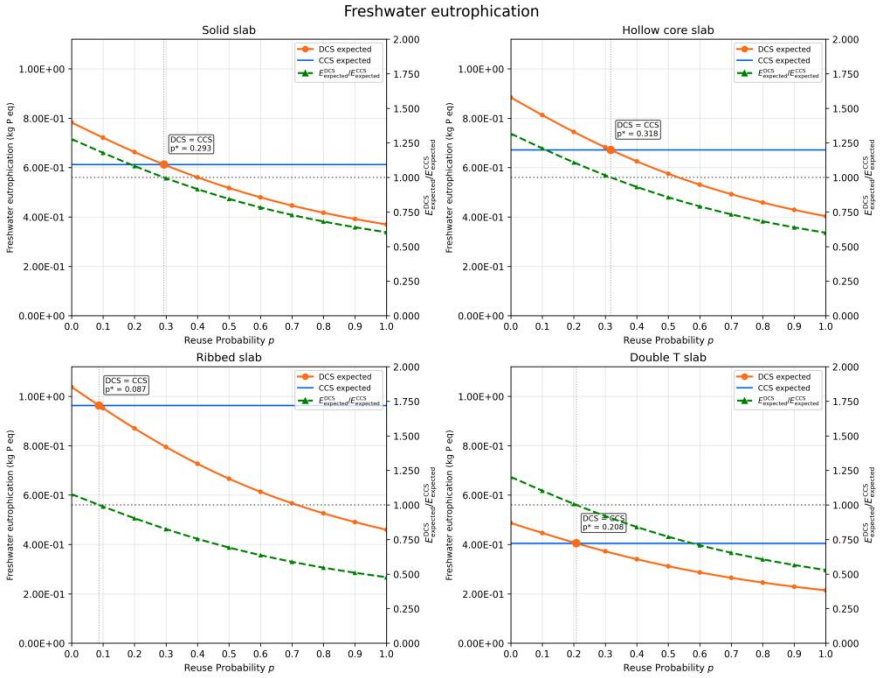


Figure D-8 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater eutrophication

Table D-9 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine eutrophication

Marine eutrophication (kg N eq)											
Project		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Solid Slab	$E^{DCS}_{expected}$	7.46E-02	6.83E-02	6.22E-02	5.66E-02	5.16E-02	4.71E-02	4.32E-02	3.97E-02	3.67E-02	3.41E-02	3.18E-02
	$E^{CCS}_{expected}$	6.28E-02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.188	1.087	0.990	0.901	0.821	0.749	0.687	0.632	0.584	0.542	0.505
Hollow Core Slab	$E^{DCS}_{expected}$	7.74E-02	7.07E-02	6.44E-02	5.86E-02	5.33E-02	4.86E-02	4.45E-02	4.09E-02	3.78E-02	3.50E-02	3.27E-02
	$E^{CCS}_{expected}$	6.07E-02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.275	1.166	1.062	0.966	0.879	0.802	0.734	0.675	0.623	0.578	0.538
Ribbed Slab	$E^{DCS}_{expected}$	8.44E-02	7.71E-02	7.02E-02	6.38E-02	5.80E-02	5.29E-02	4.84E-02	4.44E-02	4.10E-02	3.80E-02	3.53E-02
	$E^{CCS}_{expected}$	7.98E-02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.057	0.966	0.879	0.799	0.727	0.663	0.606	0.557	0.513	0.476	0.443
Double T Slab	$E^{DCS}_{expected}$	2.47E-02	2.27E-02	2.09E-02	1.91E-02	1.76E-02	1.62E-02	1.50E-02	1.40E-02	1.30E-02	1.22E-02	1.15E-02
	$E^{CCS}_{expected}$	2.15E-02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.147	1.056	0.970	0.890	0.818	0.754	0.698	0.649	0.606	0.568	0.535

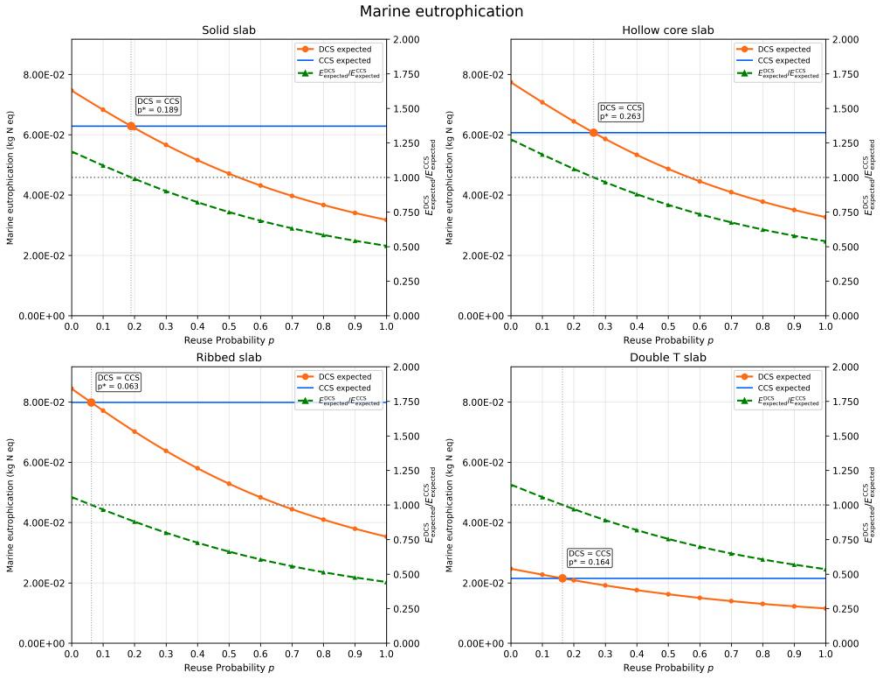


Figure D-9 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine eutrophication

Table D-10 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial ecotoxicity

Terrestrial ecotoxicity (kg 1,4-DCB)											
Project		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Solid Slab	$E^{DCS}_{expected}$	1.26E+04	1.15E+04	1.04E+04	9.40E+03	8.48E+03	7.66E+03	6.94E+03	6.30E+03	5.75E+03	5.26E+03	4.84E+03
	$E^{CCS}_{expected}$	1.02E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.243	1.130	1.023	0.923	0.833	0.753	0.682	0.619	0.565	0.517	0.475
Hollow Core Slab	$E^{DCS}_{expected}$	1.34E+04	1.22E+04	1.10E+04	9.91E+03	8.94E+03	8.07E+03	7.30E+03	6.63E+03	6.04E+03	5.52E+03	5.07E+03
	$E^{CCS}_{expected}$	1.01E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.321	1.201	1.086	0.979	0.883	0.797	0.722	0.655	0.597	0.546	0.501
Ribbed Slab	$E^{DCS}_{expected}$	1.48E+04	1.35E+04	1.22E+04	1.10E+04	9.89E+03	8.92E+03	8.07E+03	7.32E+03	6.67E+03	6.09E+03	5.59E+03
	$E^{CCS}_{expected}$	1.36E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.091	0.991	0.896	0.807	0.728	0.656	0.594	0.539	0.490	0.448	0.411
Double T Slab	$E^{DCS}_{expected}$	6.86E+03	6.24E+03	5.64E+03	5.08E+03	4.58E+03	4.13E+03	3.73E+03	3.39E+03	3.08E+03	2.82E+03	2.59E+03
	$E^{CCS}_{expected}$	5.94E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.157	1.051	0.950	0.856	0.771	0.696	0.629	0.571	0.519	0.475	0.436

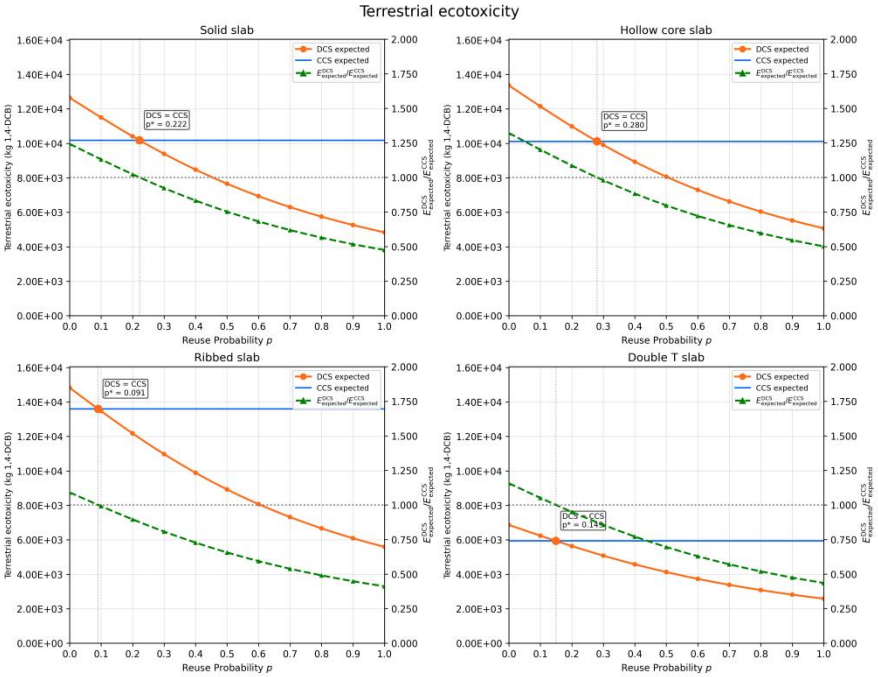


Figure D-10 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial ecotoxicity

Table D-11 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater ecotoxicity

Freshwater ecotoxicity (kg 1,4-DCB)											
Project		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Solid Slab	$E^{DCS}_{expected}$	8.20E+01	7.48E+01	6.80E+01	6.17E+01	5.60E+01	5.09E+01	4.64E+01	4.25E+01	3.91E+01	3.61E+01	3.35E+01
	$E^{CCS}_{expected}$	6.20E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.322	1.206	1.096	0.994	0.902	0.820	0.748	0.685	0.630	0.582	0.540
Hollow Core Slab	$E^{DCS}_{expected}$	9.61E+01	8.76E+01	7.94E+01	7.18E+01	6.50E+01	5.90E+01	5.36E+01	4.90E+01	4.49E+01	4.13E+01	3.82E+01
	$E^{CCS}_{expected}$	7.22E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.332	1.213	1.100	0.995	0.901	0.817	0.743	0.679	0.622	0.573	0.530
Ribbed Slab	$E^{DCS}_{expected}$	1.16E+02	1.06E+02	9.56E+01	8.64E+01	7.80E+01	7.06E+01	6.41E+01	5.84E+01	5.34E+01	4.90E+01	4.52E+01
	$E^{CCS}_{expected}$	1.07E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.084	0.986	0.892	0.806	0.728	0.659	0.598	0.545	0.498	0.457	0.422
Double T Slab	$E^{DCS}_{expected}$	5.35E+01	4.87E+01	4.40E+01	3.98E+01	3.59E+01	3.25E+01	2.95E+01	2.69E+01	2.46E+01	2.26E+01	2.08E+01
	$E^{CCS}_{expected}$	4.30E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.245	1.133	1.025	0.926	0.836	0.757	0.687	0.625	0.572	0.525	0.484

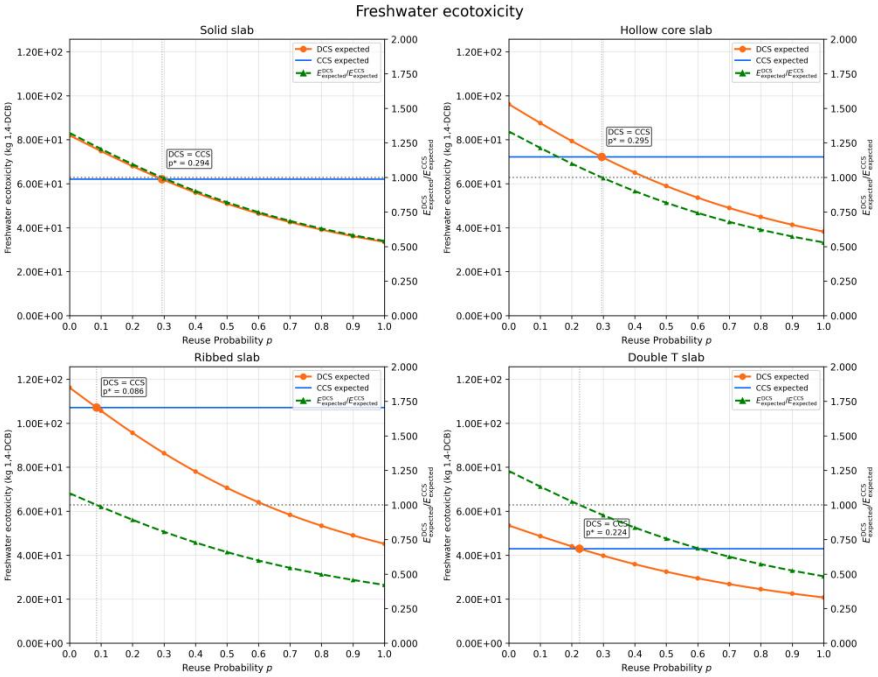


Figure D-11 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater ecotoxicity

Table D-12 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine ecotoxicity

Marine ecotoxicity (kg 1,4-DCB)											
Project		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Solid Slab	$E^{DCS}_{expected}$	1.19E+02	1.08E+02	9.83E+01	8.91E+01	8.09E+01	7.35E+01	6.70E+01	6.14E+01	5.64E+01	5.21E+01	4.83E+01
	$E^{CCS}_{expected}$	9.00E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.318	1.203	1.092	0.990	0.898	0.817	0.745	0.682	0.627	0.579	0.536
Hollow Core Slab	$E^{DCS}_{expected}$	1.39E+02	1.26E+02	1.14E+02	1.04E+02	9.37E+01	8.50E+01	7.73E+01	7.05E+01	6.46E+01	5.95E+01	5.49E+01
	$E^{CCS}_{expected}$	1.04E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.331	1.212	1.099	0.994	0.899	0.815	0.741	0.677	0.620	0.571	0.527
Ribbed Slab	$E^{DCS}_{expected}$	1.67E+02	1.52E+02	1.38E+02	1.24E+02	1.12E+02	1.02E+02	9.21E+01	8.39E+01	7.67E+01	7.04E+01	6.49E+01
	$E^{CCS}_{expected}$	1.54E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.084	0.986	0.892	0.805	0.727	0.658	0.597	0.544	0.497	0.456	0.421
Double T Slab	$E^{DCS}_{expected}$	7.73E+01	7.03E+01	6.36E+01	5.74E+01	5.18E+01	4.69E+01	4.25E+01	3.87E+01	3.54E+01	3.25E+01	2.99E+01
	$E^{CCS}_{expected}$	6.22E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.243	1.130	1.022	0.923	0.834	0.754	0.684	0.623	0.569	0.522	0.481

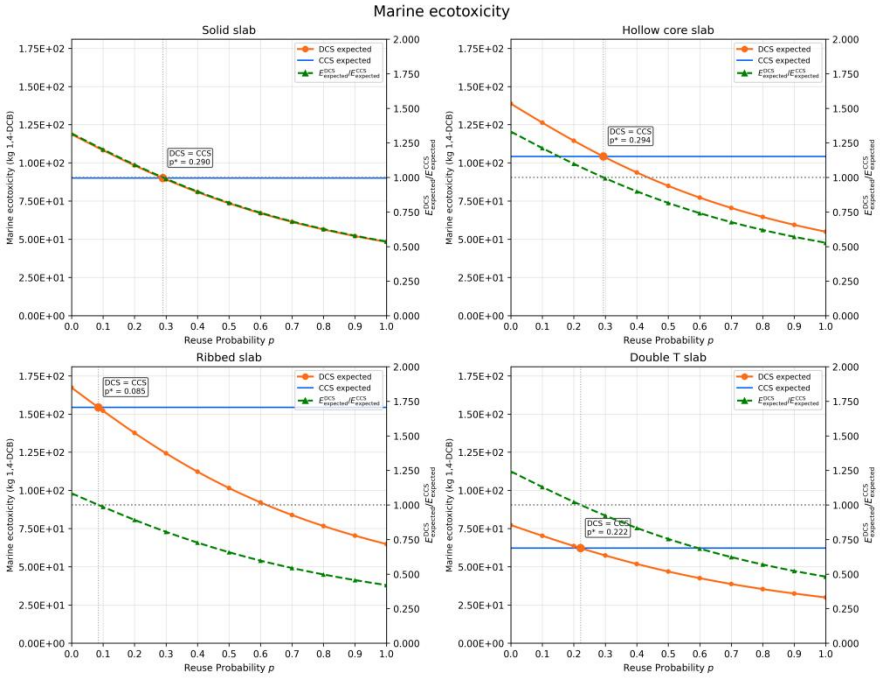


Figure D-12 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine ecotoxicity

Table D-13 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Human carcinogenic toxicity

Human carcinogenic toxicity (kg 1,4-DCB)											
Project		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Solid Slab	$E^{DCS}_{expected}$	3.08E+02	2.79E+02	2.51E+02	2.25E+02	2.02E+02	1.82E+02	1.64E+02	1.48E+02	1.34E+02	1.22E+02	1.11E+02
	$E^{CCS}_{expected}$	2.19E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.404	1.271	1.144	1.027	0.922	0.828	0.746	0.674	0.611	0.556	0.508
Hollow Core Slab	$E^{DCS}_{expected}$	3.92E+02	3.55E+02	3.19E+02	2.86E+02	2.56E+02	2.30E+02	2.07E+02	1.86E+02	1.68E+02	1.53E+02	1.39E+02
	$E^{CCS}_{expected}$	2.92E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.344	1.216	1.093	0.980	0.878	0.788	0.708	0.638	0.578	0.524	0.478
Ribbed Slab	$E^{DCS}_{expected}$	5.02E+02	4.53E+02	4.07E+02	3.65E+02	3.27E+02	2.93E+02	2.63E+02	2.37E+02	2.14E+02	1.94E+02	1.76E+02
	$E^{CCS}_{expected}$	4.63E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.083	0.979	0.879	0.787	0.705	0.632	0.567	0.511	0.461	0.418	0.381
Double T Slab	$E^{DCS}_{expected}$	2.31E+02	2.09E+02	1.88E+02	1.68E+02	1.50E+02	1.35E+02	1.21E+02	1.09E+02	9.84E+01	8.92E+01	8.12E+01
	$E^{CCS}_{expected}$	1.75E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.319	1.192	1.071	0.959	0.859	0.769	0.691	0.622	0.562	0.509	0.464

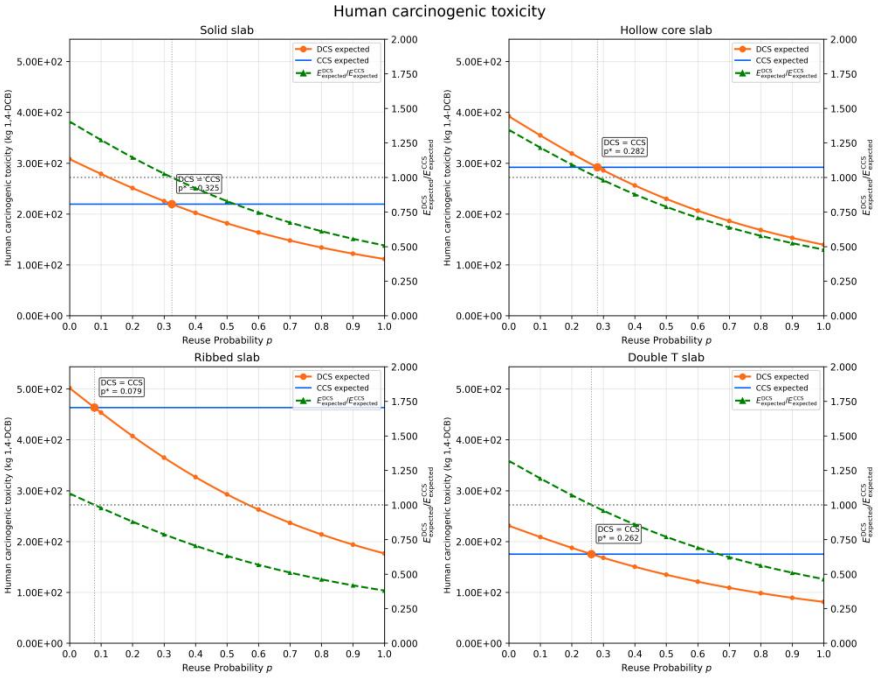


Figure D-13 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Human carcinogenic toxicity

Table D-14 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Human non-carcinogenic toxicity

Human non-carcinogenic toxicity (kg 1,4-DCB)											
Project		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Solid Slab	$E^{DCS}_{expected}$	2.50E+03	2.28E+03	2.07E+03	1.88E+03	1.70E+03	1.54E+03	1.41E+03	1.29E+03	1.18E+03	1.09E+03	1.01E+03
	$E^{CCS}_{expected}$	1.91E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.309	1.194	1.083	0.982	0.890	0.808	0.736	0.673	0.618	0.570	0.528
Hollow Core Slab	$E^{DCS}_{expected}$	2.93E+03	2.67E+03	2.42E+03	2.18E+03	1.97E+03	1.79E+03	1.62E+03	1.48E+03	1.36E+03	1.25E+03	1.15E+03
	$E^{CCS}_{expected}$	2.21E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.324	1.205	1.091	0.986	0.892	0.808	0.734	0.669	0.612	0.563	0.520
Ribbed Slab	$E^{DCS}_{expected}$	3.54E+03	3.21E+03	2.91E+03	2.62E+03	2.37E+03	2.14E+03	1.94E+03	1.76E+03	1.61E+03	1.48E+03	1.36E+03
	$E^{CCS}_{expected}$	3.28E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.078	0.980	0.886	0.800	0.722	0.653	0.592	0.538	0.491	0.451	0.415
Double T Slab	$E^{DCS}_{expected}$	1.63E+03	1.49E+03	1.34E+03	1.21E+03	1.09E+03	9.88E+02	8.96E+02	8.15E+02	7.44E+02	6.82E+02	6.28E+02
	$E^{CCS}_{expected}$	1.31E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.244	1.131	1.022	0.923	0.832	0.752	0.682	0.620	0.566	0.519	0.478

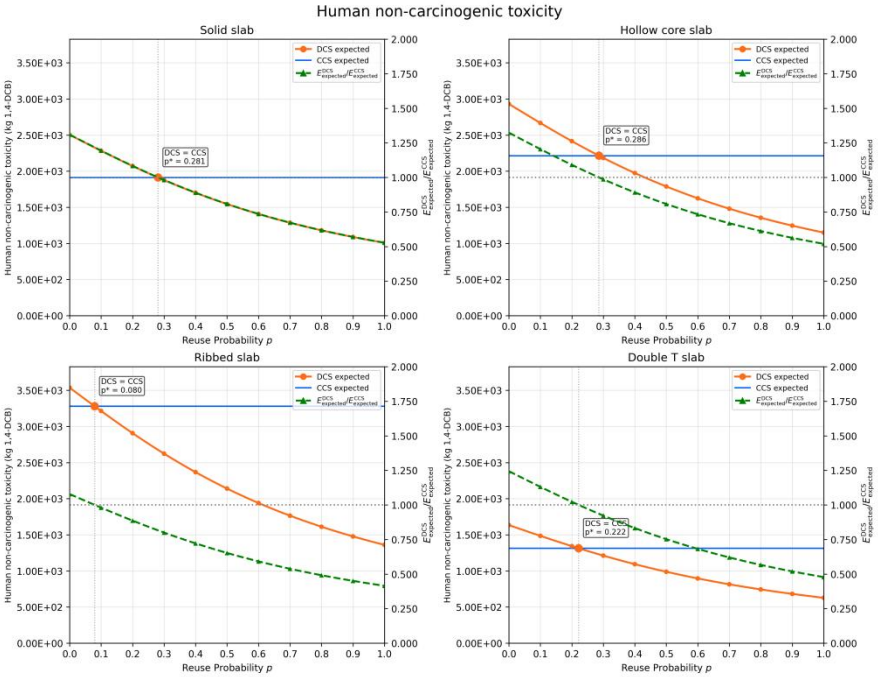


Figure D-14 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Human non-carcinogenic toxicity

Table D-15 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Land use

Land use (m^2 a crop eq)												
Project		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Solid Slab	$E^{DCS}_{expected}$	7.21E+01	6.55E+01	5.91E+01	5.33E+01	4.80E+01	4.33E+01	3.92E+01	3.56E+01	3.24E+01	2.96E+01	2.72E+01
	$E^{CCS}_{expected}$	5.88E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.225	1.113	1.005	0.906	0.816	0.737	0.666	0.605	0.551	0.504	0.463
Hollow Core Slab	$E^{DCS}_{expected}$	7.38E+01	6.70E+01	6.05E+01	5.45E+01	4.91E+01	4.43E+01	4.01E+01	3.63E+01	3.31E+01	3.03E+01	2.78E+01
	$E^{CCS}_{expected}$	5.59E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.320	1.199	1.083	0.976	0.879	0.793	0.717	0.651	0.592	0.542	0.497
Ribbed Slab	$E^{DCS}_{expected}$	7.94E+01	7.21E+01	6.51E+01	5.87E+01	5.28E+01	4.77E+01	4.31E+01	3.91E+01	3.56E+01	3.25E+01	2.98E+01
	$E^{CCS}_{expected}$	7.25E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.095	0.994	0.898	0.809	0.728	0.657	0.594	0.539	0.490	0.448	0.411
Double T Slab	$E^{DCS}_{expected}$	2.84E+01	2.58E+01	2.34E+01	2.11E+01	1.91E+01	1.72E+01	1.56E+01	1.42E+01	1.30E+01	1.19E+01	1.10E+01
	$E^{CCS}_{expected}$	2.52E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.123	1.022	0.925	0.836	0.755	0.683	0.619	0.564	0.515	0.472	0.435

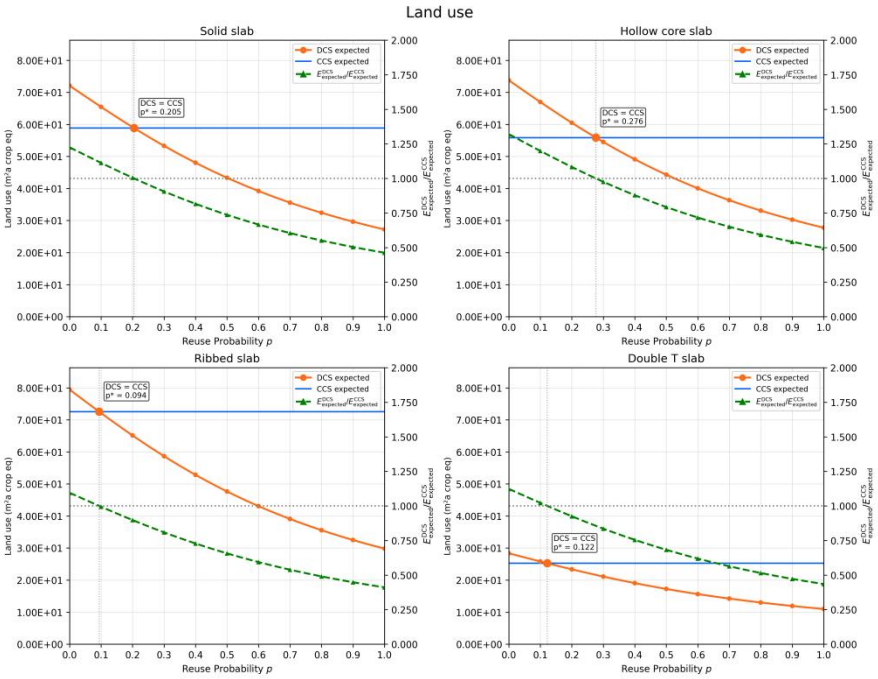


Figure D-15 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Land use

Table D-16 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Mineral resource scarcity

Mineral resource scarcity (kg Cu eq)											
Project		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Solid Slab	$E^{DCS}_{expected}$	2.64E+01	2.38E+01	2.14E+01	1.91E+01	1.71E+01	1.52E+01	1.36E+01	1.23E+01	1.10E+01	9.96E+00	9.03E+00
	$E^{CCS}_{expected}$	1.97E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.342	1.211	1.086	0.971	0.867	0.775	0.694	0.623	0.561	0.506	0.459
Hollow Core Slab	$E^{DCS}_{expected}$	3.22E+01	2.91E+01	2.60E+01	2.33E+01	2.08E+01	1.86E+01	1.66E+01	1.49E+01	1.34E+01	1.21E+01	1.10E+01
	$E^{CCS}_{expected}$	2.43E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.326	1.196	1.072	0.958	0.855	0.764	0.684	0.613	0.552	0.498	0.451
Ribbed Slab	$E^{DCS}_{expected}$	4.00E+01	3.61E+01	3.24E+01	2.89E+01	2.58E+01	2.30E+01	2.06E+01	1.85E+01	1.66E+01	1.50E+01	1.36E+01
	$E^{CCS}_{expected}$	3.72E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.076	0.970	0.869	0.777	0.693	0.619	0.554	0.497	0.447	0.403	0.365
Double T Slab	$E^{DCS}_{expected}$	1.88E+01	1.69E+01	1.52E+01	1.35E+01	1.21E+01	1.08E+01	9.66E+00	8.66E+00	7.79E+00	7.03E+00	6.36E+00
	$E^{CCS}_{expected}$	1.48E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.270	1.145	1.026	0.917	0.818	0.730	0.653	0.586	0.527	0.476	0.431

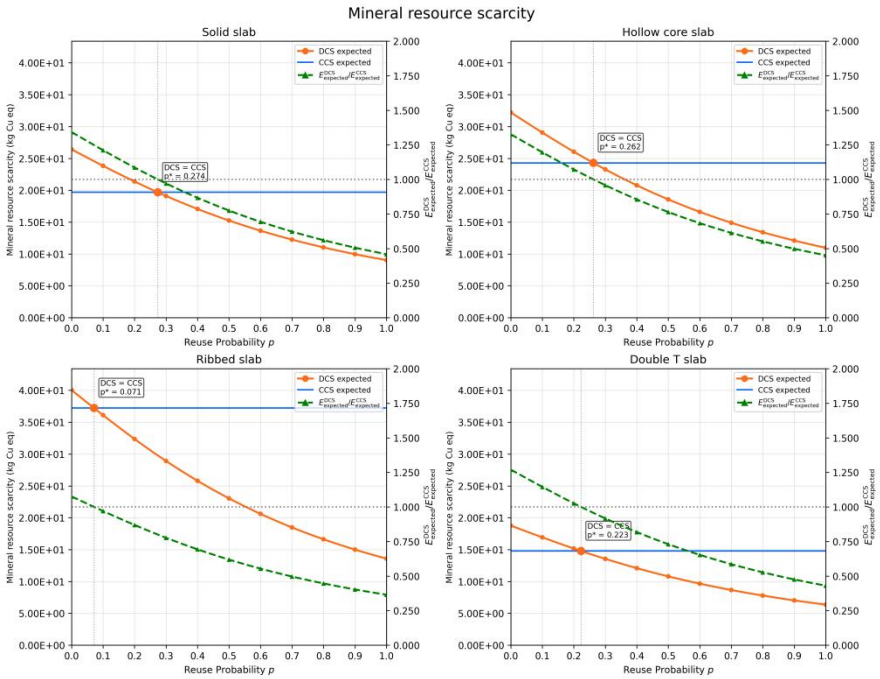


Figure D-16 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Mineral resource scarcity

Table D-17 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fossil resource scarcity

Fossil resource scarcity (kg oil eq)											
Project		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Solid Slab	$E^{DCS}_{expected}$	6.52E+02	5.99E+02	5.49E+02	5.02E+02	4.60E+02	4.23E+02	3.90E+02	3.61E+02	3.36E+02	3.14E+02	2.94E+02
	$E^{CCS}_{expected}$	4.91E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.326	1.220	1.117	1.023	0.937	0.861	0.794	0.735	0.684	0.639	0.599
Hollow Core Slab	$E^{DCS}_{expected}$	6.74E+02	6.19E+02	5.66E+02	5.18E+02	4.74E+02	4.35E+02	4.01E+02	3.71E+02	3.45E+02	3.22E+02	3.01E+02
	$E^{CCS}_{expected}$	4.75E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.417	1.302	1.192	1.090	0.998	0.916	0.844	0.781	0.725	0.677	0.634
Ribbed Slab	$E^{DCS}_{expected}$	7.32E+02	6.72E+02	6.15E+02	5.62E+02	5.14E+02	4.71E+02	4.34E+02	4.01E+02	3.72E+02	3.47E+02	3.25E+02
	$E^{CCS}_{expected}$	6.26E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.170	1.074	0.983	0.898	0.821	0.753	0.693	0.641	0.594	0.554	0.519
Double T Slab	$E^{DCS}_{expected}$	3.46E+02	3.18E+02	2.90E+02	2.65E+02	2.42E+02	2.22E+02	2.04E+02	1.89E+02	1.75E+02	1.63E+02	1.52E+02
	$E^{CCS}_{expected}$	3.09E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.119	1.027	0.939	0.857	0.784	0.718	0.660	0.610	0.565	0.526	0.493

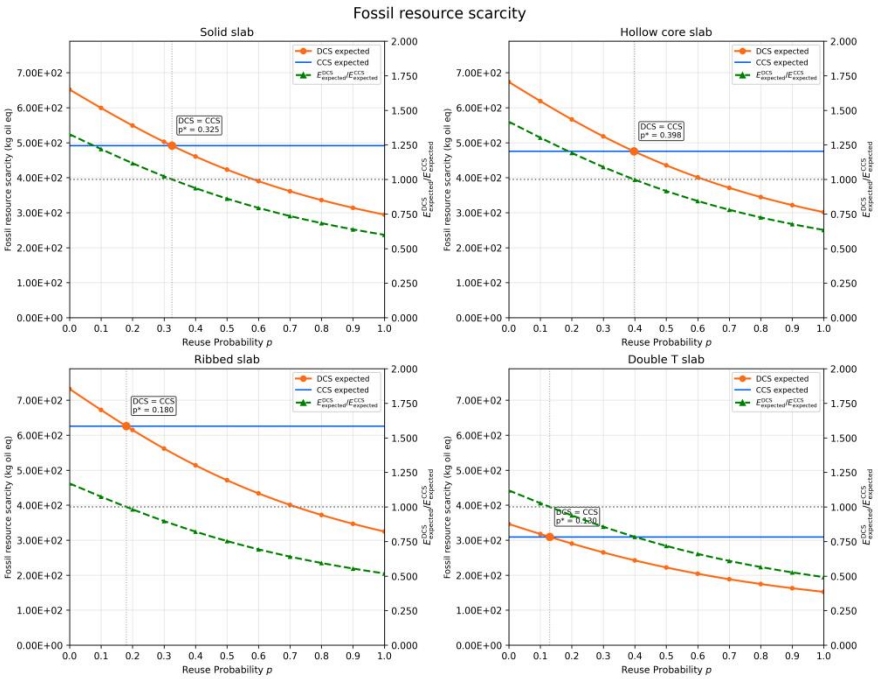


Figure D-17 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fossil resource scarcity

Table D-18 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Water consumption

Water consumption (m^3)											
Project		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Solid Slab	$E^{DCS}_{expected}$	2.39E+01	2.16E+01	1.94E+01	1.74E+01	1.56E+01	1.40E+01	1.26E+01	1.14E+01	1.03E+01	9.35E+00	8.53E+00
	$E^{CCS}_{expected}$	1.87E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.279	1.157	1.041	0.934	0.837	0.751	0.676	0.609	0.552	0.501	0.457
Hollow Core Slab	$E^{DCS}_{expected}$	2.44E+01	2.21E+01	1.99E+01	1.78E+01	1.60E+01	1.43E+01	1.29E+01	1.16E+01	1.05E+01	9.55E+00	8.71E+00
	$E^{CCS}_{expected}$	1.78E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.376	1.245	1.119	1.004	0.900	0.807	0.726	0.655	0.592	0.538	0.491
Ribbed Slab	$E^{DCS}_{expected}$	2.63E+01	2.38E+01	2.14E+01	1.92E+01	1.72E+01	1.54E+01	1.39E+01	1.25E+01	1.13E+01	1.03E+01	9.37E+00
	$E^{CCS}_{expected}$	2.31E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.140	1.031	0.927	0.831	0.745	0.668	0.601	0.542	0.490	0.445	0.406
Double T Slab	$E^{DCS}_{expected}$	1.13E+01	1.02E+01	9.21E+00	8.26E+00	7.40E+00	6.64E+00	5.98E+00	5.39E+00	4.88E+00	4.43E+00	4.04E+00
	$E^{CCS}_{expected}$	1.04E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.089	0.985	0.886	0.795	0.713	0.640	0.575	0.519	0.470	0.427	0.389

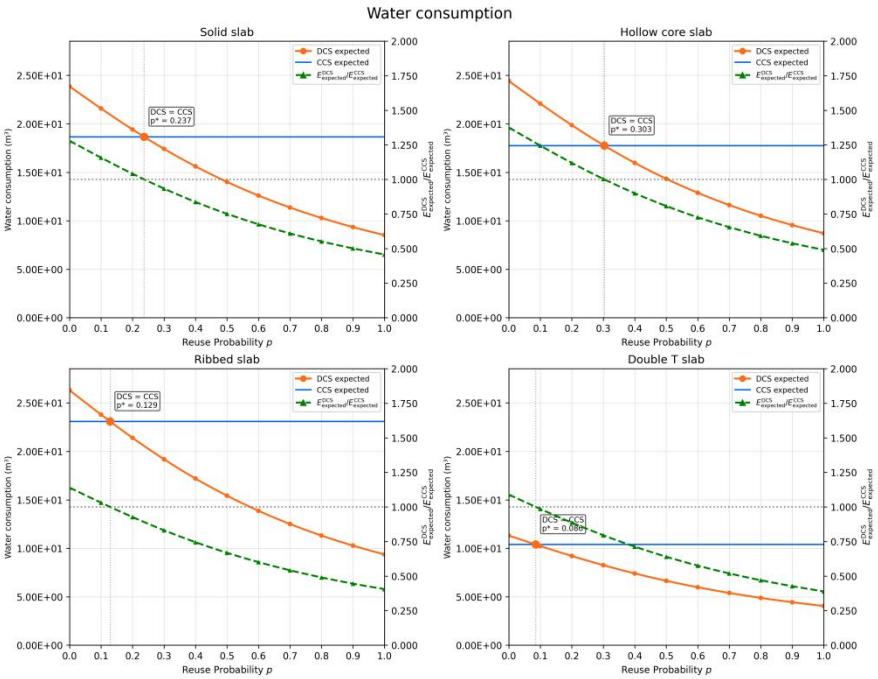


Table D-18 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Water consumption

Appendix E: p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) for the solid slab of different model lengths

Table E-1 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Global warming

Solid Slab-Global warming (kg CO_2 eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	3.09E+03	2.83E+03	2.57E+03	2.34E+03	2.13E+03	1.95E+03	1.78E+03	1.64E+03	1.51E+03	1.40E+03	1.31E+03
	$E^{DCS}_{expected}$	2.32E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.329	1.216	1.108	1.008	0.918	0.838	0.767	0.705	0.651	0.604	0.563
10.5	$E^{DCS}_{expected}$	4.03E+03	3.69E+03	3.36E+03	3.06E+03	2.78E+03	2.54E+03	2.32E+03	2.13E+03	1.97E+03	1.83E+03	1.70E+03
	$E^{DCS}_{expected}$	3.13E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.289	1.179	1.074	0.977	0.889	0.811	0.743	0.682	0.630	0.584	0.544
12.0	$E^{DCS}_{expected}$	4.57E+03	4.18E+03	3.81E+03	3.47E+03	3.15E+03	2.88E+03	2.63E+03	2.42E+03	2.23E+03	2.07E+03	1.93E+03
	$E^{DCS}_{expected}$	3.57E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.281	1.171	1.067	0.970	0.883	0.806	0.738	0.678	0.626	0.580	0.540
13.5	$E^{DCS}_{expected}$	5.12E+03	4.68E+03	4.26E+03	3.88E+03	3.53E+03	3.22E+03	2.95E+03	2.71E+03	2.50E+03	2.32E+03	2.16E+03
	$E^{DCS}_{expected}$	4.02E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.273	1.165	1.061	0.965	0.878	0.801	0.733	0.674	0.622	0.577	0.537
15.0	$E^{DCS}_{expected}$	5.66E+03	5.18E+03	4.72E+03	4.29E+03	3.90E+03	3.56E+03	3.26E+03	3.00E+03	2.77E+03	2.56E+03	2.39E+03
	$E^{DCS}_{expected}$	4.46E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.268	1.160	1.056	0.961	0.874	0.798	0.730	0.671	0.619	0.574	0.535
16.5	$E^{DCS}_{expected}$	6.97E+03	6.37E+03	5.80E+03	5.28E+03	4.80E+03	4.38E+03	4.01E+03	3.68E+03	3.39E+03	3.15E+03	2.93E+03
	$E^{DCS}_{expected}$	5.60E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.245	1.139	1.037	0.943	0.858	0.782	0.716	0.658	0.607	0.562	0.523
18.0	$E^{DCS}_{expected}$	7.58E+03	6.93E+03	6.31E+03	5.74E+03	5.22E+03	4.76E+03	4.36E+03	4.00E+03	3.69E+03	3.42E+03	3.18E+03
	$E^{DCS}_{expected}$	6.10E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.242	1.136	1.034	0.940	0.856	0.780	0.714	0.656	0.605	0.561	0.522

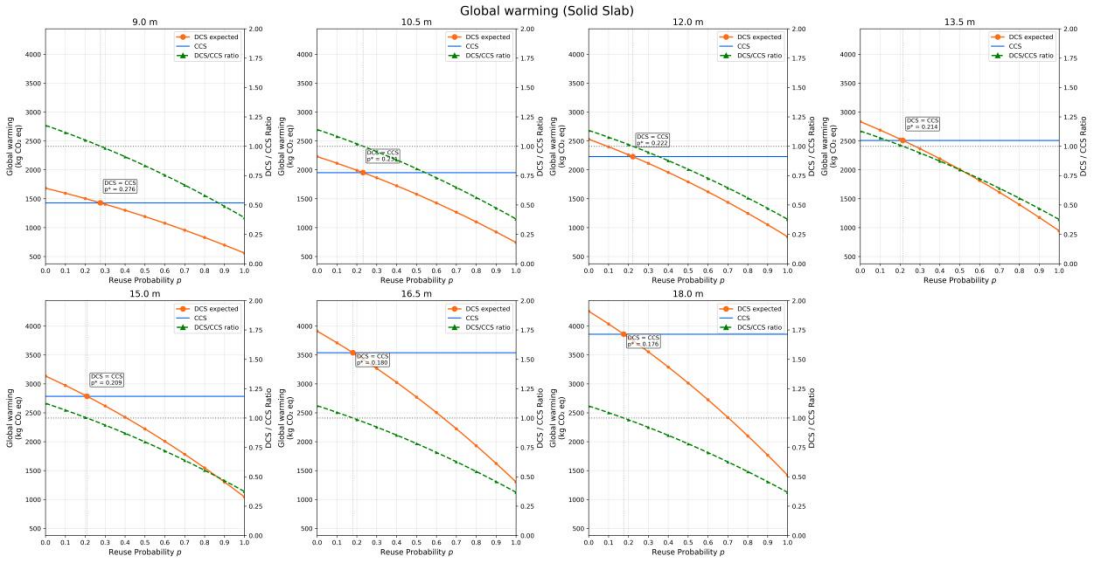


Figure E-1 p* derived from calculations of CCS ($E^{\mathrm{CCS}}_{\mathrm{expected}}$) and DCS ($E^{\mathrm{DCS}}_{\mathrm{expected}}$) of Global warming

Table E-2 p* derived from calculations of CCS ($E^{\mathrm{CCS}}_{\mathrm{expected}}$) and DCS ($E^{\mathrm{DCS}}_{\mathrm{expected}}$) of Stratospheric ozone depletion

Solid Slab-Stratospheric ozone depletion (kg CFC11 eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{\mathrm{DCS}}_{\mathrm{expected}}$	1.14E-03	1.04E-03	9.51E-04	8.67E-04	7.91E-04	7.23E-04	6.63E-04	6.10E-04	5.64E-04	5.24E-04	4.89E-04
	$E^{\mathrm{DCS}}_{\mathrm{expected}}$	7.95E-04										
	$\frac{E^{\mathrm{DCS}}_{\mathrm{expected}}}{E^{\mathrm{CCS}}_{\mathrm{expected}}}$	1.432	1.312	1.197	1.091	0.995	0.909	0.834	0.768	0.710	0.659	0.615
10.5	$E^{\mathrm{DCS}}_{\mathrm{expected}}$	1.48E-03	1.36E-03	1.24E-03	1.13E-03	1.03E-03	9.39E-04	8.61E-04	7.93E-04	7.33E-04	6.80E-04	6.34E-04
	$E^{\mathrm{DCS}}_{\mathrm{expected}}$	1.07E-03										
	$\frac{E^{\mathrm{DCS}}_{\mathrm{expected}}}{E^{\mathrm{CCS}}_{\mathrm{expected}}}$	1.388	1.272	1.160	1.057	0.964	0.880	0.807	0.743	0.687	0.637	0.594
12.0	$E^{\mathrm{DCS}}_{\mathrm{expected}}$	1.68E-03	1.54E-03	1.40E-03	1.28E-03	1.17E-03	1.07E-03	9.77E-04	8.99E-04	8.31E-04	7.71E-04	7.19E-04
	$E^{\mathrm{DCS}}_{\mathrm{expected}}$	1.22E-03										
	$\frac{E^{\mathrm{DCS}}_{\mathrm{expected}}}{E^{\mathrm{CCS}}_{\mathrm{expected}}}$	1.379	1.263	1.152	1.050	0.957	0.874	0.802	0.738	0.682	0.633	0.590
13.5	$E^{\mathrm{DCS}}_{\mathrm{expected}}$	1.88E-03	1.72E-03	1.57E-03	1.43E-03	1.30E-03	1.19E-03	1.09E-03	1.01E-03	9.30E-04	8.63E-04	8.05E-04
	$E^{\mathrm{DCS}}_{\mathrm{expected}}$	1.37E-03										
	$\frac{E^{\mathrm{DCS}}_{\mathrm{expected}}}{E^{\mathrm{CCS}}_{\mathrm{expected}}}$	1.371	1.256	1.145	1.044	0.951	0.869	0.797	0.734	0.678	0.629	0.587
15.0	$E^{\mathrm{DCS}}_{\mathrm{expected}}$	2.08E-03	1.90E-03	1.74E-03	1.58E-03	1.44E-03	1.32E-03	1.21E-03	1.11E-03	1.03E-03	9.54E-04	8.90E-04
	$E^{\mathrm{DCS}}_{\mathrm{expected}}$	1.52E-03										
	$\frac{E^{\mathrm{DCS}}_{\mathrm{expected}}}{E^{\mathrm{CCS}}_{\mathrm{expected}}}$	1.365	1.250	1.141	1.039	0.947	0.866	0.794	0.730	0.675	0.627	0.584

Solid Slab-Stratospheric ozone depletion (kg CFC11 eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
16.5	$E_{expected}^{DCS}$	2.55E-03	2.34E-03	2.13E-03	1.94E-03	1.77E-03	1.62E-03	1.48E-03	1.36E-03	1.26E-03	1.17E-03	1.09E-03
	$E_{expected}^{DCS}$	1.90E-03										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.340	1.227	1.119	1.019	0.929	0.849	0.778	0.715	0.661	0.613	0.572
18.0	$E_{expected}^{DCS}$	2.77E-03	2.54E-03	2.32E-03	2.11E-03	1.92E-03	1.76E-03	1.61E-03	1.48E-03	1.37E-03	1.27E-03	1.18E-03
	$E_{expected}^{DCS}$	2.08E-03										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.336	1.224	1.116	1.016	0.926	0.846	0.776	0.714	0.659	0.612	0.570

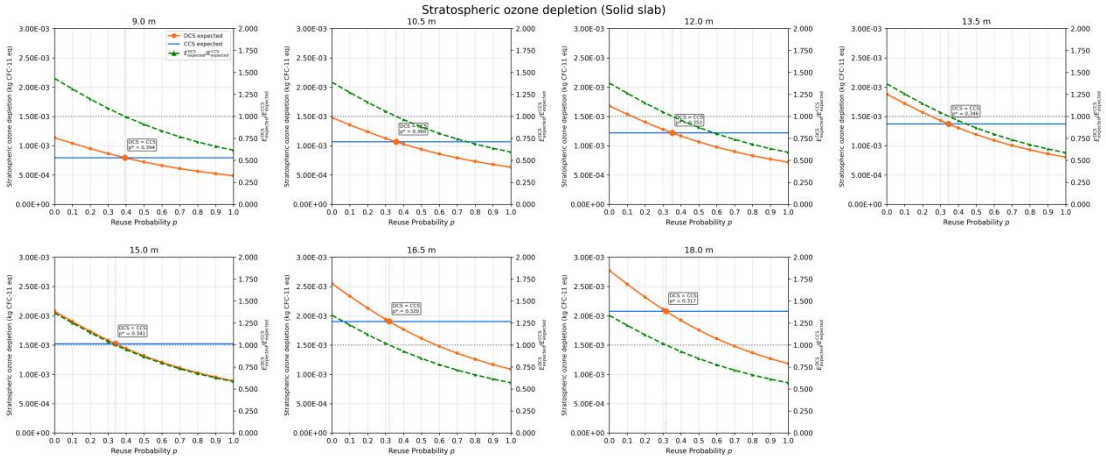


Figure E-2 p* derived from calculations of CCS ($E_{expected}^{CCS}$) and DCS ($E_{expected}^{DCS}$) of Stratospheric ozone depletion

Table E-3 p* derived from calculations of CCS ($E_{expected}^{CCS}$) and DCS ($E_{expected}^{DCS}$) of Ionizing radiation

Solid Slab-Ionizing radiation (kBq Co-60 eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E_{expected}^{DCS}$	7.50E+01	6.81E+01	6.16E+01	5.56E+01	5.01E+01	4.53E+01	4.10E+01	3.73E+01	3.41E+01	3.12E+01	2.87E+01
	$E_{expected}^{DCS}$	5.84E+01										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.283	1.166	1.054	0.951	0.858	0.775	0.703	0.639	0.583	0.534	0.492
10.5	$E_{expected}^{DCS}$	9.88E+01	8.98E+01	8.12E+01	7.32E+01	6.60E+01	5.96E+01	5.40E+01	4.91E+01	4.48E+01	4.10E+01	3.77E+01
	$E_{expected}^{DCS}$	7.96E+01										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.242	1.129	1.020	0.920	0.830	0.749	0.679	0.617	0.563	0.516	0.474
12.0	$E_{expected}^{DCS}$	1.12E+02	1.02E+02	9.21E+01	8.31E+01	7.49E+01	6.77E+01	6.13E+01	5.57E+01	5.08E+01	4.66E+01	4.28E+01
	$E_{expected}^{DCS}$	9.09E+01										

Solid Slab-Ionizing radiation (kBq Co-60 eq)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.234	1.122	1.014	0.914	0.824	0.745	0.675	0.613	0.559	0.512	0.471
13.5	$E^{DCS}_{expected}$	1.26E+02	1.14E+02	1.03E+02	9.30E+01	8.39E+01	7.58E+01	6.86E+01	6.24E+01	5.69E+01	5.21E+01	4.79E+01
	$E^{DCS}_{expected}$	1.02E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.228	1.116	1.008	0.909	0.820	0.741	0.671	0.610	0.556	0.510	0.469
15.0	$E^{DCS}_{expected}$	1.39E+02	1.26E+02	1.14E+02	1.03E+02	9.28E+01	8.38E+01	7.59E+01	6.90E+01	6.29E+01	5.76E+01	5.30E+01
	$E^{DCS}_{expected}$	1.14E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.223	1.111	1.004	0.906	0.817	0.738	0.668	0.607	0.554	0.508	0.467
16.5	$E^{DCS}_{expected}$	1.73E+02	1.57E+02	1.42E+02	1.28E+02	1.15E+02	1.04E+02	9.41E+01	8.55E+01	7.80E+01	7.14E+01	6.57E+01
	$E^{DCS}_{expected}$	1.44E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.199	1.089	0.984	0.887	0.800	0.722	0.654	0.594	0.542	0.496	0.456
18.0	$E^{DCS}_{expected}$	1.88E+02	1.70E+02	1.54E+02	1.39E+02	1.25E+02	1.13E+02	1.02E+02	9.30E+01	8.48E+01	7.77E+01	7.14E+01
	$E^{DCS}_{expected}$	1.57E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.196	1.086	0.982	0.885	0.798	0.721	0.652	0.593	0.541	0.495	0.455

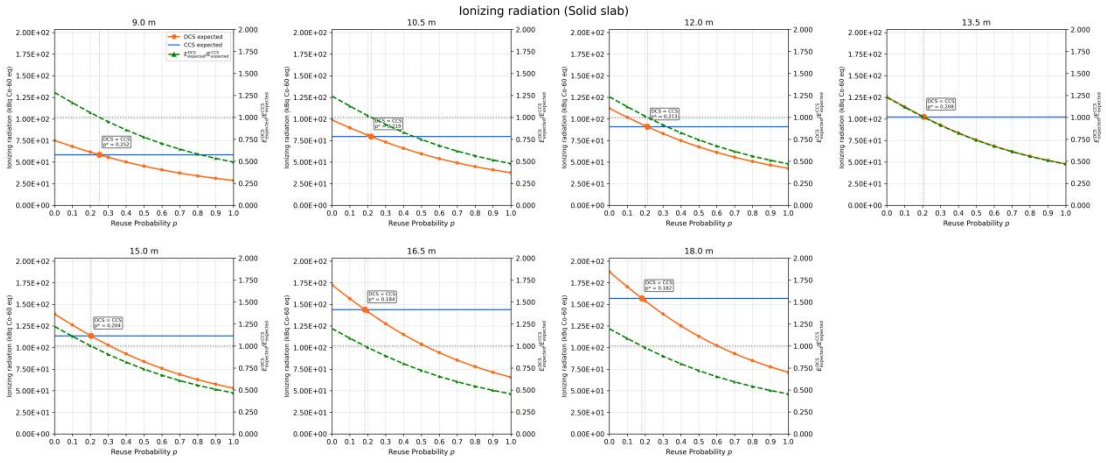


Figure E-3 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ionizing radiation

Table E-4 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation, Human health

Solid Slab-Ozone formation, Human health (kg NOx eq)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	7.54E+00	6.97E+00	6.42E+00	5.91E+00	5.45E+00	5.05E+00	4.69E+00	4.38E+00	4.10E+00	3.86E+00	3.65E+00
	$E^{DCS}_{expected}$	5.62E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.342	1.239	1.142	1.051	0.970	0.898	0.834	0.778	0.730	0.687	0.650
10.5	$E^{DCS}_{expected}$	9.89E+00	9.13E+00	8.40E+00	7.74E+00	7.13E+00	6.60E+00	6.12E+00	5.71E+00	5.35E+00	5.03E+00	4.76E+00
	$E^{DCS}_{expected}$	7.61E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.299	1.200	1.104	1.017	0.937	0.867	0.805	0.751	0.703	0.662	0.625
12.0	$E^{DCS}_{expected}$	1.12E+01	1.04E+01	9.53E+00	8.78E+00	8.09E+00	7.48E+00	6.95E+00	6.48E+00	6.07E+00	5.71E+00	5.40E+00
	$E^{DCS}_{expected}$	8.69E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.291	1.192	1.097	1.010	0.931	0.861	0.800	0.746	0.698	0.657	0.621
13.5	$E^{DCS}_{expected}$	1.26E+01	1.16E+01	1.07E+01	9.82E+00	9.06E+00	8.38E+00	7.78E+00	7.25E+00	6.79E+00	6.39E+00	6.04E+00
	$E^{DCS}_{expected}$	9.78E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.284	1.185	1.091	1.004	0.926	0.856	0.795	0.741	0.694	0.653	0.618
15.0	$E^{DCS}_{expected}$	1.39E+01	1.28E+01	1.18E+01	1.09E+01	1.00E+01	9.26E+00	8.60E+00	8.02E+00	7.51E+00	7.07E+00	6.68E+00
	$E^{DCS}_{expected}$	1.09E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.279	1.180	1.087	1.000	0.922	0.853	0.792	0.738	0.692	0.651	0.615
16.5	$E^{DCS}_{expected}$	1.72E+01	1.58E+01	1.46E+01	1.34E+01	1.23E+01	1.14E+01	1.06E+01	9.86E+00	9.23E+00	8.68E+00	8.20E+00
	$E^{DCS}_{expected}$	1.37E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.254	1.157	1.065	0.979	0.902	0.834	0.774	0.721	0.675	0.635	0.599
18.0	$E^{DCS}_{expected}$	1.87E+01	1.72E+01	1.58E+01	1.46E+01	1.34E+01	1.24E+01	1.15E+01	1.07E+01	1.00E+01	9.45E+00	8.92E+00
	$E^{DCS}_{expected}$	1.49E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.251	1.154	1.062	0.977	0.900	0.832	0.772	0.719	0.673	0.633	0.598

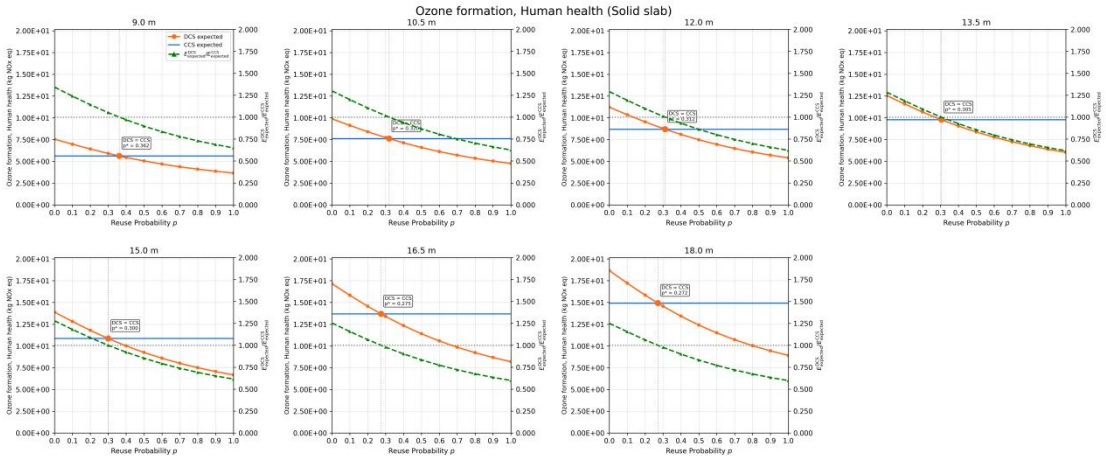


Figure E-4 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation, Human health

Table E-5 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fine particulate matter formation

Solid Slab-Fine particulate matter formation (kg PM2.5eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	4.29E+00	3.95E+00	3.62E+00	3.32E+00	3.05E+00	2.81E+00	2.60E+00	2.41E+00	2.25E+00	2.11E+00	1.98E+00
	$E^{DCS}_{expected}$	2.84E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.509	1.389	1.274	1.169	1.073	0.988	0.913	0.848	0.791	0.741	0.697
10.5	$E^{DCS}_{expected}$	5.70E+00	5.24E+00	4.80E+00	4.40E+00	4.04E+00	3.71E+00	3.43E+00	3.18E+00	2.96E+00	2.77E+00	2.61E+00
	$E^{DCS}_{expected}$	3.94E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.446	1.330	1.219	1.117	1.025	0.943	0.871	0.807	0.752	0.704	0.662
12.0	$E^{DCS}_{expected}$	6.46E+00	5.94E+00	5.45E+00	4.99E+00	4.58E+00	4.21E+00	3.89E+00	3.61E+00	3.36E+00	3.15E+00	2.96E+00
	$E^{DCS}_{expected}$	4.50E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.438	1.323	1.212	1.111	1.019	0.937	0.866	0.803	0.748	0.700	0.658
13.5	$E^{DCS}_{expected}$	7.24E+00	6.66E+00	6.10E+00	5.59E+00	5.13E+00	4.72E+00	4.36E+00	4.04E+00	3.76E+00	3.52E+00	3.31E+00
	$E^{DCS}_{expected}$	5.06E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.431	1.316	1.206	1.105	1.014	0.932	0.861	0.798	0.744	0.696	0.654
15.0	$E^{DCS}_{expected}$	8.01E+00	7.37E+00	6.75E+00	6.18E+00	5.67E+00	5.22E+00	4.82E+00	4.47E+00	4.16E+00	3.90E+00	3.66E+00
	$E^{DCS}_{expected}$	5.62E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.426	1.311	1.202	1.101	1.010	0.929	0.858	0.795	0.741	0.693	0.652
16.5	$E^{DCS}_{expected}$	1.00E+01	9.20E+00	8.43E+00	7.72E+00	7.07E+00	6.50E+00	6.00E+00	5.55E+00	5.17E+00	4.83E+00	4.54E+00

Solid Slab-Fine particulate matter formation (kg PM2.5eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E^{DCS}_{expected}$	7.23E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.385	1.273	1.166	1.067	0.978	0.899	0.829	0.768	0.715	0.668	0.628
18.0	$E^{DCS}_{expected}$	1.09E +01	1.00E +01	9.17E +00	8.39E +00	7.69E +00	7.07E +00	6.52E +00	6.04E +00	5.62E +00	5.26E +00	4.94E +00
	$E^{DCS}_{expected}$	7.88E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.382	1.270	1.163	1.065	0.976	0.897	0.827	0.766	0.713	0.667	0.626

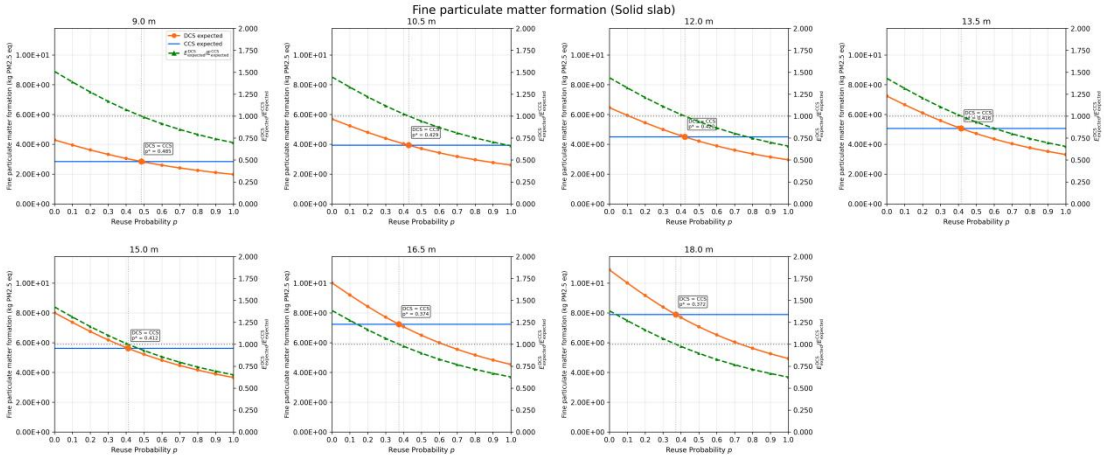


Figure E-5 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fine particulate matter formation

Table E-6 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation, Terrestrial ecosystem

Solid Slab-Ozone formation, Terrestrial ecosystem (kg NOx eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	7.71E +00	7.12E +00	6.56E +00	6.04E +00	5.57E +00	5.15E +00	4.78E +00	4.46E +00	4.18E +00	3.94E +00	3.72E +00
	$E^{DCS}_{expected}$	5.75E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.340	1.237	1.139	1.049	0.968	0.895	0.831	0.776	0.727	0.684	0.647
10.5	$E^{DCS}_{expected}$	1.01E +01	9.33E +00	8.59E +00	7.90E +00	7.28E +00	6.73E +00	6.25E +00	5.83E +00	5.46E +00	5.13E +00	4.85E +00
	$E^{DCS}_{expected}$	7.79E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.297	1.197	1.102	1.014	0.935	0.864	0.802	0.748	0.700	0.659	0.622
12.0	$E^{DCS}_{expected}$	1.15E +01	1.06E +01	9.74E +00	8.97E +00	8.26E +00	7.64E +00	7.09E +00	6.61E +00	6.19E +00	5.82E +00	5.50E +00
	$E^{DCS}_{expected}$	8.90E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.289	1.190	1.095	1.007	0.928	0.858	0.797	0.743	0.695	0.654	0.618
13.5	$E^{DCS}_{expected}$	1.28E	1.19E	1.09E	1.00E	9.25E	8.55E	7.94E	7.40E	6.93E	6.52E	6.16E

Solid Slab-Ozone formation, Terrestrial ecosystem (kg NOx eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
		+01	+01	+01	+01	+00	+00	+00	+00	+00	+00	+00
	$E^{DCS}_{expected}$	1.00E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.282	1.183	1.089	1.002	0.923	0.854	0.792	0.738	0.691	0.650	0.615
15.0	$E^{DCS}_{expected}$	1.42E +01	1.31E +01	1.21E +01	1.11E +01	1.02E +01	9.46E +00	8.78E +00	8.18E +00	7.66E +00	7.21E +00	6.81E +00
	$E^{DCS}_{expected}$	1.11E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.277	1.178	1.084	0.998	0.920	0.850	0.789	0.735	0.689	0.648	0.612
16.5	$E^{DCS}_{expected}$	1.76E +01	1.62E +01	1.49E +01	1.37E +01	1.26E +01	1.17E +01	1.08E +01	1.01E +01	9.42E +00	8.86E +00	8.36E +00
	$E^{DCS}_{expected}$	1.40E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.252	1.155	1.062	0.977	0.900	0.831	0.771	0.718	0.672	0.631	0.596
18.0	$E^{DCS}_{expected}$	1.91E +01	1.76E +01	1.62E +01	1.49E +01	1.37E +01	1.27E +01	1.18E +01	1.10E +01	1.02E +01	9.63E +00	9.10E +00
	$E^{DCS}_{expected}$	1.53E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.248	1.152	1.059	0.974	0.897	0.829	0.769	0.716	0.670	0.630	0.595

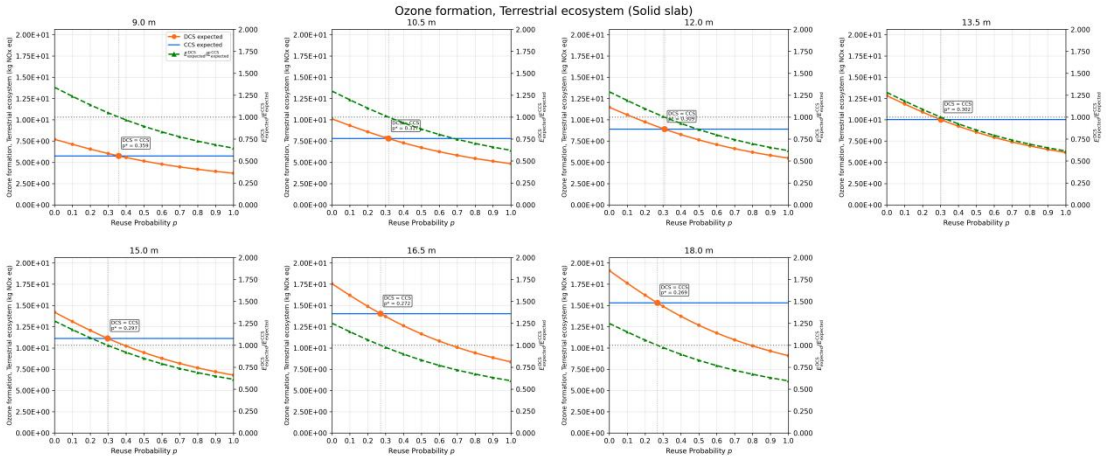


Figure E-6 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation, Terrestrial ecosystem

Table E-7 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial acidification

Solid Slab-Terrestrial acidification (kg SO2 eq)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	9.81E+00	9.05E+00	8.33E+00	7.66E+00	7.06E+00	6.53E+00	6.06E+00	5.65E+00	5.29E+00	4.97E+00	4.70E+00
	$E^{DCS}_{expected}$	6.29E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.561	1.440	1.325	1.219	1.124	1.039	0.964	0.898	0.841	0.791	0.747
10.5	$E^{DCS}_{expected}$	1.29E+01	1.19E+01	1.09E+01	1.00E+01	9.23E+00	8.53E+00	7.91E+00	7.37E+00	6.89E+00	6.48E+00	6.12E+00
	$E^{DCS}_{expected}$	8.55E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.504	1.387	1.276	1.173	1.080	0.998	0.926	0.862	0.806	0.758	0.716
12.0	$E^{DCS}_{expected}$	1.46E+01	1.35E+01	1.24E+01	1.14E+01	1.05E+01	9.68E+00	8.97E+00	8.36E+00	7.82E+00	7.35E+00	6.94E+00
	$E^{DCS}_{expected}$	9.76E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.494	1.378	1.268	1.166	1.073	0.992	0.920	0.856	0.801	0.753	0.711
13.5	$E^{DCS}_{expected}$	1.63E+01	1.51E+01	1.38E+01	1.27E+01	1.17E+01	1.08E+01	1.00E+01	9.35E+00	8.75E+00	8.22E+00	7.76E+00
	$E^{DCS}_{expected}$	1.10E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.486	1.371	1.261	1.159	1.067	0.986	0.914	0.851	0.797	0.749	0.707
15.0	$E^{DCS}_{expected}$	1.81E+01	1.67E+01	1.53E+01	1.41E+01	1.30E+01	1.20E+01	1.11E+01	1.03E+01	9.68E+00	9.09E+00	8.58E+00
	$E^{DCS}_{expected}$	1.22E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.480	1.365	1.256	1.154	1.063	0.982	0.911	0.848	0.793	0.746	0.704
16.5	$E^{DCS}_{expected}$	2.23E+01	2.06E+01	1.89E+01	1.74E+01	1.60E+01	1.48E+01	1.37E+01	1.27E+01	1.19E+01	1.12E+01	1.05E+01
	$E^{DCS}_{expected}$	1.54E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.445	1.332	1.224	1.125	1.035	0.956	0.886	0.824	0.771	0.724	0.683
18.0	$E^{DCS}_{expected}$	2.43E+01	2.24E+01	2.06E+01	1.89E+01	1.74E+01	1.60E+01	1.49E+01	1.38E+01	1.29E+01	1.22E+01	1.15E+01
	$E^{DCS}_{expected}$	1.68E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.441	1.329	1.221	1.122	1.033	0.953	0.883	0.822	0.769	0.722	0.681

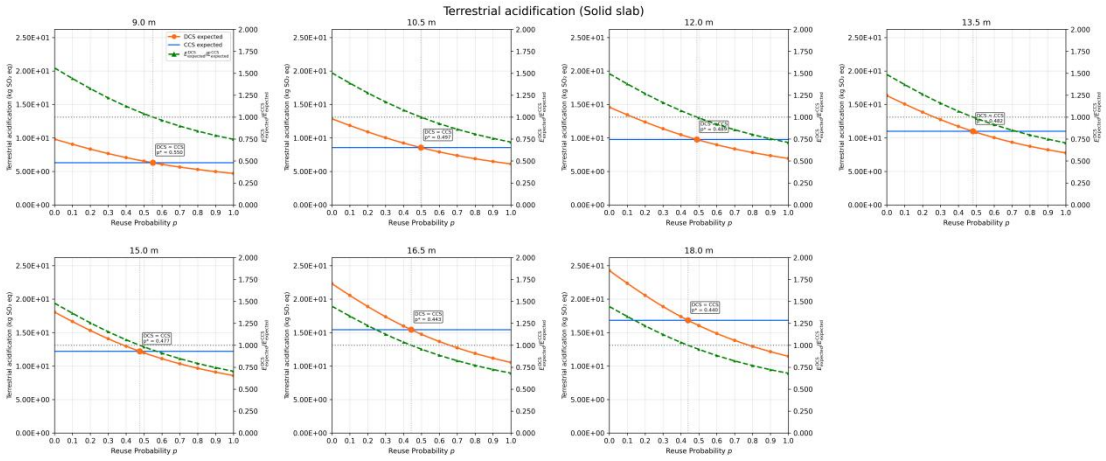


Figure E-7 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial acidification

Table E-8 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater eutrophication

Solid Slab-Freshwater eutrophication (kg P eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	7.83E-01	7.22E-01	6.63E-01	6.09E-01	5.61E-01	5.17E-01	4.79E-01	4.46E-01	4.17E-01	3.92E-01	3.70E-01
	$E^{DCS}_{expected}$	6.13E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.278	1.178	1.082	0.994	0.915	0.845	0.783	0.728	0.681	0.640	0.603
10.5	$E^{DCS}_{expected}$	1.08E+00	9.93E-01	9.11E-01	8.35E-01	7.67E-01	7.07E-01	6.53E-01	6.07E-01	5.66E-01	5.30E-01	4.99E-01
	$E^{DCS}_{expected}$	8.84E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.220	1.123	1.030	0.945	0.868	0.799	0.739	0.686	0.640	0.600	0.565
12.0	$E^{DCS}_{expected}$	1.23E+00	1.13E+00	1.03E+00	9.49E-01	8.71E-01	8.02E-01	7.42E-01	6.89E-01	6.43E-01	6.02E-01	5.67E-01
	$E^{DCS}_{expected}$	1.01E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.215	1.119	1.026	0.941	0.864	0.796	0.736	0.683	0.637	0.597	0.562
13.5	$E^{DCS}_{expected}$	1.38E+00	1.27E+00	1.16E+00	1.06E+00	9.77E-01	9.00E-01	8.32E-01	7.73E-01	7.21E-01	6.75E-01	6.35E-01
	$E^{DCS}_{expected}$	1.14E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.211	1.114	1.022	0.937	0.861	0.793	0.733	0.680	0.634	0.594	0.559
15.0	$E^{DCS}_{expected}$	1.52E+00	1.40E+00	1.28E+00	1.18E+00	1.08E+00	9.96E-01	9.21E-01	8.55E-01	7.97E-01	7.47E-01	7.03E-01
	$E^{DCS}_{expected}$	1.26E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.208	1.112	1.020	0.935	0.858	0.790	0.731	0.678	0.633	0.593	0.558
16.5	$E^{DCS}_{expected}$	1.97E+00	1.81E+00	1.66E+00	1.52E+00	1.39E+00	1.28E+00	1.18E+00	1.09E+00	1.02E+00	9.51E-01	8.93E-01

Solid Slab-Freshwater eutrophication (kg P eq)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E^{DCS}_{expected}$	1.68E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.172	1.077	0.986	0.903	0.827	0.760	0.702	0.650	0.605	0.566	0.531
18.0	$E^{DCS}_{expected}$	2.14E+00	1.97E+00	1.80E+00	1.65E+00	1.51E+00	1.39E+00	1.28E+00	1.19E+00	1.11E+00	1.03E+00	9.72E-01
	$E^{DCS}_{expected}$	1.83E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.170	1.075	0.984	0.901	0.826	0.759	0.700	0.649	0.604	0.565	0.530

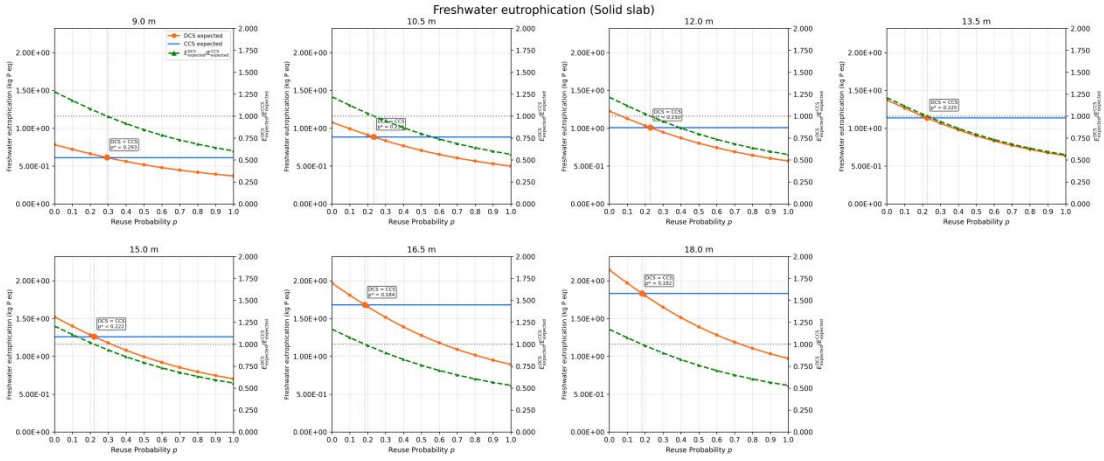


Figure E-8 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater eutrophication

Table E-9 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine eutrophication

Solid Slab-Marine eutrophication (kg N eq)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	7.46E-02	6.83E-02	6.22E-02	5.66E-02	5.16E-02	4.71E-02	4.32E-02	3.97E-02	3.67E-02	3.41E-02	3.18E-02
	$E^{DCS}_{expected}$	6.28E-02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.188	1.087	0.990	0.901	0.821	0.749	0.687	0.632	0.584	0.542	0.505
10.5	$E^{DCS}_{expected}$	9.79E-02	8.96E-02	8.16E-02	7.42E-02	6.76E-02	6.17E-02	5.65E-02	5.19E-02	4.80E-02	4.45E-02	4.15E-02
	$E^{DCS}_{expected}$	8.49E-02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.154	1.055	0.961	0.874	0.796	0.727	0.665	0.612	0.565	0.524	0.489
12.0	$E^{DCS}_{expected}$	1.11E-01	1.02E-01	9.26E-02	8.42E-02	7.67E-02	7.00E-02	6.41E-02	5.89E-02	5.44E-02	5.05E-02	4.71E-02
	$E^{DCS}_{expected}$	9.69E-02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.146	1.048	0.955	0.869	0.791	0.722	0.661	0.608	0.561	0.521	0.485
13.5	$E^{DCS}_{expected}$	1.24E-	1.14E-	1.04E-	9.43E-	8.58E-	7.83E-	7.17E-	6.60E-	6.09E-	5.65E-	5.27E-

Solid Slab-Marine eutrophication (kg N eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
		01	01	01	02	02	02	02	02	02	02	02
	$E^{DCS}_{expected}$	1.09E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.140	1.043	0.950	0.864	0.787	0.718	0.657	0.605	0.558	0.518	0.483
15.0	$E^{DCS}_{expected}$	1.38E-01	1.26E-01	1.15E-01	1.04E-01	9.49E-02	8.66E-02	7.93E-02	7.29E-02	6.74E-02	6.25E-02	5.82E-02
	$E^{DCS}_{expected}$	1.21E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.136	1.038	0.946	0.860	0.783	0.715	0.655	0.602	0.556	0.516	0.481
16.5	$E^{DCS}_{expected}$	1.70E-01	1.56E-01	1.42E-01	1.29E-01	1.17E-01	1.07E-01	9.79E-02	9.00E-02	8.30E-02	7.70E-02	7.17E-02
	$E^{DCS}_{expected}$	1.52E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.117	1.021	0.930	0.845	0.769	0.702	0.642	0.590	0.545	0.505	0.471
18.0	$E^{DCS}_{expected}$	1.85E-01	1.69E-01	1.54E-01	1.40E-01	1.28E-01	1.16E-01	1.06E-01	9.79E-02	9.03E-02	8.38E-02	7.80E-02
	$E^{DCS}_{expected}$	1.66E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.114	1.019	0.927	0.843	0.767	0.700	0.641	0.589	0.544	0.504	0.469

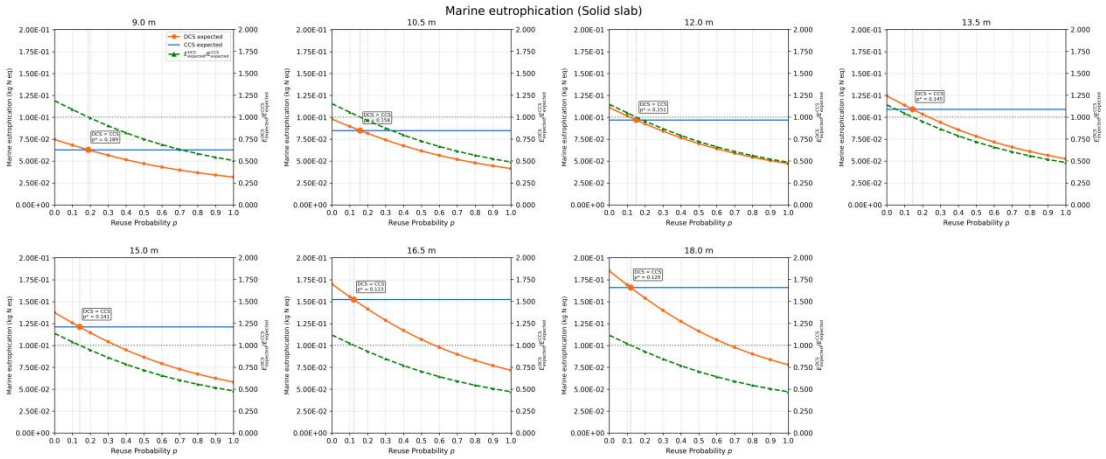


Figure E-9 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine eutrophication

Table E-10 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial ecotoxicity

Solid Slab-Terrestrial ecotoxicity (kg 1,4-DCB)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	1.26E+04	1.15E+04	1.04E+04	9.40E+03	8.48E+03	7.66E+03	6.94E+03	6.30E+03	5.75E+03	5.26E+03	4.84E+03
	$E^{DCS}_{expected}$	1.02E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.243	1.130	1.023	0.923	0.833	0.753	0.682	0.619	0.565	0.517	0.475
10.5	$E^{DCS}_{expected}$	1.68E+04	1.53E+04	1.38E+04	1.24E+04	1.12E+04	1.01E+04	9.18E+03	8.34E+03	7.60E+03	6.95E+03	6.39E+03
	$E^{DCS}_{expected}$	1.40E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.202	1.093	0.989	0.892	0.805	0.727	0.658	0.598	0.545	0.498	0.458
12.0	$E^{DCS}_{expected}$	1.90E+04	1.73E+04	1.57E+04	1.41E+04	1.27E+04	1.15E+04	1.04E+04	9.46E+03	8.62E+03	7.89E+03	7.25E+03
	$E^{DCS}_{expected}$	1.59E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.195	1.087	0.983	0.887	0.800	0.722	0.654	0.594	0.541	0.495	0.455
13.5	$E^{DCS}_{expected}$	2.13E+04	1.94E+04	1.75E+04	1.58E+04	1.43E+04	1.29E+04	1.17E+04	1.06E+04	9.66E+03	8.84E+03	8.12E+03
	$E^{DCS}_{expected}$	1.79E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.189	1.081	0.978	0.882	0.796	0.719	0.651	0.591	0.539	0.493	0.453
15.0	$E^{DCS}_{expected}$	2.36E+04	2.15E+04	1.94E+04	1.75E+04	1.58E+04	1.43E+04	1.29E+04	1.17E+04	1.07E+04	9.78E+03	8.98E+03
	$E^{DCS}_{expected}$	1.99E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.185	1.077	0.974	0.879	0.793	0.716	0.648	0.589	0.537	0.491	0.451
16.5	$E^{DCS}_{expected}$	2.95E+04	2.68E+04	2.42E+04	2.18E+04	1.97E+04	1.78E+04	1.61E+04	1.46E+04	1.33E+04	1.22E+04	1.12E+04
	$E^{DCS}_{expected}$	2.54E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.161	1.056	0.954	0.861	0.776	0.701	0.634	0.575	0.524	0.479	0.440
18.0	$E^{DCS}_{expected}$	3.21E+04	2.92E+04	2.64E+04	2.38E+04	2.14E+04	1.93E+04	1.75E+04	1.59E+04	1.45E+04	1.32E+04	1.22E+04
	$E^{DCS}_{expected}$	2.77E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.158	1.053	0.952	0.859	0.774	0.699	0.632	0.574	0.523	0.478	0.439

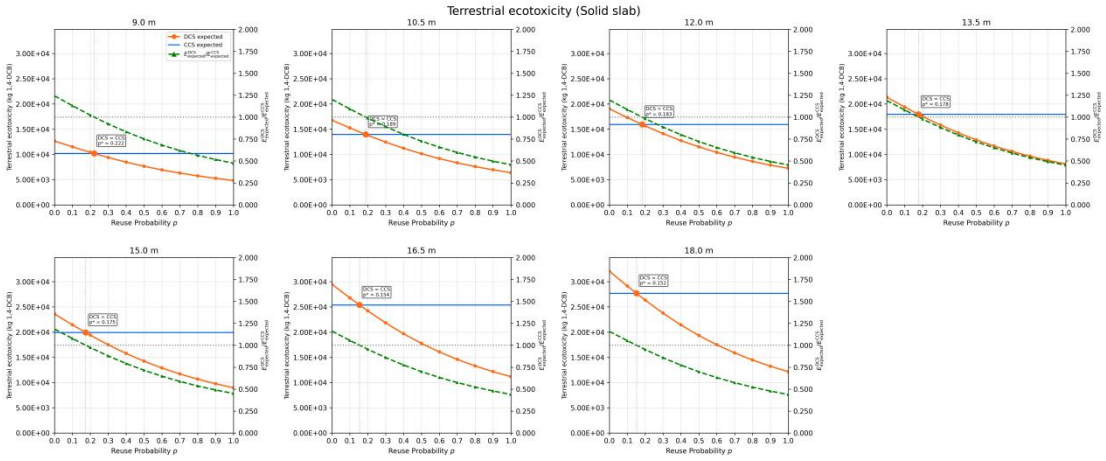


Figure E-10 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial ecotoxicity

Table E-11 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater ecotoxicity

Solid Slab-Freshwater ecotoxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	8.20E+01	7.48E+01	6.80E+01	6.17E+01	5.60E+01	5.09E+01	4.64E+01	4.25E+01	3.91E+01	3.61E+01	3.35E+01
	$E^{DCS}_{expected}$	6.20E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.322	1.206	1.096	0.994	0.902	0.820	0.748	0.685	0.630	0.582	0.540
10.5	$E^{DCS}_{expected}$	1.16E+02	1.05E+02	9.55E+01	8.65E+01	7.84E+01	7.12E+01	6.48E+01	5.93E+01	5.44E+01	5.02E+01	4.65E+01
	$E^{DCS}_{expected}$	9.25E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.249	1.138	1.033	0.936	0.848	0.770	0.701	0.641	0.588	0.542	0.502
12.0	$E^{DCS}_{expected}$	1.31E+02	1.20E+02	1.09E+02	9.83E+01	8.91E+01	8.09E+01	7.37E+01	6.73E+01	6.18E+01	5.70E+01	5.28E+01
	$E^{DCS}_{expected}$	1.05E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.246	1.135	1.030	0.933	0.845	0.767	0.699	0.639	0.587	0.541	0.501
13.5	$E^{DCS}_{expected}$	1.48E+02	1.34E+02	1.22E+02	1.10E+02	1.00E+02	9.09E+01	8.27E+01	7.56E+01	6.94E+01	6.40E+01	5.92E+01
	$E^{DCS}_{expected}$	1.19E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.242	1.132	1.027	0.930	0.843	0.765	0.697	0.637	0.584	0.539	0.499
15.0	$E^{DCS}_{expected}$	1.63E+02	1.49E+02	1.35E+02	1.22E+02	1.11E+02	1.01E+02	9.16E+01	8.37E+01	7.68E+01	7.08E+01	6.56E+01
	$E^{DCS}_{expected}$	1.32E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.240	1.130	1.025	0.928	0.841	0.764	0.695	0.636	0.583	0.538	0.498
16.5	$E^{DCS}_{expected}$	2.15E+02	1.96E+02	1.78E+02	1.61E+02	1.45E+02	1.32E+02	1.20E+02	1.09E+02	1.00E+02	9.22E+01	8.52E+01

Solid Slab-Freshwater ecotoxicity (kg 1,4-DCB)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E^{DCS}_{expected}$	1.80E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.193	1.086	0.985	0.891	0.806	0.731	0.664	0.606	0.556	0.511	0.473
18.0	$E^{DCS}_{expected}$	2.34E+02	2.13E+02	1.93E+02	1.75E+02	1.58E+02	1.43E+02	1.30E+02	1.19E+02	1.09E+02	1.00E+02	9.28E+01
	$E^{DCS}_{expected}$	1.97E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.192	1.085	0.983	0.890	0.805	0.730	0.664	0.606	0.555	0.511	0.472

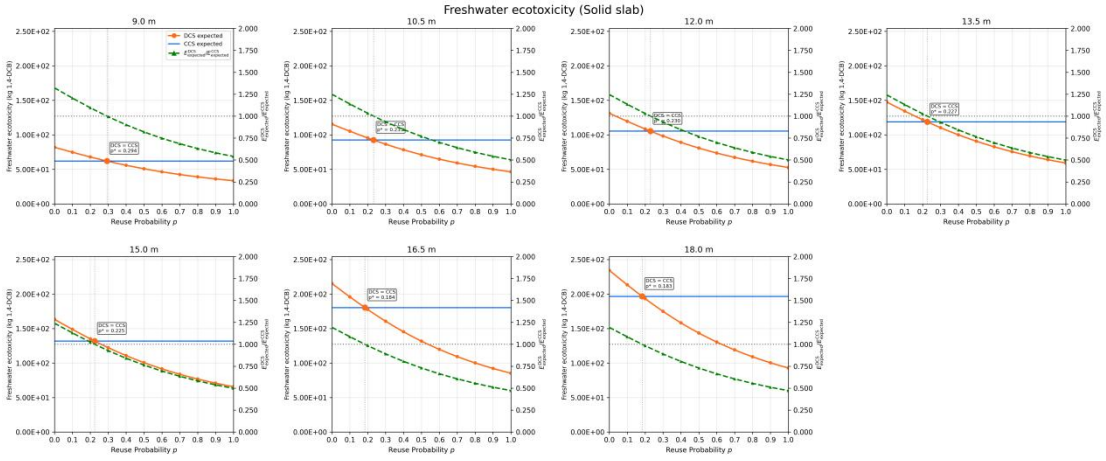


Figure E-11 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater ecotoxicity

Table E-12 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine ecotoxicity

Solid Slab-Marine ecotoxicity (kg 1,4-DCB)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	1.19E+02	1.08E+02	9.83E+01	8.91E+01	8.09E+01	7.35E+01	6.70E+01	6.14E+01	5.64E+01	5.21E+01	4.83E+01
	$E^{DCS}_{expected}$	9.00E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.318	1.203	1.092	0.990	0.898	0.817	0.745	0.682	0.627	0.579	0.536
10.5	$E^{DCS}_{expected}$	1.67E+02	1.52E+02	1.38E+02	1.25E+02	1.13E+02	1.03E+02	9.35E+01	8.54E+01	7.84E+01	7.22E+01	6.68E+01
	$E^{DCS}_{expected}$	1.34E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.247	1.136	1.031	0.933	0.845	0.767	0.699	0.638	0.586	0.540	0.499
12.0	$E^{DCS}_{expected}$	1.90E+02	1.73E+02	1.57E+02	1.42E+02	1.29E+02	1.17E+02	1.06E+02	9.71E+01	8.90E+01	8.20E+01	7.59E+01
	$E^{DCS}_{expected}$	1.53E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.243	1.133	1.028	0.930	0.843	0.765	0.696	0.636	0.584	0.538	0.498
13.5	$E^{DCS}_{expected}$	2.13E	1.94E	1.76E	1.59E	1.44E	1.31E	1.19E	1.09E	1.00E	9.21E	8.52E

Solid Slab-Marine ecotoxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
		+02	+02	+02	+02	+02	+02	+02	+02	+02	+01	+01
	$E^{DCS}_{expected}$	1.72E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.239	1.129	1.024	0.927	0.840	0.762	0.694	0.634	0.582	0.536	0.496
15.0	$E^{DCS}_{expected}$	2.36E+02	2.15E+02	1.95E+02	1.76E+02	1.60E+02	1.45E+02	1.32E+02	1.21E+02	1.11E+02	1.02E+02	9.43E+01
	$E^{DCS}_{expected}$	1.91E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.237	1.127	1.023	0.926	0.839	0.761	0.693	0.633	0.581	0.535	0.495
16.5	$E^{DCS}_{expected}$	3.10E+02	2.83E+02	2.56E+02	2.31E+02	2.09E+02	1.90E+02	1.72E+02	1.57E+02	1.44E+02	1.33E+02	1.22E+02
	$E^{DCS}_{expected}$	2.60E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.192	1.085	0.983	0.889	0.804	0.729	0.662	0.604	0.553	0.509	0.470
18.0	$E^{DCS}_{expected}$	3.38E+02	3.08E+02	2.79E+02	2.52E+02	2.28E+02	2.07E+02	1.88E+02	1.71E+02	1.57E+02	1.44E+02	1.33E+02
	$E^{DCS}_{expected}$	2.84E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.190	1.084	0.982	0.888	0.803	0.728	0.662	0.603	0.553	0.508	0.470

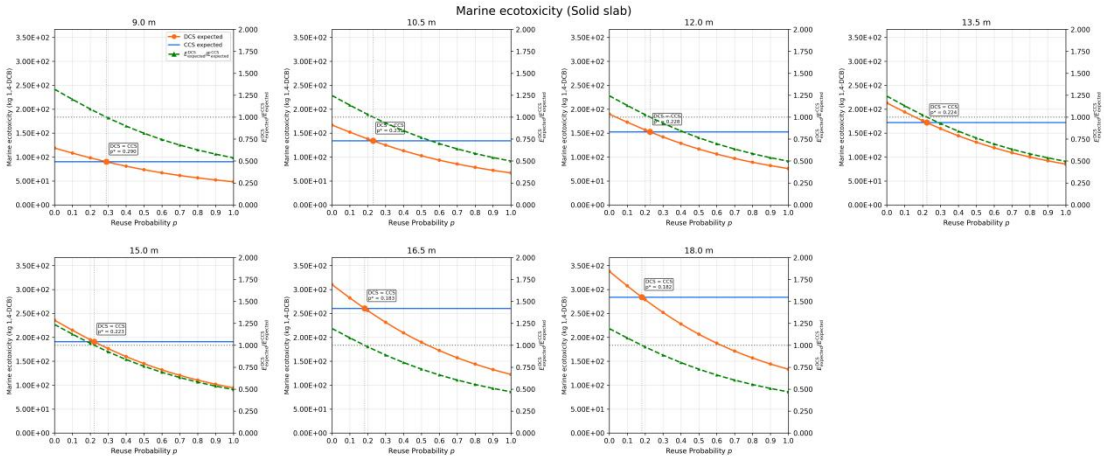


Figure E-12 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine ecotoxicity

Table E-13 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Human carcinogenic toxicity

Solid Slab-Human carcinogenic toxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	3.08E+02	2.79E+02	2.51E+02	2.25E+02	2.02E+02	1.82E+02	1.64E+02	1.48E+02	1.34E+02	1.22E+02	1.11E+02
	$E^{DCS}_{expected}$	2.19E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.404	1.271	1.144	1.027	0.922	0.828	0.746	0.674	0.611	0.556	0.508
10.5	$E^{DCS}_{expected}$	4.56E+02	4.12E+02	3.71E+02	3.33E+02	2.98E+02	2.68E+02	2.41E+02	2.17E+02	1.97E+02	1.79E+02	1.63E+02
	$E^{DCS}_{expected}$	3.53E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.292	1.169	1.051	0.943	0.846	0.759	0.683	0.616	0.558	0.507	0.463
12.0	$E^{DCS}_{expected}$	5.19E+02	4.69E+02	4.22E+02	3.79E+02	3.39E+02	3.05E+02	2.74E+02	2.47E+02	2.24E+02	2.03E+02	1.86E+02
	$E^{DCS}_{expected}$	4.02E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.292	1.169	1.051	0.943	0.845	0.759	0.683	0.616	0.558	0.507	0.462
13.5	$E^{DCS}_{expected}$	5.84E+02	5.28E+02	4.75E+02	4.26E+02	3.82E+02	3.43E+02	3.09E+02	2.78E+02	2.52E+02	2.29E+02	2.09E+02
	$E^{DCS}_{expected}$	4.53E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.290	1.167	1.050	0.941	0.844	0.758	0.681	0.615	0.557	0.506	0.461
15.0	$E^{DCS}_{expected}$	6.47E+02	5.85E+02	5.26E+02	4.72E+02	4.23E+02	3.80E+02	3.42E+02	3.08E+02	2.79E+02	2.54E+02	2.31E+02
	$E^{DCS}_{expected}$	5.02E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.290	1.167	1.049	0.941	0.844	0.757	0.681	0.615	0.557	0.506	0.461
16.5	$E^{DCS}_{expected}$	8.87E+02	8.02E+02	7.21E+02	6.46E+02	5.79E+02	5.19E+02	4.67E+02	4.21E+02	3.80E+02	3.45E+02	3.15E+02
	$E^{DCS}_{expected}$	7.27E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.220	1.103	0.992	0.889	0.797	0.714	0.642	0.579	0.523	0.475	0.433
18.0	$E^{DCS}_{expected}$	9.66E+02	8.73E+02	7.85E+02	7.04E+02	6.31E+02	5.65E+02	5.08E+02	4.58E+02	4.14E+02	3.76E+02	3.43E+02
	$E^{DCS}_{expected}$	7.92E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.220	1.103	0.992	0.889	0.796	0.714	0.642	0.579	0.523	0.475	0.433

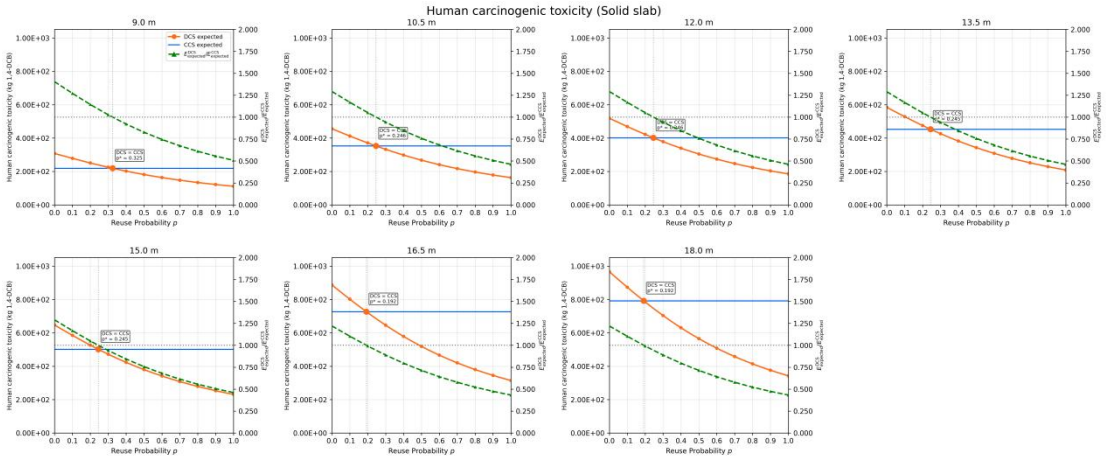


Figure E-13 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Human carcinogenic toxicity

Table E-14 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Human non-carcinogenic toxicity

Solid Slab-Human non-carcinogenic toxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	2.50E+03	2.28E+03	2.07E+03	1.88E+03	1.70E+03	1.54E+03	1.41E+03	1.29E+03	1.18E+03	1.09E+03	1.01E+03
	$E^{DCS}_{expected}$	1.91E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.309	1.194	1.083	0.982	0.890	0.808	0.736	0.673	0.618	0.570	0.528
10.5	$E^{DCS}_{expected}$	3.52E+03	3.21E+03	2.91E+03	2.63E+03	2.38E+03	2.16E+03	1.96E+03	1.79E+03	1.64E+03	1.51E+03	1.40E+03
	$E^{DCS}_{expected}$	2.84E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.239	1.129	1.023	0.926	0.838	0.760	0.691	0.631	0.578	0.532	0.492
12.0	$E^{DCS}_{expected}$	4.00E+03	3.65E+03	3.31E+03	2.99E+03	2.71E+03	2.45E+03	2.23E+03	2.04E+03	1.87E+03	1.72E+03	1.59E+03
	$E^{DCS}_{expected}$	3.24E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.236	1.125	1.020	0.923	0.835	0.757	0.689	0.629	0.576	0.530	0.490
13.5	$E^{DCS}_{expected}$	4.50E+03	4.10E+03	3.71E+03	3.36E+03	3.04E+03	2.76E+03	2.51E+03	2.29E+03	2.10E+03	1.93E+03	1.78E+03
	$E^{DCS}_{expected}$	3.65E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.232	1.122	1.017	0.920	0.832	0.755	0.686	0.626	0.574	0.528	0.488
15.0	$E^{DCS}_{expected}$	4.98E+03	4.53E+03	4.11E+03	3.72E+03	3.36E+03	3.05E+03	2.77E+03	2.53E+03	2.32E+03	2.13E+03	1.97E+03
	$E^{DCS}_{expected}$	4.05E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.230	1.120	1.015	0.918	0.831	0.753	0.685	0.625	0.573	0.527	0.487
16.5	$E^{DCS}_{expected}$	6.56E+03	5.97E+03	5.40E+03	4.88E+03	4.41E+03	3.99E+03	3.63E+03	3.31E+03	3.02E+03	2.78E+03	2.56E+03

Solid Slab-Human non-carcinogenic toxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E^{DCS}_{expected}$	5.53E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.185	1.078	0.976	0.882	0.797	0.722	0.656	0.598	0.547	0.502	0.464
18.0	$E^{DCS}_{expected}$	7.14E+03	6.50E+03	5.88E+03	5.31E+03	4.80E+03	4.35E+03	3.95E+03	3.60E+03	3.29E+03	3.03E+03	2.79E+03
	$E^{DCS}_{expected}$	6.03E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.184	1.077	0.975	0.881	0.796	0.721	0.655	0.597	0.546	0.502	0.463

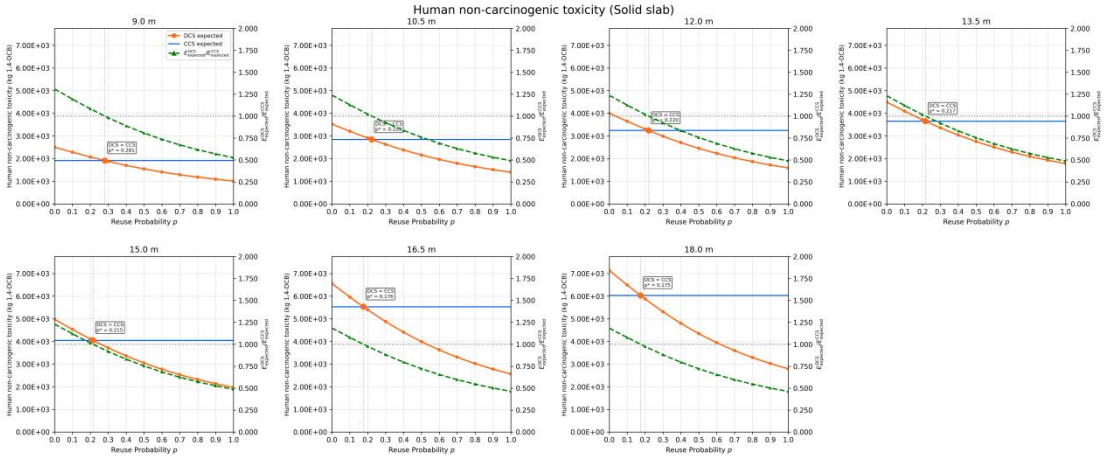


Figure E-14 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Human non-carcinogenic toxicity

Table E-15 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Land use

Solid Slab-Land use (m^2 a crop eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	7.21E+01	6.55E+01	5.91E+01	5.33E+01	4.80E+01	4.33E+01	3.92E+01	3.56E+01	3.24E+01	2.96E+01	2.72E+01
	$E^{DCS}_{expected}$	5.88E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.225	1.113	1.005	0.906	0.816	0.737	0.666	0.605	0.551	0.504	0.463
10.5	$E^{DCS}_{expected}$	9.39E+01	8.53E+01	7.70E+01	6.94E+01	6.25E+01	5.64E+01	5.10E+01	4.63E+01	4.22E+01	3.86E+01	3.54E+01
	$E^{DCS}_{expected}$	7.88E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.191	1.082	0.977	0.881	0.794	0.716	0.648	0.588	0.535	0.489	0.449
12.0	$E^{DCS}_{expected}$	1.07E+02	9.67E+01	8.74E+01	7.87E+01	7.09E+01	6.40E+01	5.79E+01	5.25E+01	4.78E+01	4.37E+01	4.02E+01
	$E^{DCS}_{expected}$	9.00E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.183	1.074	0.971	0.875	0.788	0.711	0.643	0.584	0.531	0.486	0.446
13.5	$E^{DCS}_{expected}$	1.19E+02	1.08E+02	9.78E+01	8.81E+01	7.94E+01	7.16E+01	6.48E+01	5.88E+01	5.35E+01	4.89E+01	4.49E+01

Solid Slab-Land use (m^2 a crop eq)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E^{DCS}_{expected}$	1.01E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.176	1.068	0.965	0.870	0.784	0.707	0.639	0.580	0.528	0.483	0.444
15.0	$E^{DCS}_{expected}$	1.32E+02	1.20E+02	1.08E+02	9.74E+01	8.78E+01	7.92E+01	7.16E+01	6.50E+01	5.92E+01	5.41E+01	4.97E+01
	$E^{DCS}_{expected}$	1.13E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.171	1.064	0.961	0.866	0.780	0.704	0.637	0.578	0.526	0.481	0.442
16.5	$E^{DCS}_{expected}$	1.62E+02	1.47E+02	1.33E+02	1.20E+02	1.08E+02	9.72E+01	8.79E+01	7.97E+01	7.26E+01	6.64E+01	6.09E+01
	$E^{DCS}_{expected}$	1.40E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.153	1.047	0.946	0.852	0.768	0.693	0.626	0.568	0.517	0.473	0.434
18.0	$E^{DCS}_{expected}$	1.76E+02	1.60E+02	1.44E+02	1.30E+02	1.17E+02	1.06E+02	9.56E+01	8.67E+01	7.90E+01	7.22E+01	6.63E+01
	$E^{DCS}_{expected}$	1.53E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.150	1.044	0.943	0.850	0.766	0.691	0.625	0.567	0.516	0.472	0.433

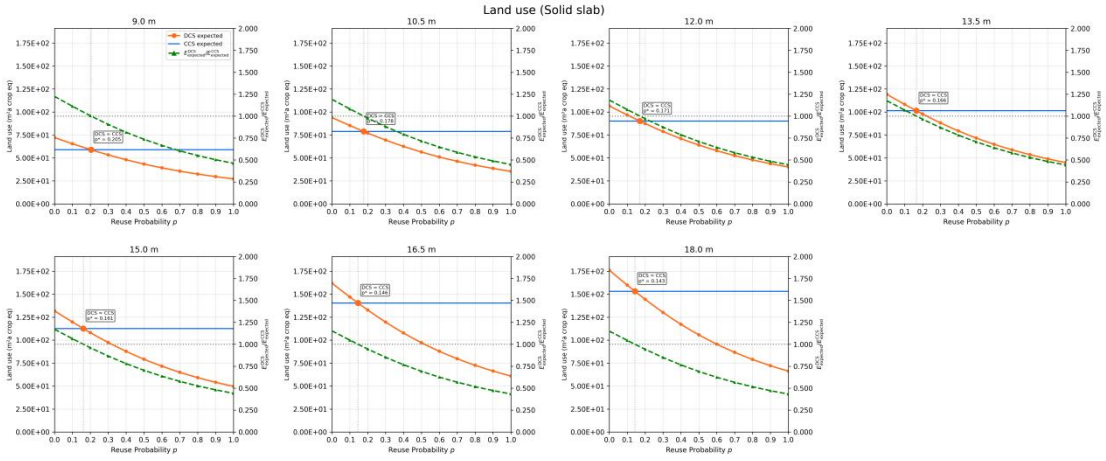


Figure E-15 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Land use

Table E-16 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Mineral resource scarcity

Solid Slab-Mineral resource scarcity (kg Cu eq)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	2.64E+01	2.38E+01	2.14E+01	1.91E+01	1.71E+01	1.52E+01	1.36E+01	1.23E+01	1.10E+01	9.96E+00	9.03E+00
	$E^{DCS}_{expected}$	1.97E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.342	1.211	1.086	0.971	0.867	0.775	0.694	0.623	0.561	0.506	0.459
10.5	$E^{DCS}_{expected}$	3.81E+01	3.44E+01	3.08E+01	2.75E+01	2.46E+01	2.20E+01	1.97E+01	1.76E+01	1.59E+01	1.43E+01	1.30E+01
	$E^{DCS}_{expected}$	3.03E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.256	1.133	1.016	0.908	0.811	0.724	0.648	0.582	0.524	0.473	0.428
12.0	$E^{DCS}_{expected}$	4.33E+01	3.91E+01	3.50E+01	3.13E+01	2.80E+01	2.50E+01	2.24E+01	2.01E+01	1.80E+01	1.63E+01	1.48E+01
	$E^{DCS}_{expected}$	3.45E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.254	1.131	1.014	0.907	0.809	0.723	0.647	0.581	0.523	0.472	0.428
13.5	$E^{DCS}_{expected}$	4.87E+01	4.39E+01	3.94E+01	3.52E+01	3.14E+01	2.81E+01	2.51E+01	2.26E+01	2.03E+01	1.83E+01	1.66E+01
	$E^{DCS}_{expected}$	3.89E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.251	1.129	1.012	0.904	0.808	0.721	0.646	0.579	0.521	0.471	0.427
15.0	$E^{DCS}_{expected}$	5.39E+01	4.87E+01	4.36E+01	3.90E+01	3.48E+01	3.11E+01	2.78E+01	2.50E+01	2.25E+01	2.03E+01	1.84E+01
	$E^{DCS}_{expected}$	4.31E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.250	1.128	1.011	0.904	0.807	0.721	0.645	0.579	0.521	0.470	0.426
16.5	$E^{DCS}_{expected}$	7.24E+01	6.53E+01	5.86E+01	5.23E+01	4.67E+01	4.17E+01	3.73E+01	3.35E+01	3.01E+01	2.72E+01	2.46E+01
	$E^{DCS}_{expected}$	6.06E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.195	1.078	0.967	0.864	0.771	0.689	0.616	0.553	0.497	0.449	0.407
18.0	$E^{DCS}_{expected}$	7.89E+01	7.12E+01	6.38E+01	5.70E+01	5.09E+01	4.55E+01	4.07E+01	3.65E+01	3.28E+01	2.96E+01	2.68E+01
	$E^{DCS}_{expected}$	6.61E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.195	1.078	0.966	0.863	0.770	0.688	0.616	0.552	0.497	0.449	0.406

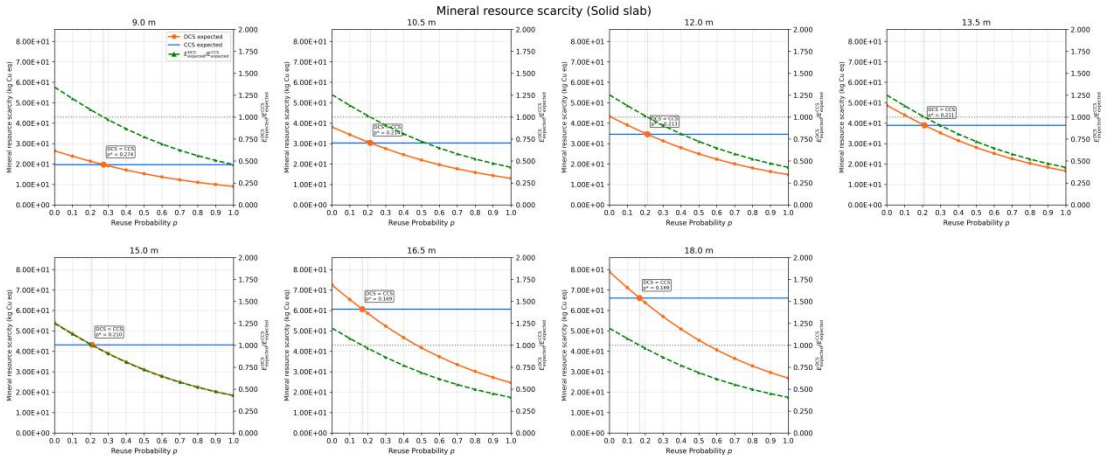


Figure E-16 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Mineral resource scarcity

Table E-17 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fossil resource scarcity

Solid Slab-Fossil resource scarcity (kg oil eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	6.52E+02	5.99E+02	5.49E+02	5.02E+02	4.60E+02	4.23E+02	3.90E+02	3.61E+02	3.36E+02	3.14E+02	2.94E+02
	$E^{DCS}_{expected}$	4.91E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.326	1.220	1.117	1.023	0.937	0.861	0.794	0.735	0.684	0.639	0.599
10.5	$E^{DCS}_{expected}$	8.54E+02	7.85E+02	7.18E+02	6.57E+02	6.02E+02	5.53E+02	5.10E+02	4.71E+02	4.38E+02	4.09E+02	3.84E+02
	$E^{DCS}_{expected}$	6.64E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.285	1.181	1.082	0.990	0.906	0.832	0.767	0.710	0.660	0.616	0.578
12.0	$E^{DCS}_{expected}$	9.69E+02	8.90E+02	8.15E+02	7.46E+02	6.83E+02	6.27E+02	5.78E+02	5.35E+02	4.97E+02	4.64E+02	4.35E+02
	$E^{DCS}_{expected}$	7.59E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.277	1.173	1.075	0.983	0.900	0.827	0.762	0.705	0.655	0.612	0.574
13.5	$E^{DCS}_{expected}$	1.08E+03	9.96E+02	9.12E+02	8.35E+02	7.65E+02	7.02E+02	6.47E+02	5.99E+02	5.56E+02	5.19E+02	4.87E+02
	$E^{DCS}_{expected}$	8.54E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.270	1.167	1.069	0.978	0.895	0.822	0.758	0.701	0.652	0.608	0.570
15.0	$E^{DCS}_{expected}$	1.20E+03	1.10E+03	1.01E+03	9.23E+02	8.45E+02	7.76E+02	7.15E+02	6.62E+02	6.15E+02	5.74E+02	5.39E+02
	$E^{DCS}_{expected}$	9.48E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.265	1.162	1.064	0.974	0.892	0.819	0.755	0.698	0.649	0.606	0.568
16.5	$E^{DCS}_{expected}$	1.48E+03	1.36E+03	1.24E+03	1.14E+03	1.04E+03	9.56E+02	8.81E+02	8.15E+02	7.57E+02	7.06E+02	6.62E+02

Solid Slab-Fossil resource scarcity (kg oil eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E^{DCS}_{expected}$	1.19E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.241	1.140	1.043	0.954	0.873	0.802	0.738	0.683	0.634	0.592	0.554
18.0	$E^{DCS}_{expected}$	1.61E+03	1.48E+03	1.35E+03	1.24E+03	1.13E+03	1.04E+03	9.58E+02	8.86E+02	8.23E+02	7.68E+02	7.20E+02
	$E^{DCS}_{expected}$	1.30E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.238	1.137	1.041	0.951	0.871	0.799	0.736	0.681	0.632	0.590	0.553

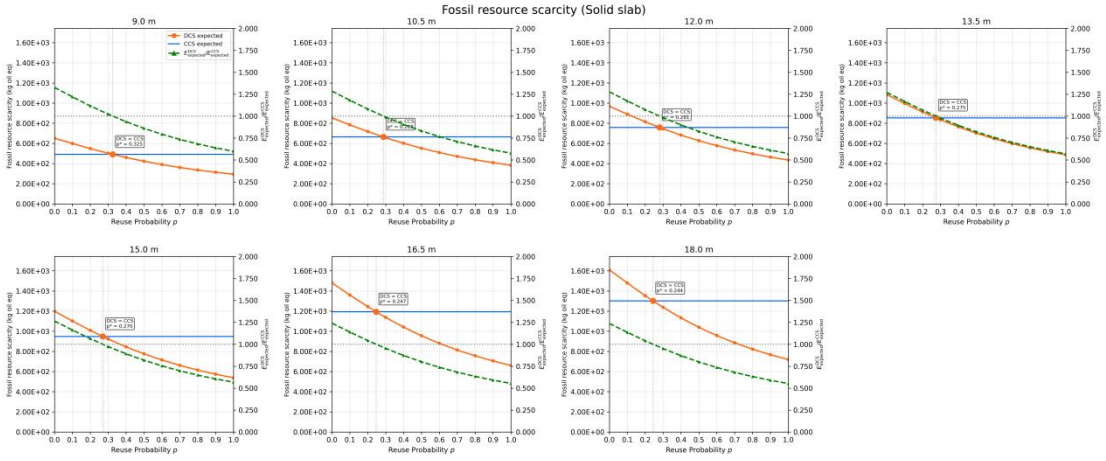


Figure E-17 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fossil resource scarcity

Table E-18 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Water consumption

Solid Slab-Water consumption (m^3)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	2.39E+01	2.16E+01	1.94E+01	1.74E+01	1.56E+01	1.40E+01	1.26E+01	1.14E+01	1.03E+01	9.35E+00	8.53E+00
	$E^{DCS}_{expected}$	1.87E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.279	1.157	1.041	0.934	0.837	0.751	0.676	0.609	0.552	0.501	0.457
10.5	$E^{DCS}_{expected}$	3.11E+01	2.81E+01	2.53E+01	2.27E+01	2.03E+01	1.82E+01	1.64E+01	1.48E+01	1.34E+01	1.22E+01	1.11E+01
	$E^{DCS}_{expected}$	2.50E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.243	1.124	1.011	0.907	0.813	0.730	0.656	0.592	0.536	0.486	0.444
12.0	$E^{DCS}_{expected}$	3.53E+01	3.19E+01	2.87E+01	2.57E+01	2.31E+01	2.07E+01	1.86E+01	1.68E+01	1.52E+01	1.38E+01	1.26E+01
	$E^{DCS}_{expected}$	2.86E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.234	1.117	1.004	0.901	0.807	0.725	0.652	0.588	0.532	0.483	0.441
13.5	$E^{DCS}_{expected}$	3.95E	3.57E	3.21E	2.88E	2.58E	2.32E	2.08E	1.88E	1.70E	1.54E	1.41E

Solid Slab-Water consumption (m^3)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
		+01	+01	+01	+01	+01	+01	+01	+01	+01	+01	+01
	$E^{DCS}_{expected}$	3.22E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.227	1.110	0.999	0.896	0.803	0.720	0.648	0.584	0.529	0.480	0.438
15.0	$E^{DCS}_{expected}$	4.36E +01	3.95E +01	3.55E +01	3.19E +01	2.85E +01	2.56E +01	2.30E +01	2.08E +01	1.88E +01	1.71E +01	1.56E +01
	$E^{DCS}_{expected}$	3.57E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.222	1.106	0.994	0.892	0.799	0.717	0.645	0.582	0.527	0.478	0.436
16.5	$E^{DCS}_{expected}$	5.36E +01	4.85E +01	4.36E +01	3.91E +01	3.51E +01	3.15E +01	2.83E +01	2.55E +01	2.31E +01	2.10E +01	1.91E +01
	$E^{DCS}_{expected}$	4.46E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.202	1.087	0.978	0.877	0.786	0.705	0.634	0.572	0.518	0.470	0.429
18.0	$E^{DCS}_{expected}$	5.83E +01	5.27E +01	4.74E +01	4.25E +01	3.81E +01	3.42E +01	3.08E +01	2.77E +01	2.51E +01	2.28E +01	2.08E +01
	$E^{DCS}_{expected}$	4.86E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.199	1.084	0.975	0.875	0.784	0.703	0.632	0.570	0.516	0.469	0.427

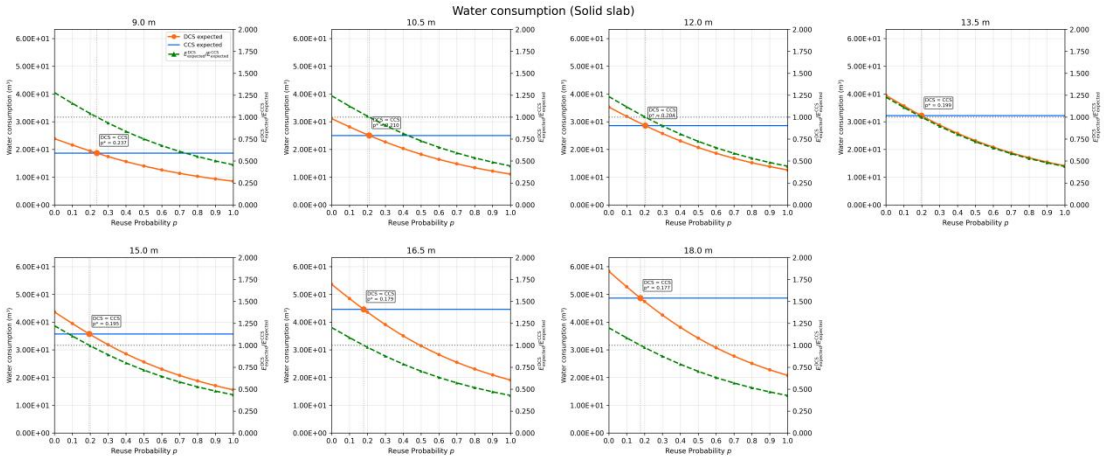


Figure E-18 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Water consumption

Appendix F: p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) for the hollow core slab of different model lengths

Table F-1 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Global warming

Hollow Core Slab-Global warming (kg CO_2 eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	3.17E+03	2.90E+03	2.64E+03	2.40E+03	2.19E+03	1.99E+03	1.82E+03	1.68E+03	1.55E+03	1.43E+03	1.33E+03
	$E^{DCS}_{expected}$	2.23E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.424	1.302	1.186	1.078	0.981	0.895	0.819	0.753	0.694	0.643	0.599
10.5	$E^{DCS}_{expected}$	3.62E+03	3.31E+03	3.01E+03	2.74E+03	2.49E+03	2.27E+03	2.08E+03	1.91E+03	1.76E+03	1.63E+03	1.52E+03
	$E^{DCS}_{expected}$	2.60E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.391	1.272	1.158	1.053	0.958	0.874	0.800	0.735	0.678	0.628	0.585
12.0	$E^{DCS}_{expected}$	4.06E+03	3.71E+03	3.38E+03	3.07E+03	2.80E+03	2.55E+03	2.33E+03	2.14E+03	1.98E+03	1.83E+03	1.71E+03
	$E^{DCS}_{expected}$	2.97E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.367	1.250	1.138	1.034	0.941	0.858	0.785	0.722	0.666	0.617	0.574
13.5	$E^{DCS}_{expected}$	5.13E+03	4.69E+03	4.27E+03	3.88E+03	3.53E+03	3.22E+03	2.94E+03	2.70E+03	2.49E+03	2.31E+03	2.15E+03
	$E^{DCS}_{expected}$	3.90E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.314	1.201	1.094	0.994	0.904	0.824	0.754	0.692	0.638	0.591	0.550
15.0	$E^{DCS}_{expected}$	5.64E+03	5.15E+03	4.69E+03	4.26E+03	3.88E+03	3.54E+03	3.23E+03	2.97E+03	2.74E+03	2.54E+03	2.36E+03
	$E^{DCS}_{expected}$	4.33E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.301	1.189	1.082	0.984	0.895	0.816	0.746	0.685	0.632	0.585	0.544
16.5	$E^{DCS}_{expected}$	6.15E+03	5.62E+03	5.12E+03	4.65E+03	4.23E+03	3.86E+03	3.53E+03	3.24E+03	2.99E+03	2.77E+03	2.57E+03
	$E^{DCS}_{expected}$	4.77E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.290	1.179	1.074	0.976	0.887	0.809	0.740	0.679	0.626	0.580	0.540
18.0	$E^{DCS}_{expected}$	7.50E+03	6.86E+03	6.24E+03	5.67E+03	5.16E+03	4.70E+03	4.30E+03	3.94E+03	3.63E+03	3.37E+03	3.13E+03
	$E^{DCS}_{expected}$	5.96E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.260	1.152	1.048	0.952	0.866	0.789	0.721	0.662	0.610	0.565	0.525

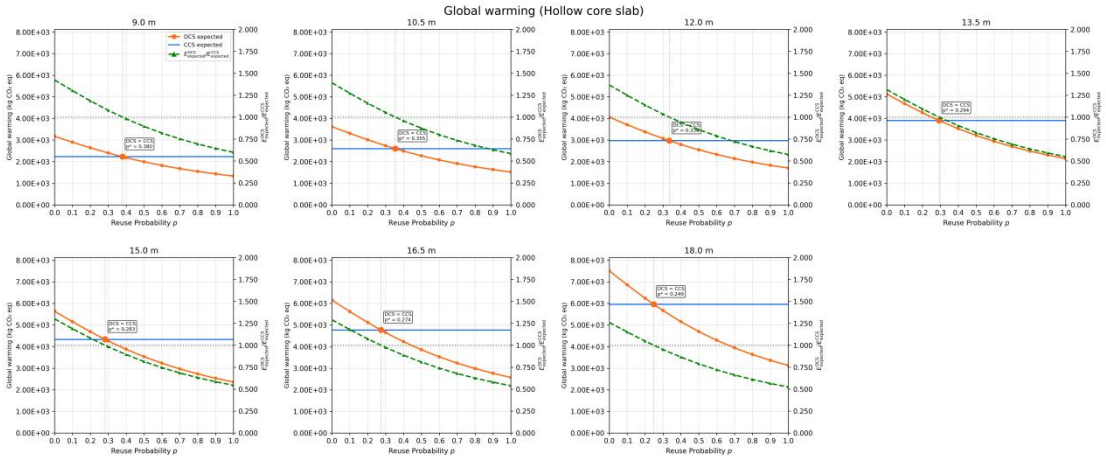


Figure F-1 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Global warming

Table F-2 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Stratospheric ozone depletion

Hollow Core Slab-Stratospheric ozone depletion (kg CFC11 eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	1.16E-03	1.06E-03	9.71E-04	8.85E-04	8.06E-04	7.37E-04	6.75E-04	6.21E-04	5.74E-04	5.33E-04	4.97E-04
	$E^{DCS}_{expected}$	7.58E-04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.534	1.405	1.281	1.167	1.064	0.972	0.891	0.820	0.757	0.703	0.655
10.5	$E^{DCS}_{expected}$	1.32E-03	1.21E-03	1.11E-03	1.01E-03	9.19E-04	8.39E-04	7.69E-04	7.08E-04	6.54E-04	6.07E-04	5.66E-04
	$E^{DCS}_{expected}$	8.85E-04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.497	1.371	1.251	1.139	1.038	0.948	0.869	0.800	0.739	0.686	0.639
12.0	$E^{DCS}_{expected}$	1.49E-03	1.36E-03	1.24E-03	1.13E-03	1.03E-03	9.41E-04	8.62E-04	7.93E-04	7.33E-04	6.80E-04	6.34E-04
	$E^{DCS}_{expected}$	1.01E-03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.470	1.346	1.228	1.118	1.019	0.931	0.853	0.785	0.725	0.673	0.627
13.5	$E^{DCS}_{expected}$	1.87E-03	1.71E-03	1.56E-03	1.42E-03	1.30E-03	1.18E-03	1.08E-03	9.97E-04	9.21E-04	8.54E-04	7.96E-04
	$E^{DCS}_{expected}$	1.32E-03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.413	1.294	1.180	1.074	0.979	0.894	0.819	0.753	0.695	0.645	0.601
15.0	$E^{DCS}_{expected}$	2.06E-03	1.88E-03	1.72E-03	1.56E-03	1.42E-03	1.30E-03	1.19E-03	1.10E-03	1.01E-03	9.39E-04	8.74E-04
	$E^{DCS}_{expected}$	1.47E-03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.398	1.280	1.167	1.063	0.968	0.884	0.810	0.745	0.688	0.638	0.595
16.5	$E^{DCS}_{expected}$	2.24E-03	2.05E-03	1.87E-03	1.70E-03	1.55E-03	1.42E-03	1.30E-03	1.19E-03	1.10E-03	1.02E-03	9.54E-04
	$E^{DCS}_{expected}$	1.62E-03										

Hollow Core Slab-Stratospheric ozone depletion (kg CFC11 eq)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.387	1.269	1.157	1.054	0.960	0.877	0.803	0.739	0.682	0.633	0.589
18.0	$E^{DCS}_{expected}$	2.73E-03	2.50E-03	2.28E-03	2.07E-03	1.89E-03	1.72E-03	1.58E-03	1.45E-03	1.34E-03	1.24E-03	1.16E-03
	$E^{DCS}_{expected}$	2.02E-03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.353	1.239	1.129	1.028	0.936	0.855	0.783	0.720	0.664	0.616	0.574

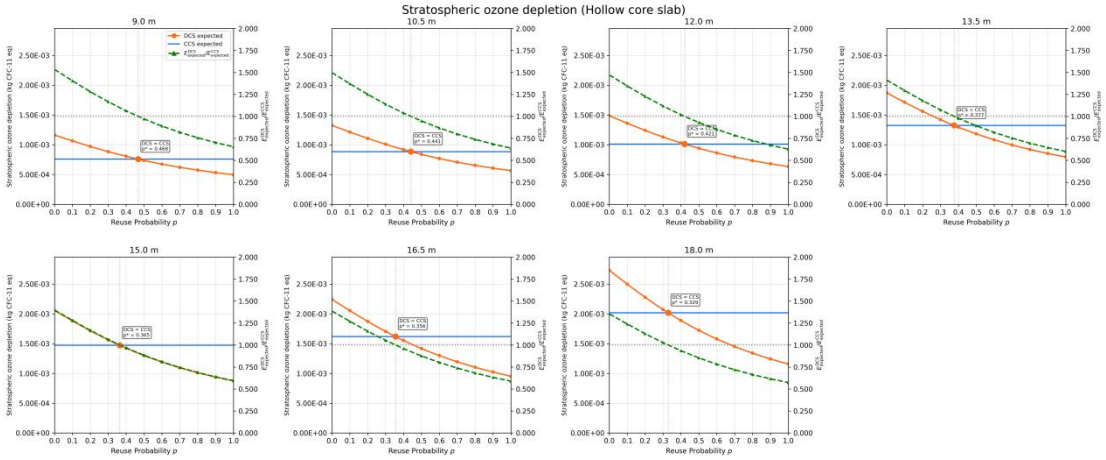


Figure F-2 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Stratospheric ozone depletion

Table F-3 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ionizing radiation

Hollow Core Slab-Ionizing radiation (kBq Co-60 eq)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	7.84E+01	7.12E+01	6.43E+01	5.80E+01	5.23E+01	4.72E+01	4.28E+01	3.89E+01	3.54E+01	3.25E+01	2.98E+01
	$E^{DCS}_{expected}$	5.73E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.367	1.242	1.122	1.012	0.912	0.824	0.746	0.678	0.618	0.566	0.521
10.5	$E^{DCS}_{expected}$	8.95E+01	8.13E+01	7.35E+01	6.63E+01	5.97E+01	5.39E+01	4.88E+01	4.44E+01	4.05E+01	3.71E+01	3.41E+01
	$E^{DCS}_{expected}$	6.69E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.338	1.215	1.098	0.990	0.892	0.806	0.730	0.663	0.605	0.554	0.509
12.0	$E^{DCS}_{expected}$	1.01E+02	9.13E+01	8.25E+01	7.44E+01	6.71E+01	6.06E+01	5.48E+01	4.98E+01	4.54E+01	4.16E+01	3.83E+01
	$E^{DCS}_{expected}$	7.64E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.316	1.195	1.080	0.974	0.878	0.793	0.718	0.652	0.594	0.544	0.501
13.5	$E^{DCS}_{expected}$	1.28E+02	1.16E+02	1.05E+02	9.48E+01	8.55E+01	7.71E+01	6.98E+01	6.34E+01	5.78E+01	5.29E+01	4.86E+01

Hollow Core Slab-Ionizing radiation (kBq Co-60 eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E^{DCS}_{expected}$	1.01E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.263	1.147	1.036	0.934	0.842	0.760	0.688	0.625	0.569	0.521	0.479
15.0	$E^{DCS}_{expected}$	1.41E+02	1.28E+02	1.16E+02	1.04E+02	9.40E+01	8.49E+01	7.68E+01	6.98E+01	6.36E+01	5.82E+01	5.35E+01
	$E^{DCS}_{expected}$	1.13E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.252	1.137	1.027	0.926	0.834	0.753	0.681	0.619	0.564	0.516	0.474
16.5	$E^{DCS}_{expected}$	1.54E+02	1.40E+02	1.26E+02	1.14E+02	1.03E+02	9.27E+01	8.39E+01	7.62E+01	6.94E+01	6.35E+01	5.84E+01
	$E^{DCS}_{expected}$	1.24E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.242	1.128	1.019	0.919	0.828	0.747	0.676	0.614	0.560	0.512	0.471
18.0	$E^{DCS}_{expected}$	1.89E+02	1.72E+02	1.55E+02	1.40E+02	1.26E+02	1.14E+02	1.03E+02	9.35E+01	8.52E+01	7.79E+01	7.16E+01
	$E^{DCS}_{expected}$	1.56E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.212	1.100	0.994	0.895	0.807	0.728	0.659	0.598	0.545	0.499	0.458

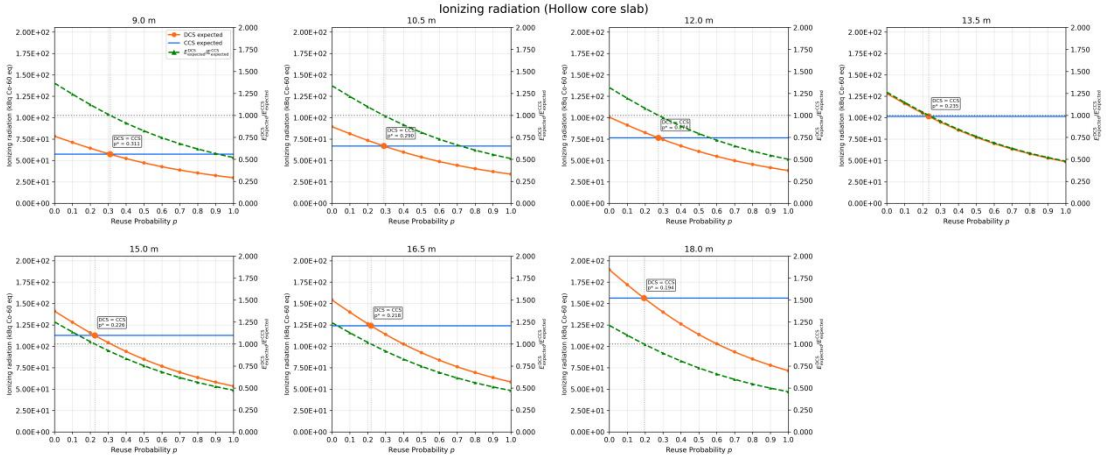


Figure F-3 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ionizing radiation

Table F-4 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation, Human health

Hollow Core Slab-Ozone formation, Human health (kg NOx eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	7.80E+00	7.20E+00	6.63E+00	6.10E+00	5.62E+00	5.19E+00	4.82E+00	4.49E+00	4.21E+00	3.96E+00	3.74E+00
	$E^{DCS}_{expected}$	5.45E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.432	1.322	1.216	1.119	1.031	0.953	0.884	0.824	0.772	0.726	0.686
10.5	$E^{DCS}_{expected}$	8.91E+00	8.22E+00	7.56E+00	6.96E+00	6.41E+00	5.92E+00	5.50E+00	5.12E+00	4.79E+00	4.51E+00	4.26E+00
	$E^{DCS}_{expected}$	6.36E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.400	1.292	1.189	1.093	1.007	0.931	0.864	0.805	0.754	0.709	0.669
12.0	$E^{DCS}_{expected}$	1.00E+01	9.22E+00	8.49E+00	7.81E+00	7.19E+00	6.65E+00	6.17E+00	5.75E+00	5.38E+00	5.06E+00	4.78E+00
	$E^{DCS}_{expected}$	7.27E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.376	1.270	1.168	1.074	0.990	0.915	0.849	0.791	0.740	0.696	0.657
13.5	$E^{DCS}_{expected}$	1.27E+01	1.17E+01	1.07E+01	9.88E+00	9.10E+00	8.40E+00	7.79E+00	7.25E+00	6.78E+00	6.37E+00	6.02E+00
	$E^{DCS}_{expected}$	9.59E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.321	1.219	1.121	1.030	0.948	0.876	0.812	0.756	0.707	0.665	0.627
15.0	$E^{DCS}_{expected}$	1.39E+01	1.29E+01	1.18E+01	1.09E+01	1.00E+01	9.24E+00	8.57E+00	7.98E+00	7.46E+00	7.01E+00	6.62E+00
	$E^{DCS}_{expected}$	1.07E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.309	1.207	1.110	1.020	0.939	0.867	0.804	0.749	0.700	0.658	0.621
16.5	$E^{DCS}_{expected}$	1.52E+01	1.40E+01	1.29E+01	1.19E+01	1.09E+01	1.01E+01	9.35E+00	8.70E+00	8.14E+00	7.65E+00	7.22E+00
	$E^{DCS}_{expected}$	1.17E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.298	1.197	1.101	1.012	0.931	0.860	0.797	0.742	0.694	0.652	0.616
18.0	$E^{DCS}_{expected}$	1.86E+01	1.72E+01	1.58E+01	1.45E+01	1.33E+01	1.23E+01	1.14E+01	1.06E+01	9.92E+00	9.31E+00	8.78E+00
	$E^{DCS}_{expected}$	1.47E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.266	1.167	1.073	0.985	0.907	0.837	0.775	0.721	0.674	0.633	0.597

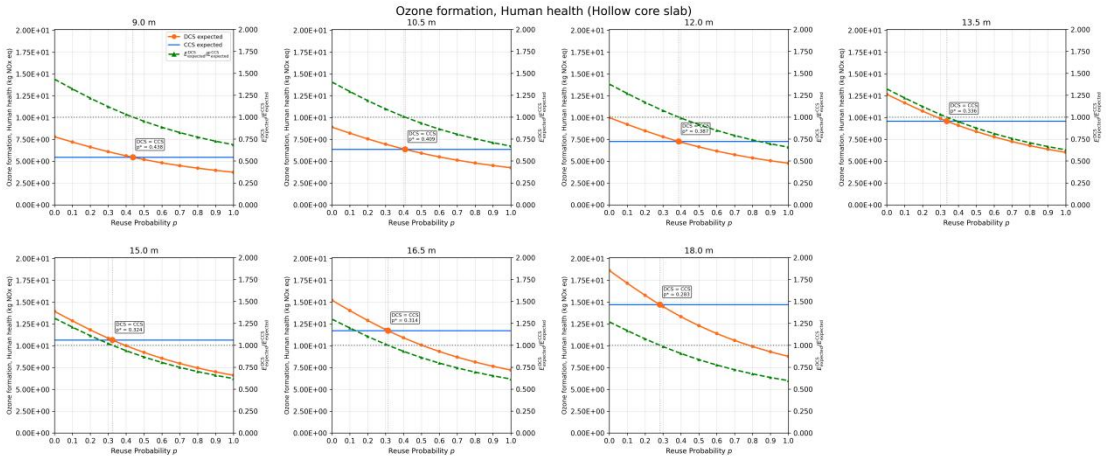


Figure F-4 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation, Human health

Table F-5 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fine particulate matter formation

Hollow Core Slab-Fine particulate matter formation (kg PM2.5eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	4.54E+00	4.17E+00	3.82E+00	3.50E+00	3.21E+00	2.95E+00	2.72E+00	2.52E+00	2.35E+00	2.20E+00	2.06E+00
	$E^{DCS}_{expected}$	2.88E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.574	1.447	1.326	1.214	1.113	1.023	0.944	0.875	0.814	0.761	0.715
10.5	$E^{DCS}_{expected}$	5.19E+00	4.77E+00	4.37E+00	4.00E+00	3.67E+00	3.37E+00	3.11E+00	2.88E+00	2.68E+00	2.51E+00	2.36E+00
	$E^{DCS}_{expected}$	3.37E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.542	1.417	1.298	1.188	1.089	1.001	0.924	0.856	0.797	0.745	0.699
12.0	$E^{DCS}_{expected}$	5.84E+00	5.36E+00	4.91E+00	4.50E+00	4.12E+00	3.79E+00	3.49E+00	3.24E+00	3.01E+00	2.82E+00	2.64E+00
	$E^{DCS}_{expected}$	3.84E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.519	1.396	1.278	1.170	1.072	0.986	0.909	0.842	0.784	0.733	0.688
13.5	$E^{DCS}_{expected}$	7.49E+00	6.88E+00	6.30E+00	5.76E+00	5.27E+00	4.84E+00	4.46E+00	4.13E+00	3.84E+00	3.59E+00	3.36E+00
	$E^{DCS}_{expected}$	5.18E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.446	1.328	1.215	1.111	1.018	0.935	0.861	0.797	0.741	0.692	0.649
15.0	$E^{DCS}_{expected}$	8.25E+00	7.58E+00	6.93E+00	6.34E+00	5.81E+00	5.33E+00	4.91E+00	4.55E+00	4.23E+00	3.95E+00	3.70E+00
	$E^{DCS}_{expected}$	5.76E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.434	1.317	1.205	1.102	1.009	0.926	0.854	0.790	0.734	0.686	0.643

Hollow Core Slab-Fine particulate matter formation (kg PM2.5eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
16.5	$E^{DCS}_{expected}$	9.02E+00	8.28E+00	7.58E+00	6.93E+00	6.34E+00	5.82E+00	5.37E+00	4.97E+00	4.61E+00	4.31E+00	4.04E+00
	$E^{DCS}_{expected}$	6.33E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.423	1.307	1.196	1.094	1.001	0.919	0.847	0.784	0.729	0.680	0.638
18.0	$E^{DCS}_{expected}$	1.11E+01	1.02E+01	9.35E+00	8.55E+00	7.82E+00	7.17E+00	6.60E+00	6.11E+00	5.67E+00	5.29E+00	4.96E+00
	$E^{DCS}_{expected}$	8.08E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.379	1.266	1.158	1.058	0.968	0.888	0.818	0.756	0.702	0.655	0.614

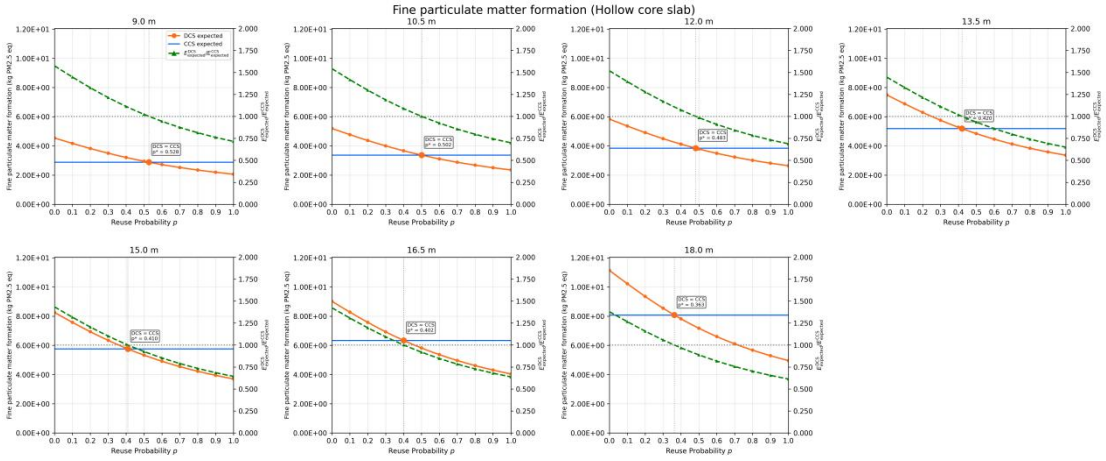


Figure F-5 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fine particulate matter formation

Table F-6 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation, Terrestrial ecosystem

Hollow Core Slab-Ozone formation, Terrestrial ecosystem (kg NOx eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	7.98E+00	7.37E+00	6.78E+00	6.23E+00	5.74E+00	5.31E+00	4.92E+00	4.58E+00	4.29E+00	4.03E+00	3.81E+00
	$E^{DCS}_{expected}$	5.59E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.429	1.319	1.213	1.116	1.028	0.950	0.881	0.821	0.768	0.722	0.682
10.5	$E^{DCS}_{expected}$	9.11E+00	8.41E+00	7.73E+00	7.11E+00	6.55E+00	6.05E+00	5.61E+00	5.23E+00	4.89E+00	4.60E+00	4.34E+00
	$E^{DCS}_{expected}$	6.52E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.397	1.289	1.186	1.090	1.004	0.928	0.861	0.802	0.750	0.705	0.666
12.0	$E^{DCS}_{expected}$	1.02E+01	9.43E+00	8.68E+00	7.98E+00	7.35E+00	6.79E+00	6.30E+00	5.87E+00	5.49E+00	5.16E+00	4.87E+00
	$E^{DCS}_{expected}$	7.45E+00										

Hollow Core Slab-Ozone formation, Terrestrial ecosystem (kg NOx eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.373	1.267	1.165	1.071	0.987	0.912	0.846	0.788	0.737	0.693	0.654
13.5	$E^{DCS}_{expected}$	1.30E+01	1.20E+01	1.10E+01	1.01E+01	9.30E+00	8.59E+00	7.96E+00	7.41E+00	6.93E+00	6.51E+00	6.14E+00
	$E^{DCS}_{expected}$	9.84E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.319	1.216	1.118	1.027	0.945	0.873	0.809	0.753	0.704	0.661	0.624
15.0	$E^{DCS}_{expected}$	1.43E+01	1.32E+01	1.21E+01	1.11E+01	1.02E+01	9.44E+00	8.75E+00	8.15E+00	7.62E+00	7.15E+00	6.75E+00
	$E^{DCS}_{expected}$	1.09E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.306	1.204	1.107	1.017	0.936	0.864	0.801	0.745	0.697	0.654	0.617
16.5	$E^{DCS}_{expected}$	1.56E+01	1.44E+01	1.32E+01	1.21E+01	1.12E+01	1.03E+01	9.55E+00	8.89E+00	8.31E+00	7.80E+00	7.36E+00
	$E^{DCS}_{expected}$	1.20E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.296	1.195	1.098	1.009	0.928	0.857	0.794	0.739	0.691	0.649	0.612
18.0	$E^{DCS}_{expected}$	1.91E+01	1.76E+01	1.62E+01	1.48E+01	1.36E+01	1.26E+01	1.17E+01	1.08E+01	1.01E+01	9.51E+00	8.97E+00
	$E^{DCS}_{expected}$	1.51E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.264	1.165	1.070	0.983	0.904	0.834	0.772	0.718	0.671	0.630	0.594

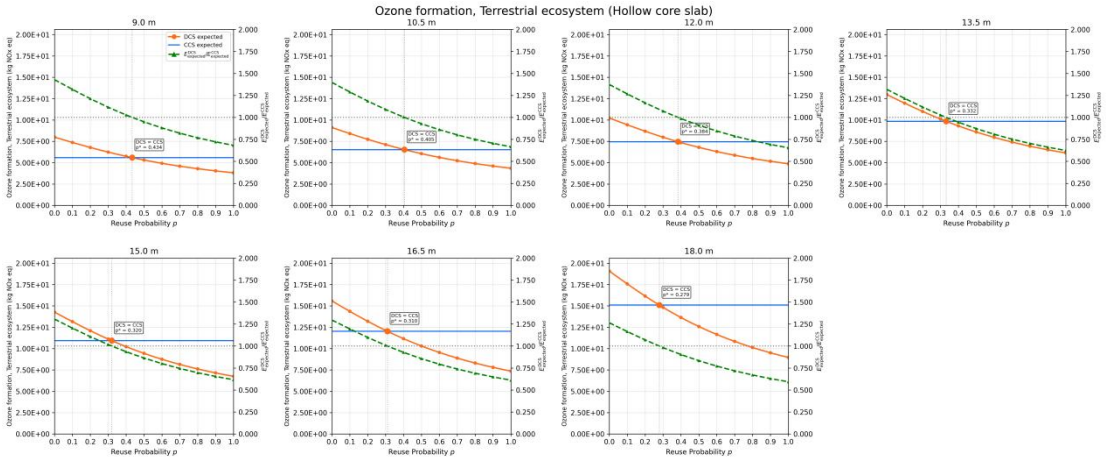


Figure F-6 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation, Terrestrial ecosystem

Table F-7 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial acidification

Hollow Core Slab-Terrestrial acidification (kg SO2 eq)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	1.01E+01	9.35E+00	8.60E+00	7.90E+00	7.27E+00	6.72E+00	6.22E+00	5.79E+00	5.42E+00	5.09E+00	4.80E+00
	$E^{DCS}_{expected}$	6.15E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.650	1.521	1.398	1.285	1.183	1.092	1.012	0.942	0.881	0.828	0.781
10.5	$E^{DCS}_{expected}$	1.16E+01	1.07E+01	9.81E+00	9.01E+00	8.30E+00	7.66E+00	7.10E+00	6.61E+00	6.18E+00	5.80E+00	5.47E+00
	$E^{DCS}_{expected}$	7.18E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.613	1.487	1.367	1.256	1.156	1.067	0.989	0.920	0.860	0.808	0.762
12.0	$E^{DCS}_{expected}$	1.30E+01	1.20E+01	1.10E+01	1.01E+01	9.31E+00	8.59E+00	7.96E+00	7.41E+00	6.93E+00	6.51E+00	6.14E+00
	$E^{DCS}_{expected}$	8.20E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.585	1.461	1.343	1.234	1.136	1.048	0.971	0.904	0.845	0.794	0.749
13.5	$E^{DCS}_{expected}$	1.65E+01	1.52E+01	1.39E+01	1.28E+01	1.18E+01	1.09E+01	1.01E+01	9.35E+00	8.74E+00	8.20E+00	7.73E+00
	$E^{DCS}_{expected}$	1.09E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.515	1.396	1.283	1.178	1.083	0.999	0.925	0.861	0.804	0.755	0.711
15.0	$E^{DCS}_{expected}$	1.81E+01	1.67E+01	1.53E+01	1.41E+01	1.30E+01	1.19E+01	1.11E+01	1.03E+01	9.61E+00	9.02E+00	8.50E+00
	$E^{DCS}_{expected}$	1.21E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.500	1.383	1.270	1.166	1.073	0.989	0.916	0.852	0.796	0.747	0.704
16.5	$E^{DCS}_{expected}$	1.98E+01	1.82E+01	1.67E+01	1.54E+01	1.41E+01	1.30E+01	1.21E+01	1.12E+01	1.05E+01	9.84E+00	9.27E+00
	$E^{DCS}_{expected}$	1.33E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.488	1.371	1.260	1.157	1.064	0.981	0.908	0.845	0.789	0.741	0.698
18.0	$E^{DCS}_{expected}$	2.42E+01	2.23E+01	2.05E+01	1.88E+01	1.73E+01	1.59E+01	1.47E+01	1.37E+01	1.28E+01	1.20E+01	1.13E+01
	$E^{DCS}_{expected}$	1.67E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.447	1.332	1.223	1.123	1.032	0.951	0.880	0.818	0.764	0.716	0.675

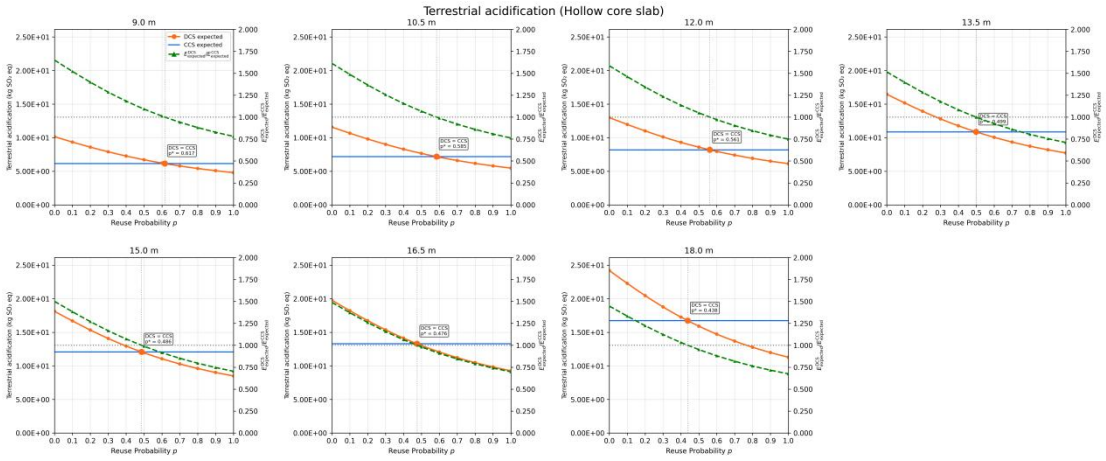


Figure Table F-7 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial acidification

Table F-8 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater eutrophication

Hollow Core Slab-Freshwater eutrophication (kg P eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	8.84E-01	8.13E-01	7.45E-01	6.82E-01	6.25E-01	5.75E-01	5.31E-01	4.92E-01	4.58E-01	4.29E-01	4.03E-01
	$E^{DCS}_{expected}$	6.72E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.317	1.210	1.109	1.016	0.931	0.856	0.791	0.733	0.683	0.639	0.600
10.5	$E^{DCS}_{expected}$	1.02E+00	9.36E-01	8.58E-01	7.85E-01	7.20E-01	6.62E-01	6.10E-01	5.66E-01	5.27E-01	4.92E-01	4.63E-01
	$E^{DCS}_{expected}$	7.85E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.298	1.193	1.092	1.000	0.917	0.843	0.778	0.721	0.671	0.627	0.589
12.0	$E^{DCS}_{expected}$	1.15E+00	1.06E+00	9.67E-01	8.85E-01	8.12E-01	7.46E-01	6.88E-01	6.38E-01	5.93E-01	5.55E-01	5.21E-01
	$E^{DCS}_{expected}$	8.95E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.284	1.180	1.081	0.989	0.907	0.833	0.769	0.712	0.663	0.620	0.582
13.5	$E^{DCS}_{expected}$	1.53E+00	1.40E+00	1.28E+00	1.17E+00	1.07E+00	9.83E-01	9.05E-01	8.37E-01	7.77E-01	7.25E-01	6.80E-01
	$E^{DCS}_{expected}$	1.25E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.224	1.123	1.027	0.938	0.858	0.788	0.725	0.671	0.623	0.581	0.545
15.0	$E^{DCS}_{expected}$	1.69E+00	1.55E+00	1.41E+00	1.29E+00	1.18E+00	1.08E+00	9.98E-01	9.23E-01	8.57E-01	8.00E-01	7.49E-01
	$E^{DCS}_{expected}$	1.39E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.217	1.116	1.021	0.933	0.853	0.783	0.721	0.666	0.619	0.577	0.541
16.5	$E^{DCS}_{expected}$	1.85E+00	1.69E+00	1.55E+00	1.41E+00	1.29E+00	1.19E+00	1.09E+00	1.01E+00	9.38E-01	8.75E-01	8.20E-01

Hollow Core Slab-Freshwater eutrophication (kg P eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E^{DCS}_{expected}$	1.52E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.211	1.111	1.016	0.928	0.849	0.779	0.717	0.662	0.615	0.574	0.538
18.0	$E^{DCS}_{expected}$	2.34E+00	2.14E+00	1.96E+00	1.79E+00	1.63E+00	1.50E+00	1.37E+00	1.27E+00	1.18E+00	1.10E+00	1.03E+00
	$E^{DCS}_{expected}$	1.99E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.174	1.076	0.983	0.897	0.819	0.751	0.690	0.637	0.591	0.550	0.515

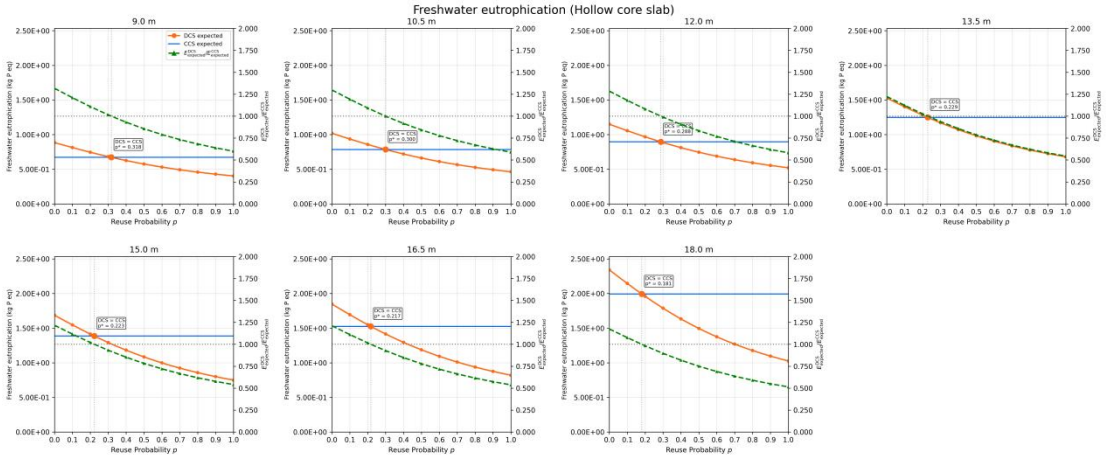


Figure F-8 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater eutrophication

Table F-9 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine eutrophication

Hollow Core Slab-Marine eutrophication (kg N eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	7.74E-02	7.07E-02	6.44E-02	5.86E-02	5.33E-02	4.86E-02	4.45E-02	4.09E-02	3.78E-02	3.50E-02	3.27E-02
	$E^{DCS}_{expected}$	6.07E-02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.275	1.166	1.062	0.966	0.879	0.802	0.734	0.675	0.623	0.578	0.538
10.5	$E^{DCS}_{expected}$	8.83E-02	8.08E-02	7.35E-02	6.69E-02	6.08E-02	5.55E-02	5.08E-02	4.67E-02	4.31E-02	4.00E-02	3.72E-02
	$E^{DCS}_{expected}$	7.08E-02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.247	1.140	1.038	0.944	0.859	0.784	0.717	0.659	0.609	0.564	0.526
12.0	$E^{DCS}_{expected}$	9.92E-02	9.06E-02	8.25E-02	7.50E-02	6.83E-02	6.23E-02	5.70E-02	5.24E-02	4.84E-02	4.48E-02	4.18E-02
	$E^{DCS}_{expected}$	8.09E-02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.226	1.121	1.020	0.928	0.844	0.770	0.705	0.648	0.598	0.554	0.516
13.5	$E^{DCS}_{expected}$	1.26E-	1.15E-	1.05E-	9.52E-	8.66E-	7.89E-	7.22E-	6.63E-	6.12E-	5.67E-	5.28E-

Hollow Core Slab-Marine eutrophication (kg N eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
		01	01	01	02	02	02	02	02	02	02	02
	$E^{DCS}_{expected}$	1.07E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.181	1.079	0.982	0.893	0.812	0.740	0.677	0.622	0.574	0.532	0.495
15.0	$E^{DCS}_{expected}$	1.38E-01	1.27E-01	1.15E-01	1.05E-01	9.52E-02	8.68E-02	7.94E-02	7.29E-02	6.73E-02	6.24E-02	5.80E-02
	$E^{DCS}_{expected}$	1.18E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.170	1.069	0.973	0.884	0.804	0.733	0.671	0.616	0.568	0.527	0.490
16.5	$E^{DCS}_{expected}$	1.51E-01	1.38E-01	1.26E-01	1.14E-01	1.04E-01	9.48E-02	8.67E-02	7.96E-02	7.34E-02	6.80E-02	6.33E-02
	$E^{DCS}_{expected}$	1.30E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.161	1.060	0.965	0.877	0.798	0.727	0.665	0.611	0.564	0.522	0.486
18.0	$E^{DCS}_{expected}$	1.85E-01	1.69E-01	1.54E-01	1.40E-01	1.27E-01	1.16E-01	1.06E-01	9.73E-02	8.97E-02	8.31E-02	7.73E-02
	$E^{DCS}_{expected}$	1.63E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.135	1.037	0.943	0.857	0.779	0.710	0.649	0.596	0.550	0.509	0.474

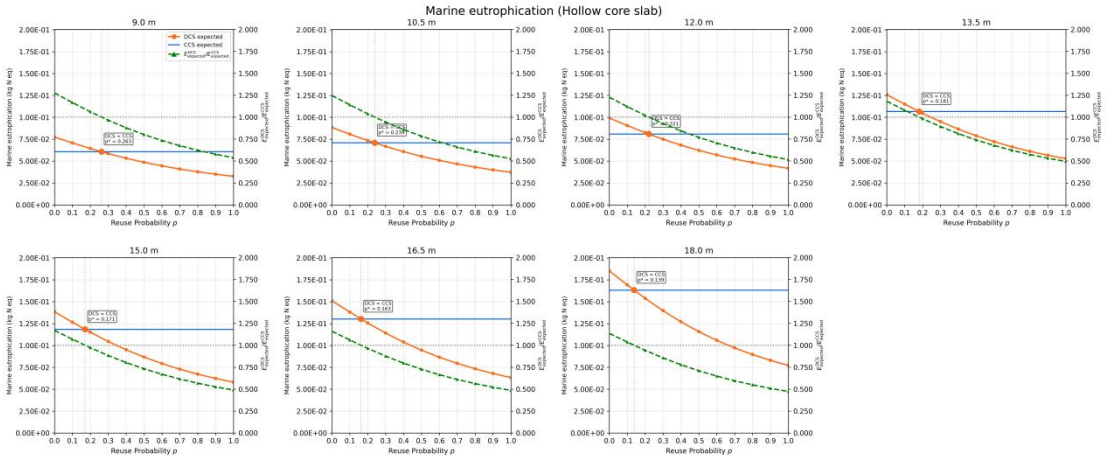


Figure F-9 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine eutrophication

Table F-10 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial ecotoxicity

Hollow Core Slab-Terrestrial ecotoxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	1.34E+04	1.22E+04	1.10E+04	9.91E+03	8.94E+03	8.07E+03	7.30E+03	6.63E+03	6.04E+03	5.52E+03	5.07E+03
	$E^{DCS}_{expected}$	1.01E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.321	1.201	1.086	0.979	0.883	0.797	0.722	0.655	0.597	0.546	0.501
10.5	$E^{DCS}_{expected}$	1.53E+04	1.39E+04	1.26E+04	1.13E+04	1.02E+04	9.22E+03	8.35E+03	7.58E+03	6.90E+03	6.31E+03	5.80E+03
	$E^{DCS}_{expected}$	1.18E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.294	1.176	1.064	0.959	0.865	0.781	0.707	0.641	0.584	0.534	0.491
12.0	$E^{DCS}_{expected}$	1.72E+04	1.56E+04	1.41E+04	1.27E+04	1.15E+04	1.04E+04	9.38E+03	8.51E+03	7.76E+03	7.09E+03	6.51E+03
	$E^{DCS}_{expected}$	1.35E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.274	1.158	1.047	0.944	0.851	0.769	0.695	0.631	0.575	0.526	0.483
13.5	$E^{DCS}_{expected}$	2.20E+04	2.00E+04	1.81E+04	1.63E+04	1.47E+04	1.33E+04	1.20E+04	1.09E+04	9.92E+03	9.07E+03	8.32E+03
	$E^{DCS}_{expected}$	1.80E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.223	1.112	1.005	0.906	0.816	0.737	0.666	0.605	0.551	0.503	0.462
15.0	$E^{DCS}_{expected}$	2.43E+04	2.21E+04	1.99E+04	1.80E+04	1.62E+04	1.46E+04	1.32E+04	1.20E+04	1.09E+04	9.98E+03	9.16E+03
	$E^{DCS}_{expected}$	2.00E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.213	1.102	0.996	0.898	0.809	0.730	0.660	0.599	0.546	0.499	0.458
16.5	$E^{DCS}_{expected}$	2.65E+04	2.41E+04	2.18E+04	1.96E+04	1.77E+04	1.60E+04	1.44E+04	1.31E+04	1.19E+04	1.09E+04	1.00E+04
	$E^{DCS}_{expected}$	2.20E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.204	1.094	0.989	0.892	0.803	0.725	0.656	0.595	0.542	0.495	0.454
18.0	$E^{DCS}_{expected}$	3.27E+04	2.98E+04	2.69E+04	2.42E+04	2.18E+04	1.97E+04	1.78E+04	1.61E+04	1.47E+04	1.34E+04	1.23E+04
	$E^{DCS}_{expected}$	2.79E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.174	1.067	0.964	0.869	0.783	0.706	0.638	0.579	0.527	0.482	0.442

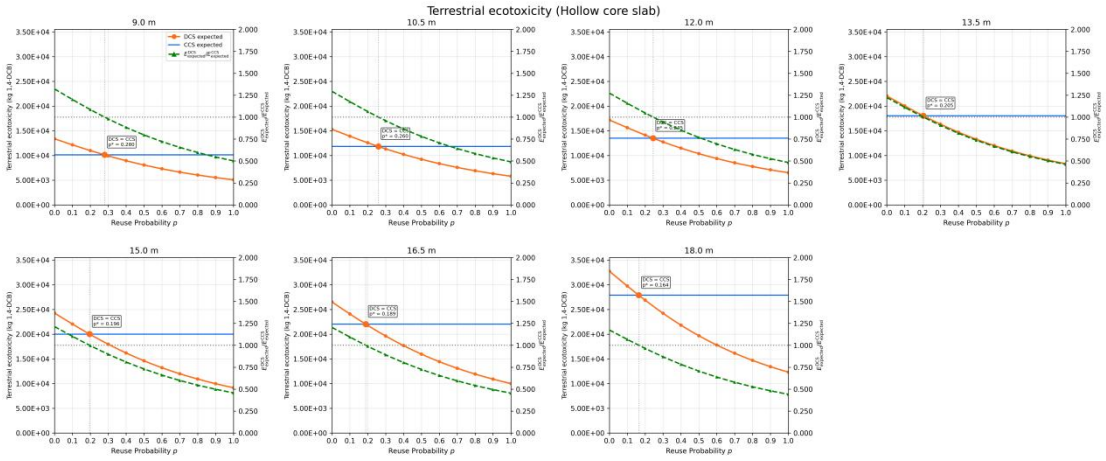


Figure F-10 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial ecotoxicity

Table F-11 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater ecotoxicity

Hollow Core Slab-Freshwater ecotoxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	9.61E+01	8.76E+01	7.94E+01	7.18E+01	6.50E+01	5.90E+01	5.36E+01	4.90E+01	4.49E+01	4.13E+01	3.82E+01
	$E^{DCS}_{expected}$	7.22E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.332	1.213	1.100	0.995	0.901	0.817	0.743	0.679	0.622	0.573	0.530
10.5	$E^{DCS}_{expected}$	1.11E+02	1.01E+02	9.18E+01	8.30E+01	7.51E+01	6.81E+01	6.19E+01	5.65E+01	5.18E+01	4.77E+01	4.41E+01
	$E^{DCS}_{expected}$	8.44E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.317	1.200	1.087	0.984	0.890	0.807	0.734	0.670	0.614	0.565	0.522
12.0	$E^{DCS}_{expected}$	1.26E+02	1.14E+02	1.04E+02	9.38E+01	8.49E+01	7.70E+01	7.00E+01	6.39E+01	5.85E+01	5.38E+01	4.98E+01
	$E^{DCS}_{expected}$	9.62E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.308	1.191	1.079	0.976	0.883	0.800	0.728	0.664	0.609	0.560	0.517
13.5	$E^{DCS}_{expected}$	1.70E+02	1.55E+02	1.40E+02	1.26E+02	1.14E+02	1.03E+02	9.39E+01	8.56E+01	7.83E+01	7.19E+01	6.64E+01
	$E^{DCS}_{expected}$	1.37E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.239	1.127	1.020	0.922	0.833	0.754	0.685	0.624	0.571	0.524	0.484
15.0	$E^{DCS}_{expected}$	1.88E+02	1.71E+02	1.55E+02	1.40E+02	1.26E+02	1.14E+02	1.04E+02	9.46E+01	8.65E+01	7.95E+01	7.33E+01
	$E^{DCS}_{expected}$	1.52E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.234	1.122	1.016	0.918	0.829	0.751	0.682	0.621	0.568	0.522	0.481
16.5	$E^{DCS}_{expected}$	2.06E+02	1.87E+02	1.70E+02	1.53E+02	1.38E+02	1.25E+02	1.14E+02	1.04E+02	9.48E+01	8.71E+01	8.03E+01

Hollow Core Slab-Freshwater ecotoxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E^{DCS}_{expected}$	1.68E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.229	1.118	1.012	0.914	0.826	0.748	0.679	0.618	0.566	0.520	0.479
18.0	$E^{DCS}_{expected}$	2.64E+02	2.40E+02	2.17E+02	1.96E+02	1.77E+02	1.60E+02	1.45E+02	1.32E+02	1.21E+02	1.11E+02	1.02E+02
	$E^{DCS}_{expected}$	2.23E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.187	1.079	0.976	0.881	0.796	0.720	0.653	0.594	0.543	0.498	0.459

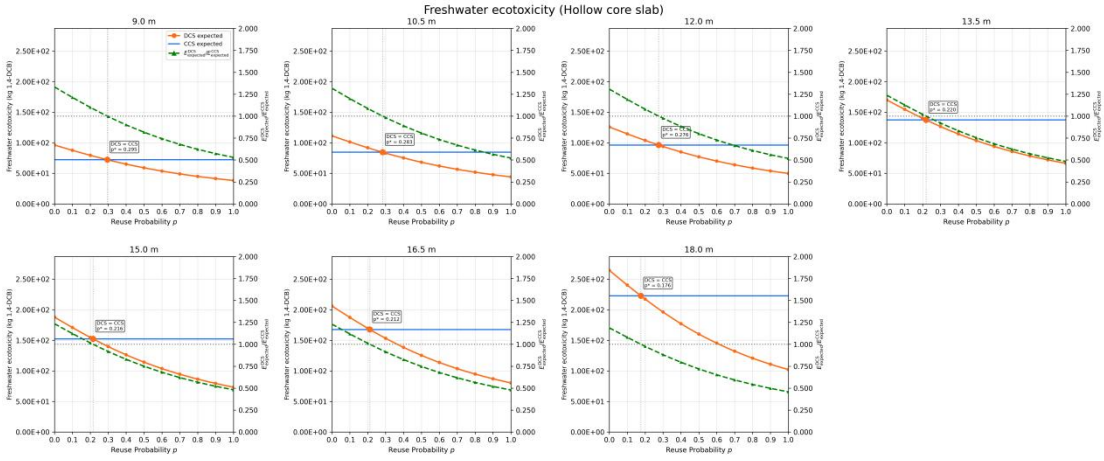


Figure F-11 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater ecotoxicity

Table F-12 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine ecotoxicity

Hollow Core Slab-Marine ecotoxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	1.39E+02	1.26E+02	1.14E+02	1.04E+02	9.37E+01	8.50E+01	7.73E+01	7.05E+01	6.46E+01	5.95E+01	5.49E+01
	$E^{DCS}_{expected}$	1.04E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.331	1.212	1.099	0.994	0.899	0.815	0.741	0.677	0.620	0.571	0.527
10.5	$E^{DCS}_{expected}$	1.60E+02	1.46E+02	1.32E+02	1.20E+02	1.08E+02	9.81E+01	8.92E+01	8.14E+01	7.45E+01	6.86E+01	6.33E+01
	$E^{DCS}_{expected}$	1.22E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.316	1.198	1.086	0.982	0.888	0.805	0.732	0.668	0.612	0.563	0.520
12.0	$E^{DCS}_{expected}$	1.81E+02	1.65E+02	1.50E+02	1.35E+02	1.22E+02	1.11E+02	1.01E+02	9.19E+01	8.42E+01	7.74E+01	7.15E+01
	$E^{DCS}_{expected}$	1.39E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.306	1.189	1.077	0.974	0.881	0.798	0.726	0.662	0.606	0.558	0.515
13.5	$E^{DCS}_{expected}$	2.45E	2.23E	2.01E	1.82E	1.64E	1.49E	1.35E	1.23E	1.13E	1.03E	9.53E

Hollow Core Slab-Marine ecotoxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
		+02	+02	+02	+02	+02	+02	+02	+02	+02	+02	+01
	$E^{DCS}_{expected}$	1.98E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.238	1.126	1.019	0.921	0.832	0.753	0.683	0.622	0.569	0.523	0.482
15.0	$E^{DCS}_{expected}$	2.70E+02	2.46E+02	2.23E+02	2.01E+02	1.82E+02	1.64E+02	1.49E+02	1.36E+02	1.24E+02	1.14E+02	1.05E+02
	$E^{DCS}_{expected}$	2.19E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.233	1.121	1.015	0.916	0.828	0.749	0.680	0.619	0.566	0.520	0.480
16.5	$E^{DCS}_{expected}$	2.97E+02	2.70E+02	2.44E+02	2.21E+02	1.99E+02	1.80E+02	1.64E+02	1.49E+02	1.36E+02	1.25E+02	1.15E+02
	$E^{DCS}_{expected}$	2.42E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.228	1.117	1.011	0.913	0.825	0.746	0.677	0.617	0.564	0.518	0.477
18.0	$E^{DCS}_{expected}$	3.80E+02	3.46E+02	3.13E+02	2.82E+02	2.55E+02	2.30E+02	2.09E+02	1.90E+02	1.73E+02	1.59E+02	1.47E+02
	$E^{DCS}_{expected}$	3.20E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.187	1.079	0.975	0.880	0.795	0.718	0.651	0.593	0.541	0.496	0.457

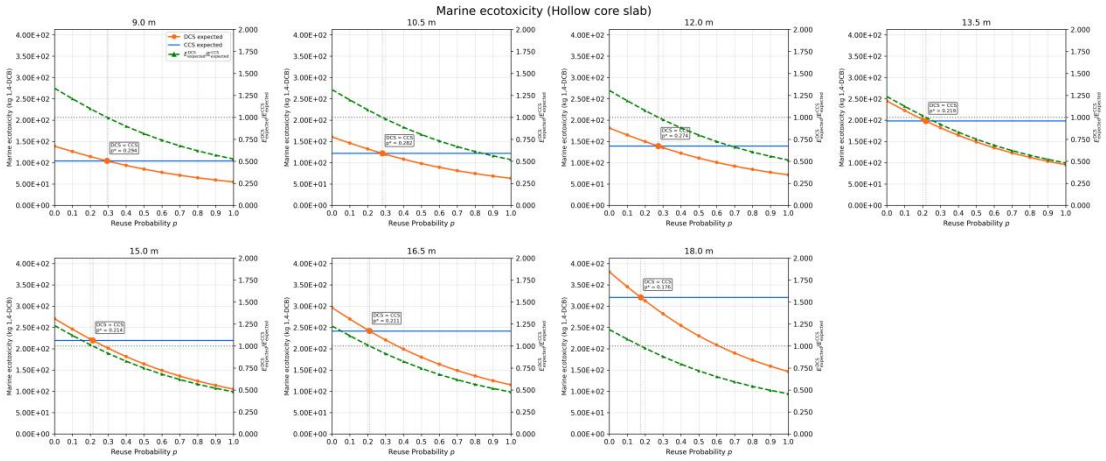


Figure F-12 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine ecotoxicity

Table F-13 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Human carcinogenic toxicity

Hollow Core Slab-Human carcinogenic toxicity (kg 1,4-DCB)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	3.92E+02	3.55E+02	3.19E+02	2.86E+02	2.56E+02	2.30E+02	2.07E+02	1.86E+02	1.68E+02	1.53E+02	1.39E+02
	$E^{DCS}_{expected}$	2.92E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.344	1.216	1.093	0.980	0.878	0.788	0.708	0.638	0.578	0.524	0.478
10.5	$E^{DCS}_{expected}$	4.57E+02	4.13E+02	3.71E+02	3.33E+02	2.98E+02	2.68E+02	2.41E+02	2.17E+02	1.96E+02	1.78E+02	1.62E+02
	$E^{DCS}_{expected}$	3.41E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.338	1.210	1.088	0.975	0.874	0.784	0.704	0.635	0.574	0.521	0.475
12.0	$E^{DCS}_{expected}$	5.19E+02	4.69E+02	4.22E+02	3.78E+02	3.39E+02	3.04E+02	2.73E+02	2.46E+02	2.23E+02	2.02E+02	1.84E+02
	$E^{DCS}_{expected}$	3.89E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.336	1.208	1.086	0.973	0.872	0.782	0.703	0.633	0.573	0.520	0.474
13.5	$E^{DCS}_{expected}$	7.26E+02	6.56E+02	5.90E+02	5.28E+02	4.73E+02	4.24E+02	3.81E+02	3.43E+02	3.10E+02	2.81E+02	2.56E+02
	$E^{DCS}_{expected}$	5.80E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.252	1.131	1.017	0.911	0.815	0.731	0.656	0.591	0.534	0.484	0.441
15.0	$E^{DCS}_{expected}$	8.05E+02	7.27E+02	6.54E+02	5.86E+02	5.24E+02	4.70E+02	4.22E+02	3.80E+02	3.43E+02	3.11E+02	2.83E+02
	$E^{DCS}_{expected}$	6.44E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.250	1.130	1.015	0.910	0.814	0.730	0.655	0.590	0.533	0.484	0.440
16.5	$E^{DCS}_{expected}$	8.85E+02	8.00E+02	7.19E+02	6.44E+02	5.76E+02	5.17E+02	4.64E+02	4.18E+02	3.77E+02	3.42E+02	3.12E+02
	$E^{DCS}_{expected}$	7.09E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.249	1.129	1.014	0.908	0.813	0.729	0.654	0.589	0.532	0.483	0.440
18.0	$E^{DCS}_{expected}$	1.16E+03	1.05E+03	9.43E+02	8.45E+02	7.56E+02	6.77E+02	6.08E+02	5.47E+02	4.94E+02	4.48E+02	4.07E+02
	$E^{DCS}_{expected}$	9.70E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.198	1.082	0.972	0.871	0.779	0.698	0.626	0.564	0.509	0.461	0.420

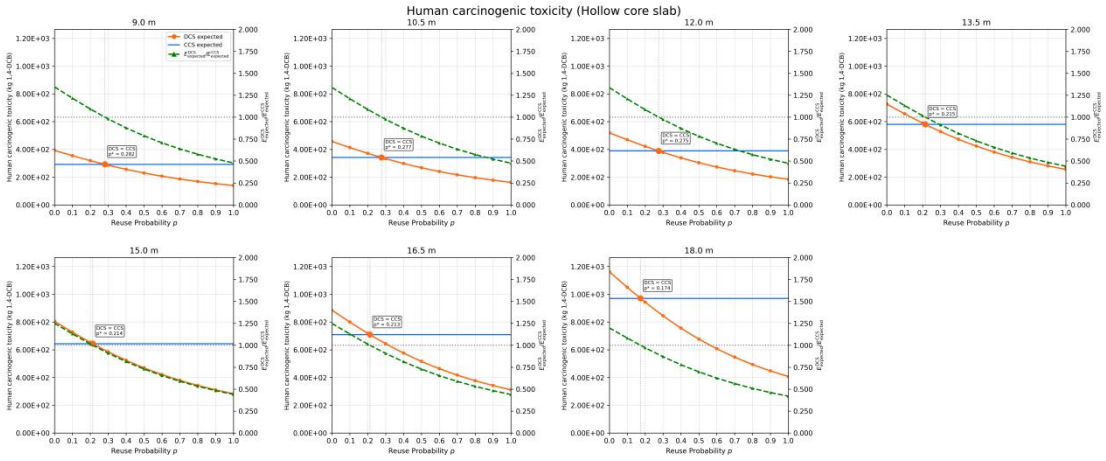


Figure F-13 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Human carcinogenic toxicity

Table F-14 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Human non-carcinogenic toxicity

Hollow Core Slab-Human non-carcinogenic toxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	2.93E+03	2.67E+03	2.42E+03	2.18E+03	1.97E+03	1.79E+03	1.62E+03	1.48E+03	1.36E+03	1.25E+03	1.15E+03
	$E^{DCS}_{expected}$	2.21E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.324	1.205	1.091	0.986	0.892	0.808	0.734	0.669	0.612	0.563	0.520
10.5	$E^{DCS}_{expected}$	3.39E+03	3.08E+03	2.79E+03	2.52E+03	2.28E+03	2.06E+03	1.88E+03	1.71E+03	1.56E+03	1.44E+03	1.33E+03
	$E^{DCS}_{expected}$	2.59E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.309	1.191	1.078	0.974	0.881	0.798	0.724	0.660	0.604	0.555	0.512
12.0	$E^{DCS}_{expected}$	3.83E+03	3.49E+03	3.16E+03	2.85E+03	2.58E+03	2.33E+03	2.12E+03	1.93E+03	1.77E+03	1.62E+03	1.50E+03
	$E^{DCS}_{expected}$	2.95E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.299	1.182	1.070	0.967	0.874	0.791	0.718	0.655	0.599	0.550	0.508
13.5	$E^{DCS}_{expected}$	5.17E+03	4.70E+03	4.25E+03	3.84E+03	3.47E+03	3.14E+03	2.84E+03	2.59E+03	2.36E+03	2.17E+03	2.00E+03
	$E^{DCS}_{expected}$	4.20E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.231	1.119	1.013	0.914	0.825	0.746	0.677	0.616	0.563	0.516	0.476
15.0	$E^{DCS}_{expected}$	5.72E+03	5.20E+03	4.70E+03	4.24E+03	3.83E+03	3.46E+03	3.14E+03	2.86E+03	2.61E+03	2.40E+03	2.21E+03
	$E^{DCS}_{expected}$	4.66E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.226	1.115	1.008	0.910	0.822	0.743	0.674	0.613	0.560	0.514	0.473
16.5	$E^{DCS}_{expected}$	6.27E+03	5.70E+03	5.16E+03	4.65E+03	4.20E+03	3.80E+03	3.44E+03	3.13E+03	2.86E+03	2.63E+03	2.42E+03

Hollow Core Slab-Human non-carcinogenic toxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	5.13E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.222	1.111	1.005	0.907	0.818	0.740	0.671	0.610	0.558	0.511	0.471
18.0	$E^{DCS}_{expected}$	8.04E+03	7.31E+03	6.61E+03	5.96E+03	5.37E+03	4.85E+03	4.40E+03	4.00E+03	3.65E+03	3.34E+03	3.07E+03
	$E^{DCS}_{expected}$	6.81E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.181	1.073	0.970	0.875	0.789	0.713	0.646	0.587	0.535	0.491	0.451

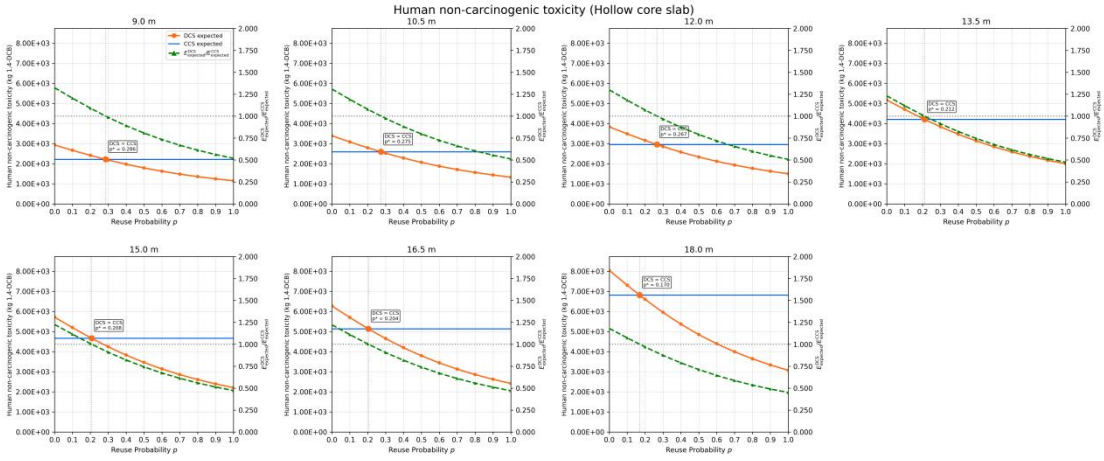


Figure F-14 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Human non-carcinogenic toxicity

Table F-15 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Land use

Hollow Core Slab-Land use (m^2 a crop eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	7.38E+01	6.70E+01	6.05E+01	5.45E+01	4.91E+01	4.43E+01	4.01E+01	3.63E+01	3.31E+01	3.03E+01	2.78E+01
	$E^{DCS}_{expected}$	5.59E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.320	1.199	1.083	0.976	0.879	0.793	0.717	0.651	0.592	0.542	0.497
10.5	$E^{DCS}_{expected}$	8.41E+01	7.63E+01	6.89E+01	6.21E+01	5.60E+01	5.05E+01	4.57E+01	4.14E+01	3.77E+01	3.45E+01	3.16E+01
	$E^{DCS}_{expected}$	6.52E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.289	1.171	1.057	0.953	0.858	0.774	0.700	0.635	0.578	0.529	0.485
12.0	$E^{DCS}_{expected}$	9.43E+01	8.56E+01	7.73E+01	6.97E+01	6.28E+01	5.66E+01	5.12E+01	4.64E+01	4.23E+01	3.87E+01	3.55E+01
	$E^{DCS}_{expected}$	7.45E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.266	1.149	1.038	0.935	0.843	0.760	0.687	0.624	0.568	0.519	0.476
13.5	$E^{DCS}_{expected}$	1.19E+02	1.08E+02	9.74E+01	8.78E+01	7.90E+01	7.13E+01	6.45E+01	5.85E+01	5.32E+01	4.86E+01	4.46E+01

Hollow Core Slab-Land use (m^2 a crop eq)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E^{DCS}_{expected}$	9.74E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.220	1.108	1.001	0.901	0.812	0.732	0.662	0.600	0.547	0.500	0.458
15.0	$E^{DCS}_{expected}$	1.31E+02	1.19E+02	1.07E+02	9.65E+01	8.69E+01	7.84E+01	7.09E+01	6.43E+01	5.85E+01	5.35E+01	4.91E+01
	$E^{DCS}_{expected}$	1.08E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.208	1.097	0.990	0.892	0.803	0.725	0.655	0.594	0.541	0.494	0.454
16.5	$E^{DCS}_{expected}$	1.42E+02	1.29E+02	1.17E+02	1.05E+02	9.48E+01	8.55E+01	7.73E+01	7.01E+01	6.38E+01	5.83E+01	5.35E+01
	$E^{DCS}_{expected}$	1.19E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.198	1.087	0.982	0.885	0.797	0.719	0.650	0.589	0.536	0.490	0.450
18.0	$E^{DCS}_{expected}$	1.74E+02	1.58E+02	1.42E+02	1.28E+02	1.15E+02	1.04E+02	9.41E+01	8.53E+01	7.76E+01	7.09E+01	6.51E+01
	$E^{DCS}_{expected}$	1.48E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.172	1.064	0.961	0.865	0.779	0.703	0.635	0.576	0.524	0.479	0.439

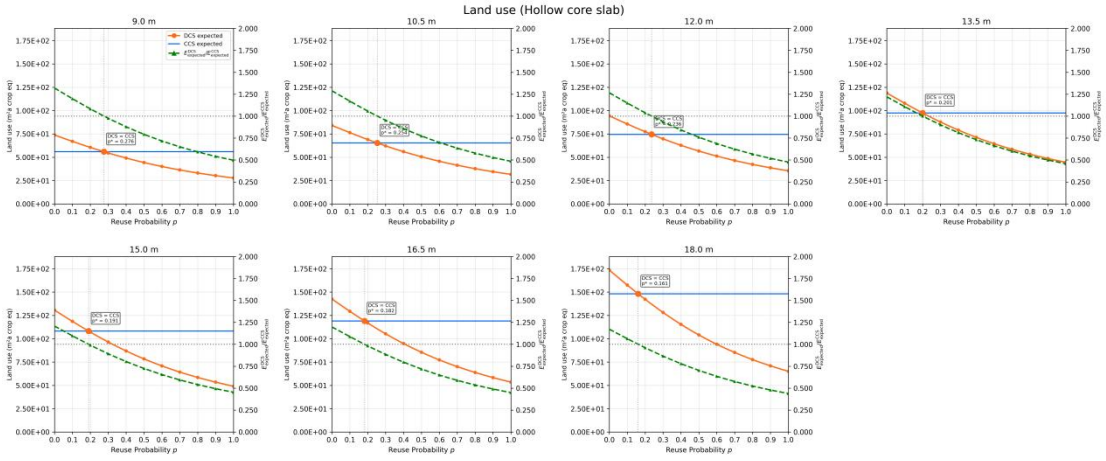


Figure F-15 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Land use

Table F-16 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Mineral resource scarcity

Hollow Core Slab-Mineral resource scarcity (kg Cu eq)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	3.22E+01	2.91E+01	2.60E+01	2.33E+01	2.08E+01	1.86E+01	1.66E+01	1.49E+01	1.34E+01	1.21E+01	1.10E+01
	$E^{DCS}_{expected}$	2.43E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.326	1.196	1.072	0.958	0.855	0.764	0.684	0.613	0.552	0.498	0.451
10.5	$E^{DCS}_{expected}$	3.74E+01	3.37E+01	3.02E+01	2.70E+01	2.41E+01	2.15E+01	1.93E+01	1.73E+01	1.56E+01	1.40E+01	1.27E+01
	$E^{DCS}_{expected}$	2.84E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.316	1.187	1.064	0.951	0.849	0.758	0.678	0.608	0.547	0.494	0.448
12.0	$E^{DCS}_{expected}$	4.24E+01	3.82E+01	3.43E+01	3.06E+01	2.73E+01	2.44E+01	2.18E+01	1.96E+01	1.76E+01	1.59E+01	1.44E+01
	$E^{DCS}_{expected}$	3.24E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.309	1.181	1.059	0.946	0.845	0.754	0.675	0.605	0.545	0.492	0.445
13.5	$E^{DCS}_{expected}$	5.82E+01	5.25E+01	4.71E+01	4.21E+01	3.75E+01	3.35E+01	3.00E+01	2.69E+01	2.42E+01	2.18E+01	1.98E+01
	$E^{DCS}_{expected}$	4.71E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.235	1.114	0.999	0.892	0.796	0.711	0.636	0.571	0.513	0.463	0.419
15.0	$E^{DCS}_{expected}$	6.45E+01	5.82E+01	5.21E+01	4.66E+01	4.16E+01	3.71E+01	3.32E+01	2.98E+01	2.68E+01	2.42E+01	2.19E+01
	$E^{DCS}_{expected}$	5.23E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.232	1.111	0.996	0.890	0.794	0.709	0.634	0.569	0.512	0.462	0.418
16.5	$E^{DCS}_{expected}$	7.08E+01	6.39E+01	5.72E+01	5.11E+01	4.56E+01	4.08E+01	3.65E+01	3.27E+01	2.94E+01	2.65E+01	2.40E+01
	$E^{DCS}_{expected}$	5.76E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.229	1.109	0.994	0.888	0.792	0.707	0.633	0.568	0.511	0.461	0.417
18.0	$E^{DCS}_{expected}$	9.19E+01	8.29E+01	7.43E+01	6.64E+01	5.92E+01	5.29E+01	4.73E+01	4.24E+01	3.81E+01	3.44E+01	3.12E+01
	$E^{DCS}_{expected}$	7.76E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.184	1.068	0.957	0.855	0.763	0.681	0.609	0.546	0.491	0.443	0.401

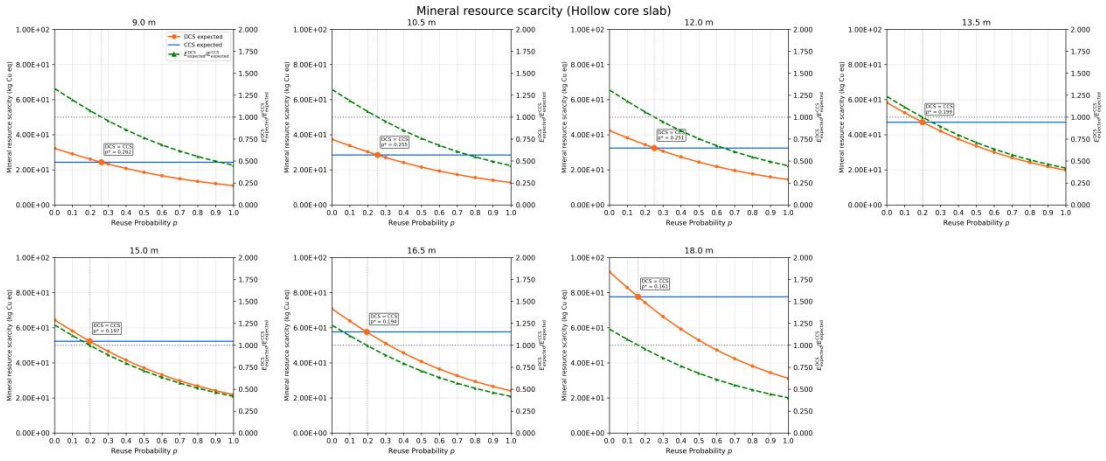


Figure F-16 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Mineral resource scarcity

Table F-17 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fossil resource scarcity

Hollow Core Slab-Fossil resource scarcity (kg oil eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	6.74E+02	6.19E+02	5.66E+02	5.18E+02	4.74E+02	4.35E+02	4.01E+02	3.71E+02	3.45E+02	3.22E+02	3.01E+02
	$E^{DCS}_{expected}$	4.75E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.417	1.302	1.192	1.090	0.998	0.916	0.844	0.781	0.725	0.677	0.634
10.5	$E^{DCS}_{expected}$	7.68E+02	7.06E+02	6.46E+02	5.91E+02	5.41E+02	4.97E+02	4.57E+02	4.23E+02	3.93E+02	3.67E+02	3.44E+02
	$E^{DCS}_{expected}$	5.55E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.385	1.273	1.165	1.065	0.975	0.895	0.824	0.762	0.708	0.661	0.619
12.0	$E^{DCS}_{expected}$	8.62E+02	7.92E+02	7.25E+02	6.63E+02	6.07E+02	5.57E+02	5.13E+02	4.74E+02	4.41E+02	4.11E+02	3.85E+02
	$E^{DCS}_{expected}$	6.34E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.361	1.251	1.145	1.047	0.958	0.879	0.810	0.749	0.696	0.649	0.608
13.5	$E^{DCS}_{expected}$	1.09E+03	1.00E+03	9.18E+02	8.39E+02	7.68E+02	7.04E+02	6.48E+02	5.99E+02	5.56E+02	5.19E+02	4.86E+02
	$E^{DCS}_{expected}$	8.36E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.308	1.201	1.099	1.004	0.919	0.843	0.776	0.717	0.666	0.621	0.581
15.0	$E^{DCS}_{expected}$	1.20E+03	1.10E+03	1.01E+03	9.23E+02	8.44E+02	7.74E+02	7.13E+02	6.59E+02	6.12E+02	5.70E+02	5.34E+02
	$E^{DCS}_{expected}$	9.28E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.295	1.189	1.088	0.994	0.910	0.834	0.768	0.710	0.659	0.614	0.575
16.5	$E^{DCS}_{expected}$	1.31E+03	1.20E+03	1.10E+03	1.01E+03	9.21E+02	8.45E+02	7.78E+02	7.19E+02	6.67E+02	6.22E+02	5.83E+02

Hollow Core Slab-Fossil resource scarcity (kg oil eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E^{DCS}_{expected}$	1.02E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.285	1.180	1.079	0.986	0.902	0.828	0.762	0.704	0.653	0.609	0.570
18.0	$E^{DCS}_{expected}$	1.61E+03	1.47E+03	1.35E+03	1.23E+03	1.13E+03	1.03E+03	9.49E+02	8.77E+02	8.13E+02	7.58E+02	7.10E+02
	$E^{DCS}_{expected}$	1.28E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.254	1.151	1.053	0.961	0.879	0.806	0.742	0.685	0.635	0.592	0.554

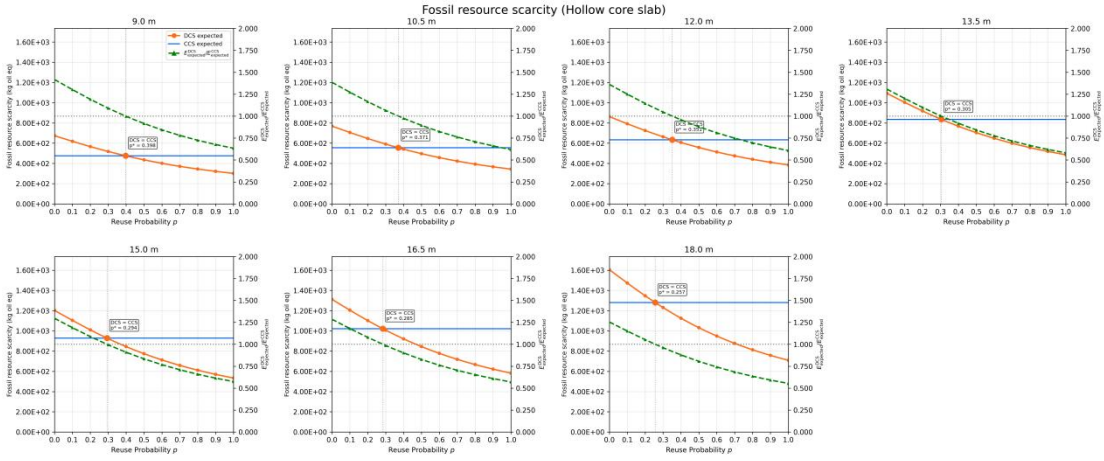


Figure F-17 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fossil resource scarcity

Table F-18 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Water consumption

Hollow Core Slab-Water consumption (m^3)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	2.44E+01	2.21E+01	1.99E+01	1.78E+01	1.60E+01	1.43E+01	1.29E+01	1.16E+01	1.05E+01	9.55E+00	8.71E+00
	$E^{DCS}_{expected}$	1.78E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.376	1.245	1.119	1.004	0.900	0.807	0.726	0.655	0.592	0.538	0.491
10.5	$E^{DCS}_{expected}$	2.78E+01	2.52E+01	2.26E+01	2.03E+01	1.82E+01	1.63E+01	1.47E+01	1.32E+01	1.20E+01	1.09E+01	9.93E+00
	$E^{DCS}_{expected}$	2.07E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.344	1.215	1.093	0.980	0.879	0.788	0.709	0.639	0.578	0.525	0.479
12.0	$E^{DCS}_{expected}$	3.12E+01	2.82E+01	2.54E+01	2.28E+01	2.04E+01	1.83E+01	1.65E+01	1.49E+01	1.34E+01	1.22E+01	1.11E+01
	$E^{DCS}_{expected}$	2.37E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.319	1.193	1.073	0.962	0.863	0.774	0.696	0.628	0.568	0.516	0.470
13.5	$E^{DCS}_{expected}$	3.93E	3.56E	3.20E	2.87E	2.57E	2.31E	2.07E	1.87E	1.69E	1.54E	1.40E

Hollow Core Slab-Water consumption (m^3)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
		+01	+01	+01	+01	+01	+01	+01	+01	+01	+01	+01
	$E^{DCS}_{expected}$	3.10E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.271	1.149	1.034	0.927	0.831	0.745	0.670	0.604	0.547	0.496	0.452
15.0	$E^{DCS}_{expected}$	4.33E +01	3.91E +01	3.52E +01	3.16E +01	2.83E +01	2.54E +01	2.28E +01	2.06E +01	1.86E +01	1.69E +01	1.54E +01
	$E^{DCS}_{expected}$	3.44E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.258	1.137	1.023	0.917	0.822	0.737	0.663	0.598	0.541	0.491	0.448
16.5	$E^{DCS}_{expected}$	4.72E +01	4.27E +01	3.84E +01	3.44E +01	3.08E +01	2.77E +01	2.49E +01	2.24E +01	2.03E +01	1.84E +01	1.68E +01
	$E^{DCS}_{expected}$	3.78E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.247	1.128	1.014	0.910	0.815	0.731	0.657	0.593	0.536	0.487	0.444
18.0	$E^{DCS}_{expected}$	5.75E +01	5.20E +01	4.67E +01	4.19E +01	3.76E +01	3.37E +01	3.03E +01	2.73E +01	2.47E +01	2.24E +01	2.04E +01
	$E^{DCS}_{expected}$	4.71E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.219	1.103	0.992	0.889	0.797	0.715	0.642	0.579	0.524	0.476	0.434

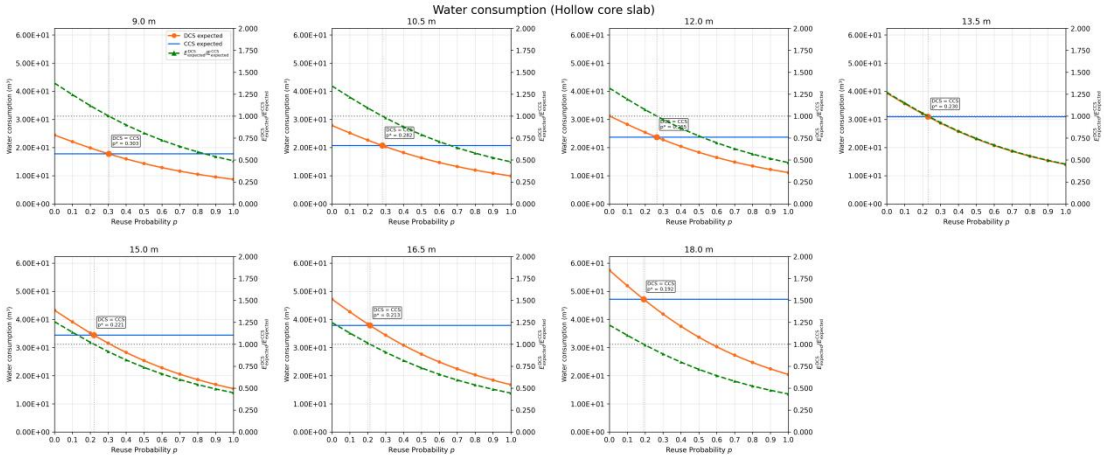


Figure F-18 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Water consumption

Appendix G: p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) for the ribbed slab of different model lengths

Table G-1 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Global warming

Ribbed Slab-Global warming (kg CO_2 eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	3.43E+03	3.14E+03	2.85E+03	2.59E+03	2.36E+03	2.15E+03	1.97E+03	1.81E+03	1.66E+03	1.54E+03	1.43E+03
	$E^{DCS}_{expected}$	2.91E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.177	1.076	0.979	0.890	0.809	0.738	0.675	0.619	0.571	0.529	0.492
10.5	$E^{DCS}_{expected}$	4.16E+03	3.80E+03	3.46E+03	3.15E+03	2.86E+03	2.61E+03	2.39E+03	2.19E+03	2.02E+03	1.87E+03	1.74E+03
	$E^{DCS}_{expected}$	3.54E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.175	1.074	0.978	0.889	0.809	0.737	0.674	0.619	0.571	0.529	0.492
12.0	$E^{DCS}_{expected}$	5.19E+03	4.74E+03	4.32E+03	3.92E+03	3.57E+03	3.25E+03	2.97E+03	2.73E+03	2.51E+03	2.33E+03	2.16E+03
	$E^{DCS}_{expected}$	4.46E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.165	1.065	0.969	0.880	0.800	0.729	0.667	0.612	0.564	0.522	0.486
13.5	$E^{DCS}_{expected}$	6.03E+03	5.51E+03	5.01E+03	4.55E+03	4.14E+03	3.77E+03	3.45E+03	3.17E+03	2.92E+03	2.70E+03	2.51E+03
	$E^{DCS}_{expected}$	5.18E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.164	1.063	0.968	0.879	0.800	0.729	0.666	0.611	0.564	0.522	0.485
15.0	$E^{DCS}_{expected}$	7.24E+03	6.62E+03	6.02E+03	5.47E+03	4.97E+03	4.53E+03	4.14E+03	3.80E+03	3.50E+03	3.24E+03	3.01E+03
	$E^{DCS}_{expected}$	6.26E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.157	1.057	0.962	0.874	0.794	0.724	0.662	0.607	0.559	0.518	0.481
16.5	$E^{DCS}_{expected}$	8.64E+03	7.89E+03	7.18E+03	6.52E+03	5.92E+03	5.40E+03	4.93E+03	4.52E+03	4.17E+03	3.86E+03	3.58E+03
	$E^{DCS}_{expected}$	7.50E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.151	1.052	0.957	0.869	0.790	0.719	0.657	0.603	0.556	0.514	0.478
18.0	$E^{DCS}_{expected}$	9.58E+03	8.75E+03	7.96E+03	7.23E+03	6.57E+03	5.99E+03	5.47E+03	5.02E+03	4.62E+03	4.28E+03	3.98E+03
	$E^{DCS}_{expected}$	8.32E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.151	1.052	0.957	0.869	0.790	0.720	0.658	0.603	0.556	0.514	0.478

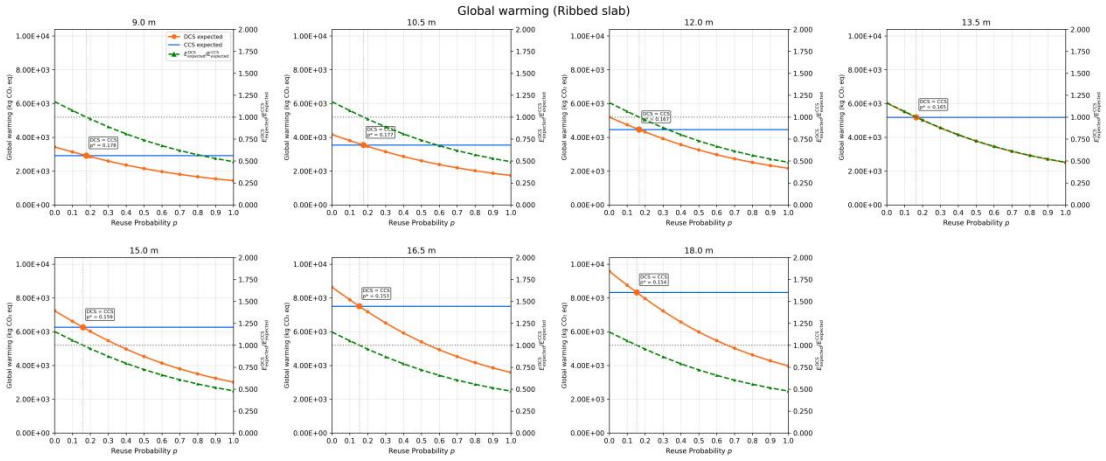


Figure G-1 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Global warming

Table G-2 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Stratospheric ozone depletion

Ribbed Slab-Stratospheric ozone depletion (kg CFC11 eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	1.25E-03	1.14E-03	1.04E-03	9.50E-04	8.66E-04	7.90E-04	7.24E-04	6.66E-04	6.15E-04	5.70E-04	5.31E-04
	$E^{DCS}_{expected}$	9.88E-04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.266	1.159	1.057	0.962	0.876	0.800	0.733	0.674	0.622	0.577	0.538
10.5	$E^{DCS}_{expected}$	1.52E-03	1.39E-03	1.27E-03	1.15E-03	1.05E-03	9.60E-04	8.79E-04	8.09E-04	7.47E-04	6.93E-04	6.46E-04
	$E^{DCS}_{expected}$	1.20E-03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.265	1.158	1.056	0.961	0.876	0.800	0.733	0.674	0.622	0.577	0.538
12.0	$E^{DCS}_{expected}$	1.89E-03	1.73E-03	1.58E-03	1.43E-03	1.31E-03	1.19E-03	1.09E-03	1.00E-03	9.27E-04	8.60E-04	8.01E-04
	$E^{DCS}_{expected}$	1.51E-03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.252	1.146	1.045	0.951	0.866	0.791	0.724	0.666	0.615	0.570	0.531
13.5	$E^{DCS}_{expected}$	2.19E-03	2.01E-03	1.83E-03	1.67E-03	1.52E-03	1.38E-03	1.27E-03	1.17E-03	1.08E-03	9.98E-04	9.30E-04
	$E^{DCS}_{expected}$	1.75E-03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.251	1.145	1.044	0.950	0.865	0.790	0.723	0.665	0.614	0.570	0.530
15.0	$E^{DCS}_{expected}$	2.63E-03	2.41E-03	2.19E-03	2.00E-03	1.82E-03	1.66E-03	1.52E-03	1.40E-03	1.29E-03	1.19E-03	1.11E-03
	$E^{DCS}_{expected}$	2.11E-03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.243	1.138	1.037	0.943	0.859	0.784	0.718	0.660	0.609	0.565	0.526
16.5	$E^{DCS}_{expected}$	3.13E-03	2.86E-03	2.61E-03	2.37E-03	2.16E-03	1.97E-03	1.81E-03	1.66E-03	1.53E-03	1.42E-03	1.32E-03
	$E^{DCS}_{expected}$	2.53E-03										

Ribbed Slab-Stratospheric ozone depletion (kg CFC11 eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.236	1.131	1.031	0.938	0.854	0.779	0.713	0.655	0.605	0.561	0.522
18.0	$E_{expected}^{DCS}$	3.47E-03	3.18E-03	2.89E-03	2.63E-03	2.40E-03	2.19E-03	2.00E-03	1.84E-03	1.70E-03	1.58E-03	1.47E-03
	$E_{expected}^{DCS}$	2.81E-03										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.236	1.131	1.031	0.938	0.854	0.779	0.714	0.656	0.605	0.561	0.522

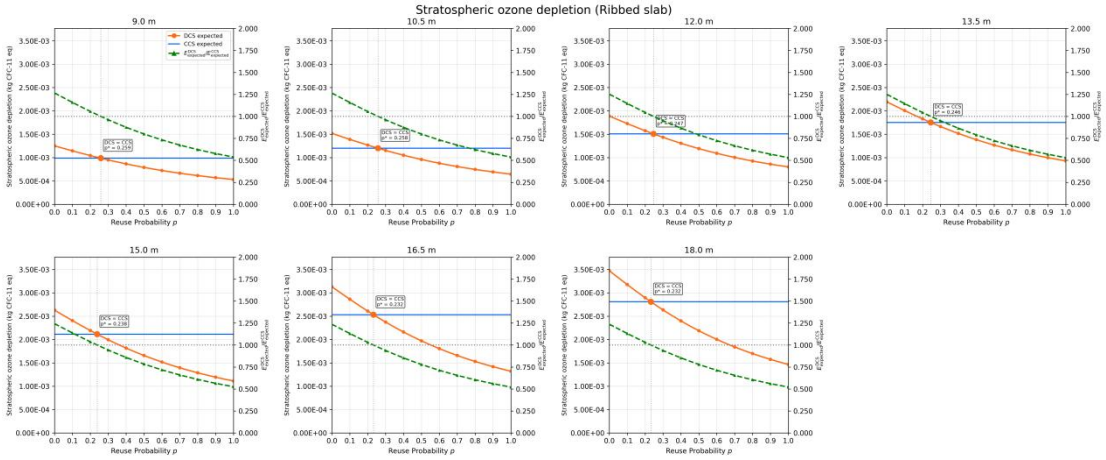


Figure G-2 p* derived from calculations of CCS ($E_{expected}^{CCS}$) and DCS ($E_{expected}^{DCS}$) of Stratospheric ozone depletion

Table G-3 p* derived from calculations of CCS ($E_{expected}^{CCS}$) and DCS ($E_{expected}^{DCS}$) of Ionizing radiation

Ribbed Slab-Ionizing radiation (kBq Co-60 eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E_{expected}^{DCS}$	8.61E+01	7.82E+01	7.06E+01	6.36E+01	5.73E+01	5.18E+01	4.68E+01	4.25E+01	3.88E+01	3.55E+01	3.26E+01
	$E_{expected}^{DCS}$	7.63E+01										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.128	1.024	0.925	0.834	0.751	0.678	0.614	0.557	0.508	0.465	0.427
10.5	$E_{expected}^{DCS}$	1.04E+02	9.46E+01	8.55E+01	7.70E+01	6.94E+01	6.27E+01	5.67E+01	5.15E+01	4.70E+01	4.30E+01	3.95E+01
	$E_{expected}^{DCS}$	9.25E+01										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.126	1.023	0.924	0.833	0.751	0.677	0.613	0.557	0.508	0.465	0.427
12.0	$E_{expected}^{DCS}$	1.31E+02	1.19E+02	1.07E+02	9.68E+01	8.73E+01	7.87E+01	7.12E+01	6.47E+01	5.89E+01	5.39E+01	4.95E+01
	$E_{expected}^{DCS}$	1.17E+02										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.116	1.013	0.915	0.825	0.743	0.671	0.607	0.551	0.502	0.459	0.422
13.5	$E_{expected}^{DCS}$	1.52E+02	1.38E+02	1.25E+02	1.12E+02	1.01E+02	9.14E+01	8.27E+01	7.50E+01	6.84E+01	6.26E+01	5.75E+01

Ribbed Slab-Ionizing radiation (kBq Co-60 eq)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E^{DCS}_{expected}$	1.36E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.115	1.012	0.914	0.824	0.742	0.670	0.606	0.550	0.501	0.459	0.421
15.0	$E^{DCS}_{expected}$	1.84E+02	1.67E+02	1.51E+02	1.36E+02	1.22E+02	1.10E+02	9.98E+01	9.06E+01	8.25E+01	7.55E+01	6.93E+01
	$E^{DCS}_{expected}$	1.66E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.108	1.006	0.909	0.819	0.737	0.665	0.602	0.546	0.498	0.455	0.418
16.5	$E^{DCS}_{expected}$	2.21E+02	2.00E+02	1.81E+02	1.63E+02	1.47E+02	1.32E+02	1.20E+02	1.09E+02	9.89E+01	9.04E+01	8.30E+01
	$E^{DCS}_{expected}$	2.00E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.103	1.001	0.904	0.814	0.733	0.661	0.598	0.543	0.494	0.452	0.415
18.0	$E^{DCS}_{expected}$	2.44E+02	2.22E+02	2.00E+02	1.80E+02	1.62E+02	1.46E+02	1.32E+02	1.20E+02	1.10E+02	1.00E+02	9.20E+01
	$E^{DCS}_{expected}$	2.21E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.103	1.001	0.904	0.814	0.733	0.662	0.598	0.543	0.495	0.452	0.415

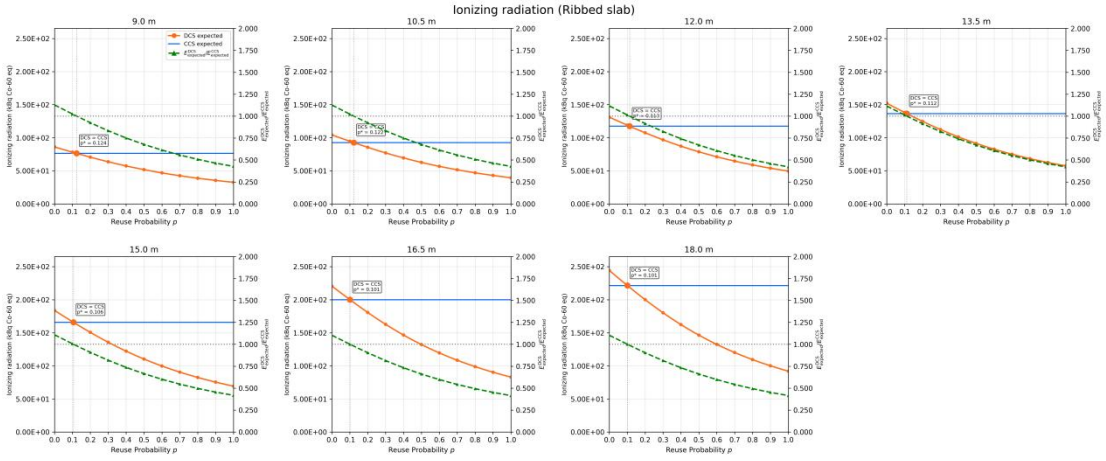


Figure G-3 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ionizing radiation

Table G-4 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation, Human health

Ribbed Slab-Ozone formation, Human health (kg NOx eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	8.49E+00	7.83E+00	7.20E+00	6.62E+00	6.09E+00	5.62E+00	5.21E+00	4.85E+00	4.54E+00	4.26E+00	4.02E+00
	$E^{DCS}_{expected}$	7.19E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.181	1.089	1.001	0.920	0.847	0.782	0.725	0.675	0.631	0.593	0.560
10.5	$E^{DCS}_{expected}$	1.03E+01	9.49E+00	8.73E+00	8.02E+00	7.39E+00	6.82E+00	6.32E+00	5.89E+00	5.51E+00	5.17E+00	4.88E+00
	$E^{DCS}_{expected}$	8.72E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.180	1.088	1.000	0.920	0.847	0.782	0.725	0.675	0.631	0.593	0.560
12.0	$E^{DCS}_{expected}$	1.29E+01	1.19E+01	1.09E+01	1.00E+01	9.23E+00	8.51E+00	7.89E+00	7.34E+00	6.86E+00	6.44E+00	6.08E+00
	$E^{DCS}_{expected}$	1.10E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.169	1.077	0.990	0.910	0.837	0.772	0.716	0.666	0.622	0.584	0.551
13.5	$E^{DCS}_{expected}$	1.50E+01	1.38E+01	1.27E+01	1.16E+01	1.07E+01	9.88E+00	9.16E+00	8.52E+00	7.97E+00	7.48E+00	7.06E+00
	$E^{DCS}_{expected}$	1.28E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.167	1.076	0.989	0.908	0.836	0.772	0.715	0.665	0.622	0.584	0.551
15.0	$E^{DCS}_{expected}$	1.80E+01	1.66E+01	1.52E+01	1.40E+01	1.29E+01	1.19E+01	1.10E+01	1.02E+01	9.56E+00	8.97E+00	8.46E+00
	$E^{DCS}_{expected}$	1.55E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.160	1.069	0.982	0.902	0.830	0.765	0.709	0.659	0.616	0.578	0.545
16.5	$E^{DCS}_{expected}$	2.15E+01	1.98E+01	1.82E+01	1.67E+01	1.54E+01	1.42E+01	1.31E+01	1.22E+01	1.14E+01	1.07E+01	1.01E+01
	$E^{DCS}_{expected}$	1.87E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.154	1.063	0.976	0.896	0.824	0.760	0.703	0.654	0.611	0.573	0.540
18.0	$E^{DCS}_{expected}$	2.39E+01	2.20E+01	2.02E+01	1.85E+01	1.70E+01	1.57E+01	1.46E+01	1.35E+01	1.26E+01	1.19E+01	1.12E+01
	$E^{DCS}_{expected}$	2.07E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.154	1.063	0.976	0.896	0.824	0.760	0.704	0.654	0.611	0.573	0.541

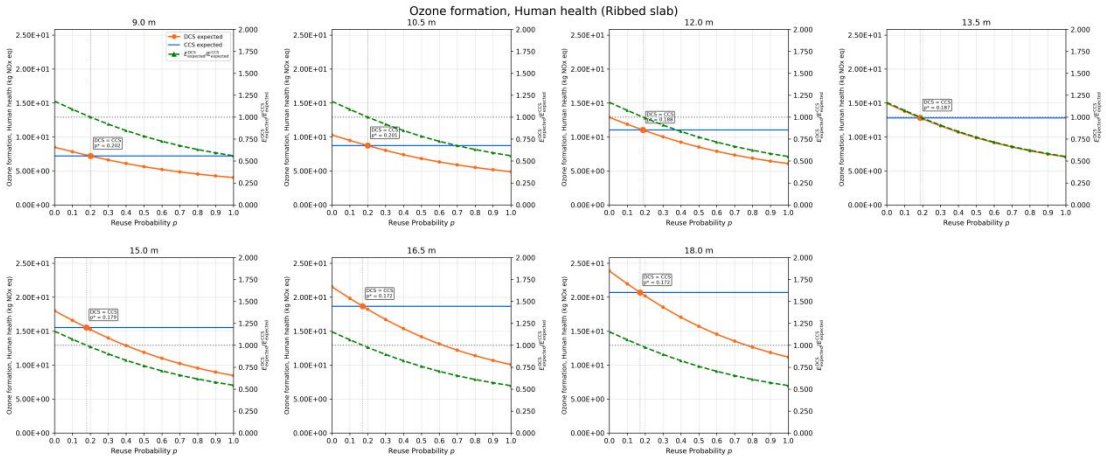


Figure G-4 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation, Human health

Table G-5 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fine particulate matter formation

Ribbed Slab-Fine particulate matter formation (kg PM2.5eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	5.04E+00	4.63E+00	4.24E+00	3.87E+00	3.55E+00	3.25E+00	3.00E+00	2.77E+00	2.58E+00	2.41E+00	2.26E+00
	$E^{DCS}_{expected}$	3.93E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.283	1.178	1.078	0.985	0.902	0.828	0.763	0.706	0.656	0.612	0.574
10.5	$E^{DCS}_{expected}$	6.09E+00	5.60E+00	5.12E+00	4.68E+00	4.29E+00	3.94E+00	3.63E+00	3.36E+00	3.12E+00	2.91E+00	2.73E+00
	$E^{DCS}_{expected}$	4.75E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.283	1.178	1.078	0.986	0.903	0.829	0.764	0.707	0.657	0.614	0.576
12.0	$E^{DCS}_{expected}$	7.71E+00	7.07E+00	6.47E+00	5.91E+00	5.41E+00	4.96E+00	4.57E+00	4.22E+00	3.92E+00	3.66E+00	3.43E+00
	$E^{DCS}_{expected}$	6.10E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.264	1.160	1.061	0.970	0.887	0.814	0.750	0.693	0.644	0.601	0.563
13.5	$E^{DCS}_{expected}$	8.94E+00	8.21E+00	7.51E+00	6.86E+00	6.28E+00	5.76E+00	5.30E+00	4.90E+00	4.55E+00	4.25E+00	3.98E+00
	$E^{DCS}_{expected}$	7.08E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.263	1.159	1.060	0.969	0.887	0.813	0.749	0.693	0.643	0.600	0.562
15.0	$E^{DCS}_{expected}$	1.08E+01	9.95E+00	9.10E+00	8.31E+00	7.60E+00	6.97E+00	6.41E+00	5.93E+00	5.50E+00	5.13E+00	4.80E+00
	$E^{DCS}_{expected}$	8.67E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.251	1.148	1.049	0.958	0.876	0.804	0.740	0.683	0.634	0.592	0.554
16.5	$E^{DCS}_{expected}$	1.31E+01	1.20E+01	1.10E+01	1.00E+01	9.14E+00	8.38E+00	7.70E+00	7.11E+00	6.60E+00	6.15E+00	5.76E+00

Ribbed Slab-Fine particulate matter formation (kg PM2.5eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E^{DCS}_{expected}$	1.05E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.240	1.137	1.039	0.949	0.867	0.795	0.731	0.675	0.626	0.584	0.546
18.0	$E^{DCS}_{expected}$	1.45E +01	1.33E +01	1.21E +01	1.11E +01	1.01E +01	9.27E +00	8.53E +00	7.88E +00	7.31E +00	6.81E +00	6.38E +00
	$E^{DCS}_{expected}$	1.16E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.241	1.138	1.040	0.950	0.868	0.796	0.732	0.676	0.627	0.585	0.548

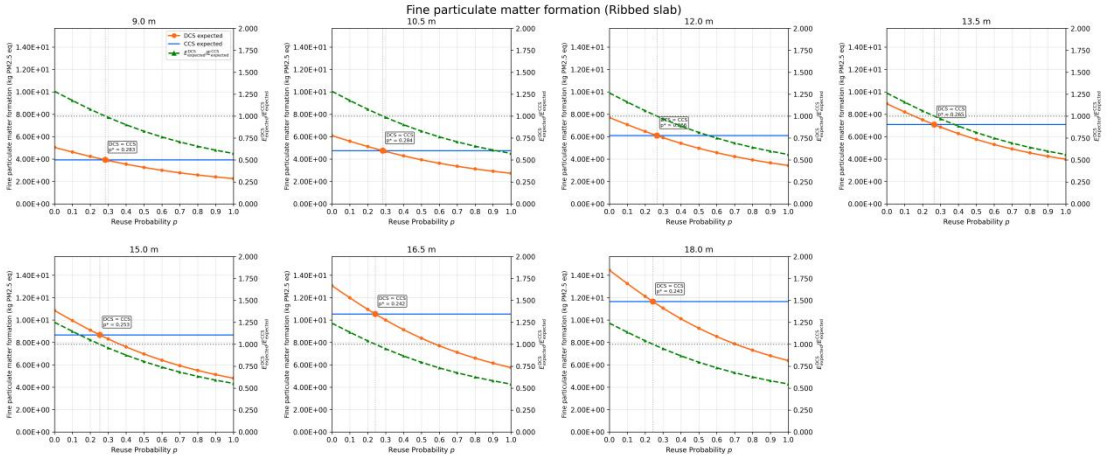


Figure G-5 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fine particulate matter formation

Table G-6 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation, Terrestrial ecosystem

Ribbed Slab-Ozone formation, Terrestrial ecosystem (kg NOx eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	8.70E +00	8.02E +00	7.37E +00	6.77E +00	6.23E +00	5.75E +00	5.33E +00	4.96E +00	4.63E +00	4.35E +00	4.10E +00
	$E^{DCS}_{expected}$	7.38E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.179	1.087	0.999	0.917	0.844	0.779	0.722	0.672	0.628	0.590	0.556
10.5	$E^{DCS}_{expected}$	1.05E +01	9.72E +00	8.93E +00	8.21E +00	7.55E +00	6.97E +00	6.46E +00	6.01E +00	5.62E +00	5.28E +00	4.98E +00
	$E^{DCS}_{expected}$	8.95E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.177	1.085	0.998	0.917	0.844	0.779	0.722	0.672	0.628	0.590	0.556
12.0	$E^{DCS}_{expected}$	1.32E +01	1.22E +01	1.12E +01	1.03E +01	9.44E +00	8.71E +00	8.06E +00	7.50E +00	7.01E +00	6.58E +00	6.20E +00
	$E^{DCS}_{expected}$	1.13E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.166	1.075	0.987	0.907	0.834	0.769	0.713	0.663	0.619	0.581	0.548
13.5	$E^{DCS}_{expected}$	1.53E	1.41E	1.30E	1.19E	1.10E	1.01E	9.36E	8.71E	8.14E	7.64E	7.20E

Ribbed Slab-Ozone formation, Terrestrial ecosystem (kg NOx eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
		+01	+01	+01	+01	+01	+01	+00	+00	+00	+00	+00
	$E^{DCS}_{expected}$	1.32E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.165	1.073	0.986	0.906	0.833	0.769	0.712	0.662	0.619	0.581	0.547
15.0	$E^{DCS}_{expected}$	1.85E+01	1.70E+01	1.56E+01	1.43E+01	1.32E+01	1.22E+01	1.13E+01	1.05E+01	9.77E+00	9.17E+00	8.64E+00
	$E^{DCS}_{expected}$	1.59E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.157	1.066	0.979	0.899	0.827	0.762	0.706	0.656	0.613	0.575	0.542
16.5	$E^{DCS}_{expected}$	2.21E+01	2.03E+01	1.87E+01	1.71E+01	1.57E+01	1.45E+01	1.34E+01	1.25E+01	1.16E+01	1.09E+01	1.03E+01
	$E^{DCS}_{expected}$	1.92E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.151	1.060	0.973	0.893	0.821	0.757	0.700	0.651	0.607	0.570	0.536
18.0	$E^{DCS}_{expected}$	2.45E+01	2.25E+01	2.07E+01	1.90E+01	1.75E+01	1.61E+01	1.49E+01	1.38E+01	1.29E+01	1.21E+01	1.14E+01
	$E^{DCS}_{expected}$	2.12E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.151	1.060	0.973	0.893	0.821	0.757	0.701	0.651	0.608	0.570	0.537

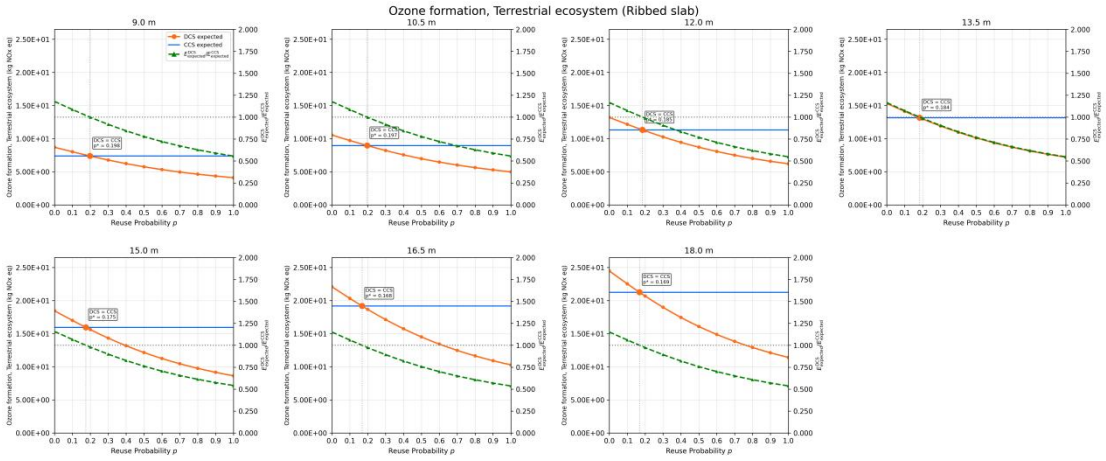


Figure G-6 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation, Terrestrial ecosystem

Table G-7 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial acidification

Ribbed Slab-Terrestrial acidification (kg SO2 eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	1.10E+01	1.02E+01	9.34E+00	8.57E+00	7.88E+00	7.27E+00	6.73E+00	6.26E+00	5.85E+00	5.48E+00	5.17E+00
	$E^{DCS}_{expected}$	8.17E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.351	1.245	1.143	1.050	0.965	0.890	0.824	0.766	0.716	0.672	0.633

Ribbed Slab-Terrestrial acidification (kg SO2 eq)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
10.5	$E^{DCS}_{expected}$	1.34E+01	1.23E+01	1.13E+01	1.04E+01	9.56E+00	8.82E+00	8.17E+00	7.59E+00	7.09E+00	6.66E+00	6.28E+00
	$E^{DCS}_{expected}$	9.90E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.351	1.245	1.143	1.050	0.966	0.891	0.825	0.767	0.716	0.672	0.634
12.0	$E^{DCS}_{expected}$	1.67E+01	1.54E+01	1.42E+01	1.30E+01	1.19E+01	1.10E+01	1.02E+01	9.47E+00	8.84E+00	8.29E+00	7.81E+00
	$E^{DCS}_{expected}$	1.26E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.333	1.228	1.127	1.035	0.951	0.877	0.811	0.754	0.704	0.660	0.622
13.5	$E^{DCS}_{expected}$	1.94E+01	1.79E+01	1.64E+01	1.51E+01	1.39E+01	1.28E+01	1.18E+01	1.10E+01	1.03E+01	9.62E+00	9.07E+00
	$E^{DCS}_{expected}$	1.46E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.332	1.227	1.126	1.034	0.950	0.876	0.811	0.753	0.703	0.660	0.622
15.0	$E^{DCS}_{expected}$	2.34E+01	2.15E+01	1.98E+01	1.81E+01	1.67E+01	1.54E+01	1.42E+01	1.32E+01	1.23E+01	1.15E+01	1.09E+01
	$E^{DCS}_{expected}$	1.77E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.320	1.216	1.116	1.024	0.941	0.867	0.802	0.745	0.695	0.652	0.614
16.5	$E^{DCS}_{expected}$	2.80E+01	2.57E+01	2.36E+01	2.17E+01	1.99E+01	1.83E+01	1.69E+01	1.57E+01	1.47E+01	1.38E+01	1.29E+01
	$E^{DCS}_{expected}$	2.14E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.310	1.206	1.106	1.015	0.932	0.858	0.794	0.737	0.687	0.644	0.606
18.0	$E^{DCS}_{expected}$	3.10E+01	2.85E+01	2.62E+01	2.40E+01	2.21E+01	2.03E+01	1.88E+01	1.75E+01	1.63E+01	1.53E+01	1.44E+01
	$E^{DCS}_{expected}$	2.36E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.311	1.207	1.107	1.016	0.933	0.859	0.795	0.738	0.689	0.645	0.607

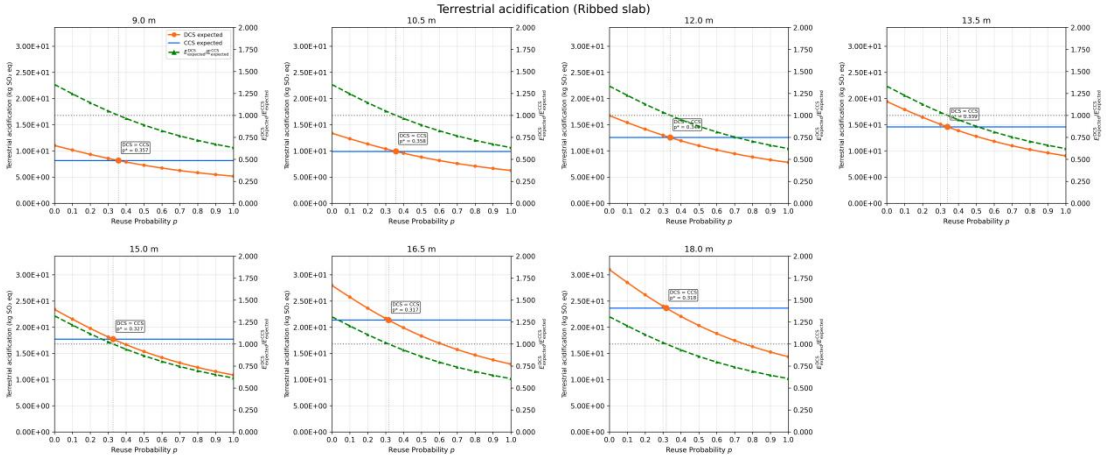


Figure G-7 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial acidification

Table G-8 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater eutrophication

Ribbed Slab-Freshwater eutrophication (kg P eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	1.04E+00	9.52E-01	8.71E-01	7.95E-01	7.27E-01	6.67E-01	6.13E-01	5.67E-01	5.26E-01	4.91E-01	4.60E-01
	$E^{DCS}_{expected}$	9.63E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.078	0.989	0.904	0.825	0.755	0.692	0.637	0.589	0.546	0.509	0.477
10.5	$E^{DCS}_{expected}$	1.24E+00	1.14E+00	1.04E+00	9.54E-01	8.73E-01	8.01E-01	7.37E-01	6.81E-01	6.33E-01	5.90E-01	5.53E-01
	$E^{DCS}_{expected}$	1.16E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.076	0.988	0.903	0.825	0.755	0.692	0.637	0.589	0.547	0.511	0.479
12.0	$E^{DCS}_{expected}$	1.62E+00	1.48E+00	1.36E+00	1.24E+00	1.13E+00	1.03E+00	9.51E-01	8.78E-01	8.14E-01	7.58E-01	7.10E-01
	$E^{DCS}_{expected}$	1.52E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.066	0.977	0.892	0.814	0.744	0.681	0.626	0.578	0.536	0.499	0.467
13.5	$E^{DCS}_{expected}$	1.88E+00	1.72E+00	1.57E+00	1.43E+00	1.31E+00	1.20E+00	1.10E+00	1.02E+00	9.44E-01	8.79E-01	8.23E-01
	$E^{DCS}_{expected}$	1.76E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.064	0.975	0.891	0.813	0.743	0.680	0.626	0.577	0.535	0.499	0.467
15.0	$E^{DCS}_{expected}$	2.32E+00	2.12E+00	1.94E+00	1.77E+00	1.61E+00	1.48E+00	1.36E+00	1.25E+00	1.16E+00	1.08E+00	1.01E+00
	$E^{DCS}_{expected}$	2.19E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.057	0.969	0.884	0.806	0.736	0.673	0.618	0.570	0.528	0.491	0.459
16.5	$E^{DCS}_{expected}$	2.84E+00	2.60E+00	2.38E+00	2.16E+00	1.97E+00	1.80E+00	1.66E+00	1.52E+00	1.41E+00	1.31E+00	1.22E+00
	$E^{DCS}_{expected}$	2.71E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.052	0.963	0.878	0.800	0.729	0.667	0.612	0.564	0.522	0.485	0.453
18.0	$E^{DCS}_{expected}$	3.13E+00	2.87E+00	2.62E+00	2.38E+00	2.17E+00	1.99E+00	1.82E+00	1.68E+00	1.56E+00	1.45E+00	1.35E+00
	$E^{DCS}_{expected}$	2.98E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.051	0.963	0.878	0.800	0.730	0.667	0.613	0.564	0.522	0.486	0.454

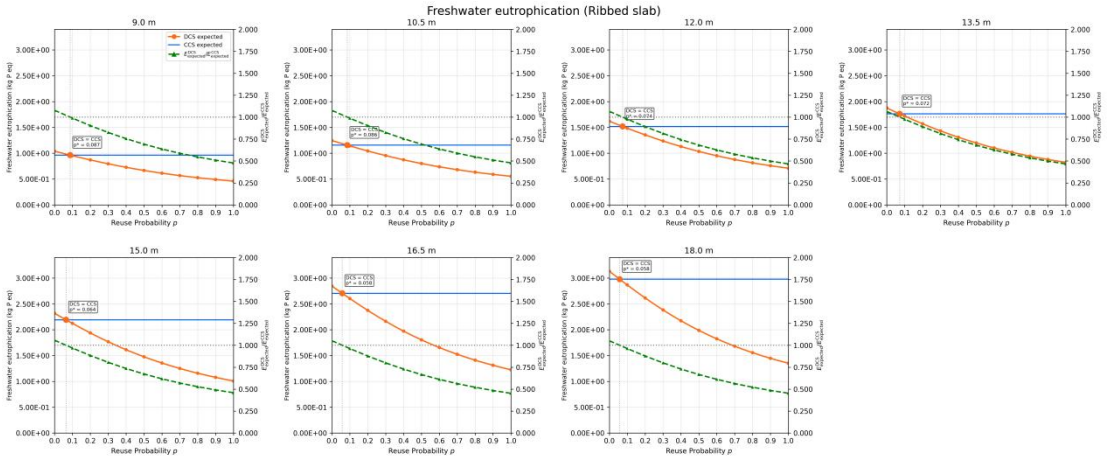


Figure G-8 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater eutrophication

Table G-9 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine eutrophication

Ribbed Slab-Marine eutrophication (kg N eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	8.44E-02	7.71E-02	7.02E-02	6.38E-02	5.80E-02	5.29E-02	4.84E-02	4.44E-02	4.10E-02	3.80E-02	3.53E-02
	$E^{DCS}_{expected}$	7.98E-02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.057	0.966	0.879	0.799	0.727	0.663	0.606	0.557	0.513	0.476	0.443
10.5	$E^{DCS}_{expected}$	1.02E-01	9.34E-02	8.50E-02	7.73E-02	7.03E-02	6.41E-02	5.86E-02	5.39E-02	4.97E-02	4.60E-02	4.29E-02
	$E^{DCS}_{expected}$	9.69E-02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.055	0.964	0.878	0.798	0.726	0.662	0.605	0.556	0.513	0.475	0.442
12.0	$E^{DCS}_{expected}$	1.28E-01	1.17E-01	1.06E-01	9.67E-02	8.80E-02	8.02E-02	7.33E-02	6.73E-02	6.20E-02	5.75E-02	5.35E-02
	$E^{DCS}_{expected}$	1.22E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.048	0.957	0.871	0.791	0.720	0.656	0.600	0.551	0.508	0.470	0.437
13.5	$E^{DCS}_{expected}$	1.49E-01	1.36E-01	1.24E-01	1.12E-01	1.02E-01	9.31E-02	8.51E-02	7.81E-02	7.20E-02	6.67E-02	6.21E-02
	$E^{DCS}_{expected}$	1.42E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.047	0.956	0.870	0.790	0.719	0.655	0.599	0.550	0.507	0.470	0.437
15.0	$E^{DCS}_{expected}$	1.79E-01	1.64E-01	1.49E-01	1.35E-01	1.23E-01	1.12E-01	1.02E-01	9.40E-02	8.66E-02	8.02E-02	7.46E-02
	$E^{DCS}_{expected}$	1.72E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.042	0.952	0.866	0.786	0.715	0.651	0.595	0.546	0.504	0.466	0.434
16.5	$E^{DCS}_{expected}$	2.14E-01	1.96E-01	1.78E-01	1.62E-01	1.47E-01	1.34E-01	1.22E-01	1.12E-01	1.03E-01	9.57E-02	8.90E-02

Ribbed Slab-Marine eutrophication (kg N eq)												
Span (m)		*p*=0.0	*p*=0.1	*p*=0.2	*p*=0.3	*p*=0.4	*p*=0.5	*p*=0.6	*p*=0.7	*p*=0.8	*p*=0.9	*p*=1.0
	$E^{DCS}_{expected}$	2.07E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.038	0.948	0.862	0.783	0.711	0.648	0.592	0.543	0.501	0.463	0.431
18.0	$E^{DCS}_{expected}$	2.38E-01	2.17E-01	1.97E-01	1.79E-01	1.63E-01	1.48E-01	1.36E-01	1.24E-01	1.15E-01	1.06E-01	9.87E-02
	$E^{DCS}_{expected}$	2.29E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.038	0.948	0.862	0.783	0.711	0.648	0.592	0.543	0.501	0.463	0.431

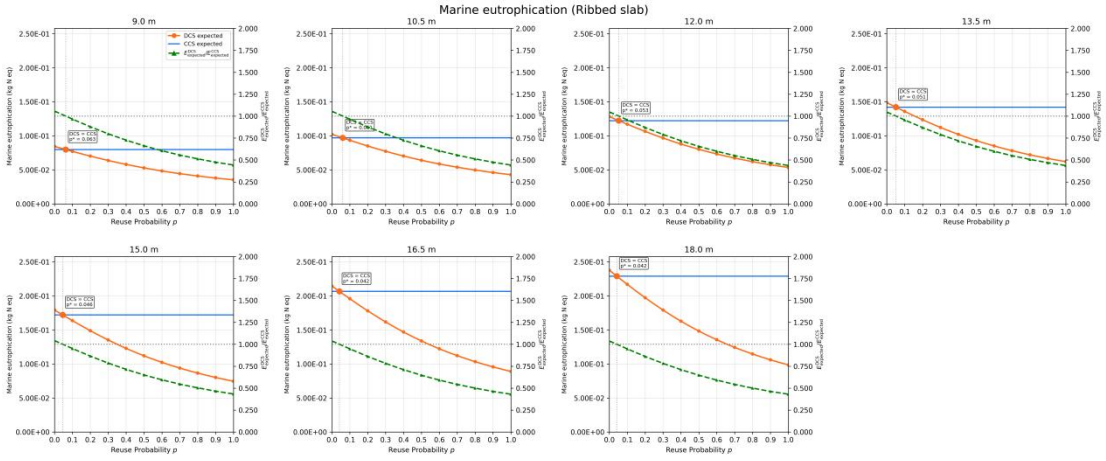


Figure G-9 *p** derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine eutrophication

Table G-10 *p** derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial ecotoxicity

Ribbed Slab-Terrestrial ecotoxicity (kg 1,4-DCB)												
Span (m)		*p*=0.0	*p*=0.1	*p*=0.2	*p*=0.3	*p*=0.4	*p*=0.5	*p*=0.6	*p*=0.7	*p*=0.8	*p*=0.9	*p*=1.0
9.0	$E^{DCS}_{expected}$	1.48E+04	1.35E+04	1.22E+04	1.10E+04	9.89E+03	8.92E+03	8.07E+03	7.32E+03	6.67E+03	6.09E+03	5.59E+03
	$E^{DCS}_{expected}$	1.36E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.091	0.991	0.896	0.807	0.728	0.656	0.594	0.539	0.490	0.448	0.411
10.5	$E^{DCS}_{expected}$	1.79E+04	1.63E+04	1.47E+04	1.33E+04	1.20E+04	1.08E+04	9.76E+03	8.86E+03	8.06E+03	7.37E+03	6.76E+03
	$E^{DCS}_{expected}$	1.65E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.089	0.989	0.894	0.806	0.727	0.656	0.593	0.538	0.490	0.448	0.411
12.0	$E^{DCS}_{expected}$	2.27E+04	2.06E+04	1.86E+04	1.68E+04	1.51E+04	1.36E+04	1.23E+04	1.12E+04	1.02E+04	9.29E+03	8.52E+03
	$E^{DCS}_{expected}$	2.10E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.080	0.981	0.886	0.799	0.720	0.649	0.587	0.532	0.485	0.443	0.406
13.5	$E^{DCS}_{expected}$	2.63E	2.39E	2.16E	1.94E	1.75E	1.58E	1.43E	1.30E	1.18E	1.08E	9.89E

Ribbed Slab-Terrestrial ecotoxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
		+04	+04	+04	+04	+04	+04	+04	+04	+04	+04	+03
	$E^{DCS}_{expected}$	2.44E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.078	0.980	0.885	0.798	0.719	0.648	0.586	0.532	0.484	0.442	0.406
15.0	$E^{DCS}_{expected}$	3.19E +04	2.90E +04	2.62E +04	2.36E +04	2.12E +04	1.91E +04	1.73E +04	1.57E +04	1.43E +04	1.30E +04	1.20E +04
	$E^{DCS}_{expected}$	2.97E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.072	0.974	0.880	0.793	0.714	0.644	0.582	0.528	0.481	0.439	0.403
16.5	$E^{DCS}_{expected}$	3.84E +04	3.49E +04	3.15E +04	2.84E +04	2.55E +04	2.30E +04	2.08E +04	1.89E +04	1.72E +04	1.57E +04	1.44E +04
	$E^{DCS}_{expected}$	3.60E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.068	0.970	0.876	0.789	0.710	0.641	0.579	0.525	0.477	0.436	0.400
18.0	$E^{DCS}_{expected}$	4.25E +04	3.86E +04	3.48E +04	3.14E +04	2.83E +04	2.55E +04	2.30E +04	2.09E +04	1.90E +04	1.74E +04	1.59E +04
	$E^{DCS}_{expected}$	3.98E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.067	0.969	0.876	0.789	0.711	0.641	0.579	0.525	0.478	0.436	0.400

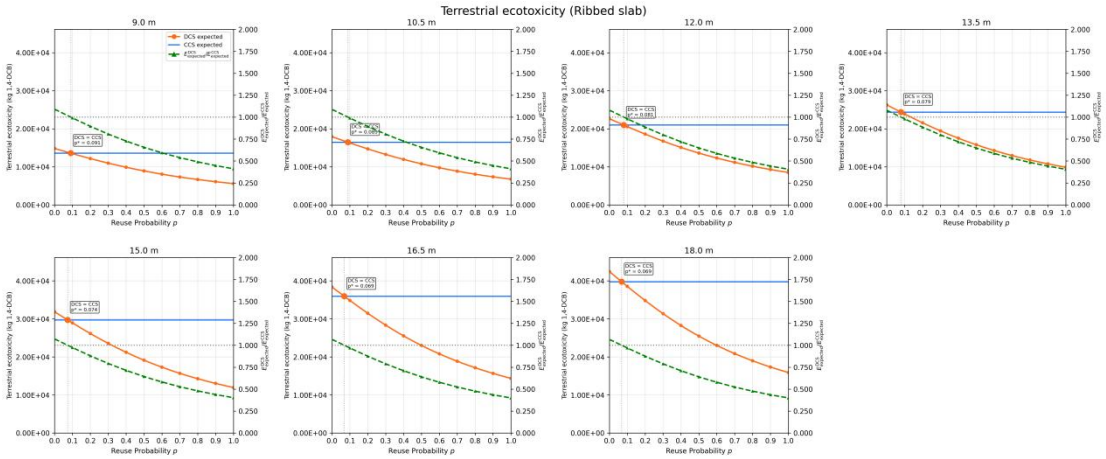


Figure G-10 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial ecotoxicity

Table G-11 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater ecotoxicity

Ribbed Slab-Freshwater ecotoxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	1.16E +02	1.06E +02	9.56E +01	8.64E +01	7.80E +01	7.06E +01	6.41E +01	5.84E +01	5.34E +01	4.90E +01	4.52E +01
	$E^{DCS}_{expected}$	1.07E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.084	0.986	0.892	0.806	0.728	0.659	0.598	0.545	0.498	0.457	0.422
10.5	$E^{DCS}_{expected}$	1.39E +02	1.26E +02	1.14E +02	1.03E +02	9.33E +01	8.44E +01	7.66E +01	6.98E +01	6.39E +01	5.87E +01	5.41E +01

Ribbed Slab-Freshwater ecotoxicity (kg 1,4-DCB)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E^{DCS}_{expected}$	1.28E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.083	0.985	0.892	0.806	0.728	0.659	0.598	0.545	0.499	0.458	0.423
12.0	$E^{DCS}_{expected}$	1.83E+02	1.66E+02	1.50E+02	1.36E+02	1.23E+02	1.11E+02	1.01E+02	9.15E+01	8.36E+01	7.67E+01	7.06E+01
	$E^{DCS}_{expected}$	1.71E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.071	0.974	0.881	0.795	0.718	0.649	0.589	0.536	0.489	0.449	0.414
13.5	$E^{DCS}_{expected}$	2.12E+02	1.92E+02	1.74E+02	1.57E+02	1.42E+02	1.28E+02	1.16E+02	1.06E+02	9.68E+01	8.88E+01	8.18E+01
	$E^{DCS}_{expected}$	1.98E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.069	0.972	0.879	0.794	0.717	0.648	0.588	0.535	0.489	0.448	0.413
15.0	$E^{DCS}_{expected}$	2.64E+02	2.40E+02	2.17E+02	1.96E+02	1.77E+02	1.60E+02	1.45E+02	1.32E+02	1.20E+02	1.10E+02	1.01E+02
	$E^{DCS}_{expected}$	2.49E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.062	0.965	0.872	0.787	0.710	0.642	0.582	0.529	0.483	0.443	0.408
16.5	$E^{DCS}_{expected}$	3.27E+02	2.97E+02	2.68E+02	2.42E+02	2.18E+02	1.97E+02	1.78E+02	1.62E+02	1.48E+02	1.36E+02	1.25E+02
	$E^{DCS}_{expected}$	3.10E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.055	0.958	0.866	0.781	0.704	0.636	0.576	0.523	0.478	0.438	0.402
18.0	$E^{DCS}_{expected}$	3.59E+02	3.26E+02	2.95E+02	2.66E+02	2.40E+02	2.17E+02	1.96E+02	1.78E+02	1.63E+02	1.49E+02	1.37E+02
	$E^{DCS}_{expected}$	3.40E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.055	0.958	0.866	0.781	0.704	0.636	0.576	0.524	0.478	0.438	0.403

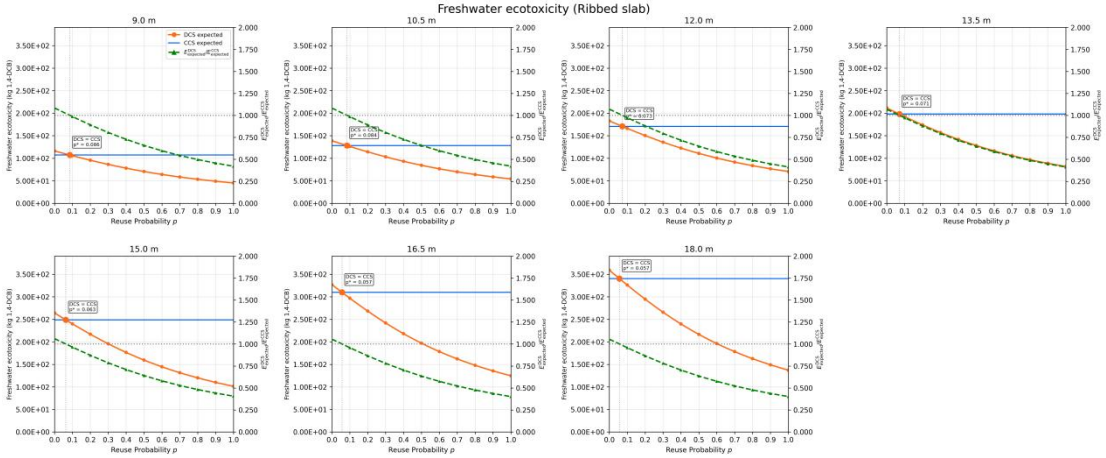


Figure G-11 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater ecotoxicity

Table G-12 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine ecotoxicity

Ribbed Slab-Marine ecotoxicity (kg 1,4-DCB)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	1.67E+02	1.52E+02	1.38E+02	1.24E+02	1.12E+02	1.02E+02	9.21E+01	8.39E+01	7.67E+01	7.04E+01	6.49E+01
	$E^{DCS}_{expected}$	1.54E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.084	0.986	0.892	0.805	0.727	0.658	0.597	0.544	0.497	0.456	0.421
10.5	$E^{DCS}_{expected}$	2.00E+02	1.82E+02	1.64E+02	1.49E+02	1.34E+02	1.21E+02	1.10E+02	1.00E+02	9.18E+01	8.43E+01	7.77E+01
	$E^{DCS}_{expected}$	1.85E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.083	0.985	0.891	0.805	0.727	0.658	0.597	0.544	0.497	0.457	0.421
12.0	$E^{DCS}_{expected}$	2.63E+02	2.39E+02	2.16E+02	1.95E+02	1.76E+02	1.59E+02	1.44E+02	1.31E+02	1.20E+02	1.10E+02	1.01E+02
	$E^{DCS}_{expected}$	2.46E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.071	0.973	0.880	0.794	0.717	0.648	0.588	0.535	0.488	0.448	0.413
13.5	$E^{DCS}_{expected}$	3.05E+02	2.77E+02	2.50E+02	2.26E+02	2.04E+02	1.84E+02	1.67E+02	1.52E+02	1.39E+02	1.27E+02	1.17E+02
	$E^{DCS}_{expected}$	2.85E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.069	0.972	0.879	0.793	0.716	0.647	0.587	0.534	0.488	0.447	0.412
15.0	$E^{DCS}_{expected}$	3.80E+02	3.45E+02	3.12E+02	2.81E+02	2.54E+02	2.29E+02	2.08E+02	1.89E+02	1.72E+02	1.58E+02	1.45E+02
	$E^{DCS}_{expected}$	3.58E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.062	0.964	0.872	0.786	0.709	0.641	0.581	0.528	0.482	0.442	0.407
16.5	$E^{DCS}_{expected}$	4.70E+02	4.26E+02	3.85E+02	3.47E+02	3.13E+02	2.83E+02	2.56E+02	2.33E+02	2.12E+02	1.94E+02	1.79E+02
	$E^{DCS}_{expected}$	4.45E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.055	0.958	0.866	0.781	0.704	0.635	0.575	0.523	0.477	0.437	0.402
18.0	$E^{DCS}_{expected}$	5.16E+02	4.69E+02	4.23E+02	3.82E+02	3.44E+02	3.11E+02	2.82E+02	2.56E+02	2.33E+02	2.14E+02	1.97E+02
	$E^{DCS}_{expected}$	4.89E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.055	0.958	0.866	0.781	0.704	0.636	0.576	0.523	0.477	0.437	0.402

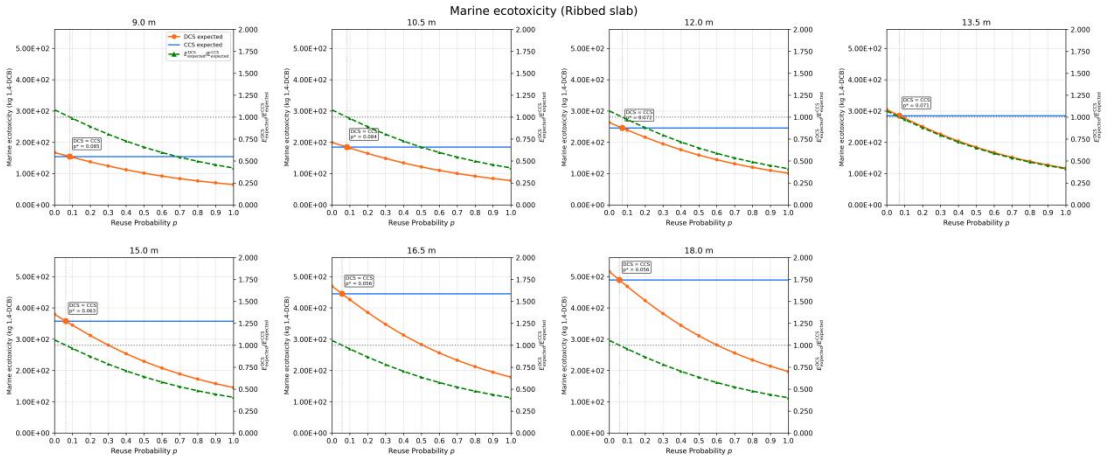


Figure G-12 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine ecotoxicity

Table G-13 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Human carcinogenic toxicity

Ribbed Slab-Human carcinogenic toxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	5.02E+02	4.53E+02	4.07E+02	3.65E+02	3.27E+02	2.93E+02	2.63E+02	2.37E+02	2.14E+02	1.94E+02	1.76E+02
	$E^{DCS}_{expected}$	4.63E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.083	0.979	0.879	0.787	0.705	0.632	0.567	0.511	0.461	0.418	0.381
10.5	$E^{DCS}_{expected}$	5.94E+02	5.37E+02	4.83E+02	4.32E+02	3.87E+02	3.47E+02	3.12E+02	2.81E+02	2.54E+02	2.30E+02	2.09E+02
	$E^{DCS}_{expected}$	5.50E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.082	0.978	0.878	0.787	0.704	0.631	0.567	0.511	0.461	0.418	0.381
12.0	$E^{DCS}_{expected}$	8.04E+02	7.27E+02	6.53E+02	5.85E+02	5.23E+02	4.69E+02	4.21E+02	3.79E+02	3.42E+02	3.10E+02	2.82E+02
	$E^{DCS}_{expected}$	7.53E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.068	0.965	0.867	0.776	0.695	0.622	0.558	0.503	0.454	0.411	0.374
13.5	$E^{DCS}_{expected}$	9.30E+02	8.40E+02	7.55E+02	6.76E+02	6.05E+02	5.42E+02	4.86E+02	4.38E+02	3.95E+02	3.58E+02	3.26E+02
	$E^{DCS}_{expected}$	8.72E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.066	0.963	0.865	0.775	0.694	0.621	0.558	0.502	0.453	0.411	0.374
15.0	$E^{DCS}_{expected}$	1.18E+03	1.07E+03	9.57E+02	8.57E+02	7.66E+02	6.86E+02	6.16E+02	5.54E+02	5.00E+02	4.53E+02	4.12E+02
	$E^{DCS}_{expected}$	1.11E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.058	0.956	0.858	0.768	0.687	0.615	0.552	0.497	0.449	0.406	0.370
16.5	$E^{DCS}_{expected}$	1.48E+03	1.34E+03	1.20E+03	1.08E+03	9.62E+02	8.61E+02	7.73E+02	6.95E+02	6.27E+02	5.68E+02	5.16E+02

Ribbed Slab-Human carcinogenic toxicity (kg 1,4-DCB)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E_{expected}^{DCS}$	1.41E+03										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.050	0.949	0.852	0.762	0.682	0.610	0.548	0.493	0.445	0.403	0.366
18.0	$E_{expected}^{DCS}$	1.62E +03	1.46E +03	1.32E +03	1.18E +03	1.05E +03	9.43E +02	8.46E +02	7.61E +02	6.87E +02	6.22E +02	5.65E +02
	$E_{expected}^{DCS}$	1.54E+03										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.050	0.949	0.852	0.763	0.682	0.611	0.548	0.493	0.445	0.403	0.366

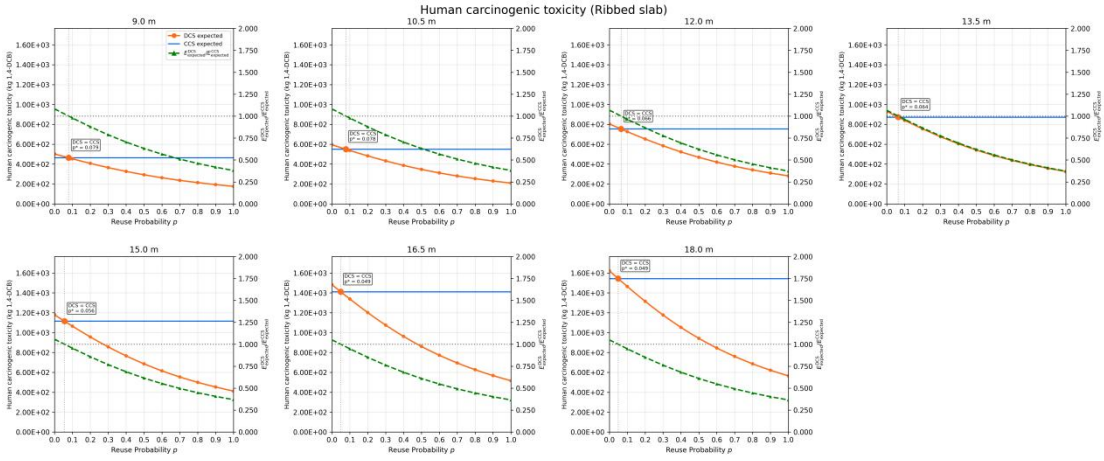


Figure G-13 p* derived from calculations of CCS ($E_{expected}^{CCS}$) and DCS ($E_{expected}^{DCS}$) of Human carcinogenic toxicity

Table G-14 p* derived from calculations of CCS ($E_{expected}^{CCS}$) and DCS ($E_{expected}^{DCS}$) of Human non-carcinogenic toxicity

Ribbed Slab-Human non-carcinogenic toxicity (kg 1,4-DCB)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E_{expected}^{DCS}$	3.54E +03	3.21E +03	2.91E +03	2.62E +03	2.37E +03	2.14E +03	1.94E +03	1.76E +03	1.61E +03	1.48E +03	1.36E +03
	$E_{expected}^{DCS}$	3.28E+03										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.078	0.980	0.886	0.800	0.722	0.653	0.592	0.538	0.491	0.451	0.415
10.5	$E_{expected}^{DCS}$	4.22E +03	3.84E +03	3.47E +03	3.13E +03	2.83E +03	2.56E +03	2.32E +03	2.11E +03	1.93E +03	1.77E +03	1.63E +03
	$E_{expected}^{DCS}$	3.92E+03										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.077	0.979	0.886	0.799	0.722	0.652	0.592	0.538	0.492	0.451	0.416
12.0	$E_{expected}^{DCS}$	5.56E +03	5.06E +03	4.57E +03	4.12E +03	3.72E +03	3.36E +03	3.04E +03	2.76E +03	2.52E +03	2.31E +03	2.13E +03
	$E_{expected}^{DCS}$	5.22E+03										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.066	0.968	0.875	0.789	0.712	0.643	0.583	0.529	0.483	0.443	0.407
13.5	$E_{expected}^{DCS}$	6.44E	5.85E	5.29E	4.77E	4.30E	3.89E	3.52E	3.20E	2.92E	2.68E	2.46E

Ribbed Slab-Human non-carcinogenic toxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
		+03	+03	+03	+03	+03	+03	+03	+03	+03	+03	+03
	$E^{DCS}_{expected}$	6.05E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.064	0.967	0.874	0.788	0.711	0.642	0.582	0.529	0.483	0.442	0.407
15.0	$E^{DCS}_{expected}$	8.03E+03	7.30E+03	6.59E+03	5.94E+03	5.36E+03	4.84E+03	4.38E+03	3.98E+03	3.63E+03	3.32E+03	3.05E+03
	$E^{DCS}_{expected}$	7.60E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.057	0.960	0.867	0.782	0.705	0.636	0.576	0.523	0.477	0.437	0.402
16.5	$E^{DCS}_{expected}$	9.94E+03	9.02E+03	8.15E+03	7.34E+03	6.61E+03	5.97E+03	5.40E+03	4.90E+03	4.47E+03	4.09E+03	3.76E+03
	$E^{DCS}_{expected}$	9.46E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.050	0.954	0.861	0.776	0.699	0.631	0.571	0.518	0.472	0.432	0.397
18.0	$E^{DCS}_{expected}$	1.09E+04	9.91E+03	8.95E+03	8.07E+03	7.27E+03	6.56E+03	5.94E+03	5.39E+03	4.91E+03	4.50E+03	4.13E+03
	$E^{DCS}_{expected}$	1.04E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.050	0.954	0.861	0.776	0.699	0.631	0.571	0.519	0.473	0.432	0.397

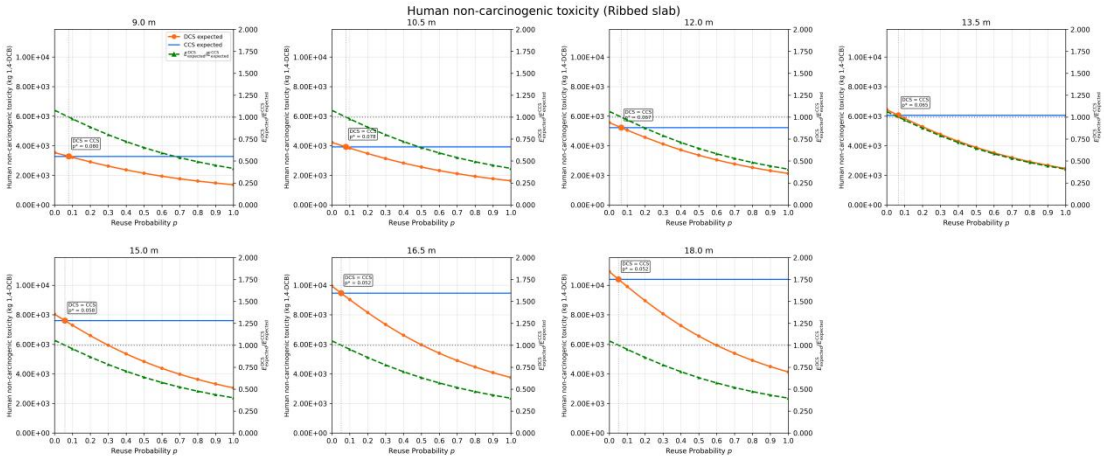


Figure G-14 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Human non-carcinogenic toxicity

Table G-15 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Land use

Ribbed Slab-Land use (m^2 a crop eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	7.94E+01	7.21E+01	6.51E+01	5.87E+01	5.28E+01	4.77E+01	4.31E+01	3.91E+01	3.56E+01	3.25E+01	2.98E+01
	$E^{DCS}_{expected}$	7.25E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.095	0.994	0.898	0.809	0.728	0.657	0.594	0.539	0.490	0.448	0.411
10.5	$E^{DCS}_{expected}$	9.64E	8.76E	7.91E	7.12E	6.42E	5.79E	5.23E	4.74E	4.32E	3.95E	3.62E

Ribbed Slab-Land use (m^2 a crop eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
		+01	+01	+01	+01	+01	+01	+01	+01	+01	+01	+01
	$E^{DCS}_{expected}$	8.82E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.093	0.992	0.896	0.807	0.727	0.656	0.593	0.538	0.490	0.447	0.411
12.0	$E^{DCS}_{expected}$	1.20E +02	1.09E +02	9.84E +01	8.86E +01	7.98E +01	7.20E +01	6.51E +01	5.90E +01	5.37E +01	4.91E +01	4.50E +01
	$E^{DCS}_{expected}$	1.11E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.085	0.985	0.890	0.801	0.722	0.651	0.588	0.533	0.485	0.444	0.407
13.5	$E^{DCS}_{expected}$	1.39E +02	1.27E +02	1.14E +02	1.03E +02	9.27E +01	8.36E +01	7.55E +01	6.85E +01	6.24E +01	5.70E +01	5.23E +01
	$E^{DCS}_{expected}$	1.29E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.084	0.984	0.889	0.800	0.721	0.650	0.587	0.533	0.485	0.443	0.406
15.0	$E^{DCS}_{expected}$	1.67E +02	1.52E +02	1.37E +02	1.23E +02	1.11E +02	1.00E +02	9.06E +01	8.21E +01	7.47E +01	6.83E +01	6.26E +01
	$E^{DCS}_{expected}$	1.55E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.079	0.979	0.884	0.796	0.717	0.647	0.584	0.530	0.482	0.441	0.404
16.5	$E^{DCS}_{expected}$	1.99E +02	1.81E +02	1.63E +02	1.47E +02	1.32E +02	1.19E +02	1.08E +02	9.77E +01	8.89E +01	8.12E +01	7.45E +01
	$E^{DCS}_{expected}$	1.85E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.075	0.976	0.881	0.793	0.714	0.644	0.582	0.528	0.480	0.438	0.402
18.0	$E^{DCS}_{expected}$	2.21E +02	2.01E +02	1.81E +02	1.63E +02	1.47E +02	1.32E +02	1.20E +02	1.08E +02	9.87E +01	9.01E +01	8.27E +01
	$E^{DCS}_{expected}$	2.06E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.074	0.975	0.881	0.793	0.714	0.644	0.582	0.527	0.480	0.438	0.402

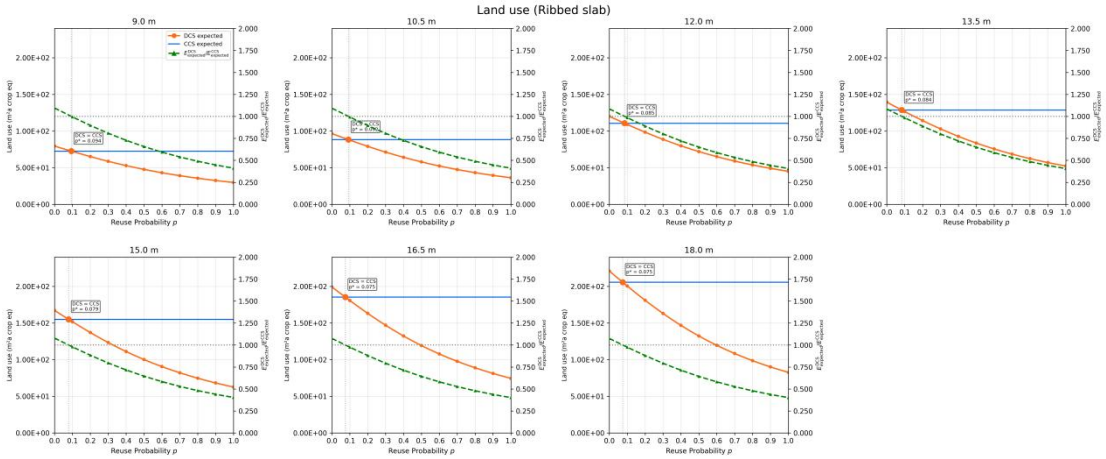


Figure G-15 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Land use

Table G-16 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Mineral resource scarcity

Ribbed Slab-Mineral resource scarcity (kg Cu eq)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	4.00E+01	3.61E+01	3.24E+01	2.89E+01	2.58E+01	2.30E+01	2.06E+01	1.85E+01	1.66E+01	1.50E+01	1.36E+01
	$E^{DCS}_{expected}$	3.72E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.076	0.970	0.869	0.777	0.693	0.619	0.554	0.497	0.447	0.403	0.365
10.5	$E^{DCS}_{expected}$	4.76E+01	4.29E+01	3.85E+01	3.44E+01	3.07E+01	2.74E+01	2.45E+01	2.20E+01	1.98E+01	1.78E+01	1.62E+01
	$E^{DCS}_{expected}$	4.43E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.074	0.969	0.868	0.776	0.692	0.618	0.553	0.496	0.446	0.403	0.365
12.0	$E^{DCS}_{expected}$	6.36E+01	5.74E+01	5.14E+01	4.59E+01	4.10E+01	3.66E+01	3.27E+01	2.93E+01	2.64E+01	2.38E+01	2.16E+01
	$E^{DCS}_{expected}$	5.99E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.062	0.958	0.859	0.767	0.684	0.611	0.547	0.490	0.441	0.398	0.360
13.5	$E^{DCS}_{expected}$	7.36E+01	6.64E+01	5.95E+01	5.31E+01	4.74E+01	4.23E+01	3.79E+01	3.39E+01	3.05E+01	2.75E+01	2.49E+01
	$E^{DCS}_{expected}$	6.94E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.060	0.956	0.857	0.766	0.683	0.610	0.546	0.489	0.440	0.397	0.359
15.0	$E^{DCS}_{expected}$	9.26E+01	8.35E+01	7.48E+01	6.68E+01	5.96E+01	5.32E+01	4.76E+01	4.27E+01	3.84E+01	3.46E+01	3.13E+01
	$E^{DCS}_{expected}$	8.79E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.053	0.949	0.851	0.760	0.678	0.605	0.541	0.485	0.437	0.394	0.357
16.5	$E^{DCS}_{expected}$	1.15E+02	1.04E+02	9.33E+01	8.33E+01	7.43E+01	6.64E+01	5.93E+01	5.32E+01	4.78E+01	4.32E+01	3.91E+01
	$E^{DCS}_{expected}$	1.10E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.046	0.943	0.845	0.755	0.674	0.601	0.538	0.482	0.434	0.391	0.354
18.0	$E^{DCS}_{expected}$	1.27E+02	1.14E+02	1.02E+02	9.13E+01	8.15E+01	7.28E+01	6.51E+01	5.83E+01	5.25E+01	4.73E+01	4.28E+01
	$E^{DCS}_{expected}$	1.21E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.046	0.943	0.845	0.755	0.674	0.601	0.538	0.482	0.434	0.391	0.354

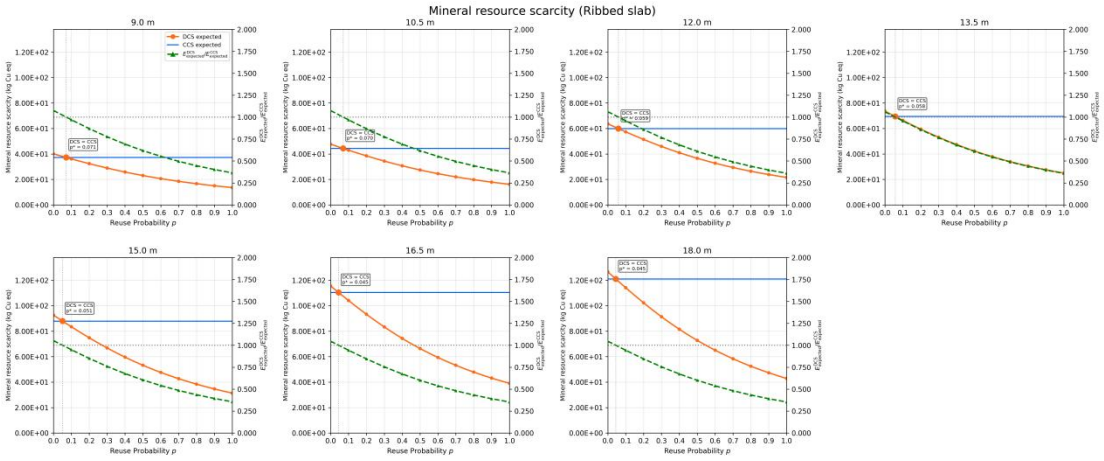


Figure G-16 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Mineral resource scarcity

Table G-17 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fossil resource scarcity

Ribbed Slab-Fossil resource scarcity (kg oil eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	7.32E+02	6.72E+02	6.15E+02	5.62E+02	5.14E+02	4.71E+02	4.34E+02	4.01E+02	3.72E+02	3.47E+02	3.25E+02
	$E^{DCS}_{expected}$	6.26E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.170	1.074	0.983	0.898	0.821	0.753	0.693	0.641	0.594	0.554	0.519
10.5	$E^{DCS}_{expected}$	8.88E+02	8.15E+02	7.46E+02	6.81E+02	6.23E+02	5.72E+02	5.26E+02	4.86E+02	4.51E+02	4.21E+02	3.94E+02
	$E^{DCS}_{expected}$	7.60E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.168	1.073	0.981	0.897	0.821	0.753	0.693	0.640	0.594	0.554	0.519
12.0	$E^{DCS}_{expected}$	1.11E+03	1.02E+03	9.32E+02	8.51E+02	7.78E+02	7.14E+02	6.57E+02	6.06E+02	5.63E+02	5.24E+02	4.91E+02
	$E^{DCS}_{expected}$	9.59E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.158	1.063	0.972	0.888	0.812	0.744	0.685	0.632	0.587	0.547	0.512
13.5	$E^{DCS}_{expected}$	1.29E+03	1.18E+03	1.08E+03	9.88E+02	9.04E+02	8.29E+02	7.62E+02	7.04E+02	6.53E+02	6.09E+02	5.70E+02
	$E^{DCS}_{expected}$	1.11E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.156	1.061	0.971	0.887	0.811	0.743	0.684	0.632	0.586	0.546	0.511
15.0	$E^{DCS}_{expected}$	1.55E+03	1.42E+03	1.30E+03	1.19E+03	1.09E+03	9.96E+02	9.16E+02	8.46E+02	7.84E+02	7.31E+02	6.84E+02
	$E^{DCS}_{expected}$	1.35E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.149	1.055	0.964	0.881	0.805	0.738	0.678	0.627	0.581	0.541	0.506
16.5	$E^{DCS}_{expected}$	1.85E+03	1.70E+03	1.55E+03	1.42E+03	1.30E+03	1.19E+03	1.09E+03	1.01E+03	9.35E+02	8.71E+02	8.14E+02

Ribbed Slab-Fossil resource scarcity (kg oil eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E_{expected}^{DCS}$	1.62E+03										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.143	1.049	0.959	0.875	0.800	0.733	0.674	0.622	0.576	0.537	0.502
18.0	$E_{expected}^{DCS}$	2.06E+03	1.89E+03	1.72E+03	1.57E+03	1.44E+03	1.32E+03	1.21E+03	1.12E+03	1.04E+03	9.66E+02	9.03E+02
	$E_{expected}^{DCS}$	1.80E+03										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.143	1.049	0.959	0.875	0.800	0.733	0.674	0.622	0.577	0.537	0.503

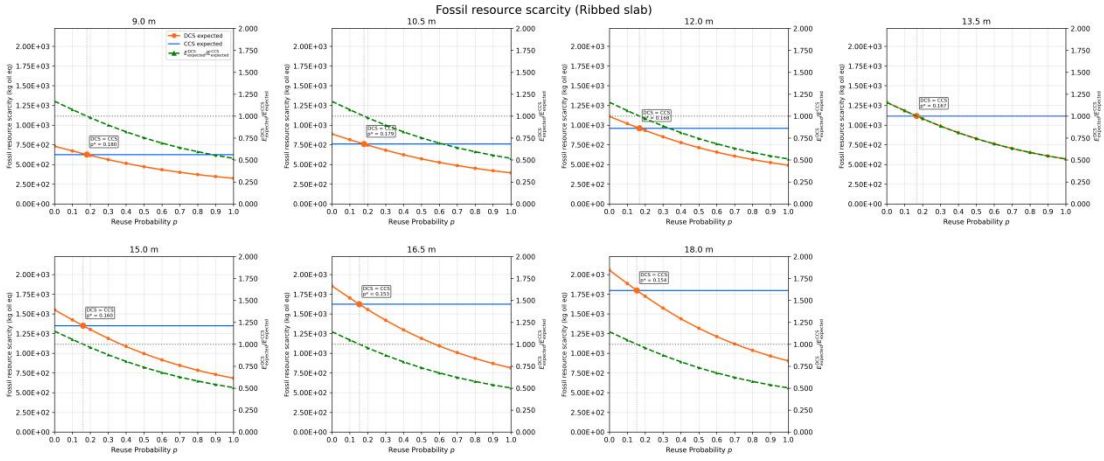


Figure G-17 p* derived from calculations of CCS ($E_{expected}^{CCS}$) and DCS ($E_{expected}^{DCS}$) of Fossil resource scarcity

Table G-18 p* derived from calculations of CCS ($E_{expected}^{CCS}$) and DCS ($E_{expected}^{DCS}$) of Water consumption

Ribbed Slab-Water consumption (m^3)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E_{expected}^{DCS}$	2.63E+01	2.38E+01	2.14E+01	1.92E+01	1.72E+01	1.54E+01	1.39E+01	1.25E+01	1.13E+01	1.03E+01	9.37E+00
	$E_{expected}^{DCS}$	2.31E+01										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.140	1.031	0.927	0.831	0.745	0.668	0.601	0.542	0.490	0.445	0.406
10.5	$E_{expected}^{DCS}$	3.19E+01	2.89E+01	2.60E+01	2.33E+01	2.09E+01	1.87E+01	1.68E+01	1.52E+01	1.37E+01	1.25E+01	1.14E+01
	$E_{expected}^{DCS}$	2.81E+01										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.138	1.029	0.925	0.830	0.744	0.667	0.600	0.541	0.489	0.444	0.405
12.0	$E_{expected}^{DCS}$	3.98E+01	3.60E+01	3.23E+01	2.90E+01	2.60E+01	2.33E+01	2.09E+01	1.89E+01	1.71E+01	1.55E+01	1.41E+01
	$E_{expected}^{DCS}$	3.52E+01										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.129	1.021	0.918	0.823	0.738	0.662	0.595	0.536	0.485	0.441	0.402
13.5	$E_{expected}^{DCS}$	4.62E	4.17E	3.75E	3.37E	3.02E	2.71E	2.43E	2.19E	1.98E	1.80E	1.64E

Ribbed Slab-Water consumption (m^3)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
		+01	+01	+01	+01	+01	+01	+01	+01	+01	+01	+01
	$E^{DCS}_{expected}$	4.09E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.127	1.020	0.917	0.822	0.737	0.661	0.594	0.536	0.485	0.440	0.401
15.0	$E^{DCS}_{expected}$	5.54E +01	5.01E +01	4.50E +01	4.04E +01	3.62E +01	3.25E +01	2.92E +01	2.63E +01	2.38E +01	2.16E +01	1.97E +01
	$E^{DCS}_{expected}$	4.94E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.121	1.014	0.912	0.818	0.733	0.657	0.591	0.533	0.482	0.437	0.399
16.5	$E^{DCS}_{expected}$	6.60E +01	5.96E +01	5.36E +01	4.81E +01	4.31E +01	3.86E +01	3.47E +01	3.13E +01	2.83E +01	2.57E +01	2.34E +01
	$E^{DCS}_{expected}$	5.91E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.117	1.010	0.908	0.814	0.729	0.654	0.588	0.530	0.479	0.435	0.397
18.0	$E^{DCS}_{expected}$	7.32E +01	6.62E +01	5.95E +01	5.33E +01	4.78E +01	4.29E +01	3.85E +01	3.47E +01	3.14E +01	2.85E +01	2.60E +01
	$E^{DCS}_{expected}$	6.55E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.116	1.010	0.908	0.814	0.729	0.654	0.588	0.530	0.479	0.435	0.397

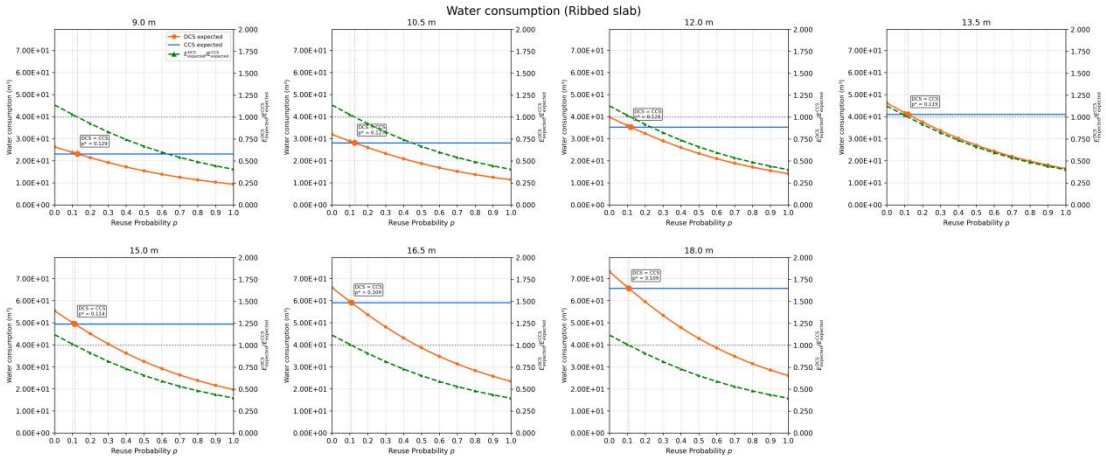


Figure G-18 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Water consumption

Appendix H: p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) for the double T slab of different model lengths

Table H-1 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Global warming

Double T Slab-Global warming (kg CO_2 eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	1.65E+03	1.51E+03	1.37E+03	1.25E+03	1.13E+03	1.03E+03	9.41E+02	8.63E+02	7.95E+02	7.35E+02	6.83E+02
	$E^{DCS}_{expected}$	1.50E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.101	1.006	0.915	0.831	0.755	0.687	0.628	0.576	0.530	0.490	0.456
10.5	$E^{DCS}_{expected}$	2.05E+03	1.87E+03	1.70E+03	1.55E+03	1.40E+03	1.28E+03	1.17E+03	1.07E+03	9.87E+02	9.13E+02	8.49E+02
	$E^{DCS}_{expected}$	1.87E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.095	1.000	0.909	0.826	0.750	0.683	0.624	0.573	0.527	0.488	0.453
12.0	$E^{DCS}_{expected}$	2.45E+03	2.24E+03	2.04E+03	1.85E+03	1.68E+03	1.53E+03	1.40E+03	1.28E+03	1.18E+03	1.09E+03	1.02E+03
	$E^{DCS}_{expected}$	2.25E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.091	0.996	0.906	0.823	0.748	0.681	0.622	0.571	0.526	0.486	0.452
13.5	$E^{DCS}_{expected}$	3.01E+03	2.75E+03	2.50E+03	2.27E+03	2.06E+03	1.88E+03	1.72E+03	1.58E+03	1.45E+03	1.34E+03	1.25E+03
	$E^{DCS}_{expected}$	2.77E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.085	0.991	0.902	0.819	0.744	0.678	0.619	0.568	0.523	0.484	0.450
15.0	$E^{DCS}_{expected}$	3.57E+03	3.27E+03	2.97E+03	2.70E+03	2.45E+03	2.23E+03	2.04E+03	1.87E+03	1.72E+03	1.60E+03	1.48E+03
	$E^{DCS}_{expected}$	3.30E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.082	0.988	0.899	0.816	0.742	0.676	0.617	0.566	0.522	0.483	0.449
16.5	$E^{DCS}_{expected}$	4.16E+03	3.80E+03	3.46E+03	3.14E+03	2.86E+03	2.60E+03	2.38E+03	2.18E+03	2.01E+03	1.86E+03	1.73E+03
	$E^{DCS}_{expected}$	3.87E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.077	0.984	0.895	0.813	0.739	0.673	0.615	0.564	0.520	0.481	0.447
18.0	$E^{DCS}_{expected}$	4.76E+03	4.35E+03	3.96E+03	3.59E+03	3.27E+03	2.97E+03	2.72E+03	2.49E+03	2.30E+03	2.12E+03	1.97E+03
	$E^{DCS}_{expected}$	4.43E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.074	0.981	0.892	0.810	0.737	0.671	0.613	0.562	0.518	0.479	0.445

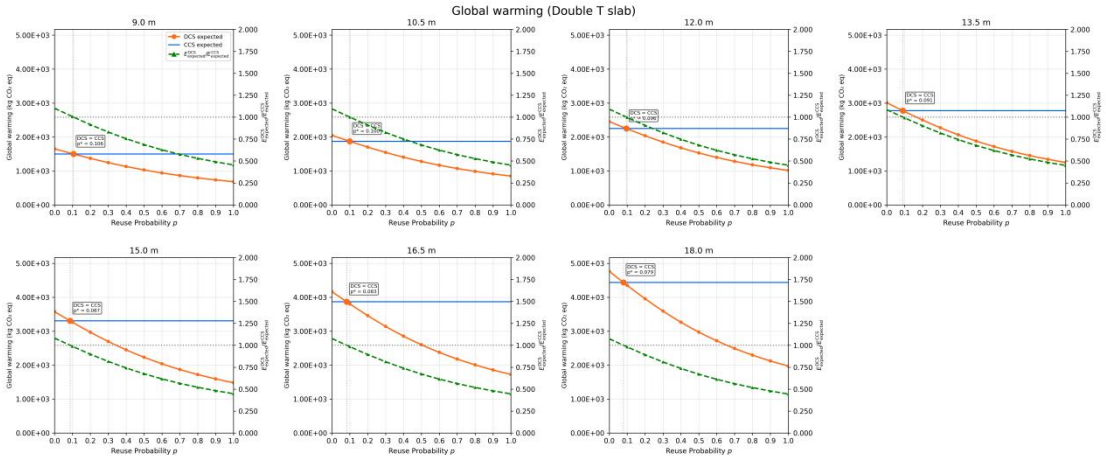


Figure H-1 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Global warming

Table H-2 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Stratospheric ozone depletion

Double T Slab-Stratospheric ozone depletion (kg CFC11 eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	5.51E-04	5.05E-04	4.61E-04	4.20E-04	3.83E-04	3.50E-04	3.21E-04	2.95E-04	2.73E-04	2.53E-04	2.36E-04
	$E^{DCS}_{expected}$	4.95E-04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.113	1.020	0.930	0.848	0.773	0.706	0.647	0.596	0.551	0.511	0.477
10.5	$E^{DCS}_{expected}$	6.85E-04	6.27E-04	5.72E-04	5.21E-04	4.75E-04	4.34E-04	3.98E-04	3.67E-04	3.39E-04	3.15E-04	2.93E-04
	$E^{DCS}_{expected}$	6.19E-04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.107	1.014	0.925	0.843	0.769	0.702	0.644	0.593	0.548	0.509	0.474
12.0	$E^{DCS}_{expected}$	8.19E-04	7.51E-04	6.85E-04	6.24E-04	5.69E-04	5.20E-04	4.77E-04	4.39E-04	4.06E-04	3.77E-04	3.51E-04
	$E^{DCS}_{expected}$	7.43E-04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.103	1.010	0.922	0.840	0.766	0.700	0.642	0.591	0.546	0.507	0.473
13.5	$E^{DCS}_{expected}$	1.01E-03	9.22E-04	8.41E-04	7.67E-04	6.99E-04	6.39E-04	5.86E-04	5.39E-04	4.99E-04	4.63E-04	4.32E-04
	$E^{DCS}_{expected}$	9.17E-04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.098	1.006	0.918	0.836	0.762	0.697	0.639	0.588	0.544	0.505	0.471
15.0	$E^{DCS}_{expected}$	1.20E-03	1.10E-03	9.99E-04	9.11E-04	8.30E-04	7.59E-04	6.96E-04	6.41E-04	5.92E-04	5.50E-04	5.13E-04
	$E^{DCS}_{expected}$	1.09E-03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.094	1.003	0.915	0.834	0.760	0.695	0.637	0.587	0.542	0.504	0.470
16.5	$E^{DCS}_{expected}$	1.39E-03	1.28E-03	1.16E-03	1.06E-03	9.67E-04	8.84E-04	8.11E-04	7.46E-04	6.90E-04	6.41E-04	5.98E-04
	$E^{DCS}_{expected}$	1.28E-03										

Double T Slab-Stratospheric ozone depletion (kg CFC11 eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.090	0.999	0.911	0.830	0.757	0.692	0.635	0.584	0.540	0.502	0.468
18.0	$E^{DCS}_{expected}$	1.59E-03	1.46E-03	1.33E-03	1.21E-03	1.11E-03	1.01E-03	9.27E-04	8.53E-04	7.89E-04	7.32E-04	6.83E-04
	$E^{DCS}_{expected}$	1.47E-03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.087	0.996	0.908	0.828	0.755	0.690	0.633	0.582	0.538	0.500	0.466

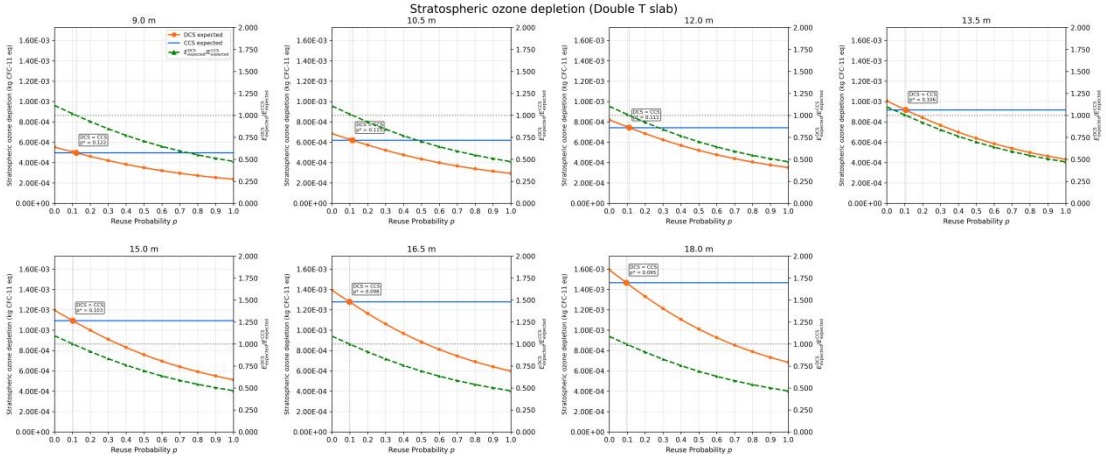


Figure H-2 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Stratospheric ozone depletion

Table H-3 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ionizing radiation

Double T Slab-Ionizing radiation (kBq Co-60 eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	4.22E+01	3.83E+01	3.46E+01	3.11E+01	2.80E+01	2.53E+01	2.29E+01	2.07E+01	1.89E+01	1.73E+01	1.59E+01
	$E^{DCS}_{expected}$	3.79E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.112	1.009	0.911	0.821	0.739	0.666	0.603	0.547	0.498	0.455	0.418
10.5	$E^{DCS}_{expected}$	5.23E+01	4.75E+01	4.29E+01	3.86E+01	3.48E+01	3.13E+01	2.83E+01	2.57E+01	2.34E+01	2.14E+01	1.97E+01
	$E^{DCS}_{expected}$	4.73E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.105	1.003	0.905	0.815	0.734	0.662	0.599	0.543	0.495	0.452	0.415
12.0	$E^{DCS}_{expected}$	6.25E+01	5.68E+01	5.12E+01	4.62E+01	4.16E+01	3.75E+01	3.39E+01	3.08E+01	2.80E+01	2.56E+01	2.35E+01
	$E^{DCS}_{expected}$	5.69E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.100	0.998	0.901	0.812	0.731	0.659	0.596	0.541	0.493	0.450	0.414
13.5	$E^{DCS}_{expected}$	7.66E+01	6.95E+01	6.28E+01	5.65E+01	5.09E+01	4.59E+01	4.15E+01	3.77E+01	3.43E+01	3.14E+01	2.88E+01

Double T Slab-Ionizing radiation (kBq Co-60 eq)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E^{DCS}_{expected}$	7.01E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.093	0.992	0.896	0.807	0.727	0.656	0.593	0.538	0.490	0.448	0.411
15.0	$E^{DCS}_{expected}$	9.08E+01	8.25E+01	7.44E+01	6.71E+01	6.04E+01	5.45E+01	4.93E+01	4.47E+01	4.07E+01	3.72E+01	3.42E+01
	$E^{DCS}_{expected}$	8.35E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.088	0.988	0.892	0.804	0.724	0.653	0.590	0.536	0.488	0.446	0.410
16.5	$E^{DCS}_{expected}$	1.06E+02	9.61E+01	8.68E+01	7.82E+01	7.04E+01	6.35E+01	5.74E+01	5.21E+01	4.75E+01	4.34E+01	3.99E+01
	$E^{DCS}_{expected}$	9.78E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.083	0.983	0.888	0.800	0.720	0.650	0.588	0.533	0.486	0.444	0.408
18.0	$E^{DCS}_{expected}$	1.21E+02	1.10E+02	9.94E+01	8.95E+01	8.06E+01	7.27E+01	6.58E+01	5.97E+01	5.43E+01	4.97E+01	4.56E+01
	$E^{DCS}_{expected}$	1.12E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.079	0.980	0.884	0.797	0.718	0.647	0.585	0.531	0.484	0.442	0.406

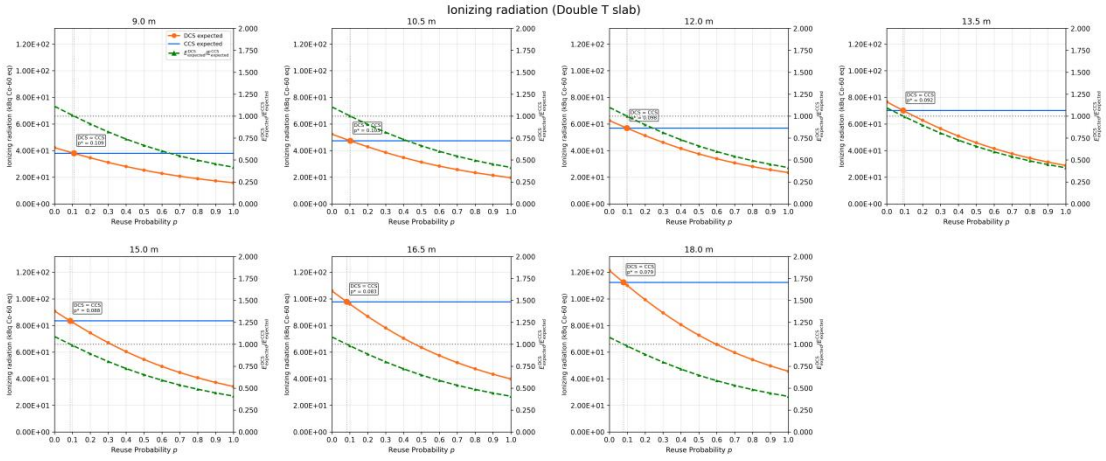


Figure H-3 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ionizing radiation

Table H-4 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation, Human health

Double T Slab-Ozone formation, Human health (kg NOx eq)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	4.02E+00	3.71E+00	3.41E+00	3.13E+00	2.88E+00	2.65E+00	2.46E+00	2.28E+00	2.13E+00	2.00E+00	1.89E+00
	$E^{DCS}_{expected}$	3.64E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.106	1.019	0.936	0.859	0.790	0.729	0.675	0.628	0.586	0.550	0.519
10.5	$E^{DCS}_{expected}$	4.99E	4.60E	4.23E	3.88E	3.57E	3.29E	3.05E	2.84E	2.65E	2.49E	2.34E

Double T Slab-Ozone formation, Human health (kg NOx eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
		+00	+00	+00	+00	+00	+00	+00	+00	+00	+00	+00
	$E^{DCS}_{expected}$	4.54E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.099	1.013	0.930	0.854	0.786	0.725	0.671	0.624	0.583	0.547	0.516
12.0	$E^{DCS}_{expected}$	5.97E +00	5.50E +00	5.06E +00	4.64E +00	4.27E +00	3.94E +00	3.65E +00	3.39E +00	3.17E +00	2.97E +00	2.80E +00
	$E^{DCS}_{expected}$	5.46E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.095	1.009	0.927	0.851	0.783	0.722	0.669	0.622	0.581	0.545	0.514
13.5	$E^{DCS}_{expected}$	7.32E +00	6.75E +00	6.20E +00	5.70E +00	5.24E +00	4.83E +00	4.48E +00	4.16E +00	3.89E +00	3.65E +00	3.45E +00
	$E^{DCS}_{expected}$	6.73E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.089	1.003	0.922	0.847	0.779	0.719	0.666	0.619	0.579	0.543	0.512
15.0	$E^{DCS}_{expected}$	8.69E +00	8.01E +00	7.36E +00	6.76E +00	6.22E +00	5.74E +00	5.32E +00	4.95E +00	4.62E +00	4.34E +00	4.09E +00
	$E^{DCS}_{expected}$	8.01E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.085	1.000	0.919	0.844	0.776	0.716	0.663	0.617	0.577	0.542	0.511
16.5	$E^{DCS}_{expected}$	1.01E +01	9.33E +00	8.58E +00	7.88E +00	7.25E +00	6.69E +00	6.19E +00	5.76E +00	5.38E +00	5.05E +00	4.77E +00
	$E^{DCS}_{expected}$	9.38E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.080	0.995	0.914	0.840	0.773	0.713	0.660	0.614	0.574	0.539	0.508
18.0	$E^{DCS}_{expected}$	1.16E +01	1.07E +01	9.81E +00	9.01E +00	8.29E +00	7.65E +00	7.08E +00	6.59E +00	6.16E +00	5.78E +00	5.45E +00
	$E^{DCS}_{expected}$	1.08E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.076	0.992	0.911	0.837	0.770	0.710	0.658	0.612	0.572	0.537	0.506

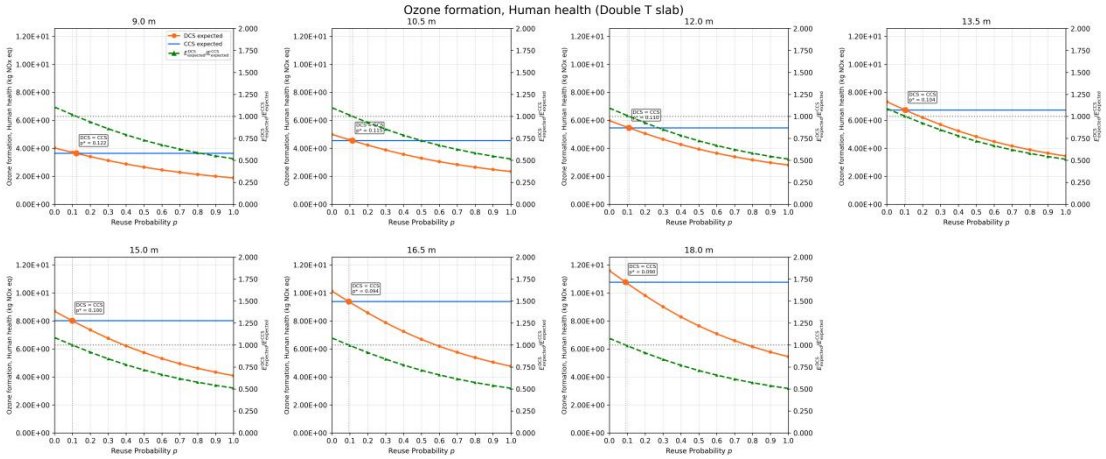


Figure H-4 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation, Human health

Table H-5 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fine particulate matter formation

Double T Slab-Fine particulate matter formation (kg PM2.5eq)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	2.36E+00	2.17E+00	1.98E+00	1.81E+00	1.66E+00	1.52E+00	1.40E+00	1.29E+00	1.20E+00	1.12E+00	1.05E+00
	$E^{DCS}_{expected}$	2.08E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.133	1.040	0.951	0.869	0.795	0.730	0.672	0.621	0.577	0.538	0.504
10.5	$E^{DCS}_{expected}$	2.92E+00	2.68E+00	2.45E+00	2.24E+00	2.05E+00	1.88E+00	1.73E+00	1.60E+00	1.49E+00	1.39E+00	1.30E+00
	$E^{DCS}_{expected}$	2.59E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.125	1.033	0.945	0.863	0.790	0.725	0.668	0.617	0.573	0.535	0.502
12.0	$E^{DCS}_{expected}$	3.49E+00	3.20E+00	2.93E+00	2.68E+00	2.45E+00	2.25E+00	2.07E+00	1.92E+00	1.78E+00	1.66E+00	1.56E+00
	$E^{DCS}_{expected}$	3.12E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.119	1.028	0.940	0.859	0.786	0.721	0.664	0.614	0.571	0.533	0.499
13.5	$E^{DCS}_{expected}$	4.26E+00	3.91E+00	3.58E+00	3.27E+00	3.00E+00	2.75E+00	2.53E+00	2.34E+00	2.18E+00	2.03E+00	1.91E+00
	$E^{DCS}_{expected}$	3.84E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.111	1.020	0.934	0.854	0.781	0.717	0.661	0.611	0.568	0.530	0.497
15.0	$E^{DCS}_{expected}$	5.05E+00	4.64E+00	4.24E+00	3.88E+00	3.55E+00	3.26E+00	3.00E+00	2.78E+00	2.58E+00	2.41E+00	2.26E+00
	$E^{DCS}_{expected}$	4.57E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.106	1.015	0.929	0.850	0.778	0.714	0.658	0.609	0.566	0.528	0.495
16.5	$E^{DCS}_{expected}$	5.89E+00	5.41E+00	4.95E+00	4.53E+00	4.14E+00	3.80E+00	3.50E+00	3.24E+00	3.01E+00	2.81E+00	2.64E+00
	$E^{DCS}_{expected}$	5.36E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.100	1.010	0.924	0.845	0.773	0.710	0.654	0.605	0.562	0.525	0.492
18.0	$E^{DCS}_{expected}$	6.75E+00	6.20E+00	5.67E+00	5.19E+00	4.75E+00	4.36E+00	4.01E+00	3.71E+00	3.45E+00	3.22E+00	3.02E+00
	$E^{DCS}_{expected}$	6.17E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.095	1.005	0.920	0.841	0.770	0.706	0.651	0.602	0.559	0.522	0.489

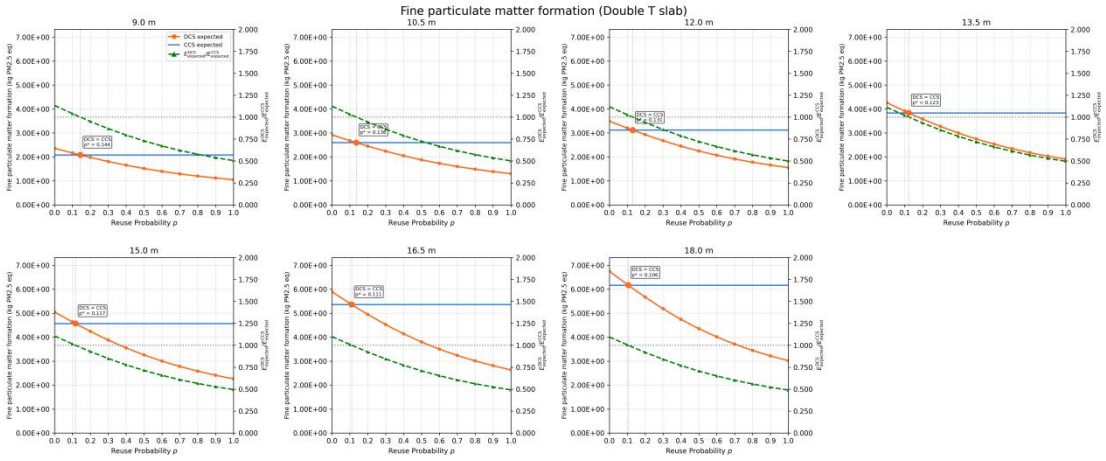


Figure H-5 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fine particulate matter formation

Table H-6 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation, Terrestrial ecosystem

Double T Slab-Ozone formation, Terrestrial ecosystem (kg NOx eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	4.12E+00	3.79E+00	3.48E+00	3.20E+00	2.94E+00	2.71E+00	2.51E+00	2.33E+00	2.18E+00	2.04E+00	1.92E+00
	$E^{DCS}_{expected}$	3.72E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.108	1.020	0.937	0.860	0.791	0.729	0.675	0.627	0.585	0.549	0.517
10.5	$E^{DCS}_{expected}$	5.11E+00	4.71E+00	4.32E+00	3.97E+00	3.65E+00	3.36E+00	3.11E+00	2.89E+00	2.70E+00	2.54E+00	2.39E+00
	$E^{DCS}_{expected}$	4.64E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.101	1.014	0.931	0.855	0.786	0.725	0.671	0.624	0.582	0.546	0.515
12.0	$E^{DCS}_{expected}$	6.11E+00	5.63E+00	5.17E+00	4.75E+00	4.36E+00	4.02E+00	3.72E+00	3.46E+00	3.23E+00	3.03E+00	2.86E+00
	$E^{DCS}_{expected}$	5.57E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.096	1.010	0.927	0.851	0.783	0.722	0.668	0.621	0.580	0.544	0.513
13.5	$E^{DCS}_{expected}$	7.49E+00	6.90E+00	6.34E+00	5.82E+00	5.35E+00	4.94E+00	4.57E+00	4.25E+00	3.97E+00	3.73E+00	3.51E+00
	$E^{DCS}_{expected}$	6.87E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.090	1.004	0.922	0.847	0.779	0.718	0.665	0.619	0.578	0.542	0.511
15.0	$E^{DCS}_{expected}$	8.89E+00	8.19E+00	7.52E+00	6.91E+00	6.35E+00	5.86E+00	5.43E+00	5.05E+00	4.72E+00	4.43E+00	4.17E+00
	$E^{DCS}_{expected}$	8.19E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.086	1.001	0.919	0.844	0.776	0.716	0.663	0.617	0.576	0.541	0.510
16.5	$E^{DCS}_{expected}$	1.04E+01	9.55E+00	8.77E+00	8.05E+00	7.40E+00	6.83E+00	6.32E+00	5.88E+00	5.49E+00	5.16E+00	4.86E+00

Double T Slab-Ozone formation, Terrestrial ecosystem (kg NOx eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E^{DCS}_{expected}$	9.58E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.081	0.996	0.915	0.840	0.773	0.713	0.660	0.614	0.573	0.538	0.507
18.0	$E^{DCS}_{expected}$	1.19E +01	1.09E +01	1.00E +01	9.21E +00	8.47E +00	7.81E +00	7.23E +00	6.73E +00	6.28E +00	5.89E +00	5.56E +00
	$E^{DCS}_{expected}$	1.10E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.077	0.993	0.912	0.837	0.770	0.710	0.657	0.611	0.571	0.536	0.505

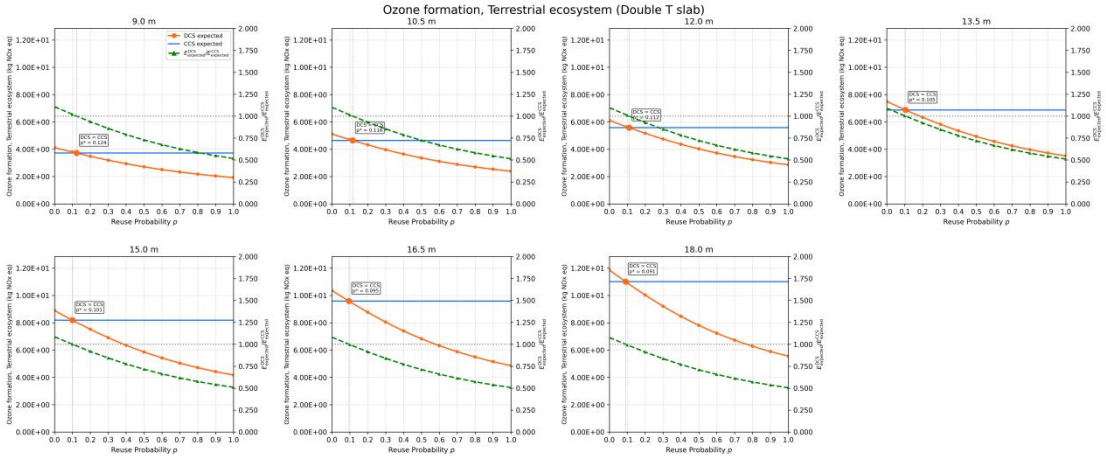


Figure H-6 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation, Terrestrial ecosystem

Table H-7 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial acidification

Double T Slab-Terrestrial acidification (kg SO2 eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	5.16E +00	4.75E +00	4.36E +00	4.00E +00	3.68E +00	3.39E +00	3.14E +00	2.91E +00	2.72E +00	2.55E +00	2.40E +00
	$E^{DCS}_{expected}$	4.65E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.109	1.021	0.938	0.860	0.791	0.729	0.674	0.626	0.585	0.548	0.516
10.5	$E^{DCS}_{expected}$	6.40E +00	5.89E +00	5.41E +00	4.96E +00	4.56E +00	4.21E +00	3.89E +00	3.62E +00	3.38E +00	3.17E +00	2.98E +00
	$E^{DCS}_{expected}$	5.80E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.102	1.015	0.932	0.855	0.786	0.725	0.670	0.623	0.582	0.545	0.514
12.0	$E^{DCS}_{expected}$	7.65E +00	7.05E +00	6.47E +00	5.94E +00	5.46E +00	5.03E +00	4.66E +00	4.33E +00	4.04E +00	3.79E +00	3.57E +00
	$E^{DCS}_{expected}$	6.97E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.098	1.011	0.928	0.852	0.783	0.722	0.668	0.621	0.579	0.543	0.512
13.5	$E^{DCS}_{expected}$	9.38E	8.64E	7.94E	7.28E	6.70E	6.17E	5.71E	5.31E	4.96E	4.65E	4.38E

Double T Slab-Terrestrial acidification (kg SO2 eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
		+00	+00	+00	+00	+00	+00	+00	+00	+00	+00	+00
	$E^{DCS}_{expected}$	8.60E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.092	1.005	0.923	0.847	0.779	0.718	0.665	0.618	0.577	0.541	0.510
15.0	$E^{DCS}_{expected}$	1.11E +01	1.03E +01	9.42E +00	8.65E +00	7.95E +00	7.33E +00	6.78E +00	6.31E +00	5.89E +00	5.53E +00	5.21E +00
	$E^{DCS}_{expected}$	1.02E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.087	1.002	0.920	0.844	0.776	0.716	0.663	0.616	0.575	0.540	0.508
16.5	$E^{DCS}_{expected}$	1.30E +01	1.20E +01	1.10E +01	1.01E +01	9.26E +00	8.54E +00	7.91E +00	7.35E +00	6.86E +00	6.44E +00	6.07E +00
	$E^{DCS}_{expected}$	1.20E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.083	0.997	0.916	0.841	0.773	0.713	0.659	0.613	0.572	0.537	0.506
18.0	$E^{DCS}_{expected}$	1.48E +01	1.37E +01	1.26E +01	1.15E +01	1.06E +01	9.77E +00	9.04E +00	8.40E +00	7.85E +00	7.36E +00	6.93E +00
	$E^{DCS}_{expected}$	1.38E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.079	0.994	0.913	0.838	0.770	0.710	0.657	0.611	0.570	0.535	0.504

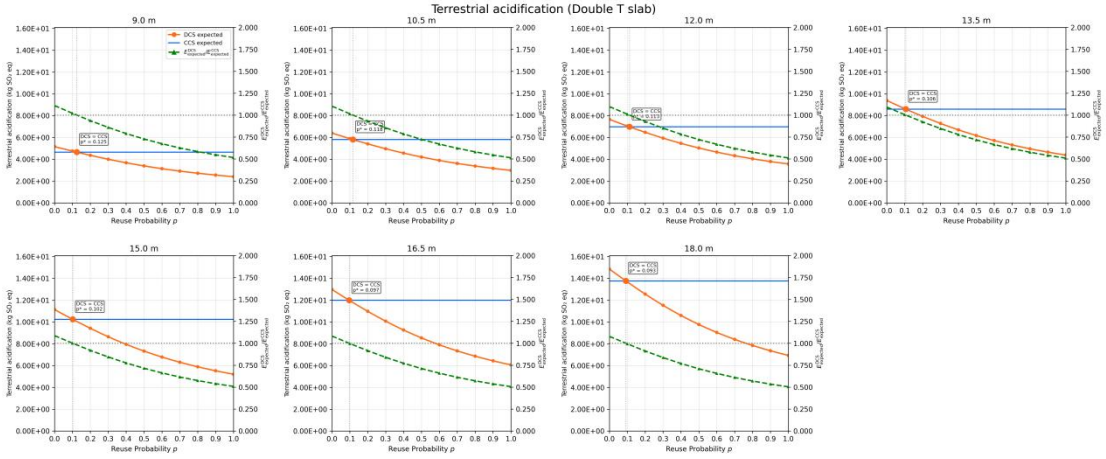


Figure H-7 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial acidification

Table H-8 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater eutrophication

Double T Slab-Freshwater eutrophication (kg P eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	4.87E-01	4.47E-01	4.08E-01	3.72E-01	3.40E-01	3.12E-01	2.87E-01	2.65E-01	2.46E-01	2.29E-01	2.14E-01
	$E^{DCS}_{expected}$	4.05E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.203	1.103	1.007	0.920	0.840	0.770	0.708	0.654	0.607	0.565	0.529

Double T Slab-Freshwater eutrophication (kg P eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
10.5	$E^{DCS}_{expected}$	5.99E-01	5.49E-01	5.02E-01	4.58E-01	4.19E-01	3.84E-01	3.53E-01	3.26E-01	3.03E-01	2.82E-01	2.64E-01
	$E^{DCS}_{expected}$	5.03E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.191	1.092	0.998	0.911	0.833	0.763	0.702	0.649	0.602	0.561	0.526
12.0	$E^{DCS}_{expected}$	7.14E-01	6.55E-01	5.99E-01	5.47E-01	5.00E-01	4.58E-01	4.22E-01	3.90E-01	3.62E-01	3.37E-01	3.16E-01
	$E^{DCS}_{expected}$	6.05E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.182	1.084	0.991	0.904	0.827	0.758	0.698	0.644	0.598	0.558	0.522
13.5	$E^{DCS}_{expected}$	8.65E-01	7.93E-01	7.25E-01	6.63E-01	6.06E-01	5.56E-01	5.12E-01	4.73E-01	4.39E-01	4.10E-01	3.84E-01
	$E^{DCS}_{expected}$	7.40E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.169	1.072	0.981	0.896	0.819	0.751	0.692	0.639	0.594	0.554	0.519
15.0	$E^{DCS}_{expected}$	1.02E+00	9.36E-01	8.56E-01	7.82E-01	7.16E-01	6.57E-01	6.04E-01	5.59E-01	5.19E-01	4.84E-01	4.54E-01
	$E^{DCS}_{expected}$	8.80E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.159	1.064	0.973	0.889	0.813	0.746	0.687	0.635	0.590	0.550	0.516
16.5	$E^{DCS}_{expected}$	1.19E+00	1.09E+00	1.00E+00	9.14E-01	8.36E-01	7.67E-01	7.06E-01	6.53E-01	6.06E-01	5.66E-01	5.30E-01
	$E^{DCS}_{expected}$	1.04E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.149	1.054	0.964	0.881	0.806	0.739	0.680	0.629	0.584	0.545	0.511
18.0	$E^{DCS}_{expected}$	1.37E+00	1.26E+00	1.15E+00	1.05E+00	9.62E-01	8.82E-01	8.12E-01	7.51E-01	6.97E-01	6.50E-01	6.09E-01
	$E^{DCS}_{expected}$	1.20E+00										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.140	1.046	0.957	0.874	0.799	0.733	0.675	0.623	0.579	0.540	0.506

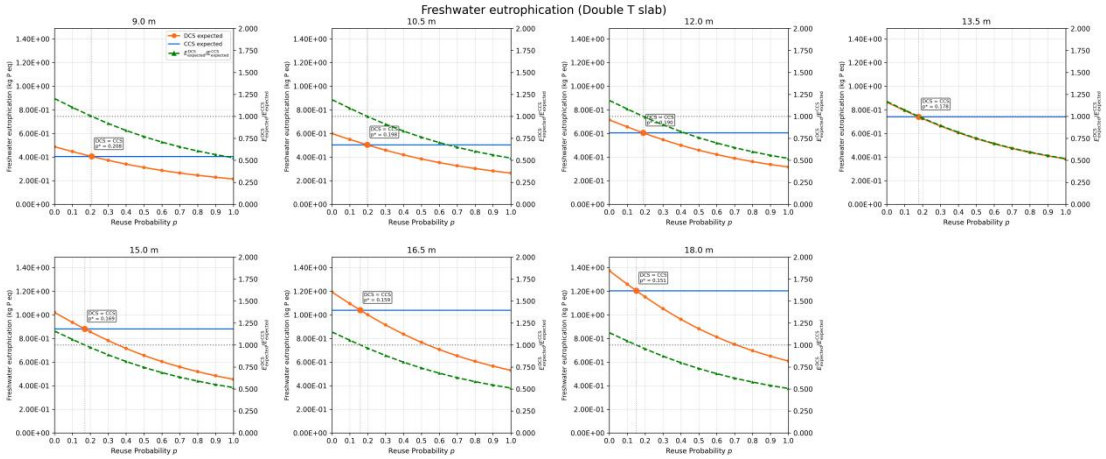


Figure H-8 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater eutrophication

Table H-9 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine eutrophication

Double T Slab-Marine eutrophication (kg N eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	2.47E-02	2.27E-02	2.09E-02	1.91E-02	1.76E-02	1.62E-02	1.50E-02	1.40E-02	1.30E-02	1.22E-02	1.15E-02
	$E^{DCS}_{expected}$	2.15E-02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.147	1.056	0.970	0.890	0.818	0.754	0.698	0.649	0.606	0.568	0.535
10.5	$E^{DCS}_{expected}$	3.05E-02	2.81E-02	2.58E-02	2.37E-02	2.18E-02	2.01E-02	1.86E-02	1.73E-02	1.61E-02	1.51E-02	1.43E-02
	$E^{DCS}_{expected}$	2.68E-02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.138	1.048	0.963	0.884	0.812	0.749	0.693	0.645	0.602	0.565	0.532
12.0	$E^{DCS}_{expected}$	3.64E-02	3.36E-02	3.08E-02	2.83E-02	2.60E-02	2.40E-02	2.22E-02	2.06E-02	1.93E-02	1.81E-02	1.71E-02
	$E^{DCS}_{expected}$	3.22E-02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.132	1.043	0.957	0.879	0.808	0.745	0.690	0.641	0.599	0.562	0.530
13.5	$E^{DCS}_{expected}$	4.44E-02	4.09E-02	3.76E-02	3.45E-02	3.18E-02	2.93E-02	2.71E-02	2.52E-02	2.36E-02	2.21E-02	2.09E-02
	$E^{DCS}_{expected}$	3.96E-02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.123	1.035	0.950	0.873	0.803	0.741	0.686	0.638	0.596	0.559	0.527
15.0	$E^{DCS}_{expected}$	5.26E-02	4.85E-02	4.45E-02	4.09E-02	3.76E-02	3.47E-02	3.22E-02	2.99E-02	2.80E-02	2.62E-02	2.48E-02
	$E^{DCS}_{expected}$	4.71E-02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.117	1.029	0.945	0.868	0.799	0.737	0.683	0.635	0.593	0.557	0.525
16.5	$E^{DCS}_{expected}$	6.14E-02	5.66E-02	5.20E-02	4.77E-02	4.39E-02	4.05E-02	3.75E-02	3.49E-02	3.26E-02	3.06E-02	2.89E-02
	$E^{DCS}_{expected}$	5.53E-02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.110	1.023	0.939	0.863	0.794	0.732	0.678	0.631	0.589	0.553	0.522
18.0	$E^{DCS}_{expected}$	7.04E-02	6.49E-02	5.96E-02	5.47E-02	5.03E-02	4.64E-02	4.30E-02	4.00E-02	3.74E-02	3.51E-02	3.31E-02
	$E^{DCS}_{expected}$	6.38E-02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.104	1.018	0.935	0.858	0.789	0.728	0.674	0.627	0.586	0.550	0.518

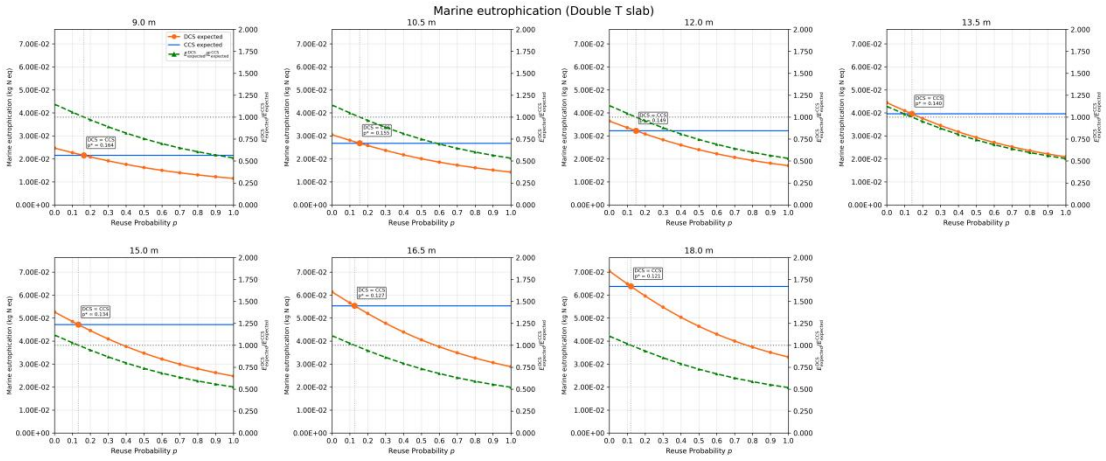


Figure H-9 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine eutrophication

Table H-10 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial ecotoxicity

Double T Slab-Terrestrial ecotoxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	6.86E+03	6.24E+03	5.64E+03	5.08E+03	4.58E+03	4.13E+03	3.73E+03	3.39E+03	3.08E+03	2.82E+03	2.59E+03
	$E^{DCS}_{expected}$	5.94E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.157	1.051	0.950	0.856	0.771	0.696	0.629	0.571	0.519	0.475	0.436
10.5	$E^{DCS}_{expected}$	8.49E+03	7.72E+03	6.98E+03	6.29E+03	5.67E+03	5.11E+03	4.62E+03	4.19E+03	3.82E+03	3.49E+03	3.20E+03
	$E^{DCS}_{expected}$	7.40E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.148	1.043	0.943	0.850	0.766	0.691	0.625	0.567	0.516	0.472	0.433
12.0	$E^{DCS}_{expected}$	1.02E+04	9.23E+03	8.34E+03	7.52E+03	6.77E+03	6.11E+03	5.53E+03	5.01E+03	4.56E+03	4.17E+03	3.83E+03
	$E^{DCS}_{expected}$	8.89E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.142	1.038	0.938	0.846	0.762	0.687	0.622	0.564	0.514	0.469	0.431
13.5	$E^{DCS}_{expected}$	1.24E+04	1.13E+04	1.02E+04	9.19E+03	8.28E+03	7.47E+03	6.76E+03	6.13E+03	5.58E+03	5.10E+03	4.68E+03
	$E^{DCS}_{expected}$	1.09E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.134	1.031	0.932	0.840	0.757	0.683	0.618	0.561	0.510	0.467	0.428
15.0	$E^{DCS}_{expected}$	1.47E+04	1.34E+04	1.21E+04	1.09E+04	9.81E+03	8.85E+03	8.01E+03	7.27E+03	6.62E+03	6.05E+03	5.55E+03
	$E^{DCS}_{expected}$	1.30E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.129	1.026	0.927	0.836	0.753	0.680	0.615	0.558	0.508	0.464	0.426
16.5	$E^{DCS}_{expected}$	1.71E+04	1.56E+04	1.41E+04	1.27E+04	1.14E+04	1.03E+04	9.34E+03	8.47E+03	7.72E+03	7.05E+03	6.47E+03

Double T Slab-Terrestrial ecotoxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E^{DCS}_{expected}$	1.53E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.122	1.020	0.922	0.831	0.749	0.676	0.611	0.555	0.505	0.462	0.424
18.0	$E^{DCS}_{expected}$	1.97E +04	1.79E +04	1.61E +04	1.46E +04	1.31E +04	1.18E +04	1.07E +04	9.71E +03	8.84E +03	8.08E +03	7.41E +03
	$E^{DCS}_{expected}$	1.76E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.117	1.015	0.918	0.827	0.746	0.673	0.608	0.552	0.503	0.459	0.422

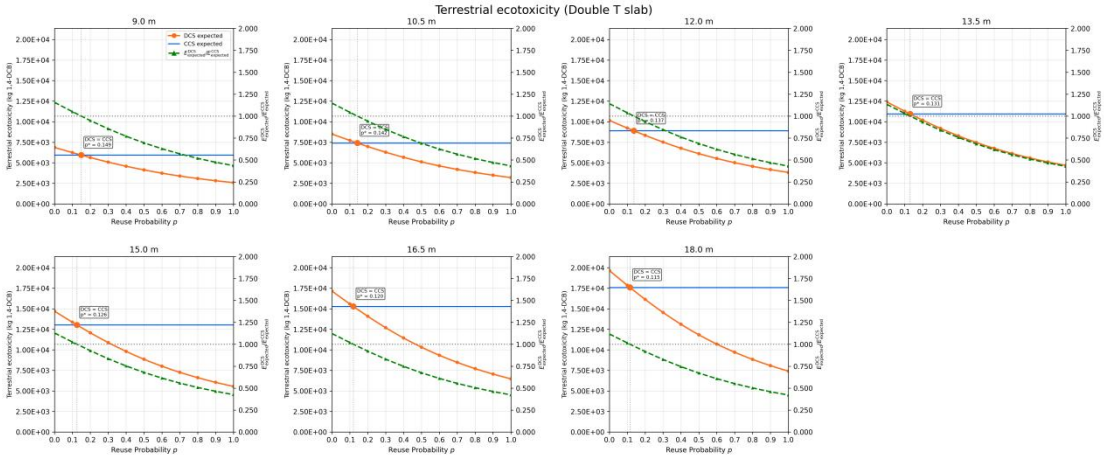


Figure H-10 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial ecotoxicity

Table H-11 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater ecotoxicity

Double T Slab-Freshwater ecotoxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	5.35E +01	4.87E +01	4.40E +01	3.98E +01	3.59E +01	3.25E +01	2.95E +01	2.69E +01	2.46E +01	2.26E +01	2.08E +01
	$E^{DCS}_{expected}$	4.30E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.245	1.133	1.025	0.926	0.836	0.757	0.687	0.625	0.572	0.525	0.484
10.5	$E^{DCS}_{expected}$	6.56E +01	5.96E +01	5.40E +01	4.88E +01	4.40E +01	3.99E +01	3.62E +01	3.30E +01	3.02E +01	2.77E +01	2.55E +01
	$E^{DCS}_{expected}$	5.32E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.231	1.120	1.014	0.916	0.827	0.749	0.680	0.619	0.566	0.520	0.480
12.0	$E^{DCS}_{expected}$	7.81E +01	7.10E +01	6.43E +01	5.81E +01	5.25E +01	4.75E +01	4.31E +01	3.93E +01	3.59E +01	3.30E +01	3.05E +01
	$E^{DCS}_{expected}$	6.40E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.221	1.110	1.005	0.908	0.820	0.742	0.674	0.614	0.562	0.516	0.476
13.5	$E^{DCS}_{expected}$	9.40E	8.56E	7.75E	7.00E	6.33E	5.73E	5.20E	4.74E	4.34E	3.99E	3.68E

Double T Slab-Freshwater ecotoxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
		+01	+01	+01	+01	+01	+01	+01	+01	+01	+01	+01
	$E^{DCS}_{expected}$	7.80E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.205	1.096	0.993	0.897	0.811	0.734	0.667	0.608	0.556	0.511	0.472
15.0	$E^{DCS}_{expected}$	1.11E +02	1.01E +02	9.12E +01	8.24E +01	7.45E +01	6.75E +01	6.13E +01	5.59E +01	5.11E +01	4.70E +01	4.34E +01
	$E^{DCS}_{expected}$	9.28E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.193	1.086	0.983	0.888	0.803	0.727	0.661	0.602	0.551	0.507	0.468
16.5	$E^{DCS}_{expected}$	1.30E +02	1.18E +02	1.07E +02	9.65E +01	8.72E +01	7.90E +01	7.17E +01	6.54E +01	5.98E +01	5.50E +01	5.08E +01
	$E^{DCS}_{expected}$	1.10E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.180	1.073	0.972	0.878	0.794	0.719	0.653	0.595	0.545	0.501	0.462
18.0	$E^{DCS}_{expected}$	1.50E +02	1.36E +02	1.23E +02	1.11E +02	1.01E +02	9.11E +01	8.27E +01	7.54E +01	6.90E +01	6.34E +01	5.85E +01
	$E^{DCS}_{expected}$	1.28E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.169	1.063	0.963	0.870	0.786	0.712	0.646	0.589	0.539	0.495	0.457

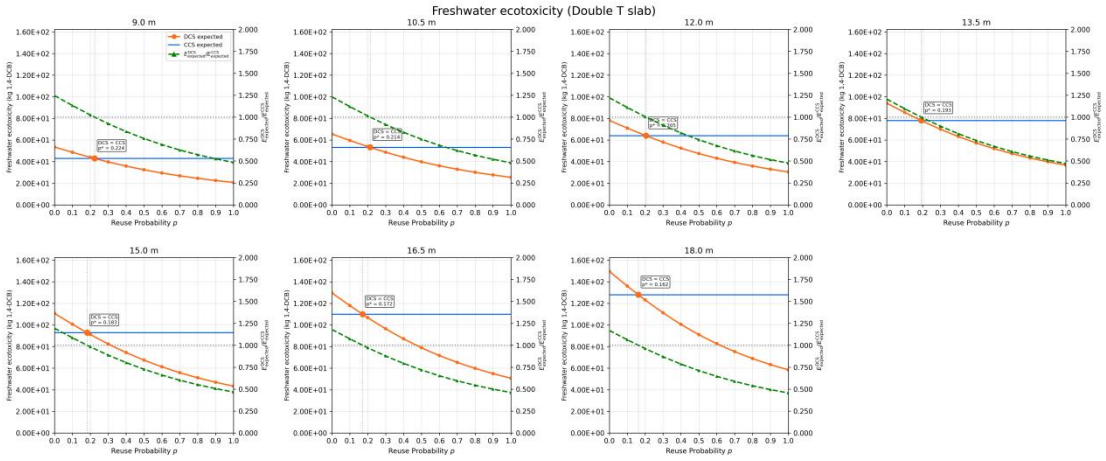


Figure H-11 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater ecotoxicity

Table H-12 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine ecotoxicity

Double T Slab-Marine ecotoxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	7.73E +01	7.03E +01	6.36E +01	5.74E +01	5.18E +01	4.69E +01	4.25E +01	3.87E +01	3.54E +01	3.25E +01	2.99E +01
	$E^{DCS}_{expected}$	6.22E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.243	1.130	1.022	0.923	0.834	0.754	0.684	0.623	0.569	0.522	0.481

Double T Slab-Marine ecotoxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
10.5	$E^{DCS}_{expected}$	9.47E+01	8.62E+01	7.80E+01	7.04E+01	6.36E+01	5.75E+01	5.22E+01	4.75E+01	4.35E+01	3.99E+01	3.68E+01
	$E^{DCS}_{expected}$	7.71E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.229	1.117	1.011	0.913	0.825	0.746	0.677	0.617	0.564	0.518	0.477
12.0	$E^{DCS}_{expected}$	1.13E+02	1.03E+02	9.29E+01	8.39E+01	7.58E+01	6.86E+01	6.22E+01	5.67E+01	5.18E+01	4.76E+01	4.39E+01
	$E^{DCS}_{expected}$	9.27E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.218	1.108	1.002	0.905	0.818	0.740	0.671	0.612	0.559	0.513	0.473
13.5	$E^{DCS}_{expected}$	1.36E+02	1.24E+02	1.12E+02	1.01E+02	9.14E+01	8.27E+01	7.51E+01	6.84E+01	6.26E+01	5.75E+01	5.30E+01
	$E^{DCS}_{expected}$	1.13E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.203	1.094	0.990	0.895	0.808	0.732	0.664	0.605	0.554	0.508	0.469
15.0	$E^{DCS}_{expected}$	1.60E+02	1.46E+02	1.32E+02	1.19E+02	1.08E+02	9.75E+01	8.85E+01	8.06E+01	7.38E+01	6.78E+01	6.25E+01
	$E^{DCS}_{expected}$	1.34E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.191	1.083	0.981	0.886	0.801	0.725	0.658	0.600	0.549	0.504	0.465
16.5	$E^{DCS}_{expected}$	1.87E+02	1.71E+02	1.54E+02	1.39E+02	1.26E+02	1.14E+02	1.04E+02	9.44E+01	8.63E+01	7.93E+01	7.31E+01
	$E^{DCS}_{expected}$	1.59E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.178	1.072	0.970	0.876	0.792	0.717	0.651	0.593	0.542	0.498	0.460
18.0	$E^{DCS}_{expected}$	2.16E+02	1.97E+02	1.78E+02	1.61E+02	1.45E+02	1.31E+02	1.19E+02	1.09E+02	9.94E+01	9.13E+01	8.42E+01
	$E^{DCS}_{expected}$	1.85E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.167	1.062	0.961	0.868	0.784	0.710	0.644	0.587	0.537	0.493	0.455

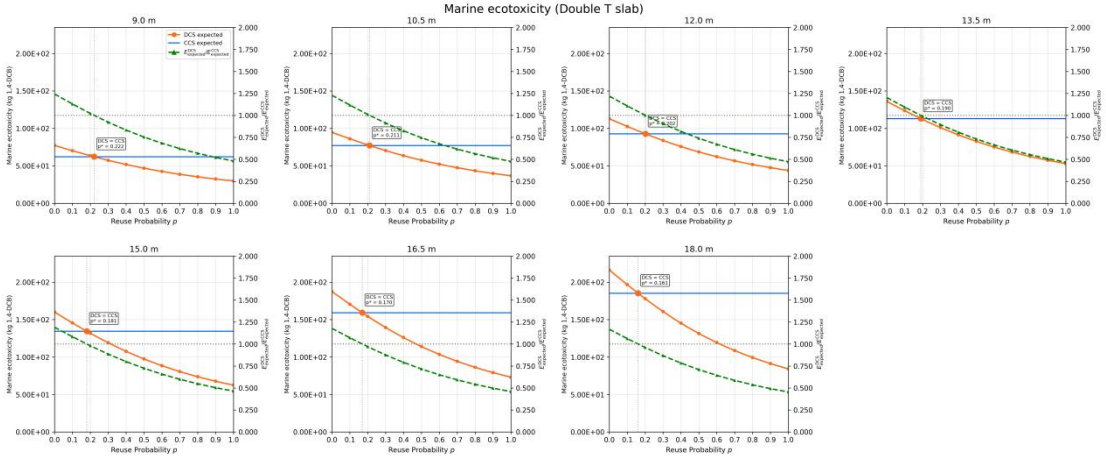


Figure H-12 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine ecotoxicity

Table H-13 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Human carcinogenic toxicity

Double T Slab-Human carcinogenic toxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	2.31E+02	2.09E+02	1.88E+02	1.68E+02	1.50E+02	1.35E+02	1.21E+02	1.09E+02	9.84E+01	8.92E+01	8.12E+01
	$E^{DCS}_{expected}$	1.75E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.319	1.192	1.071	0.959	0.859	0.769	0.691	0.622	0.562	0.509	0.464
10.5	$E^{DCS}_{expected}$	2.81E+02	2.54E+02	2.28E+02	2.05E+02	1.83E+02	1.64E+02	1.47E+02	1.33E+02	1.20E+02	1.09E+02	9.90E+01
	$E^{DCS}_{expected}$	2.16E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.302	1.176	1.057	0.947	0.848	0.760	0.682	0.614	0.555	0.503	0.458
12.0	$E^{DCS}_{expected}$	3.34E+02	3.02E+02	2.72E+02	2.43E+02	2.18E+02	1.95E+02	1.75E+02	1.58E+02	1.43E+02	1.29E+02	1.18E+02
	$E^{DCS}_{expected}$	2.60E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.287	1.163	1.045	0.936	0.838	0.751	0.674	0.607	0.549	0.498	0.453
13.5	$E^{DCS}_{expected}$	3.99E+02	3.61E+02	3.24E+02	2.90E+02	2.60E+02	2.33E+02	2.09E+02	1.89E+02	1.70E+02	1.55E+02	1.41E+02
	$E^{DCS}_{expected}$	3.15E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.267	1.145	1.029	0.922	0.826	0.740	0.665	0.599	0.541	0.491	0.447
15.0	$E^{DCS}_{expected}$	4.68E+02	4.23E+02	3.80E+02	3.40E+02	3.05E+02	2.73E+02	2.45E+02	2.21E+02	2.00E+02	1.81E+02	1.65E+02
	$E^{DCS}_{expected}$	3.74E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.250	1.130	1.015	0.910	0.815	0.730	0.656	0.591	0.534	0.485	0.441
16.5	$E^{DCS}_{expected}$	5.49E+02	4.96E+02	4.46E+02	3.99E+02	3.58E+02	3.21E+02	2.88E+02	2.59E+02	2.34E+02	2.13E+02	1.94E+02
	$E^{DCS}_{expected}$	4.46E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.231	1.113	1.000	0.896	0.802	0.719	0.646	0.582	0.526	0.477	0.434
18.0	$E^{DCS}_{expected}$	6.36E+02	5.75E+02	5.16E+02	4.63E+02	4.14E+02	3.71E+02	3.33E+02	3.00E+02	2.71E+02	2.46E+02	2.24E+02
	$E^{DCS}_{expected}$	5.23E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.215	1.098	0.987	0.884	0.791	0.709	0.637	0.574	0.519	0.470	0.428

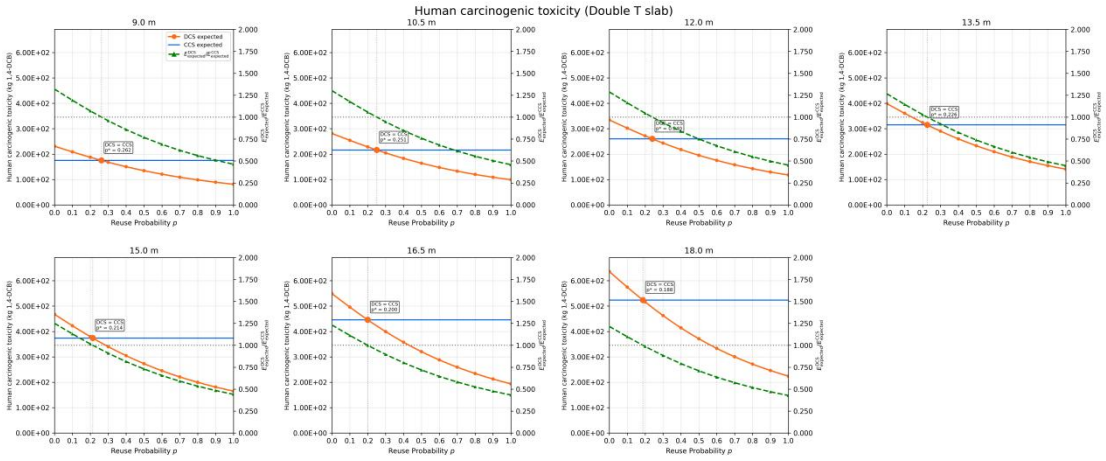


Figure H-13 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Human carcinogenic toxicity

Table H-14 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Human non-carcinogenic toxicity

Double T Slab-Human non-carcinogenic toxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	1.63E+03	1.49E+03	1.34E+03	1.21E+03	1.09E+03	9.88E+02	8.96E+02	8.15E+02	7.44E+02	6.82E+02	6.28E+02
	$E^{DCS}_{expected}$	1.31E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.244	1.131	1.022	0.923	0.832	0.752	0.682	0.620	0.566	0.519	0.478
10.5	$E^{DCS}_{expected}$	2.00E+03	1.82E+03	1.65E+03	1.49E+03	1.34E+03	1.21E+03	1.10E+03	1.00E+03	9.13E+02	8.38E+02	7.71E+02
	$E^{DCS}_{expected}$	1.63E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.230	1.118	1.011	0.913	0.824	0.745	0.675	0.614	0.561	0.514	0.474
12.0	$E^{DCS}_{expected}$	2.39E+03	2.17E+03	1.96E+03	1.77E+03	1.60E+03	1.45E+03	1.31E+03	1.19E+03	1.09E+03	9.99E+02	9.20E+02
	$E^{DCS}_{expected}$	1.96E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.219	1.108	1.002	0.905	0.817	0.738	0.669	0.609	0.556	0.510	0.470
13.5	$E^{DCS}_{expected}$	2.87E+03	2.61E+03	2.36E+03	2.13E+03	1.93E+03	1.74E+03	1.58E+03	1.44E+03	1.31E+03	1.21E+03	1.11E+03
	$E^{DCS}_{expected}$	2.39E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.204	1.095	0.990	0.894	0.807	0.730	0.662	0.603	0.551	0.505	0.466
15.0	$E^{DCS}_{expected}$	3.38E+03	3.08E+03	2.78E+03	2.51E+03	2.27E+03	2.05E+03	1.86E+03	1.70E+03	1.55E+03	1.42E+03	1.31E+03
	$E^{DCS}_{expected}$	2.84E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.192	1.084	0.981	0.886	0.800	0.723	0.656	0.597	0.546	0.501	0.462
16.5	$E^{DCS}_{expected}$	3.96E+03	3.60E+03	3.26E+03	2.94E+03	2.66E+03	2.40E+03	2.18E+03	1.98E+03	1.81E+03	1.66E+03	1.53E+03

Double T Slab-Human non-carcinogenic toxicity (kg 1,4-DCB)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E^{DCS}_{expected}$	3.36E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.179	1.072	0.970	0.876	0.791	0.715	0.649	0.590	0.539	0.495	0.456
18.0	$E^{DCS}_{expected}$	4.57E+03	4.16E+03	3.76E+03	3.39E+03	3.06E+03	2.77E+03	2.51E+03	2.29E+03	2.09E+03	1.92E+03	1.77E+03
	$E^{DCS}_{expected}$	3.91E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.168	1.062	0.961	0.867	0.783	0.708	0.642	0.584	0.534	0.490	0.451

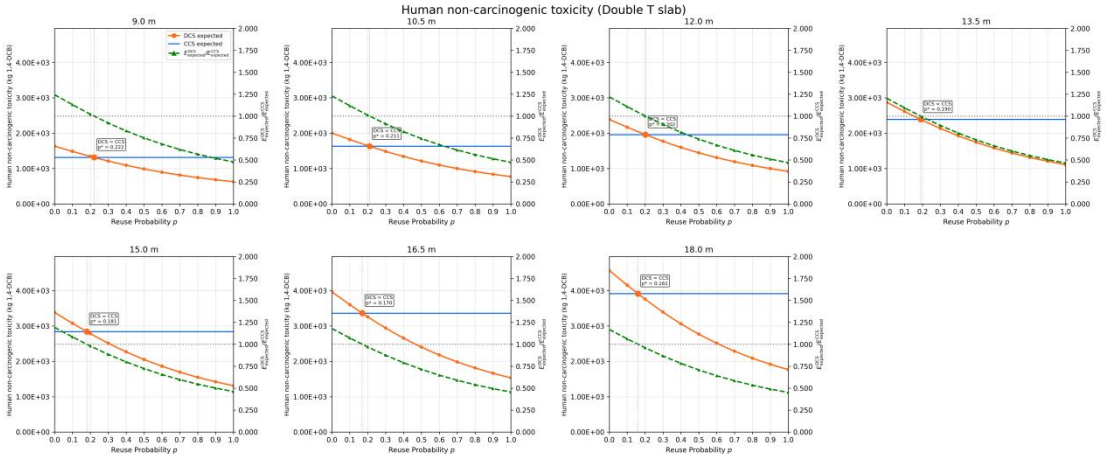


Figure H-14 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Human non-carcinogenic toxicity

Table H-15 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Land use

Double T Slab-Land use (m^2 a crop eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	2.84E+01	2.58E+01	2.34E+01	2.11E+01	1.91E+01	1.72E+01	1.56E+01	1.42E+01	1.30E+01	1.19E+01	1.10E+01
	$E^{DCS}_{expected}$	2.52E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.123	1.022	0.925	0.836	0.755	0.683	0.619	0.564	0.515	0.472	0.435
10.5	$E^{DCS}_{expected}$	3.52E+01	3.20E+01	2.90E+01	2.62E+01	2.36E+01	2.14E+01	1.94E+01	1.77E+01	1.61E+01	1.48E+01	1.36E+01
	$E^{DCS}_{expected}$	3.15E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.116	1.016	0.919	0.831	0.750	0.679	0.616	0.560	0.512	0.470	0.433
12.0	$E^{DCS}_{expected}$	4.21E+01	3.83E+01	3.47E+01	3.13E+01	2.83E+01	2.56E+01	2.32E+01	2.11E+01	1.93E+01	1.77E+01	1.63E+01
	$E^{DCS}_{expected}$	3.79E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.111	1.011	0.916	0.827	0.747	0.676	0.613	0.558	0.510	0.468	0.431
13.5	$E^{DCS}_{expected}$	5.16E+01	4.69E+01	4.25E+01	3.84E+01	3.47E+01	3.14E+01	2.85E+01	2.59E+01	2.37E+01	2.17E+01	2.00E+01

Double T Slab-Land use (m^2 a crop eq)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E_{expected}^{DCS}$	4.67E+01										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.105	1.006	0.911	0.823	0.743	0.672	0.610	0.555	0.507	0.466	0.429
15.0	$E_{expected}^{DCS}$	6.12E+01	5.57E+01	5.04E+01	4.56E+01	4.12E+01	3.72E+01	3.38E+01	3.08E+01	2.81E+01	2.58E+01	2.38E+01
	$E_{expected}^{DCS}$	5.56E+01										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.101	1.002	0.907	0.820	0.741	0.670	0.608	0.553	0.506	0.464	0.427
16.5	$E_{expected}^{DCS}$	7.13E+01	6.49E+01	5.88E+01	5.31E+01	4.80E+01	4.34E+01	3.94E+01	3.58E+01	3.28E+01	3.00E+01	2.77E+01
	$E_{expected}^{DCS}$	6.51E+01										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.096	0.998	0.903	0.816	0.737	0.667	0.605	0.551	0.503	0.462	0.425
18.0	$E_{expected}^{DCS}$	8.16E+01	7.43E+01	6.73E+01	6.08E+01	5.49E+01	4.97E+01	4.50E+01	4.10E+01	3.75E+01	3.44E+01	3.17E+01
	$E_{expected}^{DCS}$	7.47E+01										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.092	0.994	0.900	0.813	0.735	0.665	0.603	0.549	0.501	0.460	0.424

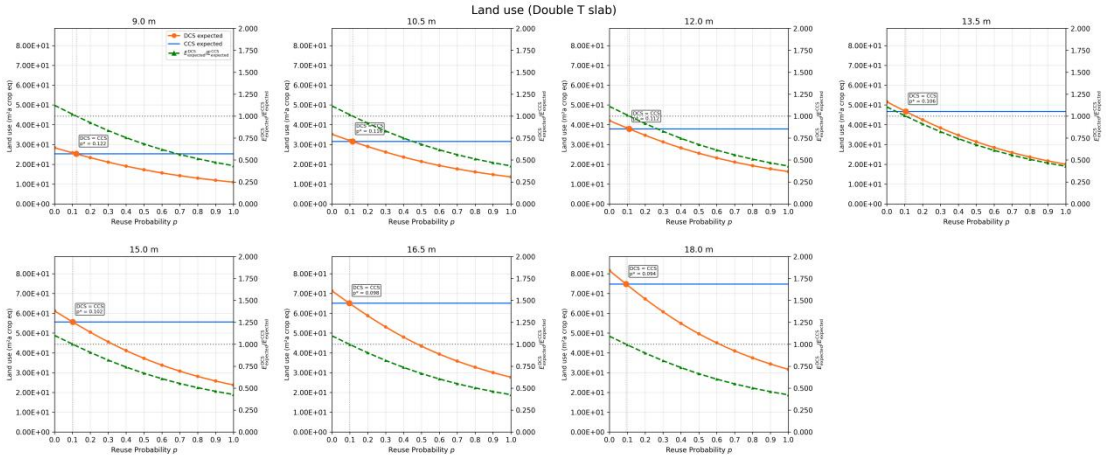


Figure H-15 p* derived from calculations of CCS ($E_{expected}^{CCS}$) and DCS ($E_{expected}^{DCS}$) of Land use

Table H-16 p* derived from calculations of CCS ($E_{expected}^{CCS}$) and DCS ($E_{expected}^{DCS}$) of Mineral resource scarcity

Double T Slab-Mineral resource scarcity (kg Cu eq)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E_{expected}^{DCS}$	1.88E+01	1.69E+01	1.52E+01	1.35E+01	1.21E+01	1.08E+01	9.66E+00	8.66E+00	7.79E+00	7.03E+00	6.36E+00
	$E_{expected}^{DCS}$	1.48E+01										
	$\frac{E_{expected}^{DCS}}{E_{expected}^{CCS}}$	1.270	1.145	1.026	0.917	0.818	0.730	0.653	0.586	0.527	0.476	0.431
10.5	$E_{expected}^{DCS}$	2.29E+01	2.07E+01	1.85E+01	1.66E+01	1.48E+01	1.32E+01	1.18E+01	1.06E+01	9.52E+00	8.59E+00	7.78E+00

Double T Slab-Mineral resource scarcity (kg Cu eq)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E^{DCS}_{expected}$	1.83E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.254	1.131	1.014	0.906	0.808	0.722	0.646	0.579	0.521	0.470	0.425
12.0	$E^{DCS}_{expected}$	2.73E+01	2.46E+01	2.21E+01	1.97E+01	1.76E+01	1.57E+01	1.41E+01	1.26E+01	1.13E+01	1.02E+01	9.26E+00
	$E^{DCS}_{expected}$	2.20E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.241	1.120	1.004	0.897	0.800	0.714	0.639	0.573	0.516	0.465	0.421
13.5	$E^{DCS}_{expected}$	3.27E+01	2.95E+01	2.65E+01	2.36E+01	2.11E+01	1.88E+01	1.69E+01	1.51E+01	1.36E+01	1.23E+01	1.11E+01
	$E^{DCS}_{expected}$	2.67E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.224	1.104	0.990	0.884	0.789	0.705	0.630	0.565	0.508	0.459	0.416
15.0	$E^{DCS}_{expected}$	3.85E+01	3.47E+01	3.11E+01	2.78E+01	2.48E+01	2.21E+01	1.98E+01	1.78E+01	1.60E+01	1.44E+01	1.31E+01
	$E^{DCS}_{expected}$	3.18E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.210	1.091	0.978	0.874	0.780	0.697	0.623	0.559	0.503	0.454	0.411
16.5	$E^{DCS}_{expected}$	4.51E+01	4.07E+01	3.64E+01	3.26E+01	2.91E+01	2.60E+01	2.32E+01	2.08E+01	1.87E+01	1.69E+01	1.53E+01
	$E^{DCS}_{expected}$	3.77E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.195	1.077	0.966	0.863	0.770	0.688	0.615	0.552	0.496	0.448	0.406
18.0	$E^{DCS}_{expected}$	5.21E+01	4.70E+01	4.21E+01	3.76E+01	3.36E+01	3.00E+01	2.68E+01	2.41E+01	2.16E+01	1.95E+01	1.77E+01
	$E^{DCS}_{expected}$	4.41E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.182	1.066	0.955	0.853	0.762	0.680	0.608	0.546	0.491	0.443	0.401

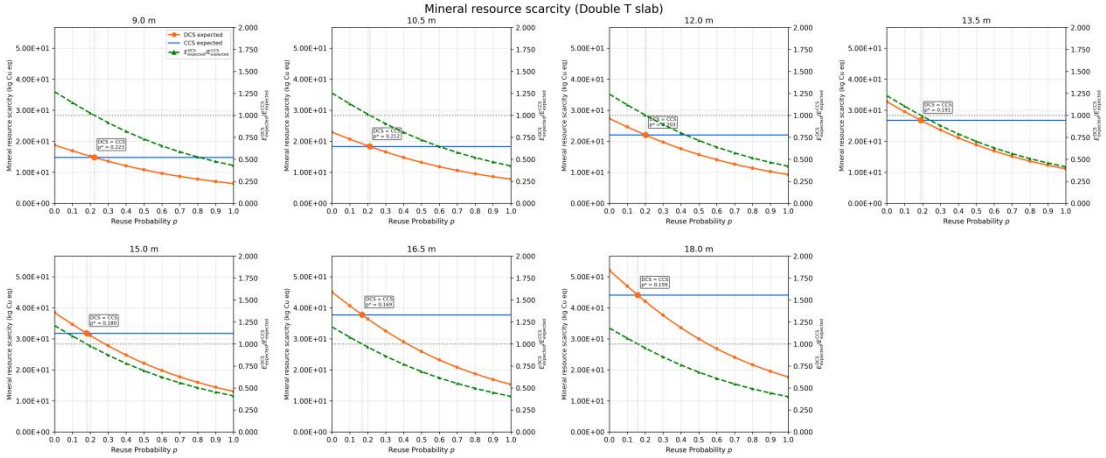


Figure H-16 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Mineral resource scarcity

Table H-17 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fossil resource scarcity

Double T Slab-Fossil resource scarcity (kg oil eq)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	3.46E+02	3.18E+02	2.90E+02	2.65E+02	2.42E+02	2.22E+02	2.04E+02	1.89E+02	1.75E+02	1.63E+02	1.52E+02
	$E^{DCS}_{expected}$	3.09E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.119	1.027	0.939	0.857	0.784	0.718	0.660	0.610	0.565	0.526	0.493
10.5	$E^{DCS}_{expected}$	4.29E+02	3.94E+02	3.60E+02	3.29E+02	3.01E+02	2.76E+02	2.53E+02	2.34E+02	2.17E+02	2.02E+02	1.89E+02
	$E^{DCS}_{expected}$	3.86E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.113	1.021	0.933	0.852	0.779	0.714	0.657	0.606	0.562	0.524	0.490
12.0	$E^{DCS}_{expected}$	5.14E+02	4.71E+02	4.31E+02	3.94E+02	3.60E+02	3.30E+02	3.03E+02	2.80E+02	2.60E+02	2.42E+02	2.26E+02
	$E^{DCS}_{expected}$	4.64E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.108	1.017	0.930	0.849	0.776	0.711	0.654	0.604	0.560	0.522	0.488
13.5	$E^{DCS}_{expected}$	6.30E+02	5.78E+02	5.29E+02	4.83E+02	4.41E+02	4.05E+02	3.72E+02	3.44E+02	3.19E+02	2.97E+02	2.78E+02
	$E^{DCS}_{expected}$	5.72E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.102	1.011	0.925	0.845	0.772	0.708	0.651	0.601	0.558	0.520	0.486
15.0	$E^{DCS}_{expected}$	7.48E+02	6.86E+02	6.27E+02	5.73E+02	5.24E+02	4.80E+02	4.42E+02	4.08E+02	3.79E+02	3.53E+02	3.30E+02
	$E^{DCS}_{expected}$	6.81E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.098	1.008	0.922	0.842	0.770	0.705	0.649	0.599	0.556	0.518	0.485
16.5	$E^{DCS}_{expected}$	8.71E+02	8.00E+02	7.31E+02	6.68E+02	6.11E+02	5.60E+02	5.15E+02	4.75E+02	4.41E+02	4.11E+02	3.85E+02
	$E^{DCS}_{expected}$	7.97E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.093	1.003	0.917	0.838	0.766	0.702	0.646	0.597	0.553	0.516	0.483
18.0	$E^{DCS}_{expected}$	9.97E+02	9.15E+02	8.37E+02	7.64E+02	6.99E+02	6.40E+02	5.89E+02	5.44E+02	5.04E+02	4.70E+02	4.40E+02
	$E^{DCS}_{expected}$	9.15E+02										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.090	1.000	0.914	0.835	0.763	0.700	0.644	0.594	0.551	0.514	0.481

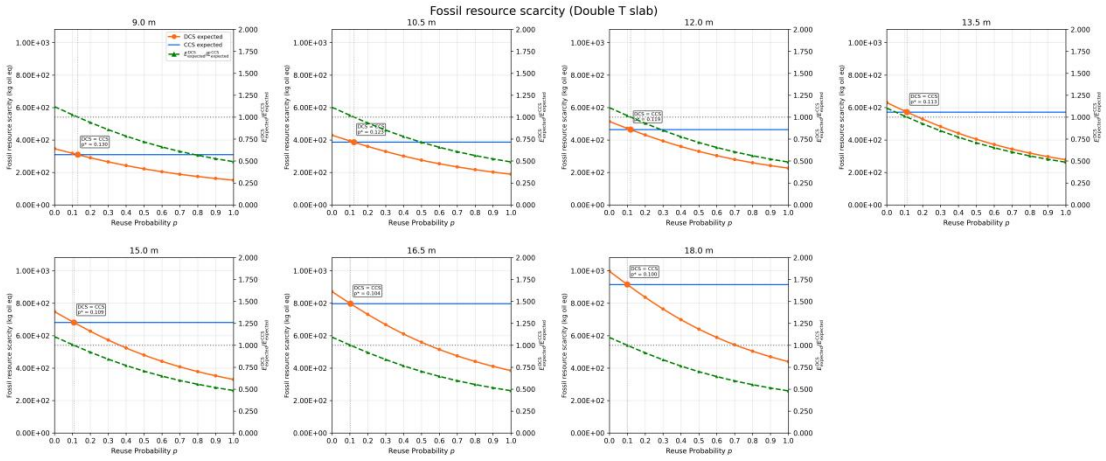


Figure H-17 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fossil resource scarcity

Table H-18 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Water consumption

Double T Slab-Water consumption (m^3)												
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
9.0	$E^{DCS}_{expected}$	1.13E+01	1.02E+01	9.21E+00	8.26E+00	7.40E+00	6.64E+00	5.98E+00	5.39E+00	4.88E+00	4.43E+00	4.04E+00
	$E^{DCS}_{expected}$	1.04E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.089	0.985	0.886	0.795	0.713	0.640	0.575	0.519	0.470	0.427	0.389
10.5	$E^{DCS}_{expected}$	1.40E+01	1.27E+01	1.14E+01	1.03E+01	9.19E+00	8.25E+00	7.42E+00	6.69E+00	6.06E+00	5.51E+00	5.02E+00
	$E^{DCS}_{expected}$	1.30E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.083	0.980	0.881	0.790	0.709	0.636	0.572	0.516	0.467	0.424	0.387
12.0	$E^{DCS}_{expected}$	1.68E+01	1.52E+01	1.37E+01	1.23E+01	1.10E+01	9.87E+00	8.88E+00	8.01E+00	7.25E+00	6.59E+00	6.01E+00
	$E^{DCS}_{expected}$	1.56E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.079	0.976	0.878	0.787	0.706	0.634	0.570	0.514	0.465	0.423	0.386
13.5	$E^{DCS}_{expected}$	2.06E+01	1.87E+01	1.68E+01	1.51E+01	1.35E+01	1.21E+01	1.09E+01	9.83E+00	8.90E+00	8.09E+00	7.38E+00
	$E^{DCS}_{expected}$	1.92E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.073	0.971	0.874	0.784	0.703	0.631	0.567	0.512	0.463	0.421	0.384
15.0	$E^{DCS}_{expected}$	2.45E+01	2.22E+01	1.99E+01	1.79E+01	1.60E+01	1.44E+01	1.29E+01	1.17E+01	1.06E+01	9.61E+00	8.76E+00
	$E^{DCS}_{expected}$	2.29E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.070	0.968	0.871	0.781	0.700	0.628	0.565	0.510	0.462	0.420	0.383
16.5	$E^{DCS}_{expected}$	2.85E+01	2.58E+01	2.32E+01	2.08E+01	1.87E+01	1.68E+01	1.51E+01	1.36E+01	1.23E+01	1.12E+01	1.02E+01

Double T Slab-Water consumption (m^3)											
Span (m)		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
	$E^{DCS}_{expected}$	2.68E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.065	0.964	0.867	0.778	0.697	0.626	0.563	0.508	0.460	0.418	0.381
18.0	$E^{DCS}_{expected}$	3.26E+01	2.95E+01	2.66E+01	2.38E+01	2.14E+01	1.92E+01	1.72E+01	1.56E+01	1.41E+01	1.28E+01	1.17E+01
	$E^{DCS}_{expected}$	3.07E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.062	0.961	0.865	0.775	0.695	0.624	0.561	0.506	0.458	0.417	0.380

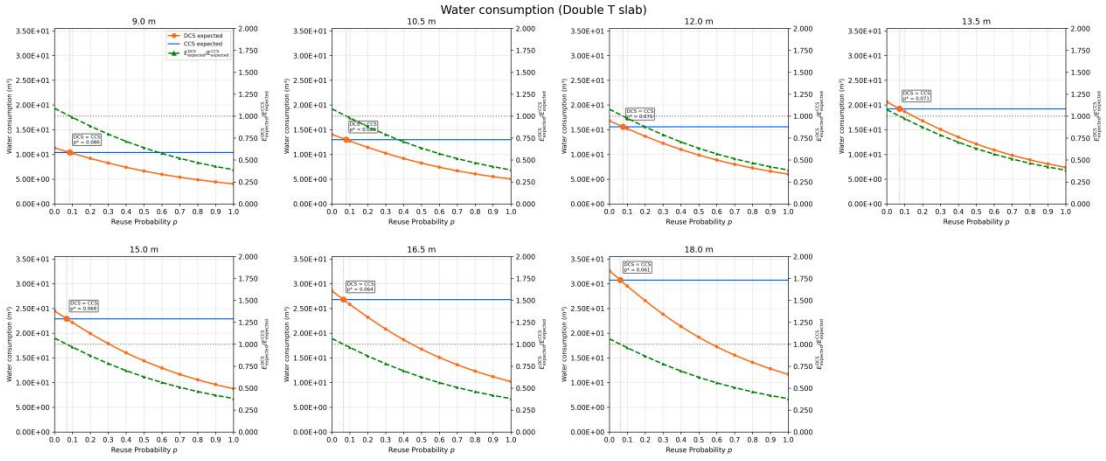


Figure H-18 p* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Water consumption

Appendix I: Connections of the structural level

(1) Bolt selection and load capacity verification

The bolt specifications shall be consistent with the embedded sleeves in the floor slab; high-strength M12 bolts of grade 10.9 shall be used, with an effective cross-sectional area A_s=84.3mm^2, yield strength f_{yb}=940MPa, and ultimate tensile strength f_{ub}=1040MPa. According to the arrangement of embedded sleeves at the ends of the ribbed plate described in Chapter 3 (two M12 sleeves at each end), two M12 high-strength bolts shall be installed at each end of the steel beam connection node. The total number of bolts for the entire building is 221×4=884; the weight of a single M12 bolt is approximately 0.10 kg; the total steel consumption for bolts in the entire building is 884×0.10=88.400kg. The design shear bearing capacity of the bolts shall be calculated in accordance with Section 3.6 of EN 1993-1-8 for bearing-type connections.

$$F_{v,\,Rd}=\frac{0.6\times A_s\times f_{ub}}{\gamma_{M2}}$$

Taking γ_{M2}=1.25 , the design value of the transverse shear force, Rd, is calculated as: $F_{v,\,Rd}=\frac{0.6\times 84.3\times 1040}{1.25\times 1000}$=42.1kN. The design longitudinal shear force V_{Ed} per floor slab is determined by combining the self-weight of the slab with the live load; the self-weight of the ribbed slab is G_k=104.6kN, and the live load is taken as 2.5kN/m^2. Consequently, V_{Ed}=130kN. Each node shall bear half of this shear force, i.e., 65 kN. The combined resistance of the two bolts within the node is 2×42.1=84.2kN>65kN , which exceeds 65 kN. The safety factor is therefore $\frac{84.2}{65}$=1.29 , satisfying the requirements of EN 1993-1-8.

(2) Design of flange connection plates

Welded connection plates shall be installed on the top surface of the steel beam flange to facilitate bolt installation and alignment with the embedded steel plates of the floor slab. The thickness of the connection plates shall be determined in accordance with clause 6.2.6.4 of EN 1993-1-8 using the equivalent T-shaped component method, with a specified thickness of 10 mm (not less than 0.8 times the bolt diameter). For each steel beam segment, two connection plates shall be provided—one at each end of the plate—with plan dimensions of 200 mm × 90 mm, ensuring compliance with the required edge clearances from the bolt holes. The volume of a single connection plate is calculated as 0.2×0.09×0.01=0.00018m^3; given the steel density of 7850 kg/m³, the weight of one plate is approximately 1.41 kg. Since each floor slab incorporates two connection plates, the steel consumption per plate is 2.82 kg. The total steel consumption for all connection plates in the entire building is therefore 221×2.82=623.220kg.

(3) Stiffening rib design

To prevent local buckling of the flange on the steel beam beneath the flange connection plates, stiffening ribs are welded to the corresponding positions on both sides of the steel beam web. Each node is equipped with two stiffening ribs—one on each side of the web—with dimensions of height 120 mm, width 60 mm, and thickness 8 mm. The volume of each rib is 0.12×0.06×0.008=0.0000576m^3; based on the steel density of 7850 kg/m³, the weight of each rib

is approximately 0.452 kg, and the total weight for two ribs per node is 0.904 kg. Since each floor slab has two nodes, the steel consumption for stiffening ribs on a single slab is 1.81 kg. The total steel consumption for stiffening ribs across the entire building is 221×1.81=400.010kg.

(4) Summary of connection system material consumption

The total additional steel consumption for the steel beam connection system is denoted as:

$$M_{\text{conn,steel}}=M_{\text{bolts}}+M_{\text{plate}}+M_{\text{stiffener}}$$

The total amount is 1111.630 kg. This portion of the steel consumption will serve as an additional steel requirement for the detachable solution; it will be combined with the steel consumption of the steel frame main structure exported from ETABS in Section 4.3.5 and the steel consumption for floor slabs in Chapter 3 to determine the total steel consumption for the detachable solution.

Appendix J: Environmental impacts of 18 across 18 impact categories at the structural level

Table J-1 Specific environmental impacts of Global warming

Global warming (kg CO_2 eq)												
System	CCS				DCS							
	E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Structural level	2.04E+06	1.76E+05	1.39E+05	2.35E+06	2.15E+06	1.72E+05	9.48E+04	8.47E+04	1.42E+05	2.46E+06	2.56E+06	2.64E+06

Table J-2 Specific environmental impacts of Stratospheric ozone depletion

Stratospheric ozone depletion (kg CFC11 eq)												
System	CCS				DCS							
	E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Structural level	5.77E-01	7.79E-02	9.48E-02	7.50E-01	6.29E-01	8.13E-02	3.91E-02	3.23E-02	9.67E-02	8.07E-01	8.46E-01	8.78E-01

Table J-3 Specific environmental impacts of Ionizing radiation

Ionizing radiation (kBq Co-60 eq)												
System	CCS				DCS							
	E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Structural level	7.32E+04	2.50E+03	2.44E+03	7.82E+04	7.53E+04	2.51E+03	1.30E+03	1.12E+03	2.49E+03	8.03E+04	8.16E+04	8.27E+04

Table J-4 Specific environmental impacts of Ozone formation, Human health

Ozone formation, Human health (kg NO_x eq)												
System	CCS				DCS							
	E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Structural level	5.73E+03	6.05E+02	2.01E+02	6.54E+03	6.02E+03	5.80E+02	3.72E+02	3.57E+02	2.05E+02	6.81E+03	7.18E+03	7.54E+03

Table J-5 Specific environmental impacts of Fine particulate matter formation

Fine particulate matter formation (kg PM2.5 eq)												
System	CCS				DCS							
	E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Structural level	4.74E+03	3.11E+02	1.25E+02	5.18E+03	5.00E+03	2.90E+02	1.81E+02	1.72E+02	1.28E+02	5.42E+03	5.60E+03	5.77E+03

Table J-6 Specific environmental impacts of Ozone formation, Terrestrial ecosystem

Ozone formation, Terrestrial ecosystem (kg NO_x eq)												
System	CCS				DCS							
	E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Structural level	6.00E+03	6.14E+02	2.12E+02	6.82E+03	6.30E+03	5.89E+02	3.77E+02	3.61E+02	2.17E+02	7.10E+03	7.48E+03	7.84E+03

Table J-7 Specific environmental impacts of Terrestrial acidification

Terrestrial acidification (kg SO_2 eq)												
System	CCS				DCS							
	E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Structural level	7.13E+03	7.97E+02	2.70E+02	8.20E+03	7.80E+03	7.32E+02	4.67E+02	4.47E+02	2.76E+02	8.80E+03	9.27E+03	9.72E+03

Table J-8 Specific environmental impacts of Freshwater eutrophication

Freshwater eutrophication (kg P eq)												
System	CCS				DCS							
	E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Structural level	1.76E+03	6.04E+01	1.27E+01	1.84E+03	1.79E+03	5.31E+01	3.52E+01	3.43E+01	1.29E+01	1.85E+03	1.89E+03	1.92E+03

Table J-9 Specific environmental impacts of Marine eutrophication

Marine eutrophication (kg N eq)												
System	CCS				DCS							
	E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Structural level	6.46E+01	3.86E+00	9.61E-01	6.94E+01	6.59E+01	3.42E+00	2.24E+00	2.17E+00	9.81E-01	7.03E+01	7.25E+01	7.47E+01

Table J-10 Specific environmental impacts of Terrestrial ecotoxicity

Terrestrial ecotoxicity (kg 1,4-DCB)												
System	CCS				DCS							
	E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Structural level	1.33E+07	6.76E+05	1.50E+06	1.55E+07	1.34E+07	7.88E+05	2.65E+05	1.57E+05	1.53E+06	1.57E+07	1.60E+07	1.62E+07

Table J-11 Specific environmental impacts of Freshwater ecotoxicity

Freshwater ecotoxicity (kg 1,4-DCB)												
System	CCS				DCS							
	E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Structural level	2.40E+05	3.87E+03	2.56E+03	2.46E+05	2.42E+05	3.65E+03	2.09E+03	1.90E+03	2.62E+03	2.48E+05	2.51E+05	2.52E+05

Table J-12 Specific environmental impacts of Marine ecotoxicity

Marine ecotoxicity (kg 1,4-DCB)												
System	CCS				DCS							
	E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Structural level	3.40E+05	5.60E+03	4.25E+03	3.50E+05	3.43E+05	5.37E+03	2.98E+03	2.67E+03	4.34E+03	3.53E+05	3.56E+05	3.59E+05

Table J-13 Specific environmental impacts of Human carcinogenic toxicity

Human carcinogenic toxicity (kg 1,4-DCB)												
System	CCS				DCS							
	E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Structural level	1.39E+06	5.36E+03	3.37E+03	1.40E+06	1.40E+06	5.08E+03	2.94E+03	2.70E+03	3.44E+03	1.41E+06	1.41E+06	1.42E+06

Table J-14 Specific environmental impacts of Human non-carcinogenic toxicity

Human non-carcinogenic toxicity (kg 1,4-DCB)												
System	CCS				DCS							
	E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Structural level	7.24E+06	1.13E+05	9.28E+04	7.45E+06	7.30E+06	1.10E+05	5.96E+04	5.29E+04	9.47E+04	7.51E+06	7.57E+06	7.62E+06

Table J-15 Specific environmental impacts of Land use

Land use (m^2 a crop eq)												
System	CCS				DCS							
	E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Structural level	4.39E+04	2.81E+03	4.80E+03	5.15E+04	4.51E+04	3.07E+03	1.24E+03	8.88E+02	4.90E+03	5.30E+04	5.43E+04	5.51E+04

Table J-16 Specific environmental impacts of Mineral resource scarcity

Mineral resource scarcity (kg Cu eq)												
System	CCS				DCS							
	E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Structural level	9.79E+04	1.91E+02	3.21E+02	9.84E+04	9.86E+04	2.12E+02	8.77E+01	6.44E+01	3.28E+02	9.91E+04	9.92E+04	9.93E+04

Table J-17 Specific environmental impacts of Fossil resource scarcity

Fossil resource scarcity (kg oil eq)												
System	CCS				DCS							
	E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Structural level	4.57E+05	5.08E+04	4.76E+04	5.56E+05	4.79E+05	5.08E+04	2.67E+04	2.33E+04	4.86E+04	5.78E+05	6.05E+05	6.28E+05

Table J-18 Specific environmental impacts of Water consumption

Water consumption (m^3)												
System	CCS				DCS							
	E_0^{CCS}	E_1^{CCS}	E_W^{CCS}	E^{CCS}	E_0^{DCS}	E_1^{DCS}	E_2^{DCS}	E_3^{DCS}	E_W^{DCS}	$E_{k=1}^{DCS}$	$E_{k=2}^{DCS}$	$E_{k=3}^{DCS}$
Structural level	1.61E+04	3.84E+02	3.82E+02	1.69E+04	1.68E+04	3.86E+02	1.99E+02	1.71E+02	3.90E+02	1.76E+04	1.78E+04	1.80E+04

Appendix K: p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) st the structural level

Table K-1 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Global warming

Global warming (kg CO_2 eq)												
System		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Structural level	$E^{DCS}_{expected}$	2.46E+06	2.23E+06	2.00E+06	1.80E+06	1.61E+06	1.45E+06	1.30E+06	1.17E+06	1.06E+06	9.65E+05	8.80E+05
	$E^{DCS}_{expected}$	2.35E+06										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.047	0.947	0.852	0.764	0.685	0.615	0.553	0.499	0.452	0.410	0.374

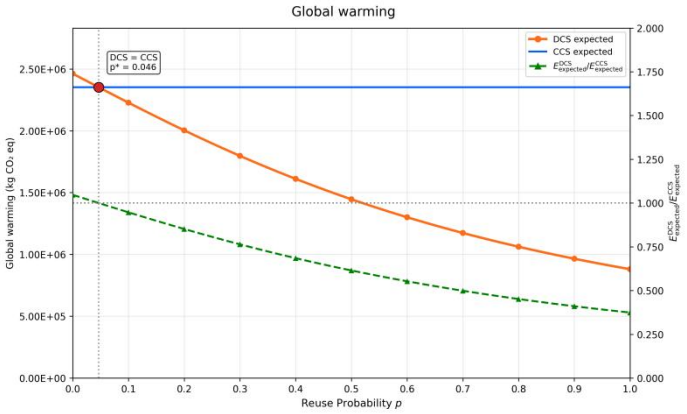


Figure K-1 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Global warming

Table K-2 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Stratospheric ozone depletion

Stratospheric ozone depletion (kg CFC11 eq)												
System		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Structural level	$E^{DCS}_{expected}$	8.07E-01	7.31E-01	6.58E-01	5.91E-01	5.31E-01	4.77E-01	4.30E-01	3.88E-01	3.52E-01	3.20E-01	2.93E-01
	$E^{DCS}_{expected}$	7.50E-01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.076	0.974	0.877	0.788	0.707	0.636	0.573	0.518	0.469	0.427	0.390

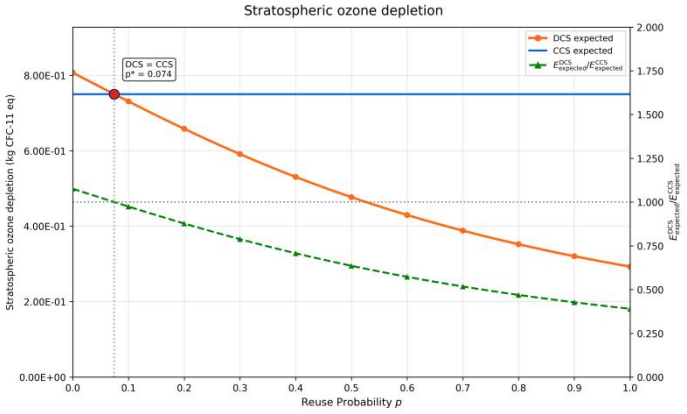


Figure K-2 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Stratospheric ozone depletion

Table K-3 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ionizing radiation

Ionizing radiation (kBq Co-60 eq)												
System		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Structural level	$E^{DCS}_{expected}$	8.03E+04	7.25E+04	6.50E+04	5.81E+04	5.19E+04	4.64E+04	4.16E+04	3.73E+04	3.36E+04	3.04E+04	2.76E+04
	$E^{DCS}_{expected}$	7.82E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.027	0.927	0.832	0.744	0.664	0.594	0.532	0.478	0.430	0.389	0.353

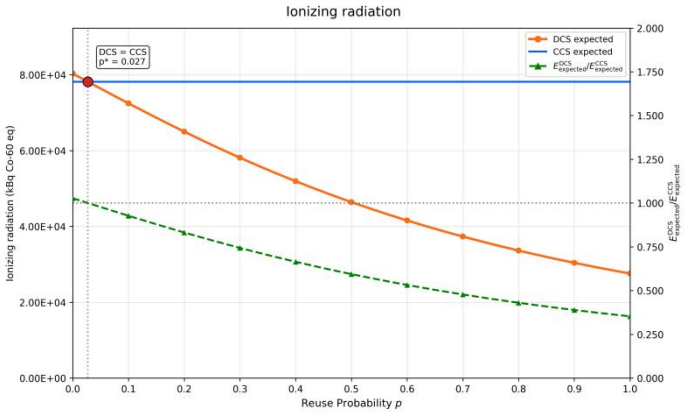


Figure K-3 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ionizing radiation

Table K-4 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation, Human health

Ozone formation, Human health (kg NO_x eq)												
System		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Structural level	$E^{DCS}_{expected}$	6.81E+03	6.17E+03	5.56E+03	5.00E+03	4.50E+03	4.05E+03	3.65E+03	3.31E+03	3.01E+03	2.74E+03	2.51E+03
	$E^{DCS}_{expected}$	6.54E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.042	0.944	0.851	0.765	0.688	0.619	0.559	0.506	0.460	0.420	0.384

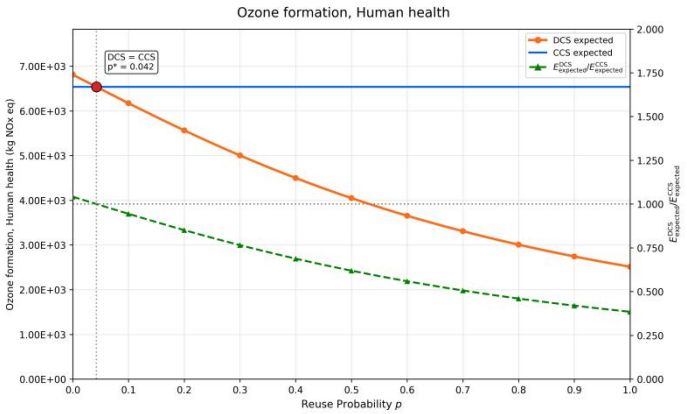


Table K-4 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation, Human health

Table K-5 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fine particulate matter formation

Fine particulate matter formation (kg PM2.5 eq)												
System		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Structural level	$E^{DCS}_{expected}$	5.42E+03	4.90E+03	4.40E+03	3.95E+03	3.54E+03	3.17E+03	2.85E+03	2.57E+03	2.32E+03	2.11E+03	1.92E+03
	$E^{DCS}_{expected}$	5.18E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.046	0.946	0.850	0.762	0.683	0.612	0.550	0.496	0.449	0.407	0.371

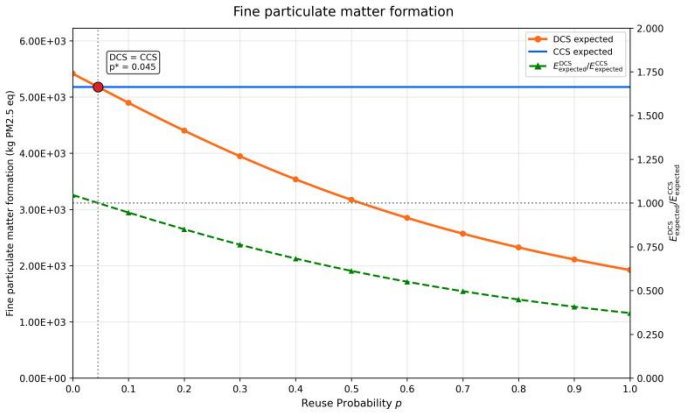


Figure K-5 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fine particulate matter formation

Table K-6 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation, Terrestrial ecosystem

Ozone formation, Terrestrial ecosystem (kg NO_x eq)												
System		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Structural level	$E^{DCS}_{expected}$	7.10E+03	6.43E+03	5.80E+03	5.21E+03	4.69E+03	4.22E+03	3.80E+03	3.44E+03	3.13E+03	2.85E+03	2.61E+03
	$E^{DCS}_{expected}$	6.82E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.040	0.943	0.850	0.764	0.687	0.618	0.557	0.505	0.458	0.418	0.383

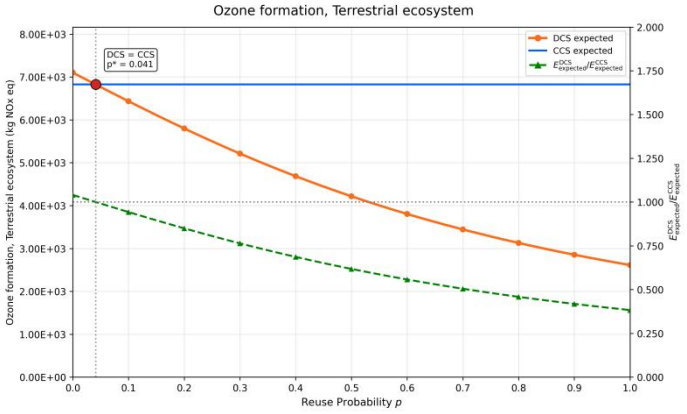


Figure K-6 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Ozone formation, Terrestrial ecosystem

Table K-7 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial acidification

Terrestrial acidification (kg SO_2 eq)												
System		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Structural level	$E^{DCS}_{expected}$	8.80E+03	7.98E+03	7.19E+03	6.46E+03	5.81E+03	5.23E+03	4.72E+03	4.27E+03	3.88E+03	3.54E+03	3.24E+03
	$E^{DCS}_{expected}$	8.20E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.074	0.973	0.877	0.788	0.709	0.638	0.575	0.521	0.473	0.431	0.395

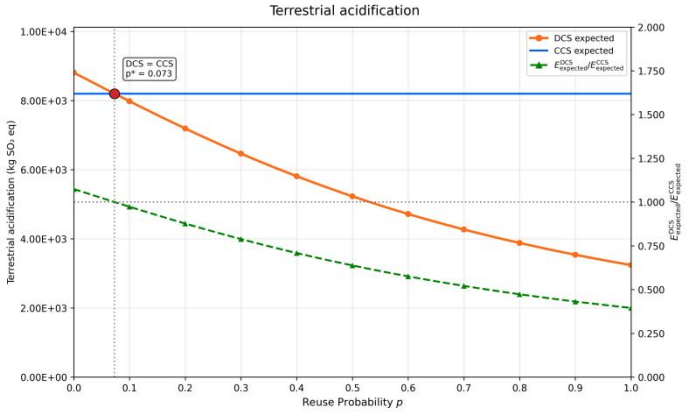


Figure K-7 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial acidification

Table K-8 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater eutrophication

Freshwater eutrophication (kg P eq)												
System		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Structural level	$E^{DCS}_{expected}$	1.85E+03	1.67E+03	1.50E+03	1.34E+03	1.20E+03	1.07E+03	9.62E+02	8.65E+02	7.80E+02	7.05E+02	6.41E+02
	$E^{DCS}_{expected}$	1.84E+03										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.008	0.910	0.817	0.731	0.653	0.584	0.524	0.471	0.424	0.384	0.349

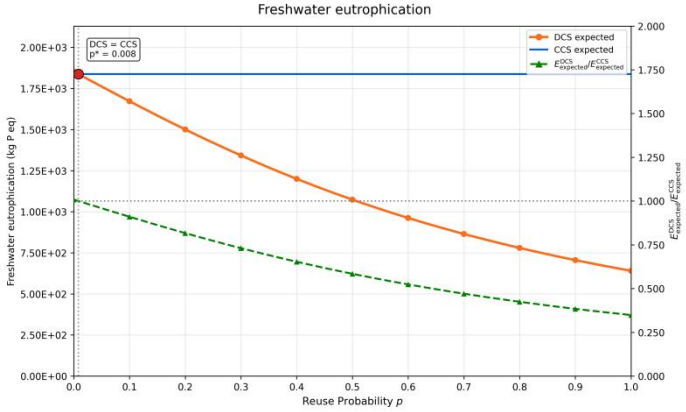


Figure K-8 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater eutrophication

Table K-9 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine eutrophication

Marine eutrophication (kg N eq)												
System		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Structural level	$E^{DCS}_{expected}$	7.03E+01	6.36E+01	5.71E+01	5.12E+01	4.59E+01	4.11E+01	3.70E+01	3.33E+01	3.01E+01	2.73E+01	2.49E+01
	$E^{DCS}_{expected}$	6.94E+01										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.013	0.915	0.823	0.737	0.661	0.592	0.532	0.480	0.434	0.394	0.359

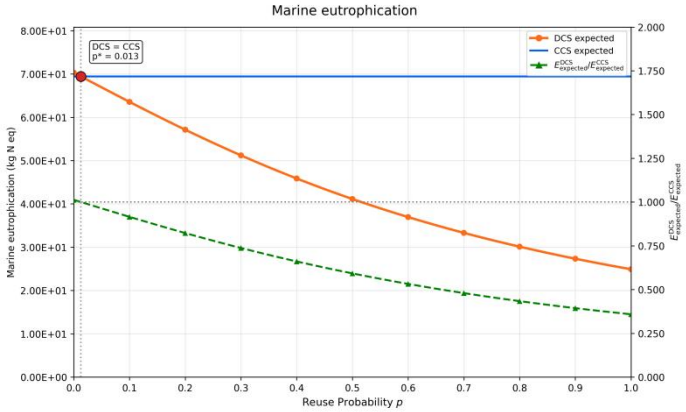


Figure K-9 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine eutrophication

Table K-10 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial ecotoxicity

Terrestrial ecotoxicity (kg 1,4-DCB)												
System		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Structural level	$E^{DCS}_{expected}$	1.57E+07	1.42E+07	1.27E+07	1.14E+07	1.02E+07	9.09E+06	8.14E+06	7.31E+06	6.58E+06	5.94E+06	5.39E+06
	$E^{DCS}_{expected}$	1.55E+07										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.018	0.919	0.824	0.737	0.658	0.588	0.527	0.473	0.426	0.384	0.348

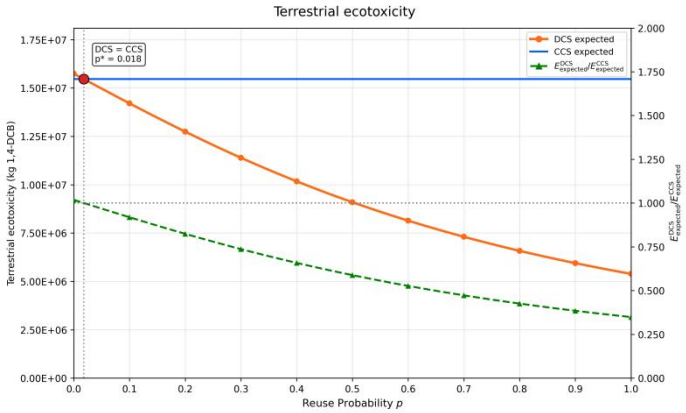


Figure K-10 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Terrestrial ecotoxicity

Table K-11 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater ecotoxicity

Freshwater ecotoxicity (kg 1,4-DCB)												
System		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Structural level	$E^{DCS}_{expected}$	2.48E+05	2.24E+05	2.01E+05	1.79E+05	1.60E+05	1.43E+05	1.28E+05	1.15E+05	1.03E+05	9.29E+04	8.41E+04
	$E^{DCS}_{expected}$	2.46E+05										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.008	0.909	0.815	0.728	0.649	0.580	0.518	0.465	0.418	0.377	0.342

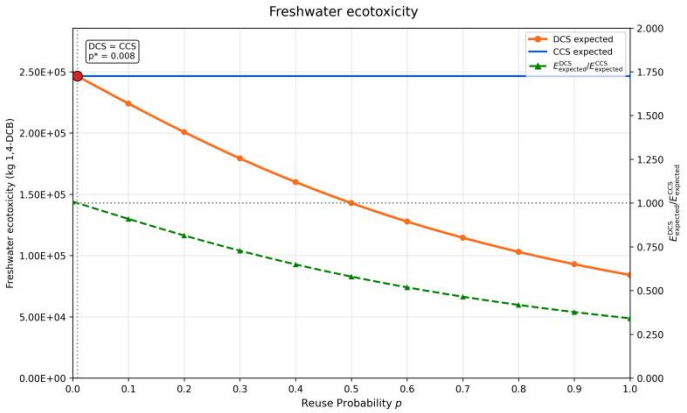


Figure K-11 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Freshwater ecotoxicity

Table K-12 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine ecotoxicity

Marine ecotoxicity (kg 1,4-DCB)												
System		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Structural level	$E^{DCS}_{expected}$	3.53E+05	3.18E+05	2.85E+05	2.55E+05	2.27E+05	2.03E+05	1.81E+05	1.63E+05	1.46E+05	1.32E+05	1.20E+05
	$E^{DCS}_{expected}$	3.50E+05										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.008	0.909	0.815	0.728	0.649	0.580	0.518	0.465	0.418	0.377	0.342

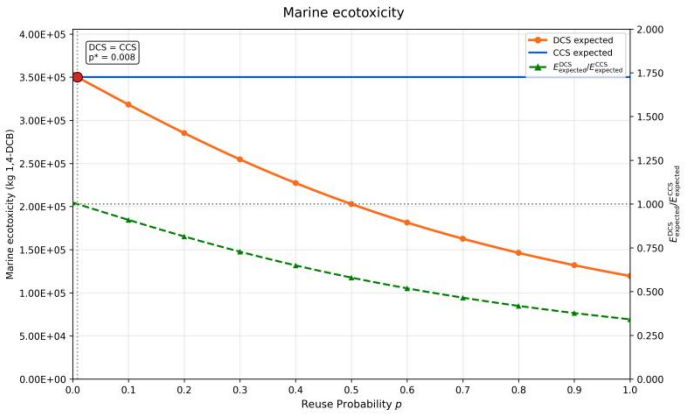


Figure K-12 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Marine ecotoxicity

Table K-13 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Human carcinogenic toxicity

Human carcinogenic toxicity (kg 1,4-DCB)												
System		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Structural level	$E^{DCS}_{expected}$	1.41E+06	1.27E+06	1.14E+06	1.02E+06	9.05E+05	8.07E+05	7.21E+05	6.46E+05	5.80E+05	5.22E+05	4.72E+05
	$E^{DCS}_{expected}$	1.40E+06										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.007	0.907	0.812	0.725	0.646	0.576	0.515	0.461	0.414	0.373	0.337

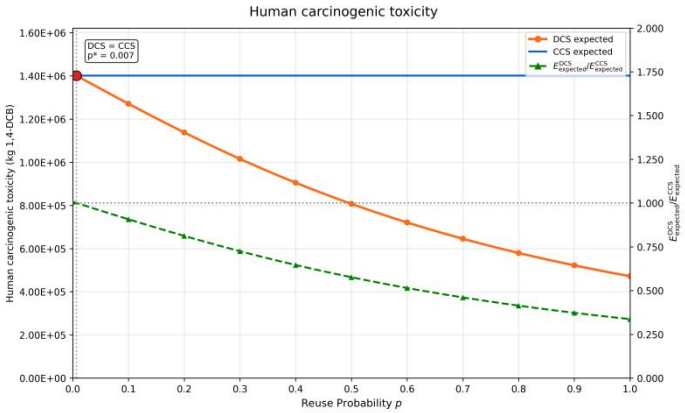


Figure K-13 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Human carcinogenic toxicity

Table K-14 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Human non-carcinogenic toxicity

Human non-carcinogenic toxicity (kg 1,4-DCB)												
System		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Structural level	$E^{DCS}_{expected}$	7.51E+06	6.77E+06	6.06E+06	5.42E+06	4.83E+06	4.31E+06	3.86E+06	3.46E+06	3.11E+06	2.81E+06	2.54E+06
	$E^{DCS}_{expected}$	7.45E+06										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.008	0.909	0.814	0.727	0.649	0.579	0.518	0.464	0.418	0.377	0.341

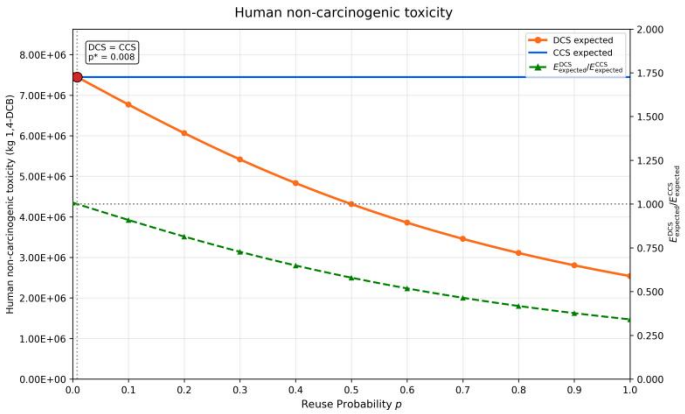


Figure K-14 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Human non-carcinogenic toxicity

Table K-15 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Land use

Land use (m² a crop eq)												
System		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Structural level	$E^{DCS}_{expected}$	5.30E+04	4.79E+04	4.30E+04	3.85E+04	3.44E+04	3.08E+04	2.76E+04	2.48E+04	2.24E+04	2.02E+04	1.84E+04
	$E^{DCS}_{expected}$	5.15E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.029	0.930	0.835	0.747	0.668	0.597	0.536	0.482	0.434	0.393	0.357

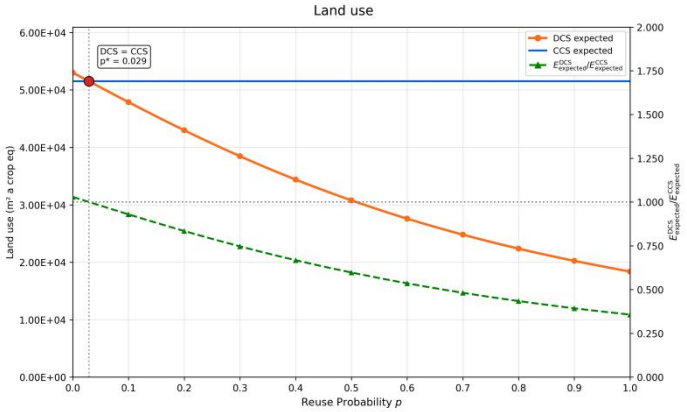


Figure K-15 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Land use

Table K-16 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Mineral resource scarcity

Mineral resource scarcity (kg Cu eq)												
System		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Structural level	$E^{DCS}_{expected}$	9.91E+04	8.93E+04	8.00E+04	7.13E+04	6.36E+04	5.67E+04	5.06E+04	4.53E+04	4.07E+04	3.66E+04	3.31E+04
	$E^{DCS}_{expected}$	9.84E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.007	0.907	0.812	0.725	0.646	0.576	0.514	0.460	0.413	0.372	0.336

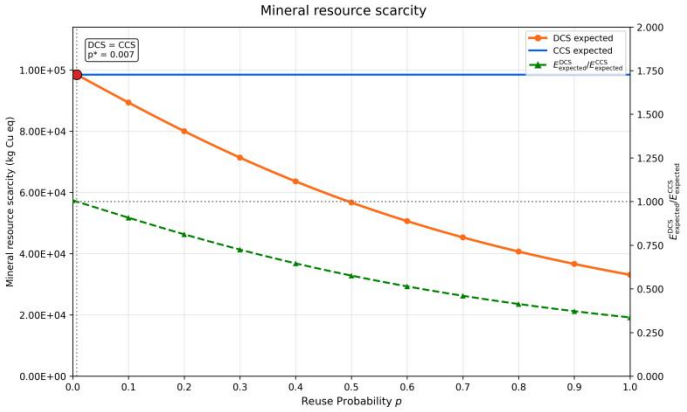


Figure K-16 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Mineral resource scarcity

Table K-17 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fossil resource scarcity

Fossil resource scarcity (kg oil eq)												
System		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Structural level	$E^{DCS}_{expected}$	5.78E+05	5.23E+05	4.71E+05	4.23E+05	3.80E+05	3.41E+05	3.07E+05	2.78E+05	2.52E+05	2.29E+05	2.09E+05
	$E^{DCS}_{expected}$	5.56E+05										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.040	0.942	0.848	0.762	0.684	0.614	0.553	0.500	0.453	0.412	0.377

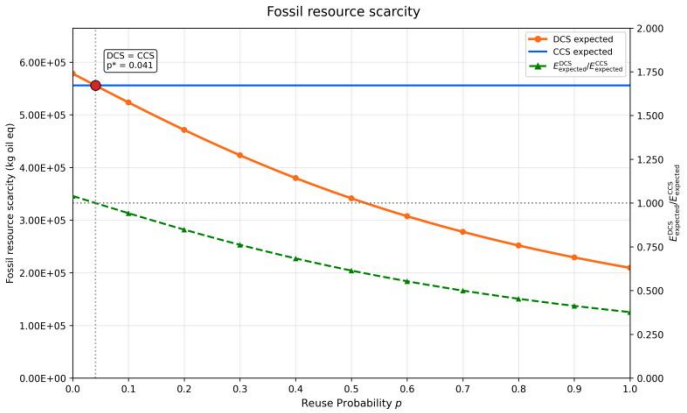


Figure K-17 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Fossil resource scarcity

Table K-18 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Water consumption

Water consumption (m³)												
System		p=0.0	p=0.1	p=0.2	p=0.3	p=0.4	p=0.5	p=0.6	p=0.7	p=0.8	p=0.9	p=1.0
Structural level	$E^{DCS}_{expected}$	1.76E+04	1.59E+04	1.42E+04	1.27E+04	1.14E+04	1.02E+04	9.08E+03	8.15E+03	7.33E+03	6.62E+03	6.00E+03
	$E^{DCS}_{expected}$	1.69E+04										
	$\frac{E^{DCS}_{expected}}{E^{CCS}_{expected}}$	1.042	0.940	0.843	0.753	0.672	0.600	0.537	0.482	0.434	0.391	0.355

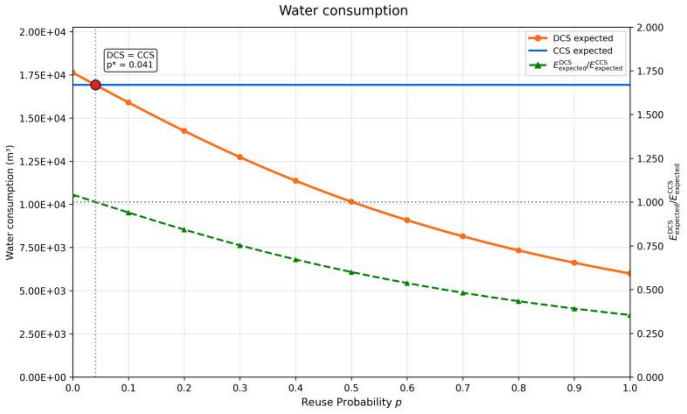


Figure K-18 p^* derived from calculations of CCS ($E^{CCS}_{expected}$) and DCS ($E^{DCS}_{expected}$) of Water consumption